



Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Undergraduate Prospectus 2022



THE IMPACT
RANKINGS 2022
SDG4 QUALITY EDUCATION
1st in Pakistan, 37th GLOBALLY

UI Green
Metric
World University Ranking

VISION

The Institute aspires for the leadership role in pursuit of excellence in engineering, sciences and technology.

MISSION

The Institute is to provide excellent teaching and research environment to produce graduates who distinguish themselves by their professional competence, research, entrepreneurship, humanistic outlook, ethical rectitude, pragmatic approach to problem solving, managerial skills and ability to respond to the challenge of socio economic development to serve as the vanguard of techno-industrial transformation of the society



The Founder



The GIK Institute is as dear to me as a child to his parents. It gives me pleasure to see that the sapling we planted in 1993 is now a flowering tree providing its cool shade to seekers of knowledge.

Ghulam Ishaq Khan



The Gulam Ishaq Khan (GIK) Institute of Engineering Sciences and Technology has earned a reputation as an institution of the excellence in the country for imparting quality education in engineering sciences and technology. Not many institutions have earned so much recognition and respect, both within the country and abroad, in such a short time as has the GIK Institute. This distinction would not have been possible without the commitment and dedication of its teachers, staff and students. Thus, all of their efforts deserve to be complimented.

Engineering sciences and Technology hold the key to the development and security of the country. Advanced technology, however, is not easily available in the international market nor can it be borrowed. We, therefore, must make a concerted effort to develop our own indigenous technologies that suit our peculiar environment and also caters to our particular needs in the industrial, economic and other sectors.

It is gratifying to note that the private sector is increasingly participating in the effort to broaden the base of scientific and engineering education in the country. I would like to commend the GIK Institute for playing a leading role in this endeavour by introducing new and emerging technologies and producing highly trained manpower capable of meeting the challenges of the modern day, complemented by a robust research program.

Institutes like GIK must play the role of a catalyst in bringing about a techno-industrial transformation. We need men and women of vision, knowledge, courage and integrity. We expect from our institutions to produce such professionals. I urge the GIK Institute to continue its quest for excellence and hope that its alumni serve the nation and contribute towards making Pakistan a highly developed and prosperous country.

May Allah Almighty be our protector and guide! Ameen

Dr. Arif Alvi
President of the Islamic Republic of Pakistan

It is a matter of great satisfaction for me to see the Ghulam Ishaq Khan (GIK) Institute of Engineering Sciences and Technology continue to take major strides in its pursuit of excellence in education and research. GIK has not only maintained its leading position in many engineering fields, but has now, further expanded its activities to provide quality education in new and innovative areas at the cutting edge of technological research, innovation and development, worldwide.



In today's digital world where information and knowledge are fast becoming the basis of human economic and social interaction and progress, GIK is determined to be in the forefront of these developments. Plans for setting up a Center for work on Artificial Intelligence (AI) and its applications to National Development are under active consideration. In this regard, GIK has forged important partnership for faculty and student exchange with a number of well-known foreign Universities and a well-known International Corporation. Our Business and Management programme also offers opportunities to students to couple their technical knowledge and skills with the practical know how of entrepreneurship and enterprise development.

GIK has always striven to select the best and the most motivated students. Selection on Merit has always been our only criteria for enrollment. Once enrolled, students can rest assured that they will be provided with all available benefits and support to facilitate them in the successful pursuit of their studies. We are very encouraged to see a quantum leap in the number of female students who are excelling in every field and hope to see larger female enrollment in the future.

We are very proud of our capable management, under the able leadership of the Rector and supported by the distinguished members of the faculty. They are, indeed, a body of competent and dedicated supervisors, teachers and support staff and an asset to the Institution. All Faculty members recognize their key role as teachers and mentors to our young students, who can rely upon them for their sympathetic guidance and support.

Finally, I would like to add that the expanding GIK facilities with spacious ground, elegant class rooms and hostels and state of the art equipment and laboratories, offer a fully supportive environment for study, research, creative activities and sports, and its tranquil and peaceful location a grand opportunity for serious scholarship

I wish you all great success.

Farid Rahman
President SOPREST

It gives me a great pleasure to welcome you at the Institute which has become an icon of quality education and is recognized at national and international levels with presence of its alumni in more than 70 countries and are playing leading role in multinational companies and organizations. Its mission of achieving excellence in education, engineering and technology and innovation through the efforts of faculty, students, and staff, to uphold this goal.

I am confident that your 4-year stay at the campus will be a unique learning experience with great achievements and learning outcomes to enable you to take the leading role in professional life which is attained through the team work of our faculty and officers and commitment to helping students and hone their critical thinking, research, multidisciplinary perspectives, practical skills and professional competence. We are committed to helping our students to become future leaders of industry and play key role in research, innovation and entrepreneurship through startup incubator.

Our curriculum is embedded with Outcome Based Education with unique experience of innovation and makers lab facilitating our students to excel and develop their competences in products, software, devices and project management with National and International recognition including Washington Accord.

The campus is located in the backdrop of picturesque mountains, where faculty and students live and learn on our self-contained, environment-friendly, clean, and green campus, which houses all the classrooms, laboratories, and recreational facilities. GIK offers a rigorous curriculum designed to improve classroom learning while imparting valuable graduate attributes. The vibrant societal culture and sports activities also allow the students to refine their collaborative, interpersonal, and teamwork skills.

GIK Institute has an extensive financial aid and outreach program, including scholarships and financial assistance. In addition, the Institute has arrangements with several of its partners, provincial Governments, GIKI Alumni Association, and many other organizations, for assistance and scholarships.

It is pertinent to laud the efforts of The GIK Alumni Association which is a source of pride for the Institute as well as a fount of inspiration for the students. The Association spearheads fundraisers, contributes scholarship funds, organizes academic and professional seminars and webinars, sponsors internships, and mentors current students.

I would also like to express our gratitude to all our partners, friends and well-wishers, particularly the Founding Fathers; past and present Members of the Board, Rectors, Pro Rectors, Deans, Directors, faculty members and all other employees; and alumni, for their immense contributions to the success of the GIK Institute.

Thank you for considering GIK Institute, and we look forward to welcoming you to our community.

**Prof. Dr. Fazal Ahmad Khalid, SI
Rector**



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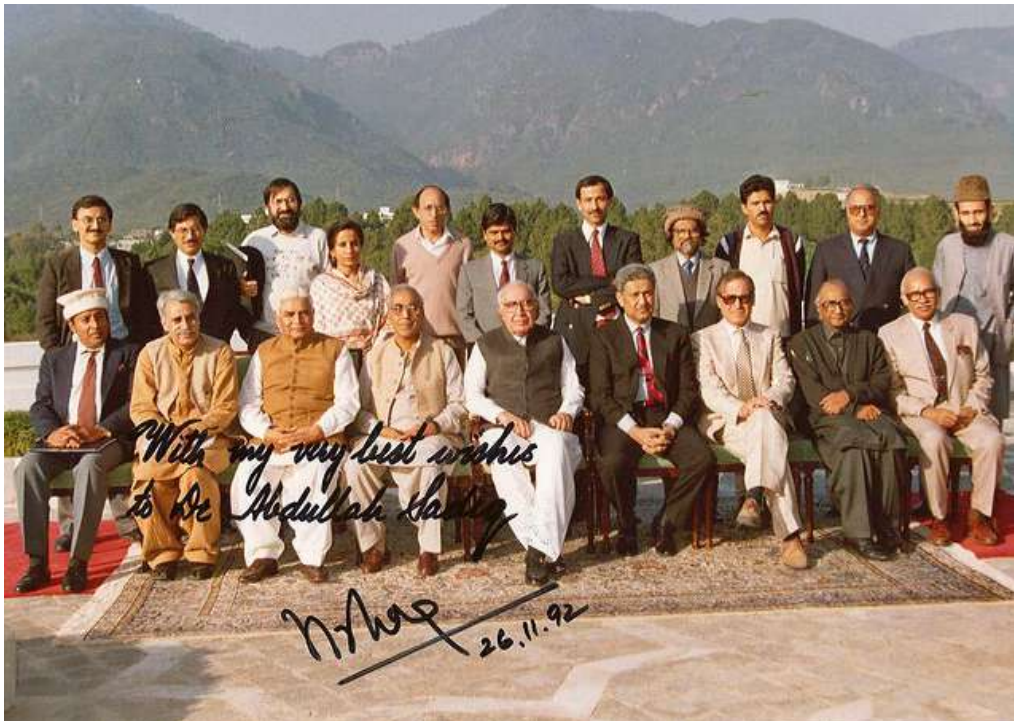
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CAMPUS LIFE



The Genesis of the Institute goes back to the early 50's when late Mr. Ghulam Ishaq Khan, during his close association with the Water and Power Development Authority and the Pakistan Industrial Development Corporation, became acutely aware of Pakistan's dependence on foreign expertise and imported technology. His frequent interaction with foreign and local experts led to the idea of establishing a center of excellence in engineering sciences and production technology whose standards of education would be comparable to those of its counterparts in the advanced countries. The transformation of this idea into a practical proposition took place in December 1985 when the Benevolent Community Care and Infaq Foundation donated Rs. 50 million for setting up an institute, and the Khyber Pakhtunkhwa Government donated 218 acres of land for its campus.

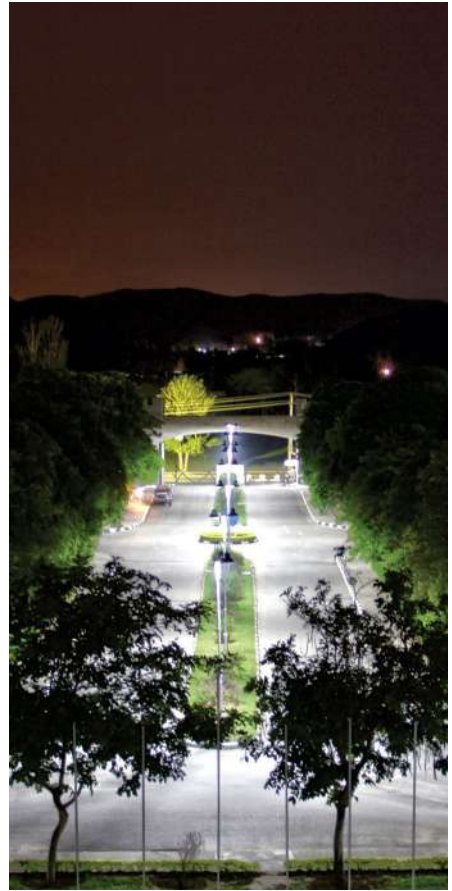
A milestone in the evolution of the Institute was the registration, in June 1988, of its parent body, namely the Society for the Promotion of Engineering Sciences and Technology in Pakistan (SOPREST). Mr. Ghulam Ishaq Khan, the then President of the Islamic Republic of Pakistan, was elected President of the Society for life and Mr. H. U. Beg appointed its honorary Executive Director. The task of conceiving and formulating the basic form and features of the Institute was entrusted to a group of eminent scientists and engineers. Civil works at the campus site were started in early 1990. An interim office of the Institute was set up in August 1992 where experienced professionals worked on the educational aims and philosophy of the Institute, its curricula, and requisite equipment for its laboratories and workshops. The ordinance for the establishment of the Institute was promulgated by the Frontier Government in March 1993 and the first batch of students entered its portals in October 1993. It is the first not-for-profit, non-governmental institute of its kind in the country and is dedicated to bring our engineering education at par with that of advanced countries.



A rare picture with an autograph by Late Ghulam Ishaq Khan founder of GIK Institute

The aim of the Institute is to pursue excellence in education and research by developing appropriate curricula and teaching practices, acquiring talented faculty /students and providing an environment conducive to teaching and learning. Its graduates are expected to possess high professional competence combined with the humanistic and moral values envisaged in its Profile of the Graduates. The educational philosophy of the Institute lays emphasis on training of the mind rather than stuffing it with an inert body of facts; on expanding the scientific imagination of the students rather than making them tread well-worn and outmoded grooves of thought. Guided by such convictions, the Institute imparts Education to its students by confronting them with real-life problems and inculcating in them a problem-solving approach. They are encouraged to explore and solve problems, to break new grounds and to cultivate leadership qualities.

Pakistan is on the threshold of a major breakthrough in the techno-industrial fields and needs professionals with ability and vision to lead the way. The Institute aims at producing such professionals with a strong base of engineering education and research. It strives to produce graduates who can upgrade existing technological activities in the country and in whom professional excellence is inseparable from a commitment to national ideals.



Meeting with Hon'ble President of Islamic Republic of Pakistan, Chancellor GIK Institute

The Board of Governors sits at the apex of the statutory pyramid of the Institute and its composition is the same as that of the General Council of the Society for the Promotion of Engineering Sciences and Technology. It has overall control of the Institute, the powers to create new components of the Institute such as a school, faculty or any other teaching or research unit, and to change the constitution of its Executive Committee and Governing Council.

PRESIDENT SOPREST

Mr. Farid Rahman

Members

Engr. Salim Saifullah Khan

Mr. M. Adil Khattak

Khwaja Zaheer Ahmad

Dr. Zaigam Habib

Mr. Osman Saifullah Khan

Dr. Nilofar Shaikh

Other Members

Mr. Abdul Razzaq Dawood

Mr. Shah Faisal Afridi

Mr. Tariq Iabal Khan

Ms. Samia Waheed

Dr. Zubair Khan

Ex-Officio Members

Chairman, WAPDA

Executive Director SOPREST

Secretary, Finance Division, Govt. of Pakistan

Chief Secretary, Khyber Pakhtunkhwa

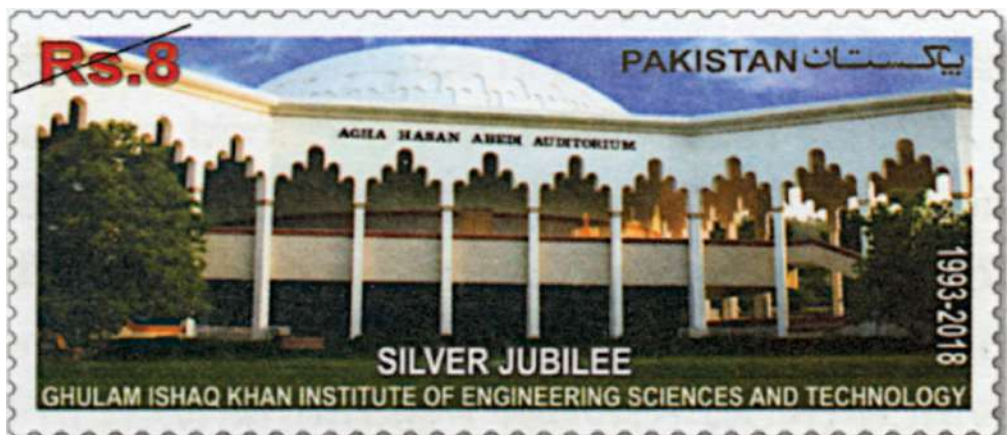
Chairman, Higher Education Commission

Secretary ST&IT, Khyber Pakhtunkhwa

Secretary Law, Khyber Pakhtunkhwa

Secretary SOPREST and BOG

Mr. Irfan Ahmad

COMMEMORATIVE POSTAGE STAMP

**SILVER JUBILEE
GHULAM ISHAQ KHAN INSTITUTE OF ENGINEERING
SCIENCES AND TECHNOLOGY
(1993-2018)**

Spread over a vast area of about 421 acres, the Ghulam Ishaq Khan Institute is located in the midst of the unspoiled and nature-rich countryside of the Khyber Pakhtunkhwa Province of Pakistan. Lying at the foot of the beautiful lake of Tarbela Dam, one of the largest earthfilled dams of the world, it is set against the picturesque backdrop of rolling hills, vast grassy fields with the mighty Indus meandering across a lush green belt.

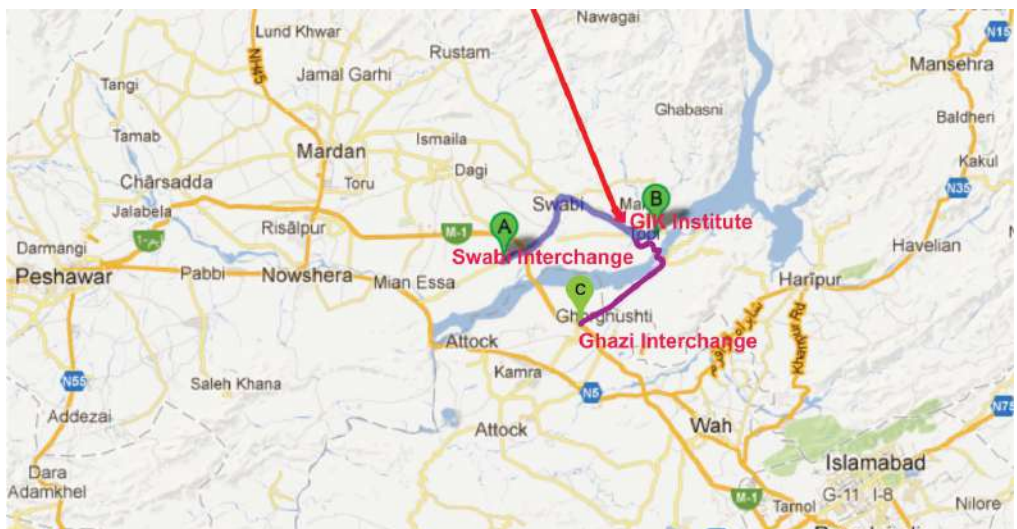
Bordering on its campus is the traditional village of Topi, the birthplace of Sahibzada Abdul Qayyum Khan, who was the pioneer of modern education in the province. Close by is the ancient village of Hund where Alexander the Great crossed the Indus. The surrounding area, once known as the land of Gandhara, is dotted profusely with archaeological sites of great cultural significance. These include the well-known sites of the ancient seats of learning, the Taxila University of the Gandhara period, and the Buddhist Monastery at Takht Bhai. To these seats of learning flocked students and scholars from all over South Asia, Central Asia and China. It is in this region that we find the sayings of Ashoka carved on rocks at Shabaz Garhi; the Naighe Gatte megaliths (stone columns) on the Swabi-Mardan road; and numerous stupas and chambers which fire the imagination of the visitors to the area with the mysteries and glories of its past. The excavated sites around Taxila, at Takht Bhai, Dir and in Swat Valley transport them back to the civilization that flourished here almost 2500 years back. Exquisite relics of that era are the treasured possessions of the museums

at Lahore, Peshawar, Karachi, Dir, Swat, and Taxila.

In addition to its great historical character, the location of the Institute offers many advantages. The Gilgit Baltistan Valley which attracts expeditions from all over the world is easily accessible from here. Perhaps the most important advantage is the invaluable opportunities it offers for establishing interaction between industry and the university and close proximity to high-quality industrial minerals. Some of the most important national industries are located quite close to the Institute. These include the Telephone Industries of Pakistan, Heavy Mechanical Complex, Heavy Electrical Complex, Heavy Industries, Pakistan, Ordnance Factories, Wah Cement Industries, Kamra Aeronautical Complex, Heavy Rebuild Factory, and Pakistan Locomotive Factory. Their proximity offers invaluable opportunities for practical training of the students.

Location of the Institute

Though away from the congestion, noise, and pollution of big cities, the Institute has easy access to Islamabad and Peshawar. Both cities are connected with the rest of the country through frequent air, train and bus services. Islamabad, the capital of Pakistan, has an international airport which provides ready access to the outside world. The Institute is located just by the river Indus, adjacent to Tarbela Dam, and near the border with Punjab. Driving time from Islamabad is just over an hour mostly along the Islamabad - Peshawar Motorway.



Each Faculty of the Institute is housed in a building of its own which has a graceful exterior and an elegant interior with all comforts and conveniences for its users, Each academic block has its teaching and research laboratories, workshops, a computer center, and offices for the faculty and staff, a well-furnished conference hall, a discussion room, three classrooms, a lecture hall, and a library for the faculty. The number of laboratories in the Institute has now risen to 74. The laboratories are equipped with the most advanced and up-to-date equipment where high-quality research is possible.

The Administration block of the Institute includes the Offices of the Rector, Pro-Rectors, and Dean (Student Affairs). Director (A&E), Director (Admin), Director (Finance), Director (Procurement), and Other allied Offices.

New Academic Block: A marvelous new addition to GIK Campus is the New Academic Block. Constructed with the gracious assistance of the Khyber Pakhtunkhwa Government. It is spread over 155,000 SFT area and has state-of-the-art classrooms, large Lecture Halls, Conference Rooms, and cafeteria and is fully air-conditioned with about one MW standby generator. It was handed over to GIK in 2019 at a ceremony presided by the Prime Minister of Pakistan, Chief Minister, Khyber Pakhtunkhwa, Speaker National Assembly, and other dignitaries. The Building hosts, the Faculty of Chemical Engineering, Department of Civil

Engineering, Faculty of Computer Sciences and Engineering, and Director of Information Technology and Student Wellness Center.

Guest House & Auditorium: The first building was constructed right at the inception of the Institute's stay at the Campus. It is situated close to the main entrance of the Institute and is centrally air-conditioned. Fully furnished, it has ten bedrooms and a big lounge for indoor functions. A spectacular auditorium occupies the central place in the campus and its dome meshes with the surrounding hills to present a breath-taking skyscape. It has a seating capacity of 535 and is a venue For conferences, seminars, and debates. declamation contests, concerts, and Other such functions. A lavishly furnished conference room, a seminar/workshop, and a service center are also parts of this block.



Business Incubator: Extending its services as a catalyst to the regional and national economic development the Institute has recently set up a business incubator. It provides managerial training, business guidance, office space, and logistic support to start-ups.



Students Accommodation: The accommodation facilities for students are entirely on-campus. There are eleven (II) hostels for boys and one separate hostel-wing

for girl students. The rooms in the hostels are equipped with modern furniture and attached bathroom. The Institute provides shared accommodation to all freshmen and sophomore students. Single rooms are usually allotted to junior and senior students on merit basis. A New Girls Hostel is under construction, which will shortly be shining along the hills standing out in silhouette offering accommodation to about 360 female students.

Civic Amenities: The campus is becoming a self-contained university town with adequate health, security, welfare, and Other civic amenities. There are three beautiful mosques on the campus, one of which is in the staff residential area, the Other near the hostels and the third one, next to the Guest House.

Parents Lodge: In view of the difficulties faced by the visiting parents/guardians for overnight stay, a parents Lodge has been set up near the students' hostels. It is a five-room fully furnished and air-conditioned accommodation. This accommodation is available to parents and guardians on first-come-first-serve basis on reasonable charges for short visits. Efforts are made to make their Stay as comfortable as possible





Health & Medical Facilities: The Institute has its own hospital in the campus that provides medical facilities to its employees, their families and students on 24/7 basis. The patients are provided free medical care which includes the ambulance service, medicines, lab investigations and other tests. The Medical Centre comprises of 13 bed hospital supported by an operation theater, pharmacy, x-rays and clinical laboratory with computerized equipment for a wide range of hematology, bio-chemistry and endocrinology tests.

Cafeteria: The Institute's cafeteria has been named as GIKafe. It offers regular meals and snacks at modest prices. It caters for both the faculty and the students. Students who normally have their regular meals in their hostel mess use this facility as an alternative. Official and private parties and numerous student functions are also held in the cafeteria.

Shopping Area: The following utility services are available at the Shopping Centre located within the premises of the Institute: General Stores, Restaurants, Barbeque Corner Coffee Corners, Stationery Shop, Fruit and Vegetable Shop, Dry Cleaning and Laundry Service, Barber Shop, Dairy Shop, Juice Bar.

Service Centre: The Centre provides photocopying, services to the staff and students on payment.

Sports Facilities: Sports Complex is located in close proximity to the students hostels. It spreads over an area of 7324 Sq. Yds. with a covered area of 31 Sq. Ft. It consists of a completely covered swimming pool of international standard with comprehensive facilities, three stand and Squash Courts and a Gymnasium. The Gymnasium consists of a main hall and an exercise room. The main hall houses the facilities to play Basketball, Volleyball and Badminton. In the exercise room, modern equipments for various physical exercises have been installed. Common facilities such as lockers, showers, storage, checkroom,





administrative offices and refreshment rooms have been provided. A separate ladies gym is also operative in the sports complex. Grounds are available for outdoor sports like tennis, basketball, volleyball, football and cricket etc. There are hiking and jogging tracks in the hills behind the main buildings of the Campus.

Transport Facilities: Transport section of the Institute has 20 vehicles in its pool comprising cars, vans, trucks and air-conditioned coaches. The Institute provides pick-and-drop services to students and staff from Rawalpindi, Islamabad and Peshawar on weekends, mid and end of semester breaks, industrial tours and picnics. Day and night emergencies are attended to by the ambulance service and

duty vehicles. In emergency, students and employees are Transported to Rawalpindi, Islamabad or Peshawar by Institute vehicles.

Faculty and Staff Residences: The Institute is fully residential. The entire faculty and staff of the Institute are accommodated in independent houses and flats on the campus.

Faculty Club: Faculty Club has been constructed on the top of a hill and presents a picturesque view of Tarbela Dam and its environs. Its building is air-conditioned and is fully furnished. The accommodation comprises four bedrooms, a spacious sitting hall and a dining room where over 100 persons can be entertained.





Library and Information Services: A stately three-story building, set against the background of brooding and austere hills of Tarbela, houses the Central Library of the Institute. Its interior design, decor, and furniture create an atmosphere of an intellectual sanctuary wherein the students and faculty can concentrate on their studies. It operates in two shifts and remains open till late in night seven days a week. It has textbooks, reference works, printed as well as online journals to meet the needs of students and faculties. To share resources through interlibrary loan and exchange of databases, it is electronically linked to all prominent libraries of the country. It also provides reprographic services.

The GIK Institute's digital library provides access to resources of HEC that include databases of journals and books to support the faculty and students community of the Institute. The digital resources include about 15000 electronic journals, 80,000 e-books, audio/video materials, IEEE databases on DVDs and other reference databases. The students and faculty members at the Institute can easily access all the resources on their desks through <http://www.digitallibrary.edu.pk/giki.html>, which provides online access to IEEE, ProQuest Dissertation & Theses, ASTM,

Ebrary, Springerlink, Taylor & Francis Journals, Wiley-Blackwell Journals.

Wireless network is available in the library and open for all users. Student society Meeting/ Discussion Room available in library on request.

Turnitin software for plagiarism detection service is also available to facilitate the students to improve the writing skills.



ACACEMIC



The founding fathers of the Institute were conscious of the fact that in spite of all the idealism one may have, new institutions tend to regress towards the existing models, and fail thereby to live up to the ideals which inspired their creation. They, therefore, took care to build monitoring devices to keep up Institute's standards of education and research. One such device is the International Advisory Board consisting of scientists, engineers, and academicians of international standing. The Board sets up international standards for the Institute in terms of the quality of education and research, caliber of the faculty, revision and review of the curricula, and adequacy of the laboratory and library facilities. It also reviews the developmental programs of the Institute and provides guidelines for its growth in future.

The present Advisory Board comprises the following:

Prof Dr. Shuichi Miyazaki

Institute of Materials Science, University of Tsukuba
Tsukuba, Japan.

Prof. Dr. Di Su

The University of Tokyo, Tokyo, Japan

Dr. Zahid Ayub

President Isotherm, Inc.,
Texas 76001, USA

Dr. Kamran Iqbal

Department of Systems Engineering
University of Arkansas at Little Rock , 2801 S.
USA

Dr. Muhammad Suhail Zubairy

Department of Physics, Texas A&M University
College Station, TX 77843-4242, USA

Prof. Dr. Samee Khan

Department of Electrical & Computer
Engineering
Mississippi State University, USA

Prof. Dr. Joseph D. Smith

Wayne and Gayle Laufer Endowed Energy Chair
Missouri University of Science and Technology,
USA

Dr. Bilal Malik

Clinical Imaging Senior Scientist, Personalized
Healthcare, Genentech, DNA Way,
South San Francisco, CA 94080, USA

Prof. Dr. Kaifeng Yang

Florida State University Tallahassee
Florida 32306-2250, USA.

Prof. Dr. Hanif Chaudhary

University of South Carolina, Columbia SC, USA

Dr. Costas Constantinou

School of Electronic Electrical and Computer
Engineering
University of Birmingham, Edgbaston.
Birmingham, UK

Dr. Talha J. Pirzada

Department of Materials
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Prof. Dr. Yan Bing

Tianjin University of Technology and Education
TUTE, China

Prof. Dr. Shahbaz Khan

Director, UNESCO, Beijing, China

Prof. Dr. Kai Sang Lock

Singapore Institute of Technology
Singapore

Prof. Dr. Manuel Andres Rodrigo Rodrigo

Faculty of Chemical Sciences and Technologies.
University of Castilla-La Mancha, Chemical
Engineering Department., Spain

Dr. Nikolettta Athanassopoulou

Institute for Manufacturing
University Of Cambridge, UK

Dr. Arsalan Ghani

Institute for Manufacturing
University Of Cambridge, UK

Prof. Dr. Jawwad Darr

Dept of Chemistry, Faculty of Maths & Physical
Sciences
University College London, UK



Muhammad Faheem Akhtar
Muhammad Waqas Malik
Zil-e-Huma
Waheed-ur-Rehman
Riaz Ahmed
Muhammad Israr
Muhammad Kamran
Gul-e-Saba
Sadia Saleem

Director
 Deputy Director
 Assistant Director
 Examination Officer
 Admission Officer
 Office Assistant
 Office Assistant
 Office Assistant
 Office Assistant

Admissions

The Institute is open to all persons who are academically qualified for admission to the courses of study offered by the Institute and no such person shall be denied the privileges of the Institute on the grounds only of sex, religion, color, creed, race, class or domicile.

The admission to the Institute is strictly on the basis of merit determined by its own admission test and earlier academic achievements. There are no special quotas, reserved seats or admission by donations nor any arbitrary age limit for the applicants, but preference will be given to fresh graduates.

Admission to the Bachelor Programs of the Institute is decided on the basis of candidate's earlier educational achievements and his/her score in the admission test. Since medium of instruction of the Institute is English, students are also assessed for their English language skills. A sample of such questions is available on Institute website. The test is held online and the timings and general instructions for the test are given on the downloadable Admit Cards. The results of the admission test are communicated to all candidates online. A former student of the Institute whose enrolment was cancelled due to unsatisfactory academic performance is also allowed to appear in the admission test. If selected, he/she will be enrolled in the first semester as a freshman. Any student who is currently on the roll of the Institute and wants to change the faculty is allowed to re-appear in the admission test. If selected,

he/she may not be given any credits for the courses passed earlier.

Applications: The admission processing fee is Rs.1,000 (US\$ 100 for applicants applying on SAT basis). Rs.2,000 in case of appearing in both admission tests. The payment can be made through Habib Bank Limited (HBL) A/C No. 00427991707703 for local applicants or equivalent amount in US Dollar for overseas applicants to Habib Bank Limited (HBL) A/C No. 00427991707703. Those applying for Financial Assistant must pay Rs.500 extra with processing fee.

Advance Standing: A person who has been enrolled for a relevant Bachelors degree program in engineering at some PEC accredited and HEC recognized Institution and has earned 15 or more transferable credits hours with a minimum CGPA of 2.5 on the scale of 4.0, may apply to this Institute for admission with advanced standing. However, the student at the GIK Institute, to qualify for a bachelor degree, must earn a minimum of 70 credits including 6 credit of senior year design project. An applicant for transfer from a local or foreign Institution is required to have passed the Institute's admission test or SAT-II, respectively, by securing equal/more marks than the minimum merit of the BS Program in which he/she seeks admission. However, acceptance of request for transfer will depend on availability of seat, and the quality of academic work already completed by the applicant. For supplementary information, please contact the Admission Office.

Basic Eligibility for BS Engineering Program

Candidates for admission must meet one of the following criteria:

1. HSSC (Pre-Engineering i.e. Mathematics, Physics and Chemistry) with 60% or above marks each in Mathematics, Physics & Overall.
2. HSSC (Pre-Medical) with Additional Mathematics and 60% or above marks each in Mathematics, Physics & Overall.
3. A-Level in three subjects Mathematics, Physics and Chemistry with D or above grade each in Mathematics & Physics and O-Level in eight subjects (English, Mathematics, Physics, Chemistry, Biology/Computer Science, Urdu, Islamic Studies & Pakistan Studies) for local applicants and in five subjects (English, Mathematics, Physics, Chemistry, Biology/Computer Science) for those applying from abroad.
4. American or Canadian High School Diploma or International Baccalaureate Diploma with Mathematics (Calculus), Physics and Chemistry with 60% or above marks, as per IBCC equivalence formula, each in Mathematics, Physics & Overall.
5. Three years Diploma of Associate Engineering (DAE) in relevant technology from a Pakistani Board of Technical Education with at least 60% marks each in Mathematics, Physics & Overall.

Note: Applicant with Mathematics, Physics and Chemistry background can apply for all Programs including Computer Engineering.

Basic Eligibility Criteria for Artificial Intelligence, Computer Science and Data Science:

Candidates for admission must meet one of the following criteria:

1. HSSC with Mathematics, Physics, and any other subject as third elective with 60% or above marks each in Mathematics, Physics & Overall.
2. A-Level in three subjects Mathematics, Physics and any other third subject, with D or above grade each in Mathematics & Physics and O-Level in eight subjects (English, Mathematics, Physics, Chemistry, Biology/Computer Science, Urdu, Islamic Studies & Pakistan Studies) for local applicants and in five subjects (English, Mathematics, Physics, Chemistry, Biology/Computer Science) for those applying from abroad.

Basic Eligibility for BS Management Sciences Program

Basic Eligibility Criteria: Candidates for Admission must one of the following criteria:

1. HSSC (Pre-Engg), HSSC (General Science), HSSC (ICS), HSSC (Pre-Medical), HSSC (Humanities) with at least 60% marks.
2. A-Level in three subjects with D's or above grades in two principal subjects and O-Level in eight subjects for local applicants and in five subjects for those applying from abroad with overall 60% or above equivalence as per IBCC formula.
3. American or Canadian High School Diploma or International Baccalaureate Diploma with overall 60% or above marks, as per IBCC equivalence formula.

Candidates are advised to carefully read above eligibility criteria before applying for admission. The admission will be cancelled if eligibility criteria is not met. The candidates are responsible for fulfilling eligibility and must immediately contact admission office if rendered ineligible upon declaration of result. The admission of ineligible candidate, upon request, can be deferred for a maximum of one year or the tuition fee will be refunded. It is mandatory for all applicants with O-level and A-level background to submit equivalence certificates from IBCC.

Comparative Assessment Criteria (Merit List)

Score in Admission Test OR SAT-II (in Mathematics and Physics for Engineering and Computer Science Programs and in any two subjects for Management Sciences Program) for those applying from outside Pakistan	85%
SSC/O-level (for Those with A-level and O-level background) / Equivalent	15%
Last completed qualification for High School diploma, IB diploma or B.Sc. or DAE	15%

Candidates, who have completed one of the above qualifications and are awaiting results, may apply for provisional admission. Confirmation of admission will, however, be subject to submission of results by the date specified in the offer letter and fulfillment of the above criteria.

HOW TO APPLY

Only Online Applications will be accepted. Complete instructions will be available on the link <http://admissions.giki.edu.pk> by June 26, 2022. The application procedure is as below:

1. Register yourself as Candidate for Admission on above link
2. Fill in and submit online admission form. Those interested in Financial Assistance (FA) and Scholarship, must fill FA and Scholarship registration form online.
3. Arrange to pay in any branch of HBL as per amount printed on bank challan.
4. Upload Payment details on admission portal
5. Download Admit Card.
6. Appear in online admission test.
7. Check your result and proceed as per online instructions.

The Institute is a not-for-profit organization and provides subsidized education. The Semester Fee (Tuition fee & Accommodation Charges), non-refundable, except in case an applicant is rendered ineligible upon declaration of results where refund is made, is as under for the 4 years for local residents and wards of expatriate Pakistanis.

S. No.	Academic Year	Engineering & Computer Sciences (Rs.)	Management Sciences (Rs.)
1.	2022-23	387,500	345,000
2.	2023-24	407,500	360,000
3.	2024-25	427,500	375,000
4.	2025-26	447,500	390,000

The annual tuition fee for foreign students is US\$ 7,500/- 5% of semester tuition fee will be collected as administrative charges against each semester. No administrative charges will be charged if student pays entire fee for two semesters in lump sum. Semester fee cannot be paid in installment. The tuition fee is payable before commencement of the semester. A non-refundable admission fee of Rs. 63,000/- for Pakistani or US \$ 680/- for foreign applicants is also required to be deposited at the time of admission. Rs. 30,000 will be charged as security, refundable at the

time of leaving the Institute subject to the clearance from relevant departments. The final year students are charged convocation fee of Rs. 17,000/-. An advance of Rs. 8,000 is to be deposited by each student as mess security.

Refund Policy:

If a freshly inducted student finds it necessary to withdraw his/her admission, he/she must inform GIK Institute in writing. The following refund policy will apply to the Semester Fee: In case an applicant is rendered ineligible upon declaration of results, 100% tuition fee refunded if applied within 10 days of declaration of result along with proof of ineligibility.

Timelines	Fee Refund (%age)
Before Joining GIK Institute	100%
Upto first seven days of commencement of classes	100 % after deduction of 10 % of semester fee as administrative charges
From 8th till 15th day of commencement of classes	50%
From 16th day of commencement of classes	No refund

Withholding Tax

Withholding Tax under section 236I of

Income Tax Ordinance 2001 is applicable @ 5% on Semester Fee, Admission Fee and Administrative Charges. However, this tax is not applicable in case:

- Fee is paid by a person appearing on the active tax payer list.
- Fee is paid by a non-resident subject to provision of the following:
- Copy of passport as evidence that during previous tax year, his stay in Pakistan was less than one hundred eighty-three days; Furnishes a certificate that he has no Pakistan-source income; and
- The fee is remitted directly from abroad through normal banking channels to the bank account of the educational institution.

Note: Tax collected under this section shall be adjustable against the tax liability of either of the parents or guardian making payment of the fee.

Free Electricity Unit

The Institute shall provide free electricity to each student in the hostels as under:

Summer 100 units per month

Winter 70 units per month

In addition to above, Free units allowed for common areas per hostel will be as under:

Summer: 3943 units per month

Winter: 534 units per month

Any excess consumption of electricity in the hostels will be charged from the students residing in respective hostels.

Academic Calendar

An academic year comprises two regular semesters of sixteen weeks each, and an eight-

week summer school. The timings of two semesters and summer school are as follows:

Fall: August to December

Spring: January to May

Summer: June to July

The last week of a semester is reserved for the final examinations. There is normally a mid-semester break in a semester.

Duration of Bachelor Studies

Students have to complete their entire degree requirements within the following time-limits:

Normal Duration: 4 years

Maximum Duration: 6 years

Financial Assistance: The Institute provides financial assistance to the needy and deserving students covering full or partial tuition fee in the form of interest free loan. Each year about 40-50 students get benefit of financial assistance. Students desirous of getting financial assistance may submit financial assistance form available at the Admission Portal GIK along with the admission form (Please add Rs. 500 added as financial assistance processing fee in addition to Rs. 1000 admission procession fee). Applicants will be informed about award of financial assistance along with admission offer.

Scholarships: Different government organizations, private companies and donors award scholarship to GIK students. The GIK Alumni Association provides scholarships to deserving students of 2nd, 3rd and 4th years. Following full or partial scholarships and financial assistance are likely to be available for those to be admitted in the academic year 2022-2023:



Scholarships	Name	Qualification/Conditions/Criteria	
	GIK Merit Scholarship	Engineering top 14 positions in admission test. Management Sciences: Top 6 positions in admissions test.	20
	GIK Alumni Association	Needy students (2nd year onwards)	55
	Chief Minister Khyber Pakhtunkhwa Scholarship	KP domiciled with annual income less than Rs.600,000	20
	DWP Foundation	KP domiciled with annual income less than Rs. 840,000	15
	Punjab Educational Endowment Fund (PEEF) Scholarship	Punjab domiciled with annual family income less than Rs. 360,000/-	10
	Khyber Pakhtunkhwa Education Foundation Scholarship	KP domiciled with annual family income less than Rs. 300,000/-	6
	Balochistan Scholarship	Balochistan domiciled	4
	Sindh Scholarship	Sindh domiciled with annual family income less than Rs. 1,200,000/-	5
	Chief Minister Educational Endowment Fund Scholarship (CMEEF)	KP domiciled with annual income less than Rs.1200,000 Faculty of Engineering Sciences with specialization in photonics	4
	Descon Endowment Fund	Eligible for Financial Assistance of GIK, rural area	1
	Fast scholarship	Electrical Engineering student with GPA 3.0 (Zakat eligible)	1
	HBL Platinum Scholarship	Need-Cum-Merit Basis	1
	ICI Scholarship	Top Female	1
	Lucky Cement Scholarship	Pakistani National, preference will be given to KP domicile holder	1
	Squeaks Foundation	Female student from Khyber Pakhtunkhwa	1
	Bestway Educational Foundation scholarship	Merit-cum-Need basis	01
Financial Assistance (Loan)	Financial Assistance by GIK Institute	Need cum merit basis	40
	GIK Financial Assistance for Top Female	Engineering top: 7 Management: top 3 females	10
	Ihsan Trust loan	need basis	20
	Habbah Trust loan	need basis	4

For further details: Please Contact Admission Office GIK Institute at Telephone: +92-938-281026, Ext: 2301, 2342, 2349, 2354, 2595.

Campus Jobs: A number of on-campus jobs are available for students with remuneration adjusted against fee and mess bills, However, qualification conditions and hours per week limitation apply.

Examinations Office

The Examinations Office works under the supervision of Director Admissions & Examinations. This office is responsible for preparing class and examination schedules, holding of semester's examinations, maintenance and compilation of results issuance of semester result reports, transcripts, certificates and degrees.

Registration Schedule

Students have to register for their courses during the period specified for the purpose before the commencement of a semester. The office of the Examinations, before the start of every semester, will notify the registration deadline. Requests for late registration for valid reasons can be entertained by the approval of Pro-Rector (Academic) till the end of the third week of a semester. However, such students are required to pay Rs. 850 per day late registration fee.

Registration in the Summer

An eight week summer session is organized each year for those students who fail to qualify in a course or they want to improve courses with D or a D+ grade. The courses offered in the summer are decided by the respective Dean's office keeping in view the number of students interested in taking a particular course. Students have to pay separately for registering in a summer course. Students cannot register in a higher level course during summer and the maximum limit for registration is 8 Credit Hours.

Double Degree Program

Graduates of the Institute desirous of obtaining a degree in a discipline other than the previously earned degree can apply afresh for a separate Double Degree Program. They would be required to spend additional two to four semesters in the Institute to complete the requirements of a double degree. The students have to do a separate project for a Double Degree. The acceptance in the Degree program and details of the requirements are worked out by the respective Dean's office and communicated to the office of the Controller of Examinations.

Attendance Rule

Although the students are expected to

attend all the lectures and laboratories work pertaining to their courses of study but are required to attend at least 80% of the total Lectures/Lab work for each course to qualify for appearance in the final examination.

Change in Courses

Once registered for a semester, students may add or drop courses only with the approval of their Deans and in conformity with the prescribed procedures and time-limits. Courses dropped during this period are not shown on the semester result report or transcript.

Withdrawal from Courses

Students may withdraw from one or more courses with the approval of their Dean between the 4th and 10th week of a semester. In such cases, a W grade appears on their transcripts. Any withdrawal after the 10th week entails award of an F grade in the course.

Incomplete (I) Grade

An I grade is given to students in a course if the outstanding requirement, in such cases, is to be met during the first two weeks of the next semester and the students themselves are responsible to make arrangement for the purpose with their instructors. Failing this, the I grade is converted to F grade. They cannot re-register for a course in which they have the I grade. The grade point average of a student for a semester is calculated excluding the I grade and it is re-calculated when a regular grade has been awarded in the course.

Repeating Courses

Courses in which students secure F grade, and which are a requirement for the degree have to be repeated in entirety. They may opt for a substitute course only if there is an alternative in the curriculum. Students can repeat courses for which they obtained F, D+ or D grade, on the condition that they repeat the courses within 3 semesters after the semesters in which they obtained these grades. In case of repeated courses, all grades achieved by students appear in their transcripts. However, only the latest grade in chronological order will be counted for the Cumulative Grade Point Average, even if it is lower than the earlier one.

Interruption of Studies

If a student interrupts his study programs

for a period longer than one semester then, upon his return, all the credits previously earned by him at the Institute are evaluated by the Dean to determine their relevance to the changes made in the curriculum, if any. He may be required to modify his degree plan to ensure conformity to the latest version of the curriculum.

Academic Advisors

All Students are assigned to academic advisors. The advisors develop plans of study for them, monitor their records, and guide them on all academic matters.

Credit Hour System

The credit hours assigned to a theory or a laboratory course are determined by the contact hours allocated to it per week throughout a semester. For a theory course one credit hour is equivalent to one contact hour of lecture per week, and for a laboratory course, three contact hours of practical work per week constitute one credit hour.

Semester Credit Load

Students can normally register in accordance with his / her degree program, 15-18 credit hours in a semester. No exception to this upper limit is allowed to freshmen. However, in later years this limit may be relaxed for students with good academic standing, with the approval of the Dean. Under all cases the maximum limit remains 21 credit hours.

Degree Requirements

For a Bachelor degree a student must earn a minimum of 132 to 136 credits, depending upon his / her faculty. At the time of graduation, the Cumulative Grade Point Average (CGPA) should not be below 2.00.

Medium of Instruction

The medium of instruction and examinations of the Institute is English. All the courses are taught throughout in English.

Curriculum Components

The major academic components of the Bachelors degree programs are described below:

Foundational Courses in Engineering Education

Courses in physics, chemistry, mathematics and introductory engineering are common for students of all faculties before they move on the major courses of their own faculty. The aim of these courses is to provide through grounding in the basic principles and analytical skills essential for studies in specialized areas of all faculties before they move on the major courses of their own faculty.

Management Sciences and Humanities Courses

Common courses in English language, social sciences and engineering management are required for all students. They are meant to inculcate in them an awareness of our history and culture, to help them cultivate aesthetic and moral dimensions of their personalities and to equip them with communicational and managerial skills.

Faculty Courses

Students are required to take a number of core and elective courses of their own faculty which are listed in the academic programs of each faculty.

Inter-faculty Courses

Students are required to select some courses offered by faculties other than their own. Such courses aim at providing broader bases to their studies, and widening their awareness of allied fields, which impinge on their areas of specialization.

Technical Electives

Students are also required to take a number of advanced technical courses. To fulfill this requirement, they may choose additional courses in their own field of specialization, select a second area of specialization, or select advanced courses from some different fields. Each faculty offers a number of advanced courses in different fields.

Senior Year Design Project

In the final year, students have to undertake a project, which is assigned 6 credits hours. They must work under direct supervision of their project advisor for the completion of the project. Students are encouraged to undertake projects, which are of interest to industry

or to government of departments. They are expected to complete their projects and present their reports by the end of the eighth semester before the final examination.

Summer Internship

Every student has to participate in a practical training program of four to eight weeks during the summer of junior year and submit a formal written report about it.

Course Codes

The courses are identified by the course numbers, which consist of two letters and three digits. The first two letters represent the major field; the first digit indicates the level of course; the next digit the broad area of the course; and the last, the sequence number of the course offered in the same area at the same level (year).

AI	Artificial Intelligence
CE	Computer Engineering
CH	Chemical Engineering
CS	Computer Science and Engineering
CV	Civil Engineering
DS	Data Science
EE	Electrical Engineering
ES	Engineering Sciences
HM	Humanities & Social Sciences
ME	Mechanical Engineering
MM	Materials Science and Engineering
MS	Management Sciences
MT	Mathematics
PH	Physics

Student Evaluation

Students are evaluated by mid-semester test, home assignments, quizzes, case studies, course project, laboratory reports, oral tests and the end-of-semester examination. The weight allocated to them depends upon the nature of the course. Usually, the end-of-semester examination carries 50% weightage of a course.

Grading System

Depending upon academic performance, students are awarded grades A, A-, B+, B-, C+, C, C-, D+, D, F, I, E and W for each course. These grades indicate the following levels of performance:

B	Good
C	Adequate
D	Minimum acceptable
F	Failure, implying that the student must repeat the course to receive any credit
I	Incomplete
E	Exemption
W	Withdrawn

Each grade is assigned Grade Points per Credit (GPC). The following table indicates the gradation from excellent to failure:

Grade	GPC	Grade	GPC
A	4.00	C	2.00
A-	3.67	C-	1.67
B+	3.33	D+	1.33
B	3.00	D	1.00
B-	2.67	F	0.00
C+	2.33		

I, E and W grades are not counted in calculation of GPA. The academic standing of a student is referred as grade point average (GPA) which is the ration of the total number of grade points earned to the total number of credits attempted. The maximum possible GPA is 4.00 The minimum semester GPA to remain is satisfactory academic standing is 2.00. Students are placed on academic probation at the end of any semester in which their semester GPA falls below 2.00. A student on probation is allowed to register only 10-13 credit hours.

A student whose semester GPA remains below 2.00 is given a warning for his/her poor performance. If his/her SGPA remains below 2.00 for two consecutive semesters (excluding summer school) his/her name is removed from the roll of the Institute. Freshmen, upon request, may start afresh with no credit transfer if two consecutive probations are experienced in first two semesters.

A	Excellent
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FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Introduction

The Faculty of Computer Science and Engineering (FCSE) is one of the five faculties at GIK Institute. FCSE offers bachelor programs in Artificial Intelligence, Computer Engineering, Computer Science, and Data Science.

FCSE employs competent faculty members qualified to accomplish the mission and goals of the Institute. When determining acceptable qualifications of its faculty, FCSE asserts primary consideration to the terminal degree in the discipline. FCSE also considers competence, effectiveness and capacity, including, as appropriate, undergraduate and graduate degrees, related work experiences in the field, professional licensure and certifications, honors and awards, continuous documented excellence in teaching, or other demonstrated competencies and achievements that contribute to effective teaching, research and student learning outcomes.

The Artificial Intelligence program, partnered by Huawei, aims to give in-depth knowledge required to transform data (including Big Data) into intelligent decisions. The theme of the program and its curriculum is primarily based on inferring the machine-based decisions from the complex inputs, such as image, text, and databases. Upon completion of the coursework students will be able to select data driven technologies to mimic or enhance human capabilities. GIK Institute joins hands with Huawei (ranked 61st on Fortune Global 500 companies), to bring the cutting-edge technology along with the latest curriculum updates and equipment for the students in Artificial Intelligence program. Huawei Pakistan supports the program by providing student internships, certification opportunities in AI, faculty training and latest AI equipment for the students, which will enhance the skillset of the students and help secure better jobs in the market in future.

The Computer Engineering focuses on knowledge of Mathematics and basic sciences, necessary for the analysis and design of computer software, hardware and systems through an understanding of the principles of computer programming, algorithms, data structures, microprocessor systems, and electronics with an understanding of the applications of computer engineering principles.

The Computer Science program offers a broad understanding of the field through introducing concepts, theory, and techniques with intensive education in core areas of Computer Science, e.g., computer architecture, compilers, software engineering, computer systems development, etc. This program encourages students to develop and use abstract models as well as apply respective technologies in practical settings.

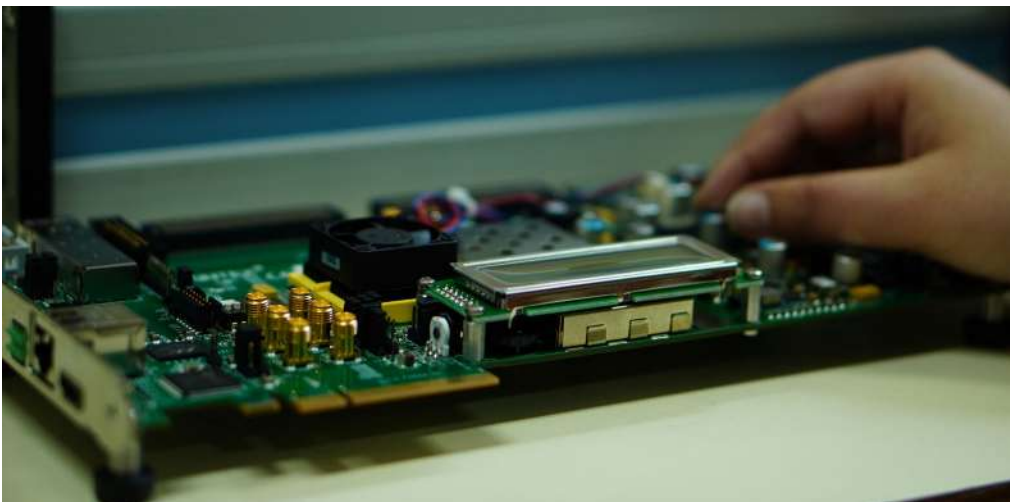
The Data Science program introduces its students to exploring the statistics hidden behind Data, through various techniques like data analytics, data visualization, and data ware housing. The program is especially meant for students, who are interested in exploring the patterns hidden in the data as well as making predictions for the future of any data-intensive industry.

All four streams focus on establishing a strong mathematical foundation, basic hardware aspects of computing, and advanced technical electives spanning areas such as computer communications, neural networks, reinforcement learning, expert systems, databases, biologically inspired computing, data and network security, signal processing and computer vision. Abundant opportunities exist for students to develop expertise in their areas of interest through enrolment in an assortment of elective courses offered by FCSE and other faculties.

Students are also accepted into various programs under the scholarship schemes offered by federal and provincial governments of Pakistan, National ICT R&D Funds and other organizations (See the Scholarship section of Prospectus).

Thrust Areas

- Artificial Intelligence and Robotics
- Algorithms and Computational Theory
- High Performance Computing
- Machine Learning & Data Mining
- Network Communications and Distributed Systems
- Signal and Image Processing
- Software and Systems Engineering
- Data Science & Data Mining
- Data Warehousing
- Data Visualization
- Cyber Security
- Penetration Testing
- Cryptography





Dean

Ahmar Rashid

PhD (Jeju National University, South Korea)

B.S. Artificial Intelligence

Raja Hashim Ali, PhD (Kungliga Tekniska Höskolan, Sweden)

Farhan Khan, PhD (Bilkent University, Turkey)

Mr. Talha Laiq, MS (Université Paris-Saclay)

Hina Ayaz (GA-4), MS (NUCES, Pakistan)

B.S. Computer Engineering

Shahabuddin Ansari, PhD (GIK Institute, Pakistan)

Muhammad Hanif, PhD (Australian National University, Australia)

Taj Muhammad Khan, PhD (University of Grenoble, France)

Farhan Khan, PhD (University of Auckland, New Zealand)

Badre Munir, MS (GIK Institute, Pakistan)

Ali Shaukat, MS (GIK Institute, Pakistan)

Mohsin Zafar, MS (PIEAS, Pakistan)

Jalees-ur-Rehman (GA-4), MS (GIK Institute, Pakistan)

Usman Haider (GA-4), MS (GIK Institute, Pakistan)

B.S. Computer Science

Zahid Halim, PhD (NUCES, Pakistan)

Syed Fawad Hussain, PhD (University of Grenoble, France)

Ghulam Abbas, PhD (University of Liverpool, UK)

Rashad Jillani, PhD (Florida Atlantic University, USA)

Usman Habib, PhD (Technical University of Vienna (TU Wien), Austria)

Ali Zeeshan Ijaz, PhD (Western Sydney University, Australia)

Ahsan Shah, MS (GIK Institute, Pakistan)

Sajid Ali, MS (NUCES, Pakistan)

Atta ur Rahman (GA-4), MS (UET, Pakistan)

B.S. Data Science

Masroor Hussain, PhD (GIK Institute, Pakistan)

Abinta Mehmood Mir, MS (UET Taxila, Pakistan)

Sania Akhtar (GA-4), MS (COMSATS, Pakistan)

Faculty on Leave for PhD

Zawar Hussain (Macquarie University, Australia)

Usman Raza (Purdue University, USA)

Lab Engineers

Mumtaz Ali Shah, MS Computer Science (VU)

Engr. Amna Arooj, BS Computer Engineering (Islamia University)

Engr. Kiran Shah, BS Computer Engineering (CIIT)

Engr. Muhammad Abu Bakr, BS Computer Engineering (UET Taxila)

Engr. Hayat Ullah, BS Electrical Engineering (UET Peshawar)

Engr. Maaz Alam, BS Computer Systems Engineering (UET Peshawar)

Zain ul Abideen, MS Computer Science (COMSATS Attock)

Graduate Assistants

Abdur Razaq (GA-F), MS Computer Science (CIIT)

Akhtar Badshah (GA-F), MS Software Engineering (UET)

Ehsan Elahi (GA-F), MS Software Engineering (CIIT)

Personal Secretary to Dean

Farhad Gul, MSc Mathematics (University of Peshawar) & MCS (Virtual University)

Faculty Mission

The Faculty of Computer Science and Engineering endeavors to produce competent scientists and engineers who can play a productive role both in the practical and research areas of computing. The faculty uses modern technologies to enhance the learning capabilities of the students and to provide them with a stimulating and challenging environment. Emphasis is put on the practical applications of computer systems to the software and hardware needs of the global industry in general and Pakistani industry in particular.

5.Objective Based Education (OBE)

5.1.Program Educational Objectives (PEOS)

a. Artificial Intelligence

The Program Educational Objectives (PEOs) of BS (AI), partnered by Huawei, are:

PEO-1: Graduates utilizing their skills and knowledge to solve complex problems in real-world settings.

PEO-2: Graduates practicing in the area of Artificial Intelligence in a socially and ethically responsible way.

PEO-3: Graduates demonstrating lifelong learning skills in Artificial Intelligence and allied disciplines.

b. Computer Engineering

The Program Educational Objectives (PEOs) of BS (CE) are:

PEO-1: Graduates responsibly practicing in a variety of computer engineering and allied disciplines.

PEO-2: Graduates utilizing their skills and knowledge to solve complex engineering problems in real-world settings.

PEO-3: Graduates demonstrating sustained learning and adapting to evolving fields through continued professional development and self-study.

c. Computer Science

Following are the Program Educational Objectives (PEOs) of BS (CS).

PEO1: Graduates excelling in professional careers while maintaining environmental, ethical and social values.

PEO2: Graduates applying and effectively communicating knowledge both individually

and in a team through state-of-the-art tools and technologies

PEO3: Graduates staying current with technological innovations through trainings, higher education, and lifelong learning.

dd. Data Science

The Program Educational Objectives (PEOs) of BS (DS) are:

PEO-1: Graduates utilizing their skills and knowledge to solve complex problems in real-world settings.

PEO-2: Graduates practicing in the area of Data Science in a socially and ethically responsible way.

PEO-3: Graduates demonstrating lifelong learning skills in Data Science and allied disciplines.

5.2.PROGRAM LEARNING OUTCOMES (PLOs)

a. Artificial Intelligence, Computer Science, & Data Science

The Student Outcomes (SOs) of BS (AI) partnered by Huawei, those of BS (CS), those of BS (CyS), and those of BS (DS) are:

1. Academic Education – Completion of an accredited program of study designed to prepare graduates as computing professionals.
2. Knowledge for Solving Computing Problems – Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3. Problem Analysis – Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4. Design/Development of Solutions – Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and

environmental considerations.

5. Modern Tool Usage – Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
6. Individual and Teamwork – Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
7. Communication – Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
8. Computing Professionalism and Society – Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
9. Ethics – Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.
10. Life-long Learning – Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

Upon completion of BS (AI) partnered by Huawei / BS (CS) / BS(DS) degree, all the students should have attained the aforementioned ten SOs.

b. Computer Engineering

The Program Learning Outcomes (PLOs) of BS (CE) are:

1. Engineering Knowledge – Ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2. Problem Analysis – Ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3. Design/Development of Solutions – Ability to design solutions for complex engineering problems and design systems, components or processes that

meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

4. Investigation – Ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.
5. Modern Tool Usage – Ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.
6. The Engineer and Society – Ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.
7. Environment and Sustainability – Ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
8. Ethics – Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
9. Individual and Teamwork – Ability to work effectively, as an individual or in a team, on multifaceted and/or multidisciplinary



settings.

10. Communication – Ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project Management – Ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.
12. Lifelong Learning – Ability to recognize importance of and pursue lifelong learning in the broader context of innovation and technological developments.

Upon completion of BS (CE) degree, all students should have attained the aforementioned twelve PLOs.

5.3. Knowledge Domains (Wks) For BS (Computer Engineering)

The engineering curriculum is the most important tool for grooming students based on the Program Learning Outcomes' 12 Graduate Attributes (GAs) (PLOs). The curriculum is based on the following knowledge profiles in order to instill different dimensions of thinking – mathematical, computational, design, and creative – among students in the Cognitive, Psychomotor, and Affective domains: The Knowledge Domains for BS (CE) program are:

- **WK1 – Natural Sciences:** A systematic

theory-based understanding of natural sciences applicable to the discipline.

- **WK2 – Mathematics and Computing:** The concept-based mathematical thinking, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline.
- **WK3 – Engineering Fundamentals:** A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.
- **WK4 – Engineering Specialization:** The knowledge of Engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.
- **WK5 – Engineering Design:** The Design Thinking Knowledge that supports engineering design in a practice area of an engineering discipline.
- **WK6 – Engineering Practice:** The Knowledge of engineering practices (technology) in different practice areas of an engineering discipline.
- **WK7 – Engineering in Society:** A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity, i.e., economic, social, cultural, environmental and sustainability.
- **WK8 – Research Literature:** Engagement with selected knowledge in the research literature of the discipline.

Upon completion of BS (CE) degree, all students should have attained the aforementioned twelve PLOs.



6. Sustainable Development Goals (SDGs)

The Sustainable Development Goals (SDGs) defined by United Nations provide global guidance for addressing the international community's global challenges. It is about better protecting the natural foundations of life and our planet for everyone everywhere, as well as preserving people's opportunities to live in dignity and prosperity across generations. The 17 objectives address all

three aspects of sustainable development: ecological, economic, and social.

The graduates of FCS&E are expected to make a difference not only in their own community, but also in the world. The Faculty is also expected to provide a well-rounded education and global awareness to its graduates. Two significant appeals are the importance of giving a systems approach lens to help students build a well-rounded understanding of how global concerns must be addressed and the capacity to ensure that governments keep their pledge to execute the Goals. The Faculty of Computer Science and Engineering aspires to provide its graduates with the best opportunities to learn about and follow the SDGs during their bachelor's education.

7. ACCREDITATION

The undergraduate programs combine the strength of Engineering, Mathematics, Statistics, and Computer Science. The courses develop a professional approach to design computer-based systems, looking at both widely applicable principles of software engineering and the evolving computing technologies in the thrust areas of the faculty. Students are admitted to a four-year BS degree in the Faculty of Computer Science & Engineering, opting for one of the following five specialized degree programs:

- Artificial Intelligence, partnered by Huawei (accredited by NCEAC)
- Computer Engineering (accredited by PEC under Washington Accord level-II, i.e., OBE)
- Computer Science (accredited by NCEAC)
- Data Science (accredited by NCEAC)

The BS program in Computer Engineering is accredited by Pakistan Engineering Council (PEC) under Washington Accord level-II, i.e., Outcome-Based Education (OBE), whereas the BS programs in Artificial Intelligence partnered by Huawei (AI), Computer Science (CS), and Data Science (DS) are accredited by the National Computing Education Accreditation Council (NCEAC).

8. Labs And Infrastructure

8.1. Common It Facilities

a. Internet/LAN

The Institute provides 500 Mbps Internet

access to all faculties and student hostels through Pakistan Education and Research Network (PERN-II) and PTCL. All rooms in student hostels are connected through Fiber Optics LAN that also provides peer-to-peer file sharing for student collaboration.

b. Video Conferencing Room

The video conferencing facility is located at the Workshop Room adjacent to the Agha Hasan Abedi auditorium. The room is equipped with LifeSize® Room 220™ system for high-definition H.323 multipoint videoconferencing and a high-resolution interactive smartboard digital projector. The room provides seating for 65 participants and is equipped with conference sound/microphone system.

8.2. Instructional, Research, And Project Laboratories

The Faculty of Computer Science and Engineering is well equipped with state-of-the-art computer systems running a wide range of applications and specialized software supporting the courses. In addition, well-equipped research laboratories are available for the use of faculty, graduate students, and senior undergraduate students. The following is a brief description of various laboratories and their functions.

a. Programming and Computing (PC) Lab

The Programming and Computing (PC) laboratory is the central computing laboratory of the Institute, providing general purpose computing facilities to all students, as well as



internet and printing facilities. It is open seven days a week from early morning till late at night. It houses 100+ Core-i3/i5 networked machines running Windows as well as Linux operating systems. DSL - WiFi facilities are also available. Introduction to Computing and Intensive Programming lab modules are conducted in this lab. Student workshops and software competitions are also held in this laboratory.

b. Software Engineering (SE) Lab

The Software Engineering (SE) laboratory focuses on providing facilities for courses such as Software Engineering, Language and Compilation Techniques, and Databases. It houses 55 networked Core-i7 machines. They are connected to database and other servers of the Institute, including the printing facilities. In addition, DSL-WiFi facilities and various software tools (e.g., Eclipse, Visual Studio, Flex, Bison, Oracle/Developer and Rational Rose) are also available in this lab. This laboratory also hosts student workshops and software competitions.

c. Operating Systems and Networking (OSN) Lab

The Operating Systems and Networking (OSN) laboratory is used mainly for lab modules for Operating Systems, Computer Communications and Networking and Systems Programming courses. This lab is equipped with 90 machines running various operating systems and network simulation software

tools, e.g., DEV C++, MATLAB, Prolog, Oracle 11G + Wamp Server, Cisco Packet Tracer and other software tools.

d. Data Structure and Algorithms (DSA) Lab

The Project and Software Development (PSD) laboratory is used for coding and simulating problems related to both academia and industry. This lab is primarily used for lab sessions for students of Data Structures and Signals as well as for conducting workshops and events organized by faculty members and different societies at GIK Institute. It is equipped with 50 Core-i5 workstation running window 7 professional, Visual Studio 2012, MATLAB, Packet Tracer and other software tools.

e. Final Year Project (FYP) Lab

The Final Year Projects (FYP) laboratory is used mainly by seniors of Computer Engineering (CE) and Computer Science (CS) programs for developing their final year projects. This lab is equipped with 10 Core-i7 workstations running various operating systems, network simulation software tools, Visual Studio 2020, and other simulation and development software. Students are allowed to add hardware and accessories according to their project requirements.

f. High-Performance Computing (HPC) Facility

High-Performance Computing (HPC) facility



was established in Faculty of Computer Science and Engineering in 2006, using an AMD Opteron-based computing cluster. This facility has been upgraded using 10 million rupees funds from Directorate of Science and Technology (DoST), Government of Khyber Pakhtunkhwa. The facility consists of 160 CPU cores, 1024 GPU cores, 640 GB main memory and 10 GB Ethernet switch interconnection.

g. Artificial Intelligence Computing (AIC) Lab

The Artificial Intelligence Computing (AIC) laboratory is the main computing facility for AI-specific lab and research tasks. It houses 55 state-of-the-art networked iMac machines with M1 processor, SSD hard disks, and 16 GB RAM. In addition, this lab is connected to high-speed computing servers with additional memory and GPU support. The lab modules of various AI courses, e.g., those of Deep Neural Networks, Operating Systems, Computer Vision, Machine Learning, and Natural Language Processing are conducted in this lab.

h. Robotics and Vision Lab

The Innovaion (Inova) laboratory is the project facility for AI-specific and Data-Science project tasks. It houses several project tools and equipment like programmable drones, programmable robots, programmable toy cars, oculus, and EEG machines. In addition, this lab contains 15 high performance computing machines with additional memory and GPU support for projects of various courses and Senior Year Design Projects for AI and DS students. The lab will also be used for conducting industrial training labs and Problem Based Learning for AI (PBL for AI) and Problem Based Learning for DS (PBL for DS) labs.

i. Data Analytics (DA) Lab

The Data Science Computing (DSC) laboratory is the main computing facility for DS-specific lab and research tasks. It houses 55 state-of-the-art Core-i7 networked All-in-One Ultra PC machines. In addition, this lab contains 10 high speed computing servers with additional memory and GPU support. Lab modules of Big Data Analytics, Data Mining, Data Visualization, and Data Warehousing and Business Intelligence are conducted in this lab.

j. Cyber Security Computing (CySeC) Lab

The Cyber Security Computing (CySeC) laboratory is the main computing facility for Cyber Security-specific lab and research tasks. It houses 55 state-of-the-art Core-i7 networked All-in-One Ultra PC machines. Lab Modules of Vulnerability Assessment, Computer Network and Communication, and Secure Software Design and Development are conducted in this lab.

k. Cyber Security Project (CySeP) Lab

The Cyber Security Project (CySeP) laboratory is the project facility for Cyber Security-specific project and research tasks. It houses a testbed for cyber security in addition to several stand-alone networked systems for testing, identifying, and stopping various types of cyber security attacks.

l. System Architecture (SA) Lab

The Hardware Project (HP) laboratory is the main hardware facility for CE-specific lab, project and research tasks. It has 20 workstations (hardware boards and tables) with seating ability for 40 students. It contains 10 state-of-the-art Core-i7 networked machines running Windows that can be used for designing hardware. In addition, the lab contains complete hardware kits for the lab modules of various Computer Engineering courses, e.g., Logic Design, Computer Organization and Assembly Language, Microprocessor Interfacing, Signal Processing, Digital System Design, and ASIC Design. In addition, it is also used for designing and manufacturing hardware components of Final



Year Projects hosted at Faculty of Computer Science and Engineering.

m. Digital Logic Design for Computing (DLDC) Lab

The Data Science Computing (DSC) laboratory is the main logic design facility for all computing programs and research tasks. It has 23 workstations (hardware boards and tables) with seating ability for 56 students. It contains 12 state-of-the-art Core-i7 networked machines running Windows that can be used for designing hardware. In addition, the lab contains complete hardware kits for the lab modules of all Computing courses, e.g., Logic Design, and Computer Organization and Assembly Language.

9. Societies

The Faculty of Computer Science and Engineering is host to many societies and technical teams.

Association for Computing Machinery (ACM) GIKI is the one of the oldest societies of GIKI Institute; it promotes computing and programming practices through seminars,

events, competitions and exhibitions and is currently advised by Dr. Raja Hashim Ali. Similarly, Netronix is a technical student society, currently advised by Dr. Ghulam Abbas, which is responsible for gaming competitions and hostel network maintenance. Several technical teams also come under the umbrella of FCS&E (Microsoft Learn Student Ambassador at GIKI, Google Developers Club, etc.), which play a crucial role in providing opportunities for managerial and technical growth and experience to students of FCS&E.

10. Careers

The graduates of FCSE faculty should be able to meet the highest standards of knowledge and training for leadership in evolving fields of artificial intelligence, computer engineering, computer science, cyber security, and data science profession, including teaching, research, development, and higher education at the national and international levels. This fact is evident from the positions and respect our graduates are enjoying at national and international universities as well as at multi-national software/IT enterprises.



Degree Requirements For Faculty Of Computer Science and Engineering

a. Degree Requirements for BS in Artificial Intelligence (136 Credit Hours)

For obtaining Bachelor of Science degree in Artificial Intelligence partnered by Huawei, a student must complete 136 credit hours with a CGPA of 2.0 or above.

In addition, every BS (AI) partnered by Huawei student is required to participate in a summer training program and submit a formal written report during the summer of Junior Year.

Moreover, every BS (AI) student will undertake a 15-hour community service activity as a degree requirement using the summers or fall semester of their Sophomore Year.

Course requirements for obtaining BS (AI) degree offered by FCS&E are given below.

Course Group (HEC Category)		Course Code	Course Title	Credit Hours	Total Credits
General Education Requirements	Computing	CS101, CS101L	Computing and AI + Lab	2+1	10
		CS205L	Computer Tools Lab	0+1	
		CS121	Computer Science	3	
		IF101	Innovation and Makers Lab I	0+1	
		IF102	Innovation and Makers Lab II	0+1	
	English Language	HM101	Communication Skills	1+2	6
		HM102	Critical Thinking and Expository Writing	2+1	
	Humanities & Management	HM211	Pakistan & Islamic Studies	3	12
		MS291	Engineering Economics	3	
		HM425	Intellectual Property and CyberLaw	3	
	Mathematics & Sciences	ES111	Probability and Statistics	3	20
		MT101	Calculus I	3	
		MT102	Differential Equations and Linear Algebra I	3	
		CH101	Applied Chemistry and Environment	3	
		PH101, PH101L	Applied Physics + Lab	3+1	
		MM101, MM141	Materials and Nanotechnology + Lab	2+1	
		CH161	Occupational Health and Safety Lab	0+1	

Core Requirements	Computing	CS112, CS112L	Object-Oriented Programming + Lab	3+1	48
		CE121, CE121L	Fundamentals Logic Design + Lab	3+1	
		CS231	Discrete Mathematics	3	
		CS221, CS221L	Data Structure & Algorithms + Lab	3+1	
		CE222, CE222L	Computer Organization & Assembly Language + Lab	3+1	
		CS232, CS232L	Database Management Systems + Lab	3+1	
		CS311, CS311L	Operating System + Lab	3+1	
		CS325	Software Engineering	3	
		CE413	Computer Networks	3	
		CS417, CS417L	Parallel Processing & Distributed Computing	3	
		CS464	Data & Networks Security	3	
		CS478	Design & Analysis of Algorithms	3	
		CS481	Senior Year Design Project (Part-I)	0+3	
	AI Domain-Specific	AI211, AI211L	Programming for Artificial Intelligence + Lab	2+1	29
		AI201, AI201L	Trends and Techniques in Artificial Intelligence + Lab	3+1	
		AI231	Knowledge Representation & Problem Solving	3	
		AI319, AI319L	Computer Vision + Lab	3+1	
		AI341, AI341L	Deep Neural Networks + Lab	3+1	
		AI351, AI351L	Machine Learning + Lab	3+1	
		AI361, AI361L	Natural Language Processing + Lab	3+1	
		AI305L	Associate in Artificial Intelligence Lab	0+1	
		AI406L	Professional in Artificial Intelligence Lab	0+1	
		AI407L	Problem Based Learning for Artificial Intelligence Lab	0+1	

Electives	Management Science Courses (Choose ONE course)	MS426	Technology Management	3	3
		CS436	Operation Research	3	
		CS492	Network Security & Cyber Ethics	3	
		MS492	Operations Management	3	
		MS493	Industrial Safety	3	
		MS494	Total Quality Management	3	
		MS496	Project Management	3	
		MS497	Procurement Management	3	
	AI Domain Breadth Courses (Choose ONE course)	AI309	Deep Reinforcement Learning & Control	3	3
		CS324	Computer Architecture	3	
		CS328	Introduction to Development Operations	3	
		AI372	Nature inspired Computing	3	
	AI Domain Specialization Courses (Choose THREE courses)	AI408	Cloud Computing	3	9
		AI410	Data Mining	3	
		CS411	Block Chain	3	
		CS412	Information Retrieval	3	
		CS413	Internet of Things	3	
		CS414	Geographic Information Systems	3	
		AI420	Medical Image Processing	3	
		CS420	Cyber Security	3	
		CE421	Computer Architecture	3	
		AI423	Object Oriented Analysis & Design	3	
		CE423	General Purpose Computing with GPU	3	
	AI425	Microprocessor Interfacing	3		
			AI426	Mobile Computing	3
CS432			Advanced Databases	3	
CS433			Computer Graphics	3	
EE436			VLSI Design	3	
CS437			Data Warehousing & Data Mining	3	
CS438			Web Mining and Social Media Analysis	3	
CS439			Data Science	3	
AI452			Techniques of Soft Computing	3	
CE453			Robotic Vision	3	
CS458			Distributed Systems	3	
CS463			Web Engineering	3	
AI473			Computational Neuroscience	3	
CS474			Bio-Informatics	3	
CS476			Computational Biology	3	
Degree total			40 courses – 112 theory credits – 24 lab credits – 136 credits		

(b) Degree Requirements for BS in Computer Engineering (136 Credit Hours)

For obtaining Bachelor of Science degree in Computer Engineering, a student has to complete 136 credit hours with a CGPA of 2.0 or above.

In addition, every BS (CE) student is required to participate in a summer training program and submit a formal written report during the summer of Junior Year.

Course requirements for obtaining BS (CE) degree offered by FCS&E are given below.

Course Group (HEC Category)		Course Code	Course Title	Credit Hours	Total Credits
General Education Requirements	Computing	CS101, CS101L	Computing and AI + Lab	2+1	6
		IF101	Innovation and Makers Lab I	0+1	
		IF102	Innovation and Makers Lab II	0+1	
	English Language	HM101	Communication Skills	1+2	6
		HM102	Critical Thinking and Expository Writing	2+1	
	Humanities & Management	HM211	Pakistan & Islamic Studies	3	12
		MS291	Engineering Economics	3	
		HM425	Intellectual Property and CyberLaw	3	
	Mathematics & Sciences	ES111	Probability and Statistics	3	20
		MT101	Calculus I	3	
		MT102	Differential Equations and Linear Algebra I	3	
		CH101	Applied Chemistry and Environment	3	
		PH101, PH101L	Applied Physics + Lab	3+1	
		MM101, MM141	Materials and Nanotechnology + Lab	2+1	3
		CH161	Occupational Health and Safety Lab	0+1	



Core Requirements	Computer Engineering	CE102L	Computer Engineering Workshop	0+1	45
		CE121, CE121L	Fundamentals of Logic Design + Lab	3+1	
		CE211, CE211L	Circuit Analysis + Lab	3+1	
		CE222, CE222L	Computer Assembly & Organization Language + Lab	3+1	
		CE231, CE231L	Electronics-I + Lab	3+1	
		CE313, CE313L	Computer Communication and Networks + Lab	3+1	
		CE324, CE324L	Microprocessor Interfacing + Lab	3+1	
		CE341, CE341L	Signals and Systems + Lab	3+1	
		CE362, CE362L	Signal Processing + Lab	3+1	
		CE408, CE408L	Cloud and Distributed Computing + Lab	3+1	
		CE436, CE436L	Digital System Design + Lab	3+1	
		CE442, CE442L	Computational Methods and Techniques + Lab	3+1	
	Computer Science	CS112, CS112L	Object-Oriented Programming + Lab	3+1	28
		CS231	Discrete Mathematics	3	
		CS221, CS221L	Data Structure and Algorithms + Lab	3+1	
		CS232, CS232L	Database Management Systems + Lab	3+1	
		CS211, CS211L	Operating Systems + Lab	3+1	
		CS425	Software Architecture and Engineering	3	
		CS481	Senior Year Design Project (Part-I)	0+3	
		CS482	Senior Year Design Project (Part-II)	0+3	
Electives	Management Science Courses (Choose any ONE course)	MS426	Technology Management	3	3
		CS436	Operation Research	3	
		CS492	Network Security & Cyber Ethics	3	
		MS492	Operations Management	3	
		MS493	Industrial Safety	3	
		MS494	Total Quality Management	3	
		MS496	Project Management	3	

Electives		MS497	Procurement Management	3	
	CE Domain Specialization Breadth Courses (Choose any TWO courses)	CS312	System Programming	3+1	8
		CS318	Introduction to Digital Image Processing	3+1	
		CE339	Introduction to Data Science	3+1	
		CE351	Linear Control Systems	3+1	
		CS351	Artificial Intelligence	3+1	
		CE361	Communication Theory	3+1	
	CE Domain Specialization Depth Courses (Choose any ONE course)	CE421	Advanced Computer Architecture	3	3
		CE423	General Processing with GPU	3	
		CE432	Digital Communication	3	
		CE453	Robotic Vision	3	
		CE465	Introduction to ASIC Design	3	
		CE475	Real Time Embedded Systems	3	
		CS323	Object-oriented Analysis and Design	3	9
		CS326	Mobile Computing	3	
		CS352	Introduction to Soft Computing	3	
		CS411	Block Chain	3	
	CS413	Internet of Things	3		
	CS414	Geographic Information Systems	3		
	CS416	Introduction to Deep Learning	3		
	CS417	Parallel Processing and Distributed Computing	3		
	CS419	Applied Image Processing	3		
	CS420	Cyber Security	3		
	CS423	Development Operations	3		
	CS425	Design Patterns	3		
	CS432	Advanced Databases	3		
	CS437	Data Warehousing and Data Mining	3		
	CS438	Web Mining and Social Media Analysis	3		
	CS439	Data Science	3		
	CS463	Web Engineering	3		
	CS472	Bio-inspired Computing	3		
	CS474	Bioinformatics	3		
	CS476	Computational Biology	3		
Degree total		42 courses – 114 theory credits – 25 lab credits – 136 credits			

(c) Degree Requirements for BS in Computer Science (134 Credit Hours)

For obtaining Bachelor of Science degree in Computer Science, a student has to complete 134 credit hours with a CGPA of 2.0 or above.

In addition, every BS (CS) student is required to participate in a summer training program and submit a formal written report during the summer of Junior Year.

Moreover, every BS (CS) student will undertake a 15-hour community service activity as a degree requirement using the summers or fall semester of their Sophomore Year.

Course requirements for obtaining BS (CS) degree offered by FCS&E are given below.

Course Group (HEC Category)		Course Code	Course Title	Credit Hours	Total Credits
	Computing	CS101, CS101L	Computing and AI + Lab	2+1	6
		CS121	Computer Science	3	
		IF101	Innovation and Makers Lab I	0+1	
		IF102	Innovation and Makers Lab II	0+1	
	English Language	HM101	Communication Skills	1+2	6
	Humanities	HM211	Pakistan & Islamic Studies	3	12
		MS291	Engineering Economics	3	
		CS391	Entrepreneurship and Technology Commercialization	3	
		HM425	Intellectual Property and CyberLaw	3	
	Mathematics	ES111	Probability and Statistics	3	20
		MT101	Calculus I	3	
		MT102	Differential Equations and Linear Algebra I	3	
		CH101	Applied Chemistry and Environment	3	
		PH101, PH101L	Applied Physics + Lab	3+1	
		MM101, MM141	Materials and Nanotechnology + Lab	2+1	
		CH161	Occupational Health and Safety Lab	0+1	
	Sciences	PH103	Fundamentals of Mechanics	2	4
		PH104	Fundamentals of Electricity and Magnetism	2	

	Computing	CS231	Discrete Mathematics	3	35
		CS221, CS221L	Data Structure and Algorithms + Lab	3+1	
		CE221, CE221L	Logic Design + Lab	3+1	
		CS225/ CS325	Software Engineering	3	
		CS232, CS232L	Database Management Systems + Lab	3+1	
		CS311, CS311L	Operating System + Lab	3+1	
		CE313, CE313L	Computer Communications & Networks + Lab	3+1	
		CS464	Data & Network Security	3	
		CS481	Senior Year Design Project (Part-I)	0+3	
	Computer Science Domain Specific	CS112, CS112L	Object-Oriented Programming + Lab	3+1	21
		CE222, CE222L	Computer Organization & Assembly Language + Lab	3+1	
		CS224	Formal Languages & Automata Theory	3	
		CS342	Numerical Analysis & Computational Methods	3	
		CS424, CS424L	Compiler Construction + Lab	3+1	
		CS478	Design & Analysis of Algorithms	3	
	Computer Science Breadth	CS312, CS312L	Systems Programming + Lab	3+1	15
		CS324	Computer Architecture	3	
		CS351, CS351L	Artificial Intelligence + Lab	3+1	
		CS417, CS417L	Parallel Processing and Distributed Computing + Lab	3+1	
	Management Science Courses (Choose any ONE course)	MS426	Technology Management	3	6
		CS436	Operation Research	3	
		CS492	Network Security & Cyber Ethics	3	
		MS492	Operations Management	3	
		MS493	Industrial Safety	3	
		MS494	Total Quality Management	3	
		MS496	Project Management	3	
		MS497	Procurement Management	3	

Electives	CS Domain Breadth Courses (Choose any ONE course)	CS323	Object-Oriented Analysis & Design	3	3
		CS326	Mobile Computing	3	
		CS327	Software Engineering II	3	
		CS329	Fundamentals of Cyber security	3	
		CS352	Introduction to Soft Computing	3	
	CS Domain Specialization Courses (Choose any THREE courses)	CS411	Block chain	3	9
		CS412	Information Retrieval	3	
		CS413	Internet of Things	3	
		CS414	Geographic Information Systems	3	
		CS416	Introduction to Deep Learning	3	
		CS418	Digital Image Processing	3	
		CS419	Applied Image Processing	3	
		AI420	Medical Image Processing	3	
		CE421	Advanced Computer Architecture	3	
		CS421	Human Computer Interaction	3	
		CS422	Professional Issues in IT	3	
		CE423	General Purpose Computing with GPU	3	
		CS423	Development Operations	3	
		CS425	Design Patterns	3	
		CS426	Software Testing and Quality Engineering	3	
		CS427	Software Design and Architecture	3	
		CS428	Design of Programming Language	3	
		CS429	Software Project Management	3	
		CS432	Advanced Databases	3	
		CS433	Computer Graphics	3	
		CS435	MIS & DSS	3	
		CS437	Data Warehousing and Data Mining	3	
		CS438	Web Mining and Social Media Analysis	3	
		CS439	Data Science	3	
		CS452	Artificial Neural Networks	3	
		CE453	Robotic Vision	3	
		CS454	Real-Time Programming	3	
		CS458	Distributed Systems	3	
		CS463	Web Engineering	3	
	CS472	Bio-Inspired Computing	3		
	CS474	Bio-Informatics	3		
	CS476	Computational Biology	3		
Degree total		41 courses – 115 theory credits – 19 lab credits – 134 credits			

(d) Degree Requirements for BS in Data Science (134 Credit Hours)

For obtaining Bachelor of Science degree in Data Science, a student has to complete 134 credit hours with a CGPA of 2.0 or above.

In addition, every BS (DS) student is required to participate in a summer training program and submit a formal written report during the summer of Junior Year.

Moreover, every BS (DS) student will undertake a 15-hour community service activity as a degree requirement using the summers or fall semester of their Sophomore Year.

Course requirements for obtaining BS (DS) degree offered by FCS&E are given below.

Course Group (HEC Category)		Course Code	Course Title	Credit Hours	Total Credits
General Education Requirements	Computing	CS103, CS103L	Computer Programming + Lab	2+1	6
		DS221	Information and Communication Technologies	2	
	English Language	HM101	Communication Skills	1+2	6
		HM102	Critical Thinking and Expository Writing	2+1	
		HM304	Advance Communication Skills	3	
	Humanities & Management	HM211	Pakistan & Islamic Studies	3	12
		MS291	Engineering Economics	3	
		CS391	Entrepreneurship and Technology Commercialization	3	
		HM425	Intellectual Property and CyberLaw	3	
	Mathematics & Physics	ES111	Probability and Statistics	3	20
		MT101	Calculus I	3	
		MT102	Differential Equations and Linear Algebra I	3	
		CH101	Applied Chemistry and Environment	3	
		PH101, PH101L	Applied Physics + Lab	3+1	
		MM101, MM141	Materials and Nanotechnology + Lab	2+1	
		CH161	Occupational Health and Safety Lab	0+1	
Core Requirements	Computing	CE213, CE213L	Computer Networks & Communications + Lab	3+1	35
		CE121, CE121L	Fundamentals of Logic Design + Lab	3+1	
		CS231	Discrete Mathematics	3	
		CS221, CS221L	Data Structure and Algorithms + Lab	3+1	
		CS232, CS232L	Database Systems + Lab	3+1	
		CS311, CS311L	Operating System + Lab	3+1	

		CS325	Software Engineering	3	
		CS464	Data & Network Security	3	
		CS481	Senior Year Design Project (Part-I)	0+3	
		CS482	Senior Year Design Project (Part-II)	0+3	
	Computer Science Domain Specific	CS112, CS112L	Object-Oriented Programming + Lab	3+1	18
		CE222, CE222L	Computer Organization & Assembly Language + Lab	3+1	
		CS251, CS251L	Artificial Intelligence	3+1	
		CS478	Design & Analysis of Algorithms	3	
		CS417	Parallel Processing & Distributed Computing	3	
	Data Science Domain Specific	DS211	Theory of Data Science	2	21
		DS221	Inferential Statistics & Applied Probability	3	
		DS331, DS331L	Data ware Housing & Business Intelligence	3+1	
		DS341, DS341L	Data Mining	2+1	
		DS351, DS351L	Data Visualization	2+1	
		DS361, DS361L	Big Data Analytics	3+1	
		DS301L	Problem Based Learning for Data Science Lab	0+1	
Electives	Management Science Courses (Choose any ONE course)	MS426	Technology Management	3	3
		CS436	Operation Research	3	
		CS492	Network Security & Cyber Ethics	3	
		MS492	Operations Management	3	
		MS493	Industrial Safety	3	
		MS494	Total Quality Management	3	
		MS496	Project Management	3	
		MS497	Procurement Management	3	
	DS Domain Breadth Courses (Choose any ONE course)	CS328	Introduction to Development Operations	3	6
		CS326	Mobile Computing	3	
		CE318	Introduction to Digital Image Processing	3	
		CS329	Fundamentals of Cyber security	3	
		CS352	Introduction to Soft Computing	3	

DS Domain Specialization Courses (Choose any THREE courses)	CS411	Block chain	3	9	
	CS412	Information Retrieval	3		
	CS413	Internet of Things	3		
	CS414	Geographic Information Systems	3		
	CS416	Introduction to Deep Learning	3		
	CS419	Applied Image Processing	3		
	AI420	Medical Image Processing	3		
Electives		CE421	Advanced Computer Architecture	3	
		CS421	Human Computer Interaction	3	
		CS422	Professional Issues in IT	3	
		CE423	General Purpose Computing with GPU	3	
		CS425	Design Patterns	3	
		CS426	Software Testing and Quality Engineering	3	
		CS427	Software Design and Architecture	3	
		CS428	Design of Programming Language	3	
		CS429	Software Project Management	3	
		CS432	Advanced Databases	3	
		CS433	Computer Graphics	3	
		CS435	MIS & DSS	3	
		CS438	Web Mining and Social Media Analysis	3	
		CS452	Artificial Neural Networks	3	
		CE453	Robotic Vision	3	
		CS454	Real-Time Programming	3	
		CS458	Distributed Systems	3	
		CS463	Web Engineering	3	
		CS472	Bio-Inspired Computing	3	
		CS474	Bio-Informatics	3	
		CS476	Computational Biology	3	
		Degree total		41 courses – 112 theory credits – 22 lab credits – 134 credits	



Semester-Wise Breakdown For Bachelor Programs

a. BS in Artificial Intelligence, Powered by Huawei (136 Credit Hours)

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
1st Semester	CH161	Occupational Health and Safety	1	0	1	None	None
	CS101	Computing and AI	2	0	2	None	CS1XXL
	CS101L	Computing and AI Lab	0	3	1	None	None
	HM101	Communication Skills	1	2	3	None	None
	IF101	Innovation and Makers Lab I	0	3	1	None	None
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	PH101L
	PH101L	Applied Physics Lab	0	3	1	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
2nd Semester	CS112	Object Oriented Programming and Design	3	0	3	CS1XX	None
	CS112L	Object Oriented Programming and Design Lab	0	3	1	CS1XXL	CS1XX
	HM102	Critical Thinking and Expository Writing	3	0	3	-	None
	MM101	Materials and Nanotechnology	2	0	2	None	None
	MM141	Materials Lab	0	3	1	None	MM1XX
	MT102	Differential Equations and Linear Algebra I	3	0	3	MT101	None
	ES111	Probability and Statistics	3	0	3	MT101	None
	IF102	Innovation and Makers Lab II	0	3	1	-	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
3rd Semester	CS203	Computer Science	3	0	3	None	None
	ES205	Advanced Linear Algebra	3	0	3	None	None
	CS231	Discrete Mathematics	3	0	3	None	None
	CE221	Logic Design	3	0	3	None	None
	CE221L	Logic Design Lab	0	3	1	None	CE2XX
	CS221	Data Structures and Algorithms	3	0	3	CS101	None
	CS221L	Data Structures and Algorithms Lab	0	3	1	None	CS221
	AI211L	Programming for AI	0	3	1	None	None

4th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	AI202	Trends and Techniques in AI	2	0	2	None	None
	DS221	Inferential Statistics and Applied Probability	3	0	3	ES1XX	None
	HM211	Pakistan and Islamic Studies	3	0	3	None	None
	CS232	Database Management Systems	3	0	3	CS1XX	None
	CS232L	Database Management Systems Lab	0	3	1	None	CS232
	CE222	Computer Organization and Assembly Language	3	0	3	CE2XX	None
	CE222L	Computer Organization and Assembly Language Lab	0	3	1	None	CE222
	CS205L	Computer Tools	0	3	1	None	None

5th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CS311	Operating Systems	3	0	3	CS221	None
	CS311L	Operating Systems Lab	0	3	1	None	CS311
	AI302	Knowledge Representation and Problem Solving	2	0	2	CS221	None
	AI341	Deep Neural Networks	3	0	3	None	None
	AI341L	Deep Neural Networks Lab	0	3	1	None	AI341
	AI351	Machine Learning	3	0	3	MT1XX	None
	AI351L	Machine Learning Lab	0	3	1	None	AI351
	MS391	Engineering Economics	3	0	3	None	None

6th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	AI361	Natural Language Processing	3	0	3	AI341	None
	AI361L	Natural Language Processing Lab	0	3	1	None	AI361
	AI319	Computer Vision	3	0	3	AI341	None
	AI319L	Computer Vision Lab	0	3	1	None	AI319
	AI3xx	AI Specialization Elective (Breadth)	3	0	3	None	None
	HM304	Advanced Communication Skills	3	0	3	None	None
	CS325	Software Engineering	3	0	3	CS221	None
	AI305L	Associate in AI Training Lab	0	3	1	AI341	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
7th Semester	AI4xx	AI Specialization Elective (Depth) I	3	0	3	None	None
	CE413	Computer Communications and Networks	3	0	3	None	None
	CE413L	Computer Communications and Networks Lab	0	3	1	None	CS413
	CS417	Parallel Processing and Distributed Computing	3	0	3	CS311	None
	CS478	Design and Analysis of Algorithms	3	0	3	CS221	None
	CS481	Senior Design Project (Part I)	0	9	3	None	None
	AI406L	Professional in AI Training Lab	0	3	1	AI305L	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
8th Semester	AI4xx	AI Specialization Elective (Depth) II	3	0	3	None	None
	CS464	Data and Network Security	3	0	3	CS221	None
	CS482	Senior Design Project (Part II)	0	9	3	None	None
	AI4xx	AI Specialization Elective (Depth) III	3	0	3	None	None
	HM425	Intellectual Property and CyberLaw	2	0	2	None	None
	AI407L	Artificial Intelligence Capstone Lab	0	3	1	None	None
	AI305L	Associate in AI Training Lab	0	3	1	AI341	-

† One course from the pool of following courses

- Computer Architecture,
- Deep Reinforcement Learning & Control,
- Development Operations,
- Nature-inspired Computing.

** means that the pre-requisites or the co-requisites may vary depending on the electives.

b. BS in Computer Engineering (136 Credit Hours)

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
1st Semester	CH161	Occupational Health and Safety	1	0	1	None	None
	CS101	Computing and AI	2	0	2	None	CS1XXL
	CS101L	Computing and AI Lab	0	3	1	None	None
	HM101	Communication Skills	1	2	3	None	None
	IF101	Innovation and Makers Lab I	0	3	1	None	None
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	PH101L
	PH101L	Applied Physics Lab	0	3	1	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
2nd Semester	CS112	Object Oriented Programming and Design	3	0	3	CS1XX	None
	CS112L	Object Oriented Programming and Design Lab	0	3	1	CS1XXL	CS1XX
	HM102	Critical Thinking and Expository Writing	3	0	3	-	None
	MM101	Materials and Nanotechnology	2	0	2	None	None
	MM141	Materials Lab	0	3	1	None	MM1XX
	MT102	Differential Equations and Linear Algebra I	3	0	3	MT101	None
	ES111	Probability and Statistics	3	0	3	MT101	None
	IF102	Innovation and Makers Lab II	0	3	1	-	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
3rd Semester	DS221	Information and Communication Technologies	2	0	2	None	None
	CE211	Circuit Analysis	3	0	3	MT101	None
	CE211L	Circuit Analysis Lab	0	3	1	None	CE211
	CS231	Discrete Mathematics	3	0	3	None	None
	ES205	Advanced Linear Algebra	3	0	3	MT102	None
	CS221	Data Structures and Algorithms	3	0	3	CS101	None
	CS221L	Data Structures and Algorithms lab	0	3	1	None	CS221
	CE222L	Computer Organization & Assembly Language Lab	0	3	1	-	CE222

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
4th Semester	CS211	Operating Systems	3	0	3	CS221	None
	CS211L	Operating Systems Lab	0	3	1	None	CS211
	CE221	Logic Design	3	0	3	None	None
	CE221L	Logic Design Lab	0	3	1	None	CE121
	CE231	Electronics – I	3	0	3	CE211	None
	CE231L	Electronics – I Lab	0	3	1	None	CE231
	CS232	Database Management Systems	3	0	3	CS231 (Discrete Mathematics)	None
	CS232L	Database Management Systems Lab	0	3	1	None	CS232
	HM211	Pakistan and Islamic Studies	3	0	3	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
5th Semester	CE322	Computer Organization and Assembly Language	3	0	3	CE221 (Logic Design)	None
	CE322L	Computer Organization and Assembly Language Lab	0	3	1	None	CE222
	CE313	Computer Communications and Networks	3	0	3	None	None
	CE313L	Computer Communications and Networks Lab	0	3	1	None	CE313
	MT203	Complex Variables & Transforms	3	0	3	MT201 (Diff. Eq. & Algebra)	None
	CE341	Signals & Systems	3	0	3	MT1XX (Diff. Eq. & Algebra)	None
	CE341L	Signals & Systems Lab	0	3	1	None	CE341
	MS391	Engineering Economics	3	0	3	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
6th Semester	CS325	Software Engineering	3	0	3	CS221	None
	CE3xx †	Specialization Elective I (Depth)	3	0	3	None	None
	CE3xxL †	Specialization Elective I Lab (Depth)	0	3	1	None	None
	CE3xx†	Specialization Elective II (Depth)	3	0	3	None	None
	CE3xxL†	Specialization Elective II Lab (Depth)	0	3	1	None	None
	CE324	Microprocessor Interfacing	3	0	3	CE3XX (COAL)	None
	CE324L	Microprocessor Interfacing Lab	0	3	1	None	CE324
	HM322	Corporate Law and Professional Practices	3	0	3	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
7th Semester	CS408	Cloud and Distributed Computing	3	0	3	CE313	None
	CS408L	Cloud and Distributed Computing Lab	0	3	1	None	CE408
	CE442	Computational Methods and Techniques + Lab	3	1	3	CS221	None
	XX4XX	IDE Elective	3	0	3	None	None
	CS481	Senior Design Project (Part-I)	0	9	3	None	None
	CE461	Digital Signal Processing	3	0	3	CS341	None
	CE461L	Digital Signal Processing Lab	0	3	1	None	CE361

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
8th Semester	CE436	Digital System Design	3	0	3	CE324	None
	CE436L	Digital System Design Lab	0	3	1	None	CE436
	CE4xx #	Specialization Elective III (Depth)	3	0	3	None	None
	CS482	Senior Design Project (Part-II)	0	9	3	None	None
	MS4xx	Management Elective	3	0	3	None	None

† Two courses and their labs from the pool of following courses

- CS312 – System Programming
- CS318 – Digital Image Processing
- CE339 – Data Science
- CS351 – Artificial Intelligence
- CE361 – Communication Theory
- EE351 – Linear Control Systems
- EE361 – Communication Systems

* Two courses from the pool of listed courses in Inter-Disciplinary Engineering Electives.

One course from the pool of listed courses in CE Domain Specialization Depth Courses.

** means that the pre-requisites or the co-requisites may vary depending on the electives.

c. BS in Computer Science (134 Credit Hours)

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
1st Semester	CH161	Occupational Health and Safety	1	0	1	None	None
	CS101	Computing and AI	2	0	2	None	CS1XXL
	CS101L	Computing and AI Lab	0	3	1	None	None
	HM101	Communication Skills	1	2	3	None	None
	IF101	Innovation and Makers Lab I	0	3	1	None	None
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	PH101L
	PH101L	Applied Physics Lab	0	3	1	None	None



2nd Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CS112	Object Oriented Programming and Design	3	0	3	CS1XX	None
	CS112L	Object Oriented Programming and Design Lab	0	3	1	CS1XXL	CS1XX
	HM102	Critical Thinking and Expository Writing	3	0	3	-	None
	MM101	Materials and Nanotechnology	2	0	2	None	None
	MM141	Materials Lab	0	3	1	None	MM1XX
	MT102	Differential Equations and Linear Algebra I	3	0	3	MT101	None
	ES111	Probability and Statistics	3	0	3	MT101	None
	IF102	Innovation and Makers Lab II	0	3	1	-	None

3rd Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CS121	Computer Science	3	0	3	None	None
	MS291	Engineering Economics	3	0	3	None	None
	CS231	Discrete Mathematics	3	0	3	None	None
	MT102	Calculus II	3	0	3	MT1XX	None
	CS221	Data Structures and Algorithms	3	0	3	CS1XX	None
	CS221L	Data Structures and Algorithms lab	0	3	1	CS1XX	CS221
	CS221L	Data Structure and Algorithms Lab	0	3	1	-	CS221

4th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	HM211	Pakistan and Islamic Studies	3	0	3	None	None
	CS224	Formal Languages & Automata Theory	3	0	3	CS2XX	None
	CS211	Operating Systems	3	0	3	CS221	None
	CS232	Database Management Systems	3	0	3	CS1XX	None
	CE221	Logic Design	3	0	3	None	None
	CS211L	Operating Systems Lab	0	3	1	None	CS2XX
	CS232L	Database Management Systems Lab	0	3	1	None	CS232
	CE221L	Logic Design Lab	0	3	1	None	CE221

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
5th Semester	CE313	Comp. Networks & Communication	3	0	3	CS2XX	None
	ES304	Linear Algebra – II	3	0	3	MT2XX	None
	CE222	Computer Organization & Assembly Language	3	0	3	CE221	None
	CS351	Artificial Intelligence	3	0	3	CS221	None
	HM321	Sociology and Human Behavior	3	0	3	None	None
	CE313L	Computer Networks & Communication Lab	0	3	1	None	CE313
	CE222L	Computer Organization & Assembly Language Lab	0	3	1	None	CE222
	CS351L	Artificial Intelligence Lab	0	3	1	None	CS351

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
6th Semester	CS342	Numerical Analysis & Computational Methods	3	0	3	MT2XX	None
	CS325	Software Engineering	3	0	3	CS221	None
	HM322	Corporate Law and Professional Ethics	3	0	3	None	None
	CS3xx	Computer Science Elective I (Breadth)	3	0	3	None	None
	CS417	Parallel Processing and Distributed Computing	3	0	3	CS2XX	None
	CE313L	Computer Communication and Networking Lab	0	3	1	-	CE313
	CS312L	Systems Programming Lab	0	3	1	-	CS312
	CS351L	Artificial Intelligence Lab	0	3	1	-	CS351

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
7th Semester	MS4xx	Management Science Elective	3	0	3	None	None
	CS478	Design and Analysis of Algorithms	3	0	3	CS221	None
	CS4xx	CS Elective – II	3	0	3	None	None
	CS4xx	CS Elective – III	3	0	3	None	None
	CS481	Senior Design Project (Part-I)	0	9	3	None	None
	CS417L	Parallel Processing Lab	0	3	1	-	CS417

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
8th Semester	CS464	Data and Network Security	3	0	3	CS2XX	None
	CS424	Compiler Construction	3	0	3	CS224	None
	CS4xx	CS Elective – IV	3	0	3	None	None
	CS4xx	CS Elective – V	3	0	3	None	None
	CS482	Senior Design Project (Part-II)	0	9	3	None	None
	CS424L	Compiler Construction Lab	0	3	1	None	CS424

† One course from the pool of following courses

- Introduction to Soft Computing,
- Mobile Computing,
- Object-oriented Analysis and Design,
- Software Engineering II,
- Fundamentals of Cyber Security

** means that the pre-requisites or the co-requisites may vary depending on the electives.

d. BS in Data Science (135 Credit Hours)

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
1st Semester	CH161	Occupational Health and Safety	1	0	1	None	None
	CS101	Computing and AI	2	0	2	None	CS1XXL
	CS101L	Computing and AI Lab	0	3	1	None	None
	HM101	Communication Skills	1	2	3	None	None
	IF101	Innovation and Makers Lab I	0	3	1	None	None
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	PH101L
	PH101L	Applied Physics Lab	0	3	1	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
2nd Semester	CS112	Object Oriented Programming and Design	3	0	3	CS1XX	None
	CS112L	Object Oriented Programming and Design Lab	0	3	1	CS1XXL	CS1XX
	HM102	Critical Thinking and Expository Writing	3	0	3	-	None
	MM101	Materials and Nanotechnology	2	0	2	None	None
	MM141	Materials Lab	0	3	1	None	MM1XX
	MT102	Differential Equations and Linear Algebra I	3	0	3	MT101	None
	ES111	Probability and Statistics	3	0	3	MT101	None
	IF102	Innovation and Makers Lab II	0	3	1	-	None

3rd Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	DS221	Information and Communication Technologies	2	0	2	None	None
	HM211	Pakistan and Islamic Studies	3	0	3	None	None
	CS231	Discrete Mathematics	3	0	3	None	None
	CE221	Logic Design	3	0	3	None	None
	CE221L	Logic Design Lab	0	3	1	None	CE221
	CS221	Data Structures and Algorithms	3	0	3	CS101	None
	CS221L	Data Structures and Algorithms Lab	0	3	1	None	CS221
	CS232L	Introduction to Databases Lab	0	3	1	-	CS232

4th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	DS221	Inferential Statistics and Applied Probability	3	0	3	ES1XX	None
	DS211	Theory of Data Science	2	0	2	CS1XX	None
	CE222	Computer Organization and Assembly Language	3	0	3	CE221	None
	CE222L	Computer Organization and Assembly Language Lab	0	3	1	None	CE222
	CS232	Database Management Systems	3	0	3	CS1XX	None
	CS232L	Database Management Systems Lab	0	3	1	None	CS232
	CS251	Artificial Intelligence	3	0	3	CS221	None
	CS251L	Artificial Intelligence Lab	0	3	1	None	CS251

5th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CS311	Operating Systems	3	0	3	CS221	None
	CS311L	Operating Systems Lab	0	3	1	None	CS311
	CE3XX	Computer Communications and Networks	3	0	3	None	None
	CE3XXL	Computer Communications and Networks Lab	0	3	1	None	CE3XX
	DS331	Data Warehousing & Business Intelligence	3	0	3	DS211	None
	DS331L	Data Warehousing & Business Intelligence Lab	0	3	1	None	DS311
	ES304	Advanced Linear Algebra	3	0	3	MT1XX	None
	HM321	Sociology and Human Behavior	3	0	3	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
6th Semester	CS378	Design & Analysis of Algorithms	3	0	3	CS221	None
	DS341	Data Mining	2	0	2	DS221	None
	DS341L	Data Mining Lab	0	3	1	None	DS341
	DS325	Software Engineering	2	0	2	CS221	None
	DS3xx	DS Elective (Breadth)	3	0	3	None	None
	HM322	Corporate Law and Professional Ethics	3	0	3	None	None
	CS417	Parallel Processing and Distributed Computing	3	0	3	CS311	None
	CS3XX	Development Lab	0	3	1	None	CS221

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
7th Semester	DS4XX	Data Visualization	2	0	2	DS341	None
	DS4XXL	Data Visualization Lab	0	3	1	None	DS4XX
	CS481	Senior Design Project (Part-I)	0	9	3	None	None
	DS461	Big Data Analytics	3	0	3	DS221	None
	DS461L	Big Data Analytics Lab	0	3	1	None	DS461
	DS4xx	DS Specialization Elective – I	3	0	3	None	None
	MS490	Engineering Economics and Management	3	0	3	None	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
8th Semester	CS464	Data & Network Security	3	0	3	CS311	None
	CS482	Senior Design Project (Part-II)	0	9	3	None	None
	DS4xx	DS Specialization Elective – II	3	0	3	None	None
	DS4xx	DS Specialization Elective – III	3	0	3	None	None
	MS4xx	Management Science Elective	3	0	3	None	None

† One course from the pool of following courses

- Introduction to Soft Computing,
- Mobile Computing,
- Introduction to Development Operations,
- Introduction to Digital Image Processing,
- Fundamentals of Cyber Security.

** means that the pre-requisites or the co-requisites may vary depending on the electives.

COURSE DESCRIPTIONS

CORE COURSES

CS101 Computing and Artificial Intelligence

(2 3 3): Computers and their applications, types of computers, history of computers, hardware and software, peripheral devices, data representation and conversion – Binary and Decimal number systems, ASCII/Unicode representation, microprocessors, primary and secondary memory, storage devices, networks, types of networks, algorithms, flowcharts, pseudocode, components of algorithms, formal specification of pseudocode, artificial intelligence – history, recent advancements, and applications, comments, assignment operators, simple and nested control structures, switch statements, repetition via loops, lists and arrays, passing arrays to functions, searching in arrays, pointers, function pointers, references, library functions, and Standard Template Library (STL).

Prerequisite(s): None

Core for: All faculties and departments

CS112 Object-Oriented Programming and Design (3 3 4):

User defined data types, structures, unions and enumerations, recursion, preprocessing in C++, bit manipulation, strings, pointers, reference and dynamic memory allocation, function pointers, ADTs and C++ classes, constructor, destructors, static data members and functions, constant data members and functions, copy constructor, inheritance, virtual functions and polymorphism, operator overloading, function and class templates, exception handling, I/O streams and file handling, graphic mode

programming, GUI programming, introduction to standard template library.

Prerequisite(s): CS101

Core for: All faculties and departments

CY201 Cyber Security Principles and Concepts (3 0 3):

Introduction to Cyber security; Networks and the Internet; cyber threat landscape; understanding security; information security Principles (Confidentiality, Integrity, Availability); Information Security Terminology; Who are the attackers; Advanced Persistent Threat (APT); Malware, types of malware; Attacks using malware; Malware Attack Lifecycle: Stages of Attack; Social engineering attacks; types of payload; Industrial Espionage in Cyberspace; Basic cryptography; Web application attacks; Database security; Cyber kill chain; Privacy and anonymity; Network security; Software security; Mobile device security; Mobile app security; Cyber Terrorism and Information Warfare; Introduction to Digital Forensics; Digital Forensics Categories.

Prerequisite(s): CS231

Core for: Cy Sec

AI202 Trends and Techniques in AI (3 0 3):

Introduction to Artificial Intelligence and its applications towards Knowledge Based Systems; Introduction to Reasoning and Knowledge Representation, Problem Solving by Searching – Informed searching, Uninformed searching, Heuristics, Local searching, Adversarial Search, Min-max algorithm, Alpha beta pruning, Game-playing; Case Studies: General Problem Solver, Eliza, Student, Macsyma; Learning from examples; Natural Language Processing; Recent trends in AI and applications, Prolog programming language to explore and illustrate various issues and techniques in Artificial Intelligence.

Prerequisite(s): None

Core for: AI

CS203 Computer Science (3 0 3): A bird-eye view of computer science, basics of computer organization and hardware, operating systems, networking and the Internet, algorithm development, software engineering, databases, use of computers in various domains, and recent and future trends in IT.

Prerequisite(s): None

Core for: AI, CS, Cy Sec



CS204 Information and Communication Technologies (2 0 2): A bird-eye view of computer science, basics of computer organization and hardware, operating systems, networking and the Internet, algorithm development, software engineering, databases, use of computers in various domains, and recent and future trends in IT.

Prerequisite(s): None

Core for: CE, DS

CS205L Computer Tools Lab (0 3 1): Use of command line: what is a shell?; how to use commands and their arguments to navigate the file system; manipulating files; IO redirection; pipes; environment variables; configuring their shell, etc. Debugging concepts and how to debug using print statements. Editing using command line editors (e.g., vim). Version control concepts and their different operations (commit, collaboration, pull, push, creating and pushing a new repository, cloning, branches, merging, and conflicts). Remotely accessing their computers using SSH and about public/private keys and about security in general.

Prerequisite(s): None

Core for: AI, Cy Sec

CS206L Intensive Programming Lab (0 3 1): IDE for the language (e.g., Jupyter Notebook or IPython), variables, expressions, operands and operators, loops, control structures, debugging, error messages, functions, strings, lists, object-oriented constructs and basic graphics in the language, writing production quality clean code in the programming language using version control (git and subversion), pertinent libraries necessary for interpreting, analyzing and plotting numerical data (e.g., NumPy, Matplotlib, Anaconda and Pandas for Python) and examples of each library using simple use cases and small case studies.

Prerequisite(s): CS112

Core for: CS

AI211L Programming for AI Lab (0 3 1): IDE for the language (e.g., Jupyter Notebook or IPython), variables, expressions, operands and operators, loops, control structures, debugging, error messages, functions, strings, lists, object-oriented constructs and basic graphics in the language, writing production quality clean code in the programming

language using version control (git and subversion), pertinent libraries necessary for interpreting, analyzing and plotting numerical data (e.g., NumPy, Matplotlib, Anaconda and Pandas for Python) and examples of each library using simple use cases and small case studies.

Prerequisite(s): CS112

Core for: AI

CE211 Circuit Analysis (3 3 4): Fundamentals of circuit analysis, Voltage, Current, Sources, Ohm's law, Develop methods and procedures (nodal/mesh analysis, network theorems) to resolve complex electric circuits, Solutions for resistive circuits followed by complex elements, e.g., capacitors, inductors and operational amplifiers, Circuits with DC sources, and Circuits with sinusoidal sources.

Prerequisite(s): MT101

Core for: CE

CS211 Operating Systems (3 3 4): History and goals, evolution of multi-user systems, process and CPU management, multithreading, kernel and user modes, protection, problems of cooperative processes, synchronization, deadlocks, memory management and virtual memory, relocation, fragmentation, paging and segmentation, secondary storage, security and protection, file systems, I/O systems, introduction to distributed operating systems, scheduling and dispatch, and introduction to concurrency.

Prerequisite(s): CS221

Core for: CE, CS

CY211 Information Security (3 0 3): Information security foundations, security design principles; security mechanisms,



symmetric and asymmetric cryptography, encryption, hash functions, digital signatures, key management, authentication and access control; software security, vulnerabilities and protections, malware, database security; network security, firewalls, intrusion detection; security policies, policy formation and enforcement, risk assessment, cybercrime, law and ethics in information security, privacy and anonymity of data.

Prerequisite(s): CS231

Core for: Cy Sec

DS211 Theory of Data Science (3 0 3):

Introduction to Big Data Analytics, Data Analytics Lifecycle, Advanced Analytical Theory and Methods: Association Rules, Regression, Classification, Time Series Analysis, Text Analysis, MapReduce and Hadoop, Setting up Python for Data Science, Cross validation and optimization, Linear Algebra, Statistics, Probability, Hypothesis and Inference.

Prerequisite(s): CS112

Core for: DS

CE213 Computer Communications (3 3 4):

Introduction to data communications, network topologies, LAN and WAN, OSI model of computer communications, communications media, data link layer, network layer, transport layer, TCP/IP protocols, switching and routing, and networking technologies.

Prerequisite(s): None

Core for: Cy Sec

CE221 Digital Logic Design (3 3 4): Deals with the basic concepts and tools used to design digital hardware consisting of both combinational and sequential logic circuits, Boolean algebra, logic gates, combinational logic design, sequential logic design, memory, programmable logic devices (PLDs), introduction to hardware description language (HDL) and their use to design the basic digital hardware.

Prerequisite(s): None

Core for: AI, CS, Cy Sec

CS221 Data Structures & Algorithms (3 3 4):

Introduction to data structures and algorithms, arrays, stacks, infix, postfix and prefix notations, recursion, backtracking, binary search, queues, linked lists, trees, graphs and operations, algorithm performance, complexity issues,

sorting algorithms, searching algorithms, hashing, dynamic memory management.

Prerequisite(s): CS112

Core for: AI, CE, CS, Cy Sec, DS

DS221 Inferential Statistics and Applied Probability (3 0 3):

Introduction to Statistics, Use of Statistics in Data Science, Experimental Design, Statistical Techniques for Forecasting, Interpolation/ Extrapolation, Introduction to Probability, Conditional Probability, Prior and Posterior Probability, Random number generation (RNG), Techniques for RNG, Correlation analysis, Chi Square Dependency tests, Diversity Index, Data Distributions Multivariate Distributions, Error estimation, Confidence Intervals, Linear transformations, Gradient Descent and Coordinate Descent, Likelihood inference, Revision of linear regression and likelihood inference, Fitting algorithms for nonlinear models and related diagnostics, Generalized linear model; exponential families; variance and link functions, Proportion and binary responses; logistic regression, Count data and Poisson responses; log-linear models, Overdispersion and quasi-likelihood; estimating functions, Mixed models, random effects, generalized additive models and penalized regression; Introduction to SPSS, Probability/ Correlation analysis/ Dependency tests/ Regression in SPSS.

Prerequisite(s): ES111

Core for: AI, DS

CE222 Computer Organization & Assembly Language (3 3 4):

Microprocessor bus structure: addressing, data and control, memory organization and structure (segmented and linear models), introduction to registers and flags, data movement, arithmetic and logic, program control, subroutines, stack and its operation, peripheral control interrupts, interfacing with high level languages, real-time applications. Objectives and perspectives of Assembly language, addressing modes, introduction to the assembler and debugger, manipulate and translate machine and assembly code, describe actions inside the processing chip, discuss operations performed by an instruction set, write documented programs, using an assembler of choice.

Prerequisite(s): CE221

Core for: AI, CS, Cy Sec, DS

CS224 Formal Languages and Automata Theory (3 0 3): Study of regular languages, regular expressions and finite state machines, deterministic and non-deterministic finite state machines, pushdown automata, context free grammar, Turing machines and applications of various kinds of finite state machines.

Prerequisite(s): CS231

Core for: CS

CE231 Electronics I (3 3 4): Introduction to basic electronics, semiconductor diode, diode applications, bipolar junction transistor, transistor configurations, DC biasing, field-effect transistor, BJT and FET small signal equivalent circuit models, design of BJT and FET amplifiers, and differential amplifiers.

Prerequisite(s): CE211

Core for: CE

CS231 Discrete Mathematics (3 0 3): Formal logic, quantifiers and predicates, tautologies, rules of inferences, proof techniques, mathematical induction, recurrence relations, set theory, counting, permutations and combinations, relations and functions, Boolean algebra, introduction to group theory and algorithms.

Prerequisite(s): None

Core for: AI, CE, CS, Cy Sec, DS

CS232 Database Management Systems (3 3 4): Introduction to databases, basic concepts and architecture, relational model, SQL, data manipulation, data definition language, methodology-driven conceptual, logical, physical database design, data modeling, entity-relationship diagrams, functional dependencies, normalization, relational database design, relational algebra, record storage and primary file organization, query processing and optimizations, transaction processing, and concurrency control.

Prerequisite(s): CS112

Core for: AI, CE, CS, Cy Sec, DS

CS251 Artificial Intelligence (3 3 4): Overview of artificial intelligence, issues and application, knowledge representation, searching techniques, pruning, heuristics, production systems, basic elements of Prolog language, expert systems, neural networks, robotics, etc.

Prerequisite(s): CS221

Core for: DS

AI302 Knowledge Representation and Problem Solving (2 0 2): Propositional Logic, First-order Logic, Horn Clauses, Description Logic, Reasoning using Description Logic, Forward and Backward Chaining in Inference Engines, Semantic Networks, Ontologies and Ontology Languages, Logical Agents, Planning, Rule-based Knowledge Representation, Reasoning Under Uncertainty, Bayesian Networks Representation, Inference in Bayesian Networks, Fuzzy Logic, Inference using Fuzzy Rules, Markov Models, Commonsense Reasoning, Explainable AI.

Prerequisite(s): CS221

Core for: AI

CS311 Operating Systems (3 3 4): History and goals, evolution of multi-user systems, process and CPU management, multithreading, kernel and user modes, protection, problems of cooperative processes, synchronization, deadlocks, memory management and virtual memory, relocation, fragmentation, paging and segmentation, secondary storage, security and protection, file systems, I/O systems, introduction to distributed operating systems, scheduling and dispatch, and introduction to concurrency.

Prerequisite(s): CS221

Core for: AI, Cy Sec, DS

CY312 Cryptography (3 0 3): Part I: Secret key cryptography (Introduction, Encryption, Stream ciphers, Block ciphers, Chosen plaintext attacks, Message integrity, Message integrity from universal hashing, Message integrity from collision resistant hashing, and Authenticated encryption). Part II: Public key cryptography



(Public key tools, Public key encryption, Chosen ciphertext secure public-key encryption, Digital signatures, Fast signatures from one-way functions, Elliptic curve cryptography and pairings, Post-quantum cryptography: lattices and isogenies, and Analysis of number theoretic assumptions). Part III: Protocols (Protocols for identification and login, Identification and signatures from sigma protocols, Proving properties in zero-knowledge, Modern proof systems, Authenticated key exchange, and Two-party and multi-party secure computation).

Prerequisite(s): MT102

Core for: Cy Sec

CE313 Computer Communications & Networks (3 3 4):

Introduction to data communications, network topologies, LAN and WAN, OSI model of computer communications, communications media, data link layer, network layer, transport layer, TCP/IP protocols, switching and routing, and networking technologies.

Prerequisite(s): None

Core for: CE, CS, DS

CS317 Parallel Processing and Distributed Computing (3 0 3):

High performance architectures and programming languages; graph concepts: control flow graph, dominance frontiers, data dependence in loops and parallel constructs; program dependence graph; loop transformations, inter-procedural transformations; concurrency analysis: synchronization, strength reduction, nested loops; vector analysis; message-passing machines; communicating sequential processes.

Prerequisite(s): CS211

Core for: CS, DS

AI319 Computer Vision (3 3 4):

Foundation of image formation, measurement, and analysis, geometric relationships between 2D images and the 3D world, object and scene recognition and categorization from images, fundamentals of image formation, camera imaging geometry, image filtering, feature detection and matching, stereo vision, motion estimation and tracking, and image classification and scene understanding.

Prerequisite(s): AI341

Core for: AI

CY321 Secure Software Design and Engineering (3 3 4):

Secure software concepts; System issues; System properties; Software Project Time Management; Software Project Costing; Software Quality Assurance; Security Concepts in the SDLC; Risk management; Security standards (e.g., coding standards, NIST standards, Federal Information Processing Standards); Best practices (e.g., OWASP development guide, OWASP code review guide, OWASP testing guide); Security methodologies (e.g., Socratic Methodology, Operationally Critical Threat, Asset, and Vulnerability Evaluation, STRIDE and DREAD, Open Source Security Testing Methodology Manual); Security frameworks (e.g., Zachman Framework, Control Objectives for Information and Related Technology, Sherwood Applied Business Security Architecture (SABSA)); Regulations, Privacy and Compliance; Security Models (e.g., BLP Confidentiality Model, Clark and Wilson Model (Access Triple Model)); Trusted Computing; Secure Software Requirements (Sources for Security Requirements, Types of Security Requirements); Secure Software Design (Design consideration, Information Technology Security Principles and Secure Design, Designing Secure Design Principles); Design Processes; Secure Software Implementation/Coding; Software Development Methodologies; Common Software Vulnerabilities and Controls; Defensive Coding Practices—Concepts and Techniques; Code Vulnerabilities and Avoiding Polymorphic Malware Attacks: Buffer overflow, Format string 92 bug, Code vulnerabilities SQL Injection, Cross-site Scripting, Cross-site Request Forgery, Session management, Replication of vulnerabilities and exploitation; Secure Software Testing; Security Testing Methodologies; Software Security Testing; Software Acceptance; Legal Protection Mechanisms; Software Deployment- Operations- Maintenance and Disposal.

Prerequisite(s): CS221

Core for: Cy Sec

CE322 Computer Organization & Assembly Language(334):

Microprocessor bus structure: addressing, data and control, memory organization and structure (segmented and linear models), introduction to registers and flags, data movement, arithmetic and logic, program control, subroutines, stack and its operation,

peripheral control interrupts, interfacing with high level languages, real-time applications. Objectives and perspectives of Assembly language, addressing modes, introduction to the assembler and debugger, manipulate and translate machine and assembly code, describe actions inside the processing chip, discuss operations performed by an instruction set, write documented programs, using an assembler of choice.

Prerequisite(s): CE221

Core for: CE, CS

CE324 Microprocessor Interfacing (3 3 4): Introduction to 16-bit microprocessor, software model, addressing modes, instruction set, assembly language programming, hardware model, read/write cycles, exception/interrupt processing, interfacing to ACIA, PIA, PI/T, DMA, A/D, D/A converters, introduction to micro-controllers and embedded systems. Prerequisite(s): CE222
Core for: CE

CS324 Computer Architecture (3 0 3): Fundamentals of computer design including performance measurements and quantitative principles, principles of Instruction Set Design, operands, addressing modes and encoding, pipelining of processors: issues and bottlenecks, exception handling features, instruction-level parallelism and dynamic handling of exceptions, memory hierarchy design, cache design, performance issues and improvements, main memory performance issues, storage systems, multiprocessors and thread level parallelism, case studies. Prerequisite(s): CE222
Core for: CE, CS
Specialization(s): AI

CS325 Software Engineering (3 0 3): Software development life cycle, software development processes, software requirement identification and specification, system analysis, software architecture, software design approaches: object-oriented and function-oriented, user interface design, program design techniques, software testing and maintenance, software technical metrics, introduction to quality assurance and project management. Hands-on Practice: using Microsoft Project, introduction to MS .Net framework, introduction to Windows form programming in

MS .Net, creating user interfaces in .Net, debugging applications, performance testing of applications.

Prerequisite(s): CS221

Core for: AI, CE, CS

CY331 Network Security and Information Assurance (3 3 4): Introduction to network security, Networking Concepts and Protocols, Network Threats and Vulnerabilities, Network Security Planning and Policy, Access Control, Defense against Network Attacks, DOS and DDOS detection and prevention, Firewalls, Intrusion Detection and Prevention Systems, Antivirus Filtering, Naming and DNS Security, DNSSEC, IP security, Secure Sockets Layer, VPN, Packet Sniffing and spoofing, Honeypot, Ethernet Security, Wireless Security, Wireless Attacks, Wireless LAN Security with 802.11i, Wireless Security Protocols, Wireless Intrusion Detection, Physical access and Security, Tor Network, Network Forensics. Defense against Network Attacks.

Introduction to (IS) Information System (Concept, Design, Functions, Architecture, Components and applications of IS); Secure System Planning and Administration; Information Security Policies and Procedures; Asset Management; Organizational and Human Security; Cyber Security Management Concepts; NIST Cyber Security Framework; Enterprise Roles and Structures; Strategic Planning; Security Plans and Policies; Contingency Planning; Laws; Laws and Regulatory Requirements; Security Standards and Controls, Risk Management Process, NIST Risk Management Framework, Security Metrics and Key Performance Indicators (KPIs); Physical Security and Environmental Events; Contingency Planning; Security Education, ISO 27001 Compliance, Training, and Awareness. Prerequisite(s): CY201

Core for: Cy Sec

DS331 Data Warehousing and Business Intelligence (3 3 4): Introduction to Data Warehouse and Business Intelligence; Necessities and essentials of Business Intelligence; DW Life Cycle and Basic Architecture; DW Architecture in SQL Server; Logical Model; Indexes; Physical Model; Optimizations; OLAP Operations, Queries and Query Optimization; Building the DW; Data visualization and reporting based on

Datawarehouse using SSAS and Tableau; Data visualization and reporting based on Cube; Reports and Dashboard management on PowerBI; Dashboard Enrichment; Business Intelligence Tools.

Prerequisite(s): DS211

Core for: DS

AI341 Deep Neural Networks (3 3 4):

Introduction to biological and artificial neurons, learning from data, artificial neural networks and non-linear activation functions, error back propagation and restricted boltzman machine algorithms, deep vs. shallow learning, data augmentation, theory of generalization, convolutional neural networks, recurrent neural networks, generative adversarial networks, deep unsupervised and reinforcement learning, parallel computing for AI (GPU computing, CuDNN etc.), and application areas of deep learning – speech recognition, images, vision, etc.

Prerequisite(s): ES205

Core for: AI

CE341 Signals & Systems (3 3 4): Introduction to continuous and discrete-time signals, concept of frequency, A/D and D/A conversion, sampling theorem, discrete-time systems: representation, classification and analysis, characteristics of LTI systems, convolution and correlation, z-transform, Fourier transform, Laplace transform, time and frequency domain analysis, introduction to digital filters.

Prerequisite(s): ES205

Core for: CE

CY341 Digital Forensics (3 3 4): An introduction to Digital Forensics; use of digital forensics; Key technical concepts; Challenges in Digital Forensics; The Difference between Computer Experts and Digital Forensics Experts; Investigative Process Methodologies; Education, Training, and Awareness; Laws, Standards, and Regulations; Ethics and Professional Conduct; Digital Evidence Management; Collecting evidence; Antiforensics; Network forensics; Mobile and Embedded Forensics; Cloud forensics; Internet Forensics; social media forensics; Investigation Methods for Collecting Digital Evidence; Digital Forensic Readiness; Digital forensics tools; Discovery of Computers and Storage Media; Discovery of Audio/ Video Evidence;

Data Visualization; Data Sources; Graphing and Charting; Analyzing Data; Data Distributions; Analysis Scenarios; Data Visualization Tools.

Prerequisite(s): CY201

Core for: Cy Sec

DS341 Data Mining (2 0 2): Introduction to data mining, related technologies – Machine learning, DBMS, OLAP, stages and techniques of data mining process, methods and applications of knowledge representation, data preprocessing, data cleaning, data transformation, data reduction, discretization, generating concept hierarchies, Weka 3 data mining system, filters, statistics and discretization in Weka, measures of interestingness, visualization techniques and experiments in Weka, attribute-oriented analysis, generalization, and relevance, class comparison, statistical generalization, and relevance, class comparison, statistical measures, algorithms and association rules, motivation and terminology, item sets, generating item sets and rules, correlation analysis, classification, basic learning/mining tasks, inferring rudimentary rules: 1R algorithm, decision trees and rules in Weka, statistical (Bayesian) classification, Bayesian networks, instance-based methods (nearest neighbor), linear models, training and testing in Weka, estimating classifier accuracy, combining multiple models (bagging, boosting, stacking), Minimum Description Length Principle (MLD), clustering, partitioning methods: k-means, expectation maximization (EM), hierarchical methods: distance-based agglomerative and divisible clustering, Cobweb, text mining: extracting attributes, structural approaches (parsing, soft parsing), Bayesian approach to classifying text, web mining: classifying web pages, extracting knowledge from the web, and data mining software and applications.

Prerequisite(s): None

Core for: DS

CS342 Numerical Analysis & Computational Methods (3 0 3):

Error and computer arithmetic, root finding for non-linear equation, interpolation and polynomial approximation, solution of system of linear equations, numerical differentiation and integration, numerical solution of ordinary differential equations.

Prerequisite(s): MT102

Core for: CS

AI351 Introduction to Machine Learning

(3 3 4): Introduction to machine learning; concept learning: General-to-specific ordering of hypotheses, Version spaces Algorithm, Candidate elimination algorithm; Supervised Learning: decision trees, Naive Bayes, Artificial Neural Networks, Support Vector Machines, Overfitting, noisy data, and pruning, Measuring Classifier Accuracy; Linear and Logistic regression; Unsupervised Learning: Hierarchical Agglomerative Clustering, k-means partitioned clustering; Self-Organizing Maps (SOM) k-Nearest-neighbor algorithm; Semi-supervised learning with EM using labeled and unlabeled data; Reinforcement Learning: Hidden Markov models, Monte Carlo some inference Exploration vs. Exploitation trade-offs, Markov Decision Processes, Ensemble Learning, Using committees of multiple hypotheses, Bagging, and Boosting.

Prerequisite(s): CS221

Core for: AI

CS351 Artificial Intelligence (3 3 4): Overview of artificial intelligence, issues and application, knowledge representation, searching techniques, pruning, heuristics, production systems, basic elements of Prolog language, expert systems, neural networks, robotics, etc.

Prerequisite(s): CS221

Core for: CS, Cy Sec

Specialization(s): CE

DS351 Data Visualization (2 3 3): Introduction of Exploratory Data Analysis and Visualization, Building Blocks and Basic Operations; Types of Exploratory Graphs, single and multi-dimensional summaries, five number summary, box plots, histogram, bar plot and others; Distributions, their representation using histograms, outliers, variance; Probability Mass Functions and their visualization; Cumulative distribution functions, percentile-based statistics, random numbers; Modelling distributions, exponential, normal, lognormal, pareto; Probability density functions, kernel density estimation; Relationship between variables, scatter plots, correlation, covariance; Estimation and Hypothesis Testing; Clustering using K-means and Hierarchical; Time series and survival analysis; Implementing concepts with R (or similar language).

Prerequisite(s): DS341

Core for: DS

AI361 Natural Language Processing (3 3 4):

Deterministic and stochastic grammars, parsing algorithms, CFGs, representing meaning/ semantics, semantic roles, temporal representations, corpus-based methods, N-grams and HMMs, smoothing and backoff, POS tagging and morphology, information retrieval, vector space model, precision and recall, information extraction, language translation, text classification, categorization, and bag of words model.

Prerequisite(s): CS221

Core for: AI

CS378 Design & Analysis of Algorithms

(3 0 3): Introduction, comparison sorting, integer sorting and selection; lower bounds, divide and conquer, master theorem, dynamic programming, graph representation, traversal, ordering, shortest paths, greedy algorithms, minimum spanning trees, string algorithms, amortized analysis, computational geometry, NP-completeness and approximation.

Prerequisite(s): CS221

Core for: DS

CY381, CY482 & CY483 Design Project – I, II & III (9 Credits) (0 27 9):

The aim of the course is to fine tune the general computing skills of the students in a specific area and exercise their communication skills. The projects chosen by students will be completed individually. It will allow students to choose a specific area of study of interest to them and to choose a method of working which is suited to their area of study. Therefore, some may adopt a research-oriented approach while others may concentrate on building specific systems to solve known problems.

Prerequisite(s): None

Core for: Cy Sec

CE408 Cloud and Distributed Computing

(3 0 3): Introduction to cloud computing, adopting the cloud, exploiting software as a service (SaaS), exploring the technical foundation for PaaS, building services with solution stacks, managing cloud storage, employing support services, deploying infrastructure as a service (IaaS), building a business case, migrating to the cloud.

Prerequisite(s): CS211

Core for: CE, Cy Sec

CE413 Computer Networks (3 0 3):

Introduction to data communications, network topologies, LAN and WAN, OSI model of computer communications, communications media, data link layer, network layer, transport layer, TCP/IP protocols, switching and routing, networking technologies.

Prerequisite(s): None

Core for: AI, CE

CS417 Parallel Processing and Distributed Computing (3 0 3):

High performance architectures and programming languages; graph concepts: control flow graph, dominance frontiers, data dependence in loops and parallel constructs; program dependence graph; loop transformations, inter-procedural transformations; concurrency analysis: synchronization, strength reduction, nested loops; vector analysis; message-passing machines; communicating sequential processes.

Prerequisite(s): CS311

Core for: AI

CS424 Compiler Construction (3 3 4):

Study and practical implementation of lexical analysis, syntax analysis using top-down and bottom-up approaches, LL, LR, and LALR parsers, semantic analysis using attributed grammars and dependency graphs, intermediate code generation using three address codes and code optimization.

Students are required to implement a small compiler using modern compiler writing tools.

Prerequisite(s): CS224

Core for: CS

Specialization(s): CE

CS425 Software Engineering (2 3 3):

Software development life cycle, software development processes, software requirement identification and specification, system analysis, software architecture, software design approaches: object-oriented and function-oriented, user interface design, program design techniques, software testing and maintenance, software technical metrics, introduction to quality assurance and project management. Hands-on Practice: using Microsoft Project, introduction to MS .Net framework, introduction to Windows form programming in MS .Net, creating user interfaces in .Net, debugging applications, performance testing of applications.

Prerequisite(s): CS112

Core for: DS

CE436 Digital Systems Design (3 0 3):

Arithmetic circuits – Data paths, arithmetic/logic unit (ALU), shifters; computer design fundamentals – introduction to computer design, Instructions set architecture (ISA); introductory HDL – 1 (Verilog or VHDL) – timing in Combinational Circuits, Hazards and Glitches,



Review of sequential logic; HDL Behavioural, Sequential coding, and ModelSim – design using flip-flop and latches, state machines, state reduction, timing issues, design of adders and subtractors, carry lookahead adders, serial adders, array multipliers, critical paths, Booth and Radix-4 encoded signed multipliers, further VHDL modeling, parameterization; FPGA Implementations – LFSR, BRM, Function Generators, Design Examples, Faults and Testability – BIST and SCAN techniques, Design for test – JTAG, Advanced HDL – Memories and Register Files, Design Examples

Prerequisite(s): CE324

Core for: CE

CE442 Computational Methods and Techniques (3 0 3):

Error analysis and interpolation, numerical differentiation and integration, methods of solution as a system of linear equations, iterative methods for linear and non-linear equations, numerical methods for IVPs and BVPs, numerical methods for computing eigenvalues, and numerical optimization.

Prerequisite(s): MT102

Core for: CE

CY451 Vulnerability Assessment (3 3 4):

Understanding the need for security assessments; Classifying vulnerabilities; Software vulnerabilities; Network vulnerabilities; Vulnerability assessment versus penetration testing; Vulnerability Assessment Tools; Vulnerability management Regulatory compliance; Calculating ROIs; Application review process; Pre-assessment; Code navigation; Codeauditing tactics; Memory corruption; understanding issues in programming languages; Steps in Reverse engineering, Common tools used for Reverse engineering; Binary Obfuscation techniques; Understanding core assembly concepts to perform malicious code analysis, Identifying key assembly logic structures with a disassembler, Malware analysis Types of malware analysis; Malware Taxonomy; Static analysis; Dynamic analysis; Malware Inspection; Malware analysis tools; Sandboxing and virtualization.

Prerequisite(s): CY201

Core for: Cy Sec

CE461 Digital Signal Processing (3 3 4):

An insight to the theory and application of DSP

and solid foundation in the basics of DSP related to both signal analysis and system analysis, analysis of design with some exposure to advanced topics in signal processing, concepts of three core areas of DSP: analysis, design and implementation, introduction to linear time-invariant systems and properties of linear time-invariant systems, analysis of linear constant-coefficient difference equations, frequency domain representation of discrete-time signals and systems, Matlab simulation exercises to understand the theories and concepts of discrete time forms of signals and systems, overview of various types of DSP processors, Fourier transforms, z-transform, sampling, filters, DFT and FFT, frequency domain representation of sampling and reconstruction of a band limited signal from its samples.

Prerequisite(s): CE341

Core for: CE

CY461 Penetration Testing (3 0 3):

Using a virtual machine-based applications that includes Kali Linux and vulnerable operating systems, you'll run through a series of practical lessons with tools like Wireshark, Nmap, and Burp Suite. As you follow along with the labs and launch attacks, you'll experience the key stages of an actual assessment—including information gathering, finding exploitable vulnerabilities, gaining access to systems, post exploitation, and more. The main contents are Crack passwords and wireless network keys with brute-forcing and wordlists, Test web applications for vulnerabilities, Use the Metasploit Framework to launch exploits and write your own Metasploit modules, Automate social-engineering attacks, Bypass antivirus software, and Turn access to one machine into total control of the enterprise in the post exploitation phase.

Prerequisite(s): CY451

Core for: Cy Sec

DS461 Big Data Analytics (3 3 4):

Introduction and Overview of Big Data Systems; Platforms for Big Data, Hadoop as a Platform, Hadoop Distributed File Systems (HDFS), MapReduce Framework, Resource Management in the cluster (YARN), Apache Scala Basic, Apache Scala Advances, Resilient Distributed Datasets (RDD), Apache Spark, Apache Spark SQL, Data analytics on Hadoop / Spark, Machine learning

on Hadoop / Spark, Spark Streaming, Other Components of Hadoop Ecosystem.

Prerequisite(s): DS221

Core for: DS

CS464 Data and Network Security (3 0 3):

Introduction, cryptology and simple cryptosystems, conventional encryption techniques, stream and block ciphers, DES, more on block ciphers, advanced encryption standard, confidentiality and message authentication: hash functions, number theory and algorithm complexity, public key encryption, RSA and discrete logarithms, elliptic curves, digital signatures, key management schemes, identification schemes, dial-up security, e-mail security, PGP, S-MIME, kerberos and directory authentication, emerging Internet security standards, SET, SSL and IPsec, VPNs, firewalls, viruses, & miscellaneous topics.

Prerequisite(s): CS311

Core for: AI, CS, DS

CS478 Design & Analysis of Algorithms (3 0 3):

Introduction, comparison sorting, integer sorting and selection; lower bounds, divide and conquer, master theorem, dynamic programming, graph representation, traversal, ordering, shortest paths, greedy algorithms, minimum spanning trees, string algorithms, amortized analysis, computational geometry, NP-completeness and approximation.

Prerequisite(s): CS221

Core for: AI, CS, Cy Sec

Specialization(s): CE

CS481 & CS482 Senior Design Project – I & II (6 Credits) (0 18 6):

The aim of the course is to fine tune the general computing skills of the

students in a specific area and exercise their communication skills. It will allow students to choose a specific area of study of interest to them and to choose a method of working which is suited to their area of study. Therefore, some may adopt a research-oriented approach while others may concentrate on building specific systems to solve known problems.

Prerequisite(s): None

Core for: AI, CE, CS, DS

Specialization Elective Courses

AI309 Deep Reinforcement Learning and Control (3 0 3):

Introduction to autonomous systems, self and reinforcement learning, core challenges and approaches to meet challenges, generalization and exploration, algorithms for control policies guided by reinforcement, demonstrations and intrinsic curiosity, evaluating complexity, generalization and generality of algorithms, examples of autonomous tasks, e.g., robotics, game playing, consumer modelling, healthcare, etc.

Prerequisite(s): AI202

Specialization(s): AI

CS312 Systems Programming (3 3 4):

Programming over Linux, gcc and associated tools, file I/O with low-level file descriptors, the standard I/O library, error reporting mechanisms, kernel statistics and parameter modifications, process creation and management system calls, signals and associated system calls, pipes, FIFOs, single & multiple reader/writers, semaphores, shared memory and message-queues, sockets, attributes and addressing schemes, multiple client connections, and connectionless socket communication.

Prerequisite(s): CS311

Specialization(s): CE, CS

CE317 Parallel Processing (3 3 4):

High performance architectures and programming languages; graph concepts: control flow graph, dominance frontiers, data dependence in loops and parallel constructs; program dependence graph; loop transformations, inter-procedural transformations; concurrency analysis: synchronization, strength reduction, nested loops; vector analysis; message-passing machines; communicating sequential processes.

Prerequisite(s): CS311



Specialization(s): CE

CE318 Digital Image Processing (3 0 3):

Introduction to digital image processing (DIP) and its applications, image sensing and acquisition, sampling and quantization; mathematical tools for DIP, intensity transformation, spatial filtering; discrete Fourier transform, filtering, image degradation and restoration; noise models, estimating degradation function, image reconstruction; color models, color image processing, smoothing, sharpening; image segmentation, morphological image processing, image compression, pattern recognition.

Prerequisite(s): CS221

Specialization(s): CE, DS

AI328 Development Operations (3 0 3):

Basics of Development Operations (DevOps), commonly used build tools, e.g., GIT and Jenkins, Build test automation, containerization using Docker, Docker commands and use cases, Kubernetes, Spinnaker, Skaffold, etc.

Prerequisite(s): CS112

Specialization(s): AI, DS

CS323 Object Oriented Analysis and Design (3 0 3):

Evolution of Object Oriented (OO) programming, OO concepts and principles, problem solving in OO paradigm, classes, methods, objects and encapsulation; constructors and destructors, operator and function overloading, virtual functions, derived classes, inheritance and polymorphism, I/O and file processing, exception handling, UML: conceptual model, use case diagrams, object models, class diagrams, system sequence diagram, object-oriented life cycle, modeling user interface requirements, designing and evaluating methods, synchronizing dependent attributes, normalizing classes with dependent data, design at the object, etc.

Prerequisite(s): CS221

Specialization(s): CS

CS326 Mobile Computing (3 0 3):

State-of-the-art of mobile computing platforms, introduction to mobile computing, architecture of android platform, using emulator, debugging and DDMS, content providers, App. networking, App. multimedia, App. 2D and 3D graphics, using sensors, publishing, designing Apps using XAML, introduction to

iPhone platform, iPhone supported development features and tools for developing mobile web applications.

Prerequisite(s): CS221

Specialization(s): CS, DS

CS327 Software Engineering II (303):

Product and process, objected oriented analysis, formal methods, algebraic approaches, verification, introduction to Z language and formal specification, function point analysis, refactoring, clean room software engineering, component bases software development, software re-engineering, architecture and estimation.

Prerequisite(s): CS325

Specialization(s): CS

CS329 Cyber Security (3 0 3):

Cybersecurity fundamentals, principles of data and technology that frame and define cybersecurity, importance of cybersecurity and role of cybersecurity professionals, cybersecurity principles, security architecture: processes and architecture, risk management, attacks, incidents: response, categories, and recovery, emerging IT and IS technology, mobile security issues, risks and vulnerabilities, cloud concepts around data and collaboration.

Prerequisite(s): CS221

Specialization(s): CS, DS

CE339 Data Science (3 0 3):

Introduction to Big Data Analytics, Data Analytics Lifecycle, Advanced Analytical Theory and Methods: Association Rules, Regression, Classification, Time Series Analysis, Text Analysis, MapReduce and Hadoop, Setting up Python for Data Science, Cross validation and optimization, Linear Algebra, Statistics, Probability, Hypothesis and Inference.

Prerequisite(s): CS221

Specialization(s): CE

CS352 Soft Computing (3 0 3):

Overview of the theoretical and the practical aspects of the soft computing paradigm, theory and applications of probabilistic graphical models and related topics, e.g., knowledge elicitation issues, belief updating in singly and multiply connected networks, simulation schemes for belief updating, parameter and structure learning of Bayesian networks, integration of time and uncertainty, models of uncertain reasoning including belief function theory and

fuzzy logic, and biologically inspired computational models (neural networks and evolutionary algorithms).

Prerequisite(s): CS232

Specialization(s): CS, DS

CE361 Communication Theory (3 0 3):

Introduction to modern analog and digital communication systems, Fourier analysis of signals and systems, signal transmission, amplitude and angle modulation techniques, sampling theorem, PCM, DPCM, and delta modulation, digital communication systems, principal of modern digital communication systems including M-ary communication, digital carrier and multiplexing, and emerging digital communication technologies.

Prerequisite(s): CE341

Specialization(s): CE

AI372 Nature Inspired Computing (3 0 3):

Introduction, characteristics of biological systems, adaptability, reactivity, distributivism, comparison with traditional human-engineered approaches to problem solving, handling complex problems using computational methods modeled after design principles encountered in nature, foundations of complex systems and theoretical biology, distributed architectures of natural complex systems, production of informatics tools with enhanced robustness, scalability, flexibility, interface with humans, principles of biology, informatics, cognitive science, robotics, cybernetics, etc., applied for computing problems.

Prerequisite(s): CS221

Specialization(s): AI

AI408 Cloud Computing (3 0 3): Introduction to cloud computing, techniques for building, deploying and maintaining images and applications, important APIs used in the Elastic Cloud Amazon and Azure, using cloud as an infrastructure for existing and new services, Opensource implementation, scalable clustering computational environments, building efficient applications, non-trivial issues of cloud, e.g., load balancing, caching, authorization management, distributed transactions etc., enterprise applications, cloud tools, RESTful Web Services, Amazon Elastic Cloud, Azure, Google App Engine, etc.

Prerequisite(s): CS221

Specialization(s): AI

AI410 Data Mining (3 0 3): Introduction to data mining, related technologies – Machine learning, DBMS, OLAP, stages and techniques of data mining process, methods and applications of knowledge representation, data preprocessing, data cleaning, data transformation, data reduction, discretization, generating concept hierarchies, Weka 3 data mining system, filters, statistics and discretization in Weka, measures of interestingness, visualization techniques and experiments in Weka, attribute-oriented analysis, generalization, and relevance, class comparison, statistical generalization, and relevance, class comparison, statistical measures, algorithms and association rules, motivation and terminology, item sets, generating item sets and rules, correlation analysis, classification, basic learning/mining tasks, inferring rudimentary rules: 1R algorithm, decision trees and rules in Weka, statistical (Bayesian) classification, Bayesian networks, instance-based methods (nearest neighbor), linear models, training and testing in Weka, estimating classifier accuracy, combining multiple models (bagging, boosting, stacking), Minimum Description Length Principle (MLD), clustering, partitioning methods: k-means, expectation maximization (EM), hierarchical methods: distance-based agglomerative and divisible clustering, Cobweb, text mining: extracting attributes, structural approaches (parsing, soft parsing), Bayesian approach to classifying text, web mining: classifying web pages, extracting knowledge from the web, and data mining software and applications.

Prerequisite(s): AI231

Specialization(s): AI

CS411 Block Chain (3 0 3): Bitcoins and Ethereum protocol, Decentralized peer-to-peer network, Immutable distributed ledger, Trust model (that defines a block chain), Basic components of a block chain (transaction, block, block header, and the chain), underlying algorithms, essentials of trust (hard fork and soft fork), hashing, cryptography foundations, block chain programming, Operations, i.e., verification, validation, and consensus model.

Prerequisite(s): CS221

Specialization(s): AI, CE, CS

CE412 Systems Programming (3 3 4):

Programming over Linux, gcc and associated tools, file I/O with low-level file descriptors,

the standard I/O library, error reporting mechanisms, kernel statistics and parameter modifications, process creation and management system calls, signals and associated system calls, pipes and FIFOs, single and multiple reader/writers, semaphores, shared memory and message-queues, sockets, attributes and addressing schemes, multiple client connections, connectionless socket communication.

Prerequisite(s): CS311

Specialization(s): CE

CS412 Information Retrieval (3 0 3):

Introduction, complications in building a modern web-scale search engine, ranking SVMs, XML, DNS, and LSI. They will also discover the seedy underworld of spam, cloaking, and doorway pages. Study MapReduce and other approaches to parallelism to go beyond megabytes and to efficiently manage petabytes.

Prerequisite(s): CS221

Specialization(s): AI, CS

CS413 Internet of Things (3 0 3):

Introduction, state-of-the-art in the Internet of Things (IoT), high-level overview of the IoT landscape, domain, architectures, principles, paradigms, building blocks, applications, technologies, development platforms, recent advances and fundamental issues around IoT, origin and enablers of IoT, M2M, architectures, physical and logical designs, communication models, components of IoT systems, IoT levels and deployment templates, technologies, standards, protocols, challenges, and security and privacy hazards.

Prerequisite(s): CS311

Specialization(s): AI, CE, CS

CS414 Geographic Information Systems (3 0 3):

Introduction to Geographical Information System (GIS), fundamental theory of Geographic Information Science, history and evolution of GIS, geo workspace environment, data acquisition, coordinate systems and geo-referencing (QGIS), data structures and models, Raster & Vector levels of measurements in GIS, Vector Data entry operator, concepts of spatial layering, mapping, modeling, management & monitoring, data downloading based on geospatial coordinates, masking, geotiff, analytics, ArcGIS, data analytics, implementing a GIS on a select topic,

geo-processing, spatial analysis, map projections and scaling, and cartography.

Prerequisite(s): CS221

Specialization(s): AI, CE, CS

CS416 Introduction to Deep Learning (3 0 3):

Introduction to neural networks, convolutional and recurrent networks, deep unsupervised and reinforcement learning, GPU computing, CuDNN, applications and case studies of Deep Learning in speech recognition, images, vision, etc.

Prerequisite(s): CS351

Specialization(s): CE, CS

CS418 Digital Image Processing (3 0 3):

Introduction to digital image processing (DIP) and its applications, image sensing and acquisition, sampling and quantization; mathematical tools for DIP, intensity transformation, spatial filtering; discrete Fourier transform, filtering, image degradation and restoration; noise models, estimating degradation function, image reconstruction; color models, color image processing, smoothing, sharpening; image segmentation, morphological image processing, image compression, pattern recognition.

Prerequisite(s): CS221

Specialization(s): CS

CS419 Applied Image Processing (3 0 3):

Review digital image processing, advanced image recognition topics like texture analysis and fractal analysis, advanced segmentation techniques using fractal dimension, clustering, watershed transform, medial axis transformation, 2D to 3D rendering, 3D visualization, stereo imaging, medical imaging applications, and signal image processing techniques.

Prerequisite(s): CE318/CS418

Specialization(s): CE, CS

AI420 Medical Image Processing (3 0 3):

Introduction, texture analysis, basic principles of medical image communication, history, terminology, algorithms, image formation and medical imaging, imaging modalities, image enhancement, image statistics, histograms, visualization of medical images, surface and volume-based approaches, illumination, image segmentation, robustness, watershed transform, active contours, live wire, active shapes, texture analysis, texture localization

and delineation, digital representation of color, image data management, standards, components of PACS, lossy and lossless compression, ImageJ.

Prerequisite(s): CS221

Specialization(s): AI, CE, CS

CS420 Cyber Security (3 0 3): Cybersecurity fundamentals, principles of data and technology that frame and define cybersecurity, importance of cybersecurity and role of cybersecurity professionals, cybersecurity principles, security architecture: processes and architecture, risk management, attacks, incidents: response, categories, and recovery, emerging IT and IS technology, mobile security issues, risks and vulnerabilities, cloud concepts around data and collaboration.

Prerequisite(s): CS221

Specialization(s): AI, CE

CE421 Advanced Computer Architecture (3 0 3): Instruction set architecture (ISA), RISC & CISC, pipelining, instruction-level parallelism, super scalar processors, VLIW architecture, parallel processing, high-speed memory systems, storage systems, interconnection networks.

Prerequisite(s): CE324/CS324

Specialization(s): AI, CE, CS

CS421 Human Computer Interaction (3 0 3): The human, computer and interaction,

usability paradigm and principles, introduction to design basics, HCI in software process, design rules, prototyping, evaluation techniques, task analysis, universal design and user support and computer supported cooperative work, introduction to specialized topics such as groupware, pervasive and ubiquitous applications.

Prerequisite(s): CS325

Specialization(s): CS

CS422 Professional Issues in IT (3 0 3):

The laws and how they are created, professional bodies in IT, the computing profession and the nature, structure and management of commercial IT organizations, financing of start-up companies, financial accounting, management accounting and the evaluation of investment proposals, human resources issues and management, discrimination and anti-discrimination legislation, social networking, spotting fake news, gender and racial issues, ethics, software contracts and liability, copyrights, piracy, intellectual property rights, freedom of information, data protection, cyber laws, peer-to-peer torrents and legislation that affects use/misuse of computers.

Prerequisite(s): None

Specialization(s): CS

AI423 Advances in Object-Oriented Analysis and Design (3 0 3):

Evolution of Object Oriented (OO) programming, OO concepts and principles, problem solving in OO paradigm, classes, methods, objects and encapsulation; constructors and destructors, operator and function overloading, virtual functions, derived classes, inheritance and polymorphism, I/O and file processing, exception handling, UML: conceptual model, use case diagrams, object models, class diagrams, system sequence diagram, object-oriented life cycle, modeling user interface requirements, designing and evaluating methods, synchronizing dependent attributes, normalizing classes with dependent data, design at the object, etc.

Prerequisite(s): CS221

Specialization(s): AI, DS

CE423 General Purpose Computing with GPU (3 0 3):

Graphics Processing Units (GPU) for computer graphics and gaming, general parallel computation, assessing performance of parallel algorithms on GPUs, measuring



Omar Kashif Majeed

(First Year BS AI &
Winner of ICPC Regional
2022)

I am grateful that GIKI gave me the opportunity to take part in ICPC. I learned different algorithms and new coding techniques while preparing for the competition. ICPC provided the opportunity for me to go out of the box and do things differently. I will try to use these skills for the benefit of the people around me.

the speedup over similar CPU algorithms, applications of signal processing, neural networks, etc., programming techniques for GPUs, NVIDIA's parallel computing language, CUDA programming model and syntax, GPU architecture, high performance computing on GPUs, parallel algorithms, CUDA libraries, applications of GPU computing, performance optimization and specific GPU applications, e.g., Machine Learning computations.

Prerequisite(s): CE324/CS324

Specialization(s): AI, CE, CS

CS423 Development Operations (3 0 3):

Basics of Development Operations (DevOps), Commonly used Build tools for DevOps, e.g., GIT and Jenkins, Build test automation, containerization using Docker, Docker commands and use cases, Kubernetes, Spinnaker, Skaflow, etc.

Prerequisite(s): CS112

Specialization(s): CE, CS

AI425 Fundamentals of Microprocessor Interfacing (3 0 3):

Introduction to 16-bit microprocessor, software model, addressing modes, instruction set, assembly language programming, hardware model, read/write cycles, exception/interrupt processing, interfacing to ACIA, PIA, PI/T, DMA, A/D, D/A converters, introduction to micro-controllers and embedded systems.

Prerequisite(s): CS324

Specialization(s): AI

CS425 Design Patterns (3 0 3): Overview of object-oriented design, software reusability, classification of design patterns, pattern description formats, design and implementation issues in: creational patterns, structural patterns, behavioral patterns; patterns in software architecture; patterns for user-interface design; pattern languages.

Prerequisite(s): CS325

Specialization(s): CE, CS, DS

AI426 Introduction to Mobile Computing (3 0 3):

State-of-the-art of mobile computing platforms, introduction to mobile computing, architecture of android platform, using emulator, debugging and DDMS, content providers, App. networking, App. multimedia, App. 2D and 3D graphics, using sensors, publishing, designing Apps using XAML, introduction to iPhone platform, iPhone supported

development features and tools for developing mobile web applications.

Prerequisite(s): CS221

Specialization(s): AI, CE

CS426 Software Testing and Quality Engineering (3 0 3):

Introduction, quality challenge, quality control v/s quality assurance, quality assurance in software projects, quality management, quality assurance and standards, quality planning and quality control, verification and validation, critical system validation, reliability validation, safety assurance, security assessment, inspections and reviews, software quality assurance (SQA), plans, SQA-organizational level initiatives, software testing, specification based test construction techniques, white-box and grey-box testing, testing techniques for SDLC, control flow oriented test construction techniques, data flow oriented test construction techniques, clean-room approach to quality assurance, product quality and process quality standards, walkthroughs and inspections, structure, checklist, audits, roles and responsibilities.

Prerequisite(s): CS325

Specialization(s): CS

CS427 Software Design and Architecture (3 0 3):

Overview of SDLC, engineering design vs. software design, design heuristics and principles, reusability, metrics and quality of design, frameworks, and architectures, framework



Muhammad Zulfiqar Ali
(First Year BS AI &
Winner of ICPC Regional
2022)

"I am grateful to GIK Institute for affording me the opportunity to participate in this competition. It was an amazing experience that allowed me to put my programming and logical prowess to the test against teams from all across Pakistan. I hope to bring further laurels to GIKI."

development approaches, service-oriented architectures, middleware architectures, design patterns, architectural patterns, design issues for distributed and real-time software, re-engineering and reverse engineering.

Prerequisite(s): CS325

Specialization(s): CS

CS428 Design of Programming Language (3 0 3):

Programming practices: program analysis and construction practices, programming language classification, data types, structured data types, subprograms, control statements scoping, and storage management.

Prerequisite(s): CS224

Specialization(s): CS

CS429 Software Project Management (3 0 3):

Project management processes and phases, resource identification, software size estimation, budgeting and costing, project planning and scheduling, customer relationship management, technical resource management, configuration management, outsourcing, team selection, risk management, software process management, process improvement framework, software release management.

Prerequisite(s): CS325

Specialization(s): CS

CE432 Digital Communication (3 0 3):

Sampling and time-division multiplexing, baseband digital signals and systems; coded pulse modulation, error control coding, digital modulation systems, information measure and source encoding, and introduction to spread spectrum communications.

Prerequisite(s): CE341

Specialization(s): CE

CS432 Advanced Databases (3 0 3): Design of data models, recently developed protocols to guarantee consistency of databases, the design of physical models, and performance analysis techniques, algorithms and data structures such as B-trees, transposed files, phantom files, and hybrid structures, distributed databases and database machines, object-oriented databases concepts.

Prerequisite(s): CS232

Specialization(s): AI, CS

CS433 Computer Graphics (3 0 3):

Computer graphics, fundamental algorithms, graphics input and output, graphics pipeline, sampling and image manipulation, three-dimensional transformations and interactive modeling, basics of modeling and animation, simple shading models and their hardware implementation, fundamental algorithms of scientific visualization, basic structure of interactive graphics systems, characteristics of various hardware devices, control of display devices, implementation of simple packages, device independence, and standard packages, distributed architectures for graphics, hidden line and hidden surfaces algorithms, representation of surfaces, 2-D graphics methods, transformations, and interactive methods, 3-D graphics, transformations, viewing geometry, object modeling, and interactive manipulation methods, basic lighting and shading, video and animation methods.

Prerequisite(s): CS221

Specialization(s): AI, CE, CS

CS435 MIS & DSS (3 0 3): Advance topics in systems analysis and software engineering, design and development of large information systems, usefulness to the management, integrated environments, application of artificial intelligence to MIS, development of expert systems and decision support systems.

Prerequisite(s): CS232

Specialization(s): CS, DS

CS437 Data Warehousing and Data Mining (3 0 3):

Concepts of data mining and data warehousing, data preparation techniques: outlier and missing data analysis, data reduction techniques, learning methods, statistical methods, cluster analysis, hierarchical, agglomerative and naïve Bayesian methods, decision trees and decision rules, association rules, other soft computing approaches, artificial neural networks, fuzzy logic, genetic algorithm, and evolutionary algorithms.

Prerequisite(s): CS232

Specialization(s): AI, CS

CS438 Web Mining and Social Media Analysis (3 0 3):

Introduction to mining data from the web and social media, state-of-the-art methods in mining heterogeneous data, association rule mining supervised and unsupervised learning with particular emphasis on web data, methods for information retrieval,

e.g., latent semantic indexing, meta-searches, search-based ranking, social media analysis such as link analysis, page rank and HITS algorithms, community discovery, etc.

Prerequisite(s): CS221

Specialization(s): AI, CS, DS

CS439 Data Science (3 0 3): Introduction to Big Data Analytics, Data Analytics Lifecycle, Advanced Analytical Theory and Methods: Association Rules, Regression, Classification, Time Series Analysis, Text Analysis, MapReduce and Hadoop, Setting up Python for Data Science, Cross validation and optimization, Linear Algebra, Statistics, Probability, Hypothesis and Inference.

Prerequisite(s): CS221

Specialization(s): AI, CS

CE444 Simulation and Modeling (3 0 3): Analysis of physical systems and industrial processes, formulation in the form of mathematical equations or inequalities, mathematical models, solution of models, use of computers for design, optimization and control of actual systems in engineering.

Prerequisite(s): MT201

Specialization(s): CE

AI452 Techniques of Soft Computing (3 0 3): Overview of the theoretical and the practical aspects of the soft computing paradigm, theory and applications of probabilistic graphical models and related topics, e.g., knowledge elicitation issues, belief updating in singly and multiply connected networks, simulation schemes for belief updating, parameter and structure learning of Bayesian networks, integration of time and uncertainty, models of uncertain reasoning including belief function theory and fuzzy logic and biologically inspired computational models (neural networks and evolutionary algorithms).

Prerequisite(s): AI231/CS351

Specialization(s): AI, CE

CS452 Artificial Neural Networks (3 0 3): Neural network basics, Hebb net, perceptron, Adaline and Madaline, Hetero-associative and auto-associative networks, discrete Hopfield network, bi-directional associative memory (BAM), backpropagation neural network (BPN), variants of BPN, simulations using backpropagation, radial basis function networks, neural

nets based on competition, self-organization aps (SOMs), learning vector quantization (LVQ), counter propagation networks, adaptive resonance theory (ART), probabilistic neural networks, temporal processing using feedforward nets, genetic algorithms, case studies.

Prerequisite(s): CS351

Specialization(s): CE, CS

CE453 Robotic Vision (3 0 3): Vision tasks and applications, Cameral models and image acquisition, image segmentation, feature detection and matching, image recognition, 3D visualization, robot perception (robot and sensors), visual navigation, localization and other topics in robotic vision.

Prerequisite(s): CS221

Specialization(s): AI, CE, CS

CS454 Real-Time Programming (3 0 3): Introduction to real-time systems, design issues, programming languages for real-time systems, fault tolerance and reliability issues, exception handling, concurrent programming, synchronization, communication, scheduling.

Prerequisite(s): CS311

Specialization(s): CS

CS458 Distributed Systems (3 0 3): Introduction to distributed systems, communication, naming and name services, processes, synchronization, fault tolerance, distributed file systems, distributed transaction processing, replication, object-based systems, document-based systems, coordination-based systems, security in distributed systems.

Prerequisite(s): CS311

Specialization(s): AI, CS

CE463 Wireless & Mobile Networks (3 0 3): Introduction to wireless environment, wireless network architectures, wireless local area networks (WLANs), wireless personal area networks, middleware for wireless and mobile networks, mobile IP, TCP in wireless environments, mobile ad-hoc networks and their routing, nomadic services, security in wireless networks, mobile data services, pervasive computing applications.

Prerequisite(s): CS311

Specialization(s): CE

CS463 Web Engineering (3 0 3): Internet technology trends, real-time data transmission,

security over Internet, introduction to Web applications development, software architecture patterns for Web Apps, MVC, Web browsers, HTTP, DOM and browser engines, client-side development with HTML, CSS & JavaScript, server-side development over Web applications framework, Web App deployment, virtualization, cloud computing, IaaS, PaaS and SaaS models.

Prerequisite(s): CS311

Specialization(s): AI, CE, CS

CE465/EE424 ASIC Design (3 0 3):

Introduction to application specific Integrated circuits (ASIC) design methodologies, design and implementation using FPGAs, design verification, digital design using hardware description language, libraries, utilities for high level description, data flow description, timing and delays, modeling techniques.

Prerequisite(s): CS222

Specialization(s): CE

CS465 Data Security and Encryption (3 0 3):

Mathematical background and principle of number theory, probability theory, primes, random numbers, modular arithmetic, cryptographic algorithms and design principles, conventional and symmetric encryption (DES, IDEA, Blowfish, Rijndael, RC-4, RC-5), public key or asymmetric encryption (RSA, Diffie-Hellman), key management, hash functions (MD5, SHA-1, RIPEMD-160, HMAC), digital signatures, and certificates, network security and authentication protocols (X.509, Kerberos), electronic mail security (S/MIME, PGP), web security and protocols for secure electronic commerce (IPSec, SSL, TLS, SET).

Prerequisite(s): CS311

Specialization(s): CE

CE471 Multimedia Systems (3 0 3):

Introduction to multimedia systems, software, hardware, various equipment, video and audio capture, annotation, storage and playback techniques, multimedia software development tools, multimedia applications, procedures to develop multimedia systems: (specification, design, testing, and prototyping), multimedia standards, Student projects - developing multimedia systems in the laboratory.

Prerequisite(s): CE341

Specialization(s): CE

CS472 Bio-Inspired Computing (3 0 3):

Introduction, characteristics of biological systems, adaptability, reactivity, distributivism, comparison with traditional human-engineered approaches to problem solving, handling complex problems using computational methods modeled after design principles encountered in nature, foundations of complex systems and theoretical biology, distributed architectures of natural complex systems, production of informatics tools with enhanced robustness, scalability, flexibility, interface with humans, principles of biology, informatics, cognitive science, robotics, cybernetics, etc., applied for computing problems.

Prerequisite(s): CS221

Specialization(s): CE, CS

AI473 Computational Neuroscience (3 0 3):

Mathematical introduction to neural coding and dynamics, convolution, correlation, linear systems, game theory, signal detection theory,



probability theory, information theory, reinforcement learning, applications to neural coding, visual system, Hodgkin-Huxley and other models of neural excitability, stochastic models of ion channels, cable theory, and models of synaptic transmission, basic computational methods for understanding nervous systems and their function, computational principles governing various aspects of vision, sensory-motor control, learning, and memory, representation of information by spiking neurons, processing of information in neural networks, and algorithms to adapt and learn, computational neuroscience: descriptive, mechanistic and interpretive models.

Prerequisite(s): AI341

Specialization(s): AI

CS474 Bio-Informatics (3 0 3): Bioinformatics, sequence analysis, microarray expression analysis, Bayesian methods, control theory, scale-free networks, and biotechnology applications, current real-world examples, actual implementations, engineering design issues, engineering issues from signal processing, network theory, machine learning, robotics and other domains, use of NCBI's Entrez, BLAST, PSI-BLAST, ClustalW, Pfam, PRINTS, BLOCKS, Prosite and PDB.

Prerequisite(s): CS121

Specialization(s): AI, CE, CS, DS

CE475 Real Time Embedded Systems (3 0 3): Introduction to real time systems, embedded systems, interrupts, performance and optimization, simple single task operating system, real time operating system and scheduling, concurrency, communication, real time benchmarks, adaptive and real time systems, real time control over the internet/remotely.

Prerequisite(s): CE324

Specialization(s): CE

CS476 Computational Biology (3 0 3): Algorithmic and machine learning foundations of computational biology, combining theory with practice, principles of algorithm design for biological datasets, and analyze influential problems and techniques, analyzing real datasets from large-scale studies in genomics and proteomics, Genomes: biological sequence analysis, hidden Markov models, gene finding, RNA folding, sequence alignment, genome assembly, networks: gene expression analysis,

regulatory motifs, graph algorithms, scale-free networks, network motifs, network evolution, evolution: comparative genomics, phylogenetics, genome duplication, genome rearrangements, evolutionary theory, rapid evolution.

Prerequisite(s): CS221

Specialization(s): AI, CE, CS, DS

MANAGEMENT ELECTIVE COURSES

CS436 Operations Research (3 0 3):

Optimization and computational efficiency in automated decision systems, art of modeling, mathematical modeling, linear programming and simplex method, duality theory and sensitivity analysis, transportation algorithm and its variants, advanced topics in linear programming, e.g., goal programming, integer programming, probabilistic models, e.g., queuing systems, simulation modeling, etc.

Prerequisite(s): MT201

Specialization(s): AI, CE, CS, DS

CS491 Entrepreneurship & Technology

Commercialization (3 0 3): Understanding the entrepreneurship process, concepts, practices and tools of the entrepreneurial world, readings, cases studies and projects covering unique environment of the entrepreneurs and new ventures, tools necessary to think creatively, to plan out whether ideas are marketable to investors, launching own business, or supporting an employer in launching and growing an entrepreneurial venture, the focus shall be on items particularly important for technology ventures.

Prerequisite(s): none

Specialization(s): AI, CE, CS, DS

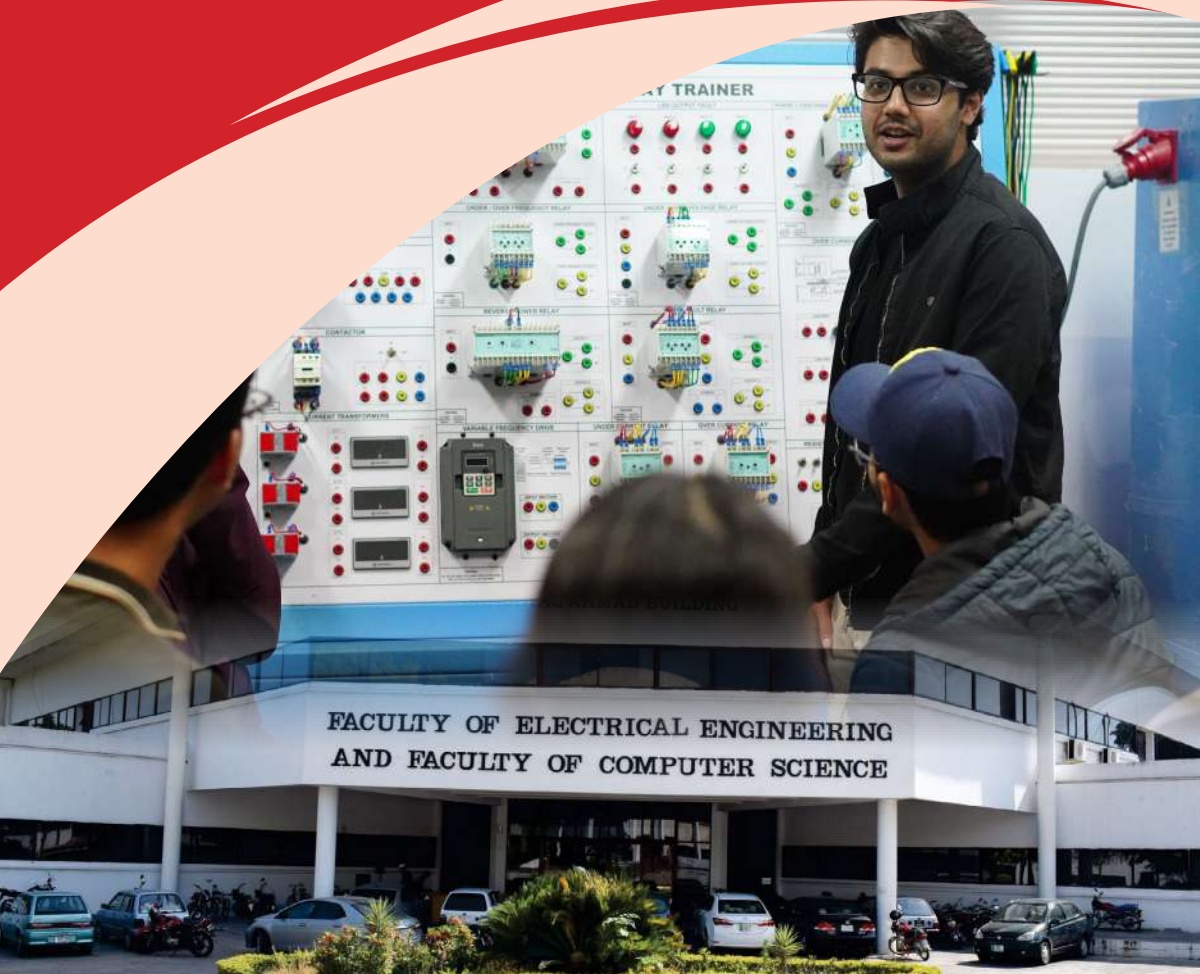
CS492 Network Security & Cyber Ethics (3

0 3): Cybercrime landscape, morality, ethics, technology, value, cyberspace infrastructure, anatomy of the security problem, enterprise security, information security protocols, best practices, security and privacy in online social networks, security in mobile system, security in the cloud, security and compliance.

Prerequisite(s): none

Specialization(s): AI, CE, CS, DS

FACULTY OF ELECTRICAL ENGINEERING



Introduction

Electrical Engineering (EE) is one of the oldest disciplines of engineering. Initially, it was confined to a few areas such as power generation, transmission and distribution; radio communication and wireless telephony. However, in the last few decades, it has seen as tremendous growth and expansion in some new areas. The diversification and expansion in new areas of Electrical Engineering have been of such a large magnitude that each area seems an independent discipline in its own. These include Electronic Engineering, Power Engineering, Telecommunication Engineering, Computer Engineering, Information Technology, and Control Systems. However, in a large part of the world, Electrical Engineering is still considered to be a mother discipline.

Electrical Power Engineering is an important and vital discipline in Pakistan due to present scenario of power sector facing shortfall of required energy needs. The main aim of this discipline is not only to study existing methods of power generation, transmission and distribution but also to study the new ways of power generation involving renewable and sustainable technologies. The Electrical Power Engineering Program covers a broad range of activities and evolving issues that are of great importance in the field of sustainable and smart power systems.

With the turn of the century, we have entered into a new era of micro and nano-fabrication technologies. With the new methods of chip design at nano scale, system-On-Chip has enabled the integration of millions of devices in small sized chips, thus exponentially enhancing capabilities offered by the electronic devices. Other micro-fabrication technologies related to the fields such as Micro-Electro-Mechanical Systems (MEMS), are now reaching a point of being able to contribute in a similar manner.

The immense development in the field of microelectronics has generated an ever-increasing demand of electronic and electrical engineers, modern engineers are expected to cope not only with the development in traditional fields like communications, robotics, digital signal processing, power and control systems but are also required to possess relevant knowledge and theoretical understanding of the emerging areas like biomedical instrumentation, security, surveillance and biometrics.

Keeping in mind the latest requirements of the national and international job markets and research trends, Faculty of Electrical Engineering has designed an updated curriculum offering high-quality courses. The program of study enables electrical engineering (EE) students to lead the teams of future young engineers in diverse working environment and to practically realize their innovative ideas. The faculty also provides students with the opportunity to learn as how research is carried out.

Thrust Areas

- Power Electronics and Electrical Drives
- Power Systems and Control
- Microelectronics and ASIC Design
- Communication and Digital Signal Processing



Dean

Muhammad Akbar

PhD, University of Tokyo, Japan

Faculty

Nisar Ahmed, PhD, ICSTM, London, UK
 Khasan Karimov, PhD, S. Petersburg, Russia
 Ziaul Haq Abbas, PhD, University of Agder, Norway
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 Husnul Maab, PhD, QAU, Islamabad, Pakistan
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 Shahid Alam, PhD, Chalmers University of Technology, Sweden
 Hadeed Ahmed Sher, PhD, King Saud University, KSA
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 Memoon Sajid, PhD, Jeju National University, South Korea
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 Mutahar Kamran Mughal, BS, NED, Karachi, Pakistan
 Shahzeb Ahmad Khan, BS, NAMAL, Pakistan

Personal Secretary to Dean

Ikram Ullah
 MA, Gomal University, Pakistan

Lab Engineers

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 Hamood Ur Rehman, MS, GIK Institute, Pakistan
 Muhammad Adeel, MS, GIK Institute, Pakistan
 Asad Khalid, BS, GIK Institute, Pakistan
 Basit Ali, BS, UET Peshawar, Pakistan
 Usman Ghani, BS, IST Islamabad, Pakistan

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 Ahsan Nadeem, MS, COMSATS, Pakistan
 Atif Mehmood, MS, IUB, Bahawalpur Pakistan
 Mujtaba Ghous, MS, NUST, Pakistan
 Irfan Ullah, MS, GIK Institute, Pakistan
 Muhammad Zaheer Saleem, MS, UET Lahore, Pakistan
 Anam Hanif, MS, Mirpur University, AJK, Pakistan
 Abdullah Numani, MS, COMSATS, Pakistan
 Haleema Sadia, MS, NUST, Pakistan
 Saba Tariq, MS, UET, Taxila, Pakistan

Mission

The Faculty of Electrical Engineering is to produce graduates equipped with broad and in-depth knowledge and relevant skills to present effective socio-economical solutions for complex problems.

Program Educational Objectives (PEOs)

The Faculty of Electrical Engineering at GIK Institute has formulated the Program Educational Objectives (PEOs) using feedback of the stakeholders. There are three PEOs of the EE program which state that the FEE graduates will become:

PEO_1 Competent Engineers: Capable of demonstrating knowledge by innovating criticizing, and finding solutions to real-world problems pertinent to the field of electrical engineering.

PEO_2 Skillful Managers: Capable of communicating, leading, making judgements, creating a collaborative environment, establishing goals, planning tasks, and meeting objectives.

PEO_3 Responsible Individuals: Appreciative of social needs with considerations of ethical, environmental, and global factors.

Program Learning Outcomes (PLOs)

There is a set of twelve Program Learning Objectives (PLOs) of Electrical Engineering which describe as what students are expected to know/perform/attain by the time they graduate. These twelve PLOs are set such that delivery of each course encompasses them.

These PLOs are described as follows:

1) Engineering Knowledge:

Ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

2) Problem Analysis:

Ability to identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

3) Design/Development of Solutions:

Ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

4) Investigation:

Ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.

5) Modern Tool Usage:

Ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering issues, with an understanding of the limitations.

6) The Engineer and Society:

Ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practices and solution to complex engineering problems.



7) Environment and Sustainability:

Ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

8) Ethics:

Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

9) Individual and Team Work:

Ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.

10) Communication:

Ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11) Project Management:

Ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

12) Lifelong Learning:

Ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments.

Knowledge Areas

WK1 Natural Sciences: A systematic theory-based understanding of natural sciences applicable to the discipline.

WK2 Mathematics and Computing: The concept-based mathematical thinking, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline.

WK3 Engineering Fundamentals: A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.

WK4 Engineering Specialization: The knowledge of engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.

WK5 Engineering Design: The Design Thinking Knowledge that supports engineering design in a practice area of an engineering

discipline.

WK6 Engineering Practice: The Knowledge of engineering practices (technology) in different practice areas of an engineering discipline.

WK7 Engineering in Society: A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity i.e. economic, social, cultural and environmental and sustainability.

WK8 Research Literature: Engagement with selected knowledge in the research literature of the discipline.

Sustainable Development Goals

- Good Health and Well-being
- Quality Education
- Affordable and Clean Energy
- Industry, Innovation and Infrastructure
- Partnerships to achieve the Goal

Undergraduate Program

The Faculty offers a four-year degree program through courses that are modular in nature and are evenly distributed across eight regular semesters. The theoretical knowledge obtained in the classroom is reinforced through laboratory work. These laboratories, which support more than 70% of the courses, are equipped with the latest pedagogical tools to illustrate important concepts, and provide their practical demonstration. The students can opt for one of the following streams:

- Electronic Engineering
- Power Engineering



FEE Laboratories

Keeping in mind the present and future needs, the Faculty of Electrical Engineering has an assortment of equipment and facilities for the students enabling them to cope up with the fast moving technology. These facilities provide them with an opportunity to learn and understand the concepts of electronic and power engineering and constructively transform them to practical use. Major laboratory facilities are summarized below:

Wave Propagation and Antennas Lab

This lab contains microwave training systems, antennas, waveguides, and transmission line demonstrators suitable for the study of generation, propagation, and reception of microwave signals. This lab is used for practical experimentation pertaining to wave propagation & antennas and microwave engineering courses.

Electric Machines Lab

In this laboratory, students augment their concepts about the fundamentals behind working of transformers and the rotating machinery. The laboratory is equipped with single and three-phase transformers, induction motors, synchronous generators and motors, DC generators and motors, DC and AC power supplies, electrical and mechanical loads, and a number of test and monitoring equipment such as watt-meters, power-factor meters, voltmeters, ammeters and frequency meters. The students also learn practically the synchronization of two electricity networks and the power flow

between them.

Digital Logic Design Lab

This lab is for the understanding of fundamental digital logic related concepts and contains 30 sets of oscilloscopes, digital trainers, digital multi meters (DMMs), function generators and support accessories. Starting with simple universal NAND/NOR gates, the students learn to design and implement different combination-al as well as sequential circuits taught in the corresponding theory class. Counter design using state machine and verilog basics are also included towards the end.

Analog Electronics Labs

There are two analog electronics laboratories in the faculty, equipped with over 60 sets of oscilloscopes, trainers, power supplies and functional generators. The labs are used for the courses of electronics devices and circuits, electronic circuit design and linear circuit analysis. The labs augment the theoretical knowledge, which the students acquire in theory classes. On the basis of experiments in these labs, the students not only verify their theoretical analysis but also learn about the limitations associated with the equipment, which are always there regardless of how sensitive and expensive the equipment is. The labs also help to enhance the students' knowledge in fundamental design concepts.

Communication Systems Lab

The faculty has a very comprehensive communication systems laboratory, which covers both



the analog and digital communication systems. The central equipment of the lab is a set of training panels, which have built-in modules ranging from phase modulation to coding of digital data. The panels are equipped with 200 kHz function generators, noise generators and spectrum analyser modules to help set up various experiments. In addition to this, the lab is also equipped with universal u-controllers and computers. Telephone switching module and optical fibre transmitter and receiver trainers are also available.

Signal Processing Simulation Lab

This lab has 50 networked core i7 PCs with various kinds of software packages installed including Matlab, PSpice, Microsoft Office, ModelSim, Xilinx ISE. Matlab is used for running exercises in the courses of signals and systems, control systems, digital communication systems, digital signal processing and digital image processing. PSpice, a simulation tool for analysing electric and electronic circuit is used in the labs of linear circuit analysis, electrical network analysis and electronics devices and circuits. Matlab and PSpice are also used in the course of power electronics.

Linear Control Systems Lab

Linear Control Systems lab equipped PCs, MATLAB software with Control Systems Toolbox for analysis, modelling, and design of Linear Time Invariant Systems. Modelling and analyzing real LTI systems help demonstrate and give practical knowledge about different theoretical concepts studied in the control systems course. Students with the concept of control system analysis and design in time and frequency domain both by classical methods and modern techniques, working with lab equipment to reinforce the concepts.

Microprocessor Interfacing Lab

The intent of this laboratory is to provide an insight into a typical microprocessor and microprocessor-based system. The lab is equipped with trainer boards designed to provide comprehensive hands-on training, employing the latest state-of-the-art technology. Easy PIC V8 microcontroller trainers with ISP and ICD options are used in this lab. The lab follows a modular approach to teach microprocessor architecture and interfacing concepts along with its applications. In addition, the laboratory

is also equipped with universal programmers for high-end controllers and advanced peripherals that can be interfaced with the controller in different applications.

ASIC Design Lab

This laboratory is equipped with VLSI and electronic design automation (EDA) tools, such as Xilinx, ModelSim, Leonardo spectrums, place and route tools, ISE web pack, Microwind and DCH. Altera and Quartus are available for ASIC design in HDL (Hardware description language), for simulation and synthesis. Moreover, the laboratory is equipped with a number of latest Xilinx/Altera FPGA development boards.

Electrical Measurement and Instrumentation Lab

This lab covers investigation of instruments, error types and characteristics of instruments, determination of dynamic behaviour of typical sensors, signal conditioning circuits such as DC and AC bridges, instrumentation amplifiers and filters, computer-based data and signal processing for different measurement systems.

Electrical Simulation Lab

This lab has 50 networked Core i7 PCs with various kinds of software packages installed; including Matlab, PSpice, Power World Simulator and Calculu. Matlab is used for running exercises in the courses of signals and systems, control systems, digital communication systems, digital signal processing, digital image processing, power system analysis and design and power distribution and utilization. PSpice, a simulation tool for analysing electric and electronic circuits is used in the labs of linear circuit analysis, electrical network analysis, and electronics devices and circuits. Power World Simulator is used for solving problems involving power flows. Calculux is used for luminance calculations in lighting systems.

Power Electronics Lab

Power electronics lab is equipped with the state-of-the-art instrumentation for design, simulation, layout, prototyping, and testing of switching/analog circuits. The experiments in the power electronics laboratory involve modeling, control, topologies, and integration of switching converters, inverters, single-phase and three-phase thyristor, power

factor correction methods active power filters and power conversion for alternative energy sources.

Industrial Automation and Controls lab

Industrial Automation and Controls lab is focused on the Programmable Logic Controllers (PLC) and Industrial Systems based on that. The total integrated automation (TIA) software is introduced as well as SIMATIC S7-Basic Human Machine Interface (HMI) and real-time interfacing of Motors, Sensors, and Actuators with PLC and HMI are covered. This Lab also provides the understanding and implementation of SCADA systems using Siemens TIA and 3rd party software for interactive GUI implementation.

Power Transmission and Energy Labs

The main focus of this lab is to introduce students with the state-of-the-art power transmission practices, connections and equipment. The lab consists of modern transmission and distribution trainers which give students a flavor of the power industry. This lab gives hands-on experience related to energy generation and power transmission.

Power Distribution and Utilization (PDU) Lab Power distribution and utilization is one of the core courses of electrical engineering. The lab consists of power distribution trainers related to both radial and mesh systems, power factor correction; as well as modern energy metering devices, batteries, illumination and earthing systems.

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Power System Protection Lab

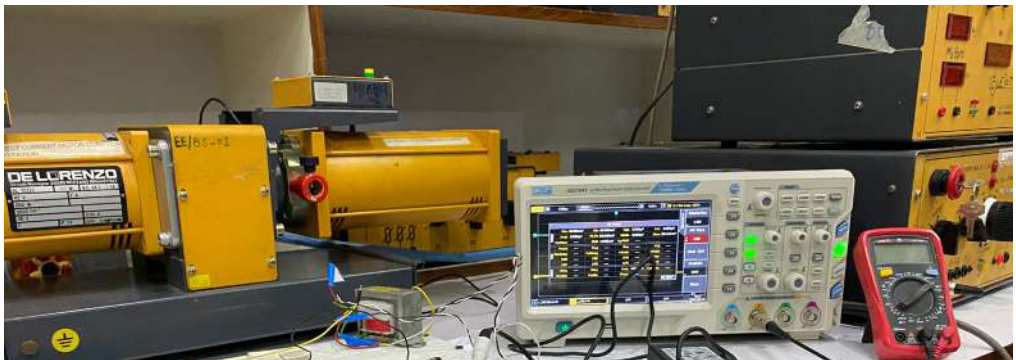
This lab has the modern protection trainers to let students perform practical work in power system protection. The equipment includes, instrument transformers (CTs, PTs) of different ratios, microprocessor based relays, and a variety of circuit breakers in modular forms.

High Voltage Engineering Lab

The laboratory is equipped with a single-stage HV kit capable of generating AC, DC and Impulse voltage up to 150 kV. The students are given the opportunity to extend their theoretical knowledge obtained in the lectures by conducting experiments. The experiments can be performed in small groups of 3-5 students under the guidance of an instructor. The idea of conducting experiments independently and in a responsible manner is particularly important with regard to the student adherence to safety measure for protecting personnel and equipment. The first part of experiments is basic in nature dealing with the generation and measurement of high voltage AC, DC and impulse. The second part of experiments relates to insulation breakdown procedure of solid, liquid & gaseous insulation systems simulating practical situations in real life condition (under voltage stress).

Accreditation

The Degree of Bachelor of Science in Electrical Engineering is accredited by the Pakistan Engineering Council (PEC), the regulatory body of engineering education.



For BS in electrical engineering degree student must complete the following requirements as details in Table a-g.

(a) General Education Requirements (54 Credit Hours)

Course Titles	Course Code	Credit Hour
Basic Engineering	PH101, PH101L, CH101, CH161, MM101, MM141L, ME233	12
Computing	CS101, CS101L, CS112, CS112L, EE222, EE421	11
English Language	HM101, HM102	6
Humanities	HM211, HM321, HM322	9
Management	MS490	3
Mathematics	MT101, MT102, MT201, MT203	12

(b) Core Requirements (51 Credit Hours)

Course Titles	Course Code	Credit Hour
Probability & Statistics	ES111	3
Digital Logic Design	EE221	3
Linear Circuit Analysis	EE211	3
Electrical Network Analysis	EE212	3
Microprocessor Systems	EE223	3
Electronic Devices and Circuits	EE231	3
Electric Machines	EE311	3
Power Distribution and Utilization	EE312	2
Electrical Instrumentation and Measurements	EE314	2
Signals and Systems	EE341	3
Linear Control Systems	EE351	3
Electromagnetic Field Theory	EE371	3
Senior Design Project (Part-I)	EE491	3
Senior Design Project (Part-II)	EE492	3
Innovation & Makers Lab-1	IF101L	1
Innovation & Makers Lab-2	IF102L	1
Digital Logic Design Lab	EE221L	1
Linear Circuit Analysis Lab	EE211L	1
Microprocessor Systems Lab	EE223L	1
Electronic Devices and Circuits Lab	EE231L	1
Electric Machines Lab	EE311L	1
Power Distribution and Utilization Lab	EE312L	1
Electrical Instrumentation and Measurements Lab	EE314L	1
Signals and Systems Lab	EE341L	1
Linear Control Systems Lab	EE351L	1
Industrial Control & Automation Lab	EE451L	1

(c) Specialization Requirement for Electrical Engineering (20*, 23 Credit Hours)**

Course Titles	Course Code	Credit Hour
Power Transmission**	EE313	2
Electronic Circuit Design*	EE331	3
Power Electronics	EE332	3
Communication Systems	EE361	3
Power System Analysis**	EE411	3
Power System Protection**	EE412	3
High Voltage Engineering**	EE413	3
Digital Signal Processing*	EE441	3
Wave Propagation and Antennas*	EE471	3
Power Transmission Lab**	EE313L	1
Electronic Circuit Design Lab*	EE331L	1
Power Electronics Lab	EE332L	1
Communication Systems Lab	EE361L	1
Power System Analysis Lab**	EE411L	1
Power System Protection Lab**	EE412L	1
High Voltage Engineering Lab**	EE413L	1
Digital Signal Processing Lab*	EE441L	1
Wave Propagation and Antennas Lab*	EE471L	1

* Electronic stream; ** Power stream

(d) Technical Elective for Electrical Engineering (09*, 06 Credit Hours)**

Course Titles	Course Code	Credit Hour
Power System Analysis*	EE411	3
Power System Protection*	EE412	3
Renewable Electrical Energy Systems	EE414	3
Electrical Machine Design and Maintenance	EE415	3
Electrical Machine Drives and Control	EE416	3
Special types of Electrical Machines	EE417	3
Introduction to ASIC Design	EE422	3
Digital Integrated Circuit Design	EE423	3
VLSI Design	EE424	3
Embedded Systems	EE425	3
Solid State Electronics	EE431	3
Industrial Electronics	EE432	3
Digital Systems Design	EE436	3
Digital Image Processing	EE442/CS418	3

Introduction to Wavelets	EE443	3
Introduction to Robotics	EE454	3
Industrial Process Control	EE452	3
Digital Control Systems	EE453	3
Communication System Design and Performance Analysis	EE461	3
Computer Communication Networks	EE462/CE313/CS313	3
Cellular Mobile Communication Systems	EE463	3
Satellite Communication Systems	EE464	3
Antenna Theory and Design	EE472	3
Microwave Engineering	EE473	3
Radar Systems	EE474	3
Electrical Estimation Installation and Planning	EE481	3
Power Generation	EE482	3
Power Plant Engineering	EE483	3
Electrical Insulation Materials	EE484	3
Power Economics and Management	EE485	3
Power System Operation and Control	EE486	3
Power System Design	EE487	3

* Electronic stream; ** Power stream

(e) Management Elective (03 Credit Hours)

Course Titles	Course Code	Credit Hour
Elective-I	MS49x	3

(f) Summer Training (Pass/Fail grade):

Every student is required to participate in a summer training program of 4-8 weeks during the summer following the junior/3rd Year. A formal written report is required at the end of the internship period.

(g) Total Credit Requirements:

A student is required to complete 137 credit hours for the Bachelor of Science degree in Electrical Engineering.



Electrical Engineering- Semester wise Breakdown

1st Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	CH101	Applied Chemistry & Environment	2	0	2	None
	CH161	Occupational Health and Safety	0	3	1	None
	CS101	Computing and AI	2	0	2	None
	CS101L	Computing and AI Lab	0	3	1	None
	HM101	English Language and Communication Skills	1	6	3	None
	MT101	Calculus I	3	0	3	None
	PH101	Applied Physics	3	0	3	None
	PH101L	Applied Physics Lab	0	3	1	None
	IF101L	Innovation and Makers Lab-1	0	3	1	None
		Total			17	

2nd Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	CS112	Object Oriented Programming and Design	2	0	2	None
	CS112L	Object Oriented Programming and Design Lab	0	6	2	None
	ES111	Probability and Statistics	3	0	3	None
	HM102	Critical Thinking and Expository Writing	2	3	3	None
	IF102L	Innovation and Makers Lab-2	0	3	1	None
	MM101	Materials and Nanotechnology	2	0	2	None
	MM141L	Materials Lab 1	0	3	1	None
	MT102	Differential Equations & Linear Algebra	3	0	3	MT101
		Total			17	

3rd Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	EE211	Linear Circuit Analysis	3	0	3	MT101
	EE221	Digital Logic Design	3	0	3	MT101
	EE222	Data Structures and Algorithms	2	0	2	CS101
	HM211	Pakistan and Islamic Studies	3	0	3	None
	ME233	Thermal Energy Engineering	2	0	2	MT101
	MT201	Calculus II	3	0	3	MT102
	EE211L	Linear Circuit Analysis Lab	0	3	1	None
	EE221L	Digital Logic Design Lab	0	3	1	None
		Total			18	

4th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	EE212	Electrical Network Analysis	3	0	3	EE211
	EE223	Microprocessor Systems	3	0	3	EE121
	EE231	Electronic Devices and Circuits	3	0	3	None
	HM321	Sociology and Human Behavior	3	0	3	None
	MT203	Complex Variables and Transforms	3	0	3	MT102
	EE223L	Microprocessor Systems Lab	0	3	1	None
	EE231L	Electronic Devices and Circuits Lab	0	3	1	None
		Total			17	

5th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	EE311	Electric Machines	3	0	3	EE212
	EE312	Power Distribution and Utilization	2	0	2	EE212
	EE331	Electronic Circuit Design*	3	0	3	EE231
	EE313	Power Transmission**	2	0	2	EE212
	EE341	Signals and Systems	3	0	3	EE212, MT201
	HM322	Cooperative law and Professional Ethics	3	0	3	None
	EE311L	Electric Machines Lab	0	3	1	None
	EE312L	Power Distribution and Utilization Lab	0	3	1	None
	EE331L	Electronic Circuit Design Lab*	0	3	1	None
	EE313L	Power Transmission Lab**	0	3	1	None
	EE341L	Signals and Systems Lab	0	3	1	None
		Total			18*,17**	

6th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	EE314	Electrical Instrumentation and Measurements	2	0	2	EE211
	EE332	Power Electronics	3	0	3	EE231, EE311
	EE351	Linear Control Systems	3	0	3	EE341
	EE361	Communication Systems	3	0	3	EE341
	EE371	Electromagnetic Field Theory	3	0	3	PH101, MT201
	EE314L	Electrical Instrumentation and Measurements Lab	0	3	1	None
	EE332L	Power Electronics Lab	0	3	1	None
	EE351L	Linear Control Systems Lab	0	3	1	None
	EE361L	Communication Systems Lab	0	3	1	None
		Total			18	

7th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	EE441	Digital Signal Processing*	3	0	3	EE341
	EE471	Wave Propagation and Antennas*	3	0	3	EE371
	EE411	Power System Analysis**	3	0	3	EE212
	EE412	Power System Protection**	3	0	3	EE311
	EE/CS/ES4xx	Technical Elective I	3	0	3	None
	EE441L	Digital Signal Processing Lab*	0	3	1	None
	EE471L	Wave Propagation and Antennas Lab*	0	3	1	None
	EE411L	Power System Analysis Lab**	0	3	1	None
	EE412L	Power System Protection Lab**	0	3	1	None
	EE451L	Industrial Control & Automation Lab	0	3	1	None
	EE491	Senior Design Project (Part-I)	0	9	3	None
	MS490	Engineering Economics & Management	3	0	3	None
		Total			18	

8th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	Credit Hours	Pre-req
	EE413	High Voltage Engineering**	3	0	3	EE312
	EE413L	High Voltage Engineering Lab**	0	3	1	None
	EE421	Programming Tools for Optimization	2	0	2	EE222
	EE492	Senior Design Project (Part-II)	0	9	3	None
	EE/CS/ES4xx	Technical Elective II	3	0	3	None
	EE/CS/ES4xx	Technical Elective III*	3	0	3	None
	MS49x	Management Elective I	3	0	3	None
		Total			14*,15**	

* Electronic stream; ** Power stream



COURSE DESCRIPTION

EE201 Applied Electrical Engineering (3 0 3):

Quantities, SI units, electric circuits, charges, current, voltage, resistance, energy and power, series/parallel circuits, KCL, KVL, review of RLC circuit and applications, integrated circuits, microprocessors and applications, AC/DC motors, AC/DC generators, transformers, AC circuits, power factor.

Pre-requisite(s): PH101

EE211 Linear Circuit Analysis (3 0 3):

System of units, circuit variables and elements, simple resistive circuits, techniques of circuit analysis, wye-delta transformation, the operational amplifier, superposition, Thevenin's and Norton's theorems, inductors and capacitors, response of first order RL and RC circuits, natural and step response of RLC circuits, sinusoidal and complex forcing functions, phasors.

Pre-requisite(s): MT101

EE212 Electrical Network Analysis (3 0 3):

Steady-state power analysis, poly-phase circuits, magnetically coupled networks, frequency characteristics, variable frequency network performance, resonant circuits, the Laplace transform, application of Laplace transform to circuit analysis, Fourier analysis techniques.

Pre-requisite(s): EE211

EE221 Digital Logic Design (3 0 3):

Number systems, operations, different codes, digital logic gates, boolean algebra and simplification of boolean functions, combinational logic, functions of combinational logic, sequential logic and state machines, flip-flops and related devices, adders, multiplexers, and applications, registers, counters, shift registers and memories.



EE222 Data Structures and Algorithms (2 0 2):

Python programming basics, object oriented programming using python, file IO, list/array, stack, queue, linked-lists, complexity and big oh notation, linear search, binary search, selection sort, merge sort, quicksort, recursion, implementation of a GUI using tinker.

Pre-requisite(s): CS101

EE223 Microprocessor Systems (3 0 3):

Register transfer and micro-operation, basic computer organization and design, programming the basic computer, pipelining and instruction scheduling, introduction to PIC microcontrollers, introduction to assembly programming language, I/O ports' programming and arithmetic/logic functions, PIC18 internal peripherals programming using embedded C language, interfacing external peripheral devices using embedded C language.

Pre-requisite(s): EE221

EE231 Electronic Devices and Circuits (3 0 3):

Introduction to semiconductors, n type and p type material, diodes, diodes equivalent circuits, types of diodes, zener regulators, light emitting diodes, load line analysis, parallel and series connections of diodes, gates, half wave and full wave rectifiers, clipper and clamper circuits, voltage doubler circuits, bipolar junction transistor, construction and operation, amplification analysis, common-base, common-emitter, common-collector amplifier configurations, cascade connections of BJT, limits of operation, fixed biasing, emitter biasing and voltage divider biasing configurations, introduction to field effect transistor, characteristics of FET, types of FET, FET applications

Pre-requisite(s): EE211

EE311 Electric Machines (3 0 3):

Introduction to electrical machines' principles, single phase transformer construction and its operational characteristics and equivalent circuits, transformer tests, auto-transformer and three phase transformers, fundamentals of electro-mechanical energy conversion, AC machinery fundamentals, synchronous machines, induction machines, DC machines, special purpose motors

Pre-requisite(s): EE212

EE312 Power Distribution and Utilization (3 0 3):

Introduction to different types of

distribution system, characteristics and estimation of load, electricity tariff and principles of its calculation, power factor and methods for its improvement, protection and grounding, heating and welding, batteries and electrochemical processes and fundamentals of illumination engineering.

Pre-requisite (s): EE212

EE313 Power Transmission (3 0 3): Percent and per-unit quantities, one-line diagram, HV, EHV and UHV systems, conductor types, resistance, skin effect, line inductance and capacitance, ferranti effect, short, medium and long transmission lines, traveling waves, surge impedance loading (SIL), mechanical design of transmission lines, corona effect, underground cables, modern trends in power transmission.

Pre-requisite (s): EE212

EE314 Electrical Instrumentation and Measurements (2 0 2): Precision measurements technologies, instrument calibration, engineering units and standards; instruments for measurement of electrical properties, signal processing and transmission; modern instrumentation techniques, instrumentation and signal conditioning circuits; data manipulation, oscilloscope, signal generators, transducers, bridges, power and energy meters; temperature and other measurements.

Pre-requisite (s): EE211

EE331 Electronic Circuit Design (3 0 3): Multistage gain calculation, impedance matching for amplifiers, hybrid model of BJT, current sources, frequency response of BJT and FET amplifiers, differential amplifiers, operational amplifiers, design op-amp circuits for various applications, instrumentation amplifier, active filters, power amplifiers, series fed and transformer coupled class A, class B, and class C amplifiers, A/D and D/A converters, phase locked loop, feedback and stability consideration for amplifiers, types of feedback in amplifier circuits, oscillators.

Pre-requisite(s): EE231

EE332 Power Electronics (3 0 3): Introduction to power electronics, power electronic devices: diode, BJT, MOSFET, IGBT, SCR, rectifier circuits: single and three phase controlled and uncontrolled rectifiers, DC-DC converters: buck, boost, buck-boost and isolated converters:

forward and flyback converters, PWM inverters, single and three phase inverters, cycloconverters, matrix converters, AC voltage regulators, power electronics for machine control, power electronics for FACTS and HVDC.

Pre-requisite(s): EE231, EE311

EE341 Signals and Systems (3 0 3): Introduction to signals, basic continuous and discrete time signals, introduction to systems, discrete time linear time invariant (DT-LTI) systems, continuous time linear time invariant (CT-LTI) systems, properties of CT-LTI systems, laplace transform and CT-LTI systems, inverse laplace transform, z-transform and discrete time LTI systems, fourier series representation of CT/DT periodic signals, the fourier transform, selected application of fourier series and transforms including sampling, filtering, communication, and control system.

Pre-requisite(s): MT102, EE212

EE351 Linear Control Systems (3 0 3): Introduction to control systems, system modeling, transient response analysis, response and pole locations, time domain specifications, effects of adding zeroes and poles, properties of feedback, disturbance rejection, tracking, steady state tracking and system type, types of controllers, stability, root-locus analysis, control systems design by root locus method, frequency response analysis, control systems design by frequency response, Nyquist stability criterion and pid control, modeling and analysis of control systems in state-space.

Pre-requisite(s): EE341

EE361 Communication Systems (3 0 3): Introduction to modern analog and digital communication systems, fourier analysis of signals and systems and signal transmission,



amplitude modulation - generation, demodulation, single side band, double side band, quadrature amplitude modulation, vestigial side band, AM receiver and other applications, angle modulation - generation, frequency modulation, demodulation, FM receiver and other applications, sampling and reconstruction, differential pulse code modulation and delta modulation, introduction to principles of digital communication systems.

Pre-requisite: EE341

EE371 Electromagnetic Field Theory (3 0 3): Vector analysis, Coulomb's law and electric field intensity, electric flux density, Gauss's law and divergence, energy and potential, electrical properties of materials, experimental mapping methods, Poisson's and Laplace's equations, the steady magnetic field and magnetic properties of materials, time-varying fields and Maxwell's equations.

Pre-requisite(s): PH101, MT102

EE411 Power System Analysis (3 0 3): Fundamentals of an electric power system, transmission line parameters, power system operation studies, load-flow studies, symmetrical components, balanced and unbalanced faults on power systems, power system stability.

Pre-requisite(s): EE212

EE412 Power System Protection (3 0 3): Types and effects of faults, power-system transients and over-voltages, principles and characteristics of protective relaying, over current protection, distance protection, differential protection of transformers, generator protection, bus bar protection, types and operation of relays and circuit-breakers, switch-gears.



Pre-requisite(s): EE311

EE413 High Voltage Engineering (3 0 3): Introduction to the subject and important properties of dielectrics and their measurements, high voltage AC, DC and impulse generation, high voltage measurement techniques, electric field distribution in insulation systems and its control, electrical breakdown of gaseous, liquid and solid insulation systems, procedures of testing high voltage equipment, non-destructive high voltage tests, insulation levels and insulation coordination.

Pre-requisite (s): EE312

EE414 Renewable Electrical Energy Systems (Elective) (3 0 3): Introduction to renewable energy systems, load interface with photovoltaics and wind energy, stand alone and grid tied systems, battery storage and standalone system design, voltage regulation, renewable energy power system architecture, safety standards and guidance regulations, maximum power point tracking for wind and solar systems

Pre-requisite (s): EE311, EE332

EE415 Electrical Machine Design and Maintenance (Elective) (3 0 3):

Part-A Machine Design:

Industrial standardization, design considerations for electrical machines, properties and applications, cooling systems of transformers and rotating machines, duty cycles, ratings and temperature-rise, mechanical design considerations, design of transformer or induction motor, introduction to CAD and CAM.

Part-B Installation, Maintenance and Troubleshooting of Machines:

Safety precautions, troubleshooting and emergency repairs, Installation, commissioning, testing, maintenance, and troubleshooting of (i) power transformers and (ii) induction motors. (iii) AC generators.

Pre-requisite (s): EE311

EE416 Electrical Machine Drives and Control (Elective) (3 0 3): Electromechanical systems, machine load characteristics, drive system elements, required drive characteristics, DC drives, induction motor drives, current-sourced inverter drives, voltage-sourced inverter drives, advanced control of voltage-sourced inverters, synchronous motor drives, induction motor

dynamics, torque (Vector) controlled drives
Pre-requisite (s): EE311, EE351

EE417 Special types of Electrical Machines (Elective) (3 0 3):

Construction and principles of special types of electrical machine, reference-frame theory, switched reluctance motors, stepper motors, theory of brushless dc machines, linearized machine equations, symmetrical and unsymmetrical 2-Phase induction machines,
Pre-requisite (s): EE311

EE421 Programming Tools for Optimization (2 0 2):

Linear programming, network programming, integer programming, stochastic programming, concepts of convexity, steepest gradient descend method for solving convex optimization problems.
Pre-requisite(s): EE222

EE422 Introduction to ASIC Design (Elective) (3 0 3):

Introduction to FPGA, ASIC technologies, design methodologies, architecture of FPGA and CPLD, design, implementation, synthesis and verification of ASIC on FPGA using verilog hardware description language (HDL), FPGA memories and programming technologies, finite state machines, design and implementation of combinational and sequential circuits on FPGA.
Pre-requisite(s): EE221

EE423 Digital Integrated Circuit Design (Elective) (3 0 3):

Introduction to VLSI: (complexity & design, concepts), logic design with MOSFETs (switches, gates, complex logic including combinational & sequential, clock & delays), physical structure of CMOS ICs (IC layers, CMOS layers, problems), fabrication of CMOS ICs (silicon processing, CMOS process flow), elements of physical design (structure layouts, cell, FET sizing), digital integrated modules design using verilog HDL.
Pre-requisite: EE221, EE331

EE424 VLSI Design (Elective) (3 0 3): CMOS devices and deep sub-micron manufacturing technology, CMOS inverters and complex gates; modeling of interconnect wires, optimization of design with respect to a number metrics: cost, reliability, performance, and power dissipation; sequential circuits, timing considerations, clocking approaches, design

of large system blocks including arithmetic, interconnect' memories, and PLA design methodologies.
Prerequisite(s): EE231

EE425 Embedded Systems (Elective) (3 0 3):

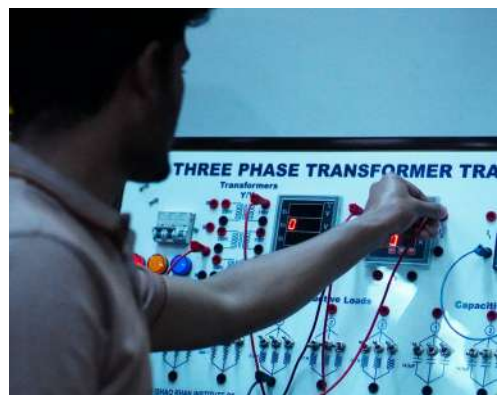
Embedded Systems and their Classification; Various design constraints and properties of Embedded Systems, Real time Embedded Systems, Scheduling Algorithm, Real time Schedulers, Bare metal programming of 32-bit ARM MCU, Introduction to the STM32F407 Discovery Kit, Using STM Cube MX and Keil as Software programming and Debugging tools, Peripherals interfacing; Universal Asynchronous Receiver Transmitter (UART), Free Real time Operating System (FreeRTOS)
Prerequisite(s): EE231

EE431 Solid State Electronics (Elective) (3 0 3):

Introduction to semiconductor materials, basic structure and properties, carrier concentration, energy band gap, carrier transport in semiconductor, pn junction, metal-semiconductor contacts, metal oxide semiconductor FET, bipolar transistors, photonic devices, solar cell, semiconductor devices growth and fabrication techniques.
Pre-requisite(s): EE231

EE432 Industrial Electronics (Elective) (3 0 3):

Electric heating: principles and applications; induction and dielectric heating; high frequency welding, spot welding control, industrial drives: speed control of DC, AC, and servo motors, process control systems, measurement of nonelectrical quantities: temperature, displacement, pressure, time, frequency; digital industrial measuring systems, ultrasonic generation and applications, photo-electric



devices, industrial control using PLCs, data acquisition for industrial processes, distributed control system in process industries, basic concepts of SCADA.

Pre-requisite (s): EE331, EE351

EE436 Digital Systems Design (Elective) (3 0 3)

Arithmetic circuits – Data paths, arithmetic/logic unit (ALU), shifters; computer design fundamentals – introduction to computer design, Instructions set architecture (ISA); introductory HDL – 1 (Verilog or VHDL) – timing in Combinational Circuits, Hazards and Glitches, Review of sequential logic; HDL Behavioral, Sequential coding, and ModelSim – design using flip-flop and latches, state machines, state reduction, timing issues, design of adders and subtractors, carry lookahead adders, serial adders, array multipliers, critical paths, Booth and Radix-4 encoded signed multipliers, further VHDL modeling, parameterization; FPGA Implementations – LFSR, BRM, Function Generators, Design Examples, Faults and Testability – BIST and SCAN techniques, Design for test – JTAG, advanced HDL – Memories and Register Files, Design Examples.

Prerequisites: EE221, EE331

EE441 Digital Signal Processing (3 0 3):



Ayza Ahmed

Three years back, as I walked into this totally new phase of my life, I was anxious about what lay ahead of me. There were all sorts of emotions ranging from excitement and optimism, to nervousness and the fear of the unknown. I reflect back on my journey, these three years have been transformative to say the least, survival in this incredibly diverse and academically challenging environment did seem daunting at first, but believing in yourself to make it through all the challenges that GIKI throws at you. Societies here are a great way to channel not only your personal growth, but also to make lifelong friendships.

Analysis and representation of discrete-time signals, discrete-time convolution, discrete-time Fourier transform, difference equations, z-transform, sampling theory, interpolation and decimation, transform analysis of LTI systems, structures for digital systems, FIR and IIR digital filters, digital filter design techniques, fast Fourier transform algorithm for computation of discrete Fourier transform, introduction to discrete stochastic processes.

Pre-requisite(s): EE341

EE442 Digital Image Processing (Elective) (3 0 3):

Introduction to image processing, 2D signals and systems, convolution and correlation, image transforms, image enhancement, image restoration and de-noising, image segmentation, image recognition, image compression, binary image processing, color-image processing.

Pre-requisite(s): EE341

EE443 Introduction to Wavelets (Elective) (3 0 3):

Analysis and synthesis of signals, time-frequency and time-scale analysis, continuous wavelet transform, multi-resolution analysis, filter banks and discrete wavelet transform, properties of the filters, scale and wavelet functions and designing wavelets.

Pre-requisite(s): EE441

EE454 Introduction to Robotics (Elective) (3 0 3):

Introduction to robotics, sensors, actuators and manipulators, introduction to computer programming for robotics, inverse kinematics, robot dynamics, robot control, trajectory planning, work-space considerations, obstacle avoidance.

Pre-requisite(s): EE351

EE452 Industrial Process Control (Elective) (3 0 3):

Introduction to process control, sensors and measurement systems, signal conditioning and processing, estimation of errors and calibration, analogue to digital conversion, pid controller design, control structures, introduction to sequence control, sequence controls system, PLCs and relay ladder logic; advanced RLL programming, control of machine tools : introduction to CNC machines; analysis of a control loop, actuators, hydraulic and pneumatic control design, introduction to advance control technique.

Pre-requisite(s): EE331, EE351

EE453 Digital Control Systems (Elective) (3 0 3): Introduction to discrete-time control systems, z-transform, z-plane analysis of discrete-time control systems, design of discrete-time control systems by conventional methods, state space analysis techniques, state space design technique, pole placement and observer based, quadratic and optimal control systems design.
Pre-requisite(s): EE351

EE461 Communication System Design and Performance Analysis (Elective) (3 0 3): Review of probability fundamentals, system noise analysis, SNR and BER calculations, behavior of analog/digital systems in presence of noise, optimal receiver design concepts, introductory information theory, coding and multiplexing techniques including source, channel, and line coding techniques, and FDM, TDM, and CDM techniques, introduction to advanced digital modulation/multiplexing techniques such as OFDM, W-OFDM, SDM.
Pre-requisite(s): ES111, EE361

EE462 Computer Communication Networks (Elective) (3 0 3): Introduction to computer networks, network models and topologies, network layering concepts and protocols, open systems interconnection (OSI) model and internet protocol (IP) and associated control protocols, end-to-end protocols, with TCP and UDP as examples, addressing schemes at link layer, network layer and transport layer, transmission media and characteristics, switching techniques, channel access techniques, MAC routing protocols and multicast, overview of application layer Protocols (HTTP, FTP, SMTP etc), multimedia protocols (RTP, RTSP, RTCP), security mechanisms and services, concepts of symmetric and asymmetric cryptography, digital signature, convergence of communication networks.
Pre-requisite(s): EE361

EE463 Cellular Mobile Communication Systems (Elective) (3 0 3): Introduction to wireless communications, basic cellular concepts, frequency reuse, channel assignment and hand-off techniques, interference and system capacity, trunking and grade of service, system capacity improvement techniques, mobile propagation models including large-scale path loss and small-scale fading models, multiple access techniques for cellular systems,

speech codes and standards, and emerging areas in mobile communications.
Pre-requisite(s): EE361, EE471

EE464 Satellite Communication Systems (Elective) (3 0 3): Introduction to satellite communication, space-segment and ground segment, orbital mechanics, geostationary and non-geostationary orbits, launching and spacecraft subsystems, look angle determination, orbital perturbations, orbital effects in communication system performance space craft and its subsystem, satellite link design, propagation characteristics of satellite links, channel modeling, access control schemes, modulation schemes, multiple access schemes, coding, system performance analysis, system design, space standards, earth station technology, satellite applications such as earth observation, weather, and communication, VSATs and network architectures, GPS, future trends.
Pre-requisite(s): EE371, EE361

EE471 Wave Propagation and Antennas (3 0 3): Wave propagation, transmission line theory, Smith chart, impedance matching and two port networks, network analysis, s-parameters, strip-type transmission line, rectangular and circular waveguides, antenna fundamental



Maaz Bin Khalid

It has been a roller-coaster ride for me, thousand miles away from home. GIKI has made me a better person in every aspect, a unique experience in which every day is a new challenge. The environment in GIKI pushes you to your limits and I believe that is what is required in your future endeavours.

parameters, radiation power density, directivity, elementary dipole antenna.

Pre-requisite (s): EE371

EE472 Antenna Theory and Design (Elective)

(3 0 3): Antenna types, applications, basic concepts, radiated power, radiation pattern, directivity, vector potentials, electric and magnetic currents, dipole antenna and loop antenna, equivalent magnetic dipole, derivation of radiated power, radiation pattern and directivity, microstrip dipole antenna, microstrip loop antennas, antenna arrays and their radiation pattern, Horn antenna, Yagi-Uda antenna, aperture antennas, reflector antennas, feed networks and impedance matching.

Pre-requisite (s): EE371

EE473 Microwave Engineering (Elective) (3 0 3):

RF behavior of passive components and RF models, chip components, distributed circuit elements, strip lines, microstrip lines, coupled striplines/coupled microstriplines, smith chart, impedance and admittance transformation, parallel and series connection, impedance matching networks, analysis of single and multiport networks using network parameters, microwave filter design, microwave amplifier design, mixers and detectors, oscillators, power dividers, directional couplers, circulators, microwave systems.

Pre-requisite (s): EE471

EE474 Radar Systems (Elective) (3 0 3):

Basic principles, wave propagation near earth's surface and atmosphere, radar antennas, radar block diagram, frequencies, radar equation, monostatic and bistatic radar cross section of various targets, radar signals and networks, pulse compression, radar resolution, probability of detection and false alarm, MTI and doppler radar systems, detection of signal in noise and clutter, microwave sources.

Pre-requisite(s): EE361, EE371

EE481 Electrical Estimation Installation and Planning (Elective) (3 0 3):

The estimating process, components of electrical system, installation, protection circuits design and testing, planning, system protection, low voltage switch boards and distribution system, grounding system, power factor correction and harmonic filtering, power cables, supply systems, electrical installation equipment and

system.

Pre-requisite (s): EE312

EE482 Power Generation (Elective) (3 0 3):

Introduction and overview of conventional power generation, review of basic thermodynamics and thermal sciences, thermal and hydroelectric power plants, reservoir based and run of the river hydroelectric projects, introduction to renewable energy systems, solar photovoltaic, wind turbine systems, geothermal energy system, biomass energy harnessing, nuclear power plants, sociological, political and economic aspects of conventional and non-conventional energy resources and sustainability analysis.

Pre-requisite (s): EE311, EE332

EE483 Power Plant Engineering (Elective)

(3 0 3): Variable load problem, gas turbine power plants, steam power plants, rankine with superheat and reheat, steam generators, fire-tube boiler, water-tube boiler, steam turbines types and efficiency, steam condensers, nuclear power plants, PWR and fast breeder reactors, hydro-electric power plant, reaction and impulse turbines, wind turbines and photovoltaics.

Pre-requisite (s): EE312

EE484 Electrical Insulation Materials

(Elective) (3 0 3): Introduction to the subject and important properties of dielectrics and their measurements, computational methods of electrical field analysis, breakdown mechanisms in gaseous, liquid and solid dielectrics, insulating materials, mechanisms of conduction and polarization in insulation media, dielectric response measurements, insulation systems in practice – organic and inorganic materials for insulation, composite insulation, outdoor insulation, high voltage diagnostic measurement techniques, pollution flashover.

Pre-requisite (s): MM102, EE371

EE485 Power Economics and Management

(Elective) (3 0 3): Principles of economics, engineering economy, economic environment, price-supply-demand relationship, elementary financial analysis, break even analysis, selection between alternatives, value engineering, linear programming, business organization, capital financing and allocation.

Pre-requisite (s): MS291, EE312

EE486 Power System Operation and Control (Elective) (3 0 3): Introduction to power system control and its importance, modes of power system operation, major tasks of operation, remote terminal unit, characteristics of power generation units, economic dispatch problems with and without consideration of losses, unit commitment, incremental fuel cost, penalty factor, economic power interchange, hydro thermal coordination, voltage, power and frequency control, evaluation of the effect of speed change on droop characteristics, SCADA system.

Pre-requisite(s): EE311, EE351

EE487 Power System Design (Elective) (3 0 3): Characteristics, performance and design of transmission lines, design of EHV transmission lines, advantages and disadvantages of HVAC and HVDC, selection of sizes and locations of generating stations and substations, designs of distribution systems, economics of distribution systems.

Pre-requisite(s): EE312, EE411

EE491 and EE492 Senior Design Project (0 18 6): The aim of the design project is to sharpen the electronic/electrical circuit/system design skills of the FEE graduating students by participating in projects that are to be identified in collaboration with the industry. Every project will be assigned a faculty advisor. The students may work independently or jointly (in small groups) on the projects. The duration of the project term is one full year. The progress will be monitored through interim presentations and reports. A final report will be due at the end of the term.

EE/CS/ES 4xx Technical Elective I/II/III (3 0 3): This is description for elective courses.

These courses are offered by the faculty in different areas of specializations to meet the changing requirements of the technology.

EE xxxL Lab Course (0 3 1): Stands for lab work associated with a theory course having the same code number. A Lab course can be registered only as a co-requisite of its associated theory course. Experiments performed in a lab course are related to those topics covered in the respective theory course.

Career in Electrical Engineering

The graduates of Faculty of Electrical Engineering have careers in three major fields:

- Communication and Digital Signal Processing
- Microelectronics and ASIC Design, and
- Electric Power, and
- Control System

These areas have been selected keeping in mind the current and future requirements of Pakistan. The first, Communication and Digital Signal Processing, is the key to the global information revolution. The second, Microelectronics and ASIC Design, targets the heart of the computer revolution, and encompasses issues related to the theory, fabrication and design of high speed, dense integrated circuits. The third, Power and Control, has special relevance to Pakistan given the level of investment in the power sector, and wide-spread applications of control systems in industry. Choosing this degree course does not restrict choice of careers. In the past, our graduates have been employed in areas as diverse as power, telecommunications, computer networks, industrial control, and VLSI/ASIC design. A significant number of graduates have opted to pursue advanced studies in the US and Europe.



FACULTY OF ENGINEERING SCIENCES



Introduction

Bachelor of Science in Engineering Sciences is a unique multidisciplinary Engineering Program in Pakistan, duly recognized by the Pakistan Engineering Council. The 4-year engineering degree encompasses some of the most modern fields of engineering, split into two specializations, namely, (i) Photonics and Microelectronics, and (ii) Modeling, Simulation, and Machine Learning. Both areas of concentration require in-depth exposure to both science and engineering. The goal of this program is to develop each student's ability to think analytically across disciplines and develop a knowledge base well-suited to tackle future technical challenges that will require a thorough understanding of a discipline in the physical sciences combined with engineering. Over the years, Engineering Sciences graduates have gone on to become successful engineers, managers, researchers, and academicians, working in fields such as medical imaging, optical communications, instrumentation, sensors, control, chip design, operations management, data science, artificial intelligence, and software solutions.

The vigorous growth of the photonic industries, lasers, semiconductor and microelectronics, instrumentation, simulation of systems, data analytics, and machine learning has created a demand for engineers who can completely cope with the present and future demands of the modern industry. Light, in particular, laser, has become the workhorse and the instrument of choice both in research and industry, covering engineering applications from communications to instrumentation. Importance of Semiconductors and Microelectronics continues to grow and the introduction of software-based design of devices as the new "fabless" industry is bringing new opportunities to Pakistan. Simulation and Machine Learning are now the key design and decision-making tools being used across the world.

The graduates of Engineering Sciences will be suitable for industry that is involved not only in production but also in research and development both within the country and abroad. Already, within the country, several organizations are pursuing R&D work and production in engineering field of technologies. At present the main power for such organizations in these fields is either trained or the assistance of foreign consultants is sought. The graduates of this faculty will be well equipped to fill this gap in national expertise and can look forward to highly rewarding careers both locally and globally.

Thrust Areas:

The undergraduate program of Faculty of Engineering Sciences is duly accredited under Washington Accord Level-II category of Outcome Based Education (OBE) system by the Pakistan Engineering Council. To complete the degree requirement, students must complete 19 credit hours in one of the specialization fields mentioned below. Students are assigned projects and suitable advanced elective courses to develop expertise in the specialized areas. Students are required to opt for the specialization in the third year (5th Semester) of their undergraduate degree plan. Following are the two specialization streams offered by FES:

- Photonics and Microelectronics
- Modeling, Simulation, and Machine Learning

Faculty Mission:

To produce capable engineers working as responsible global citizens, future leaders of society and leading practitioners of Engineering Sciences.





Dean

Engr. Dr. Naveed Razzaq Butt
PhD (LU, Sweden)

Faculty

Muhammad Hassan Sayyad, PhD (Dublin City University, Ireland)
Jameel-Un Nabi, PhD (University of Heidelberg, Germany) (On Leave)
Sirajul Haq, PhD (University of Liverpool, UK)
Muhammad Zahir Iqbal, PhD (Universitat Politècnica de Catalunya, Spain)
Muhammad Usman, PhD (Hanyang University, South Korea)
Tahseen Amin Khan Qasuria, PhD (GIK Institute, Pakistan)
Shafqat Ali, PhD (International School for Advanced Studies, Trieste Italy) (On Leave)
Sakander Hayat, PhD (University of Science and Technology, China)
Asad Mahmood, PhD (Telecom Paristech, Paris, France)
Muhammad Tayyab, PhD (University of Turin and Polytechnic University of Turin, Italy)
Mazhar Ali, PhD (Skolkovo Institute of Science and Technology, Moscow, Russia)
Fahd Sikandar Khan, PhD (University of Tokyo, Japan)
Dr. Usman Habib, PhD (University of Kent, UK)
Sheharyar Pervez, MS (Indiana University Bloomington, USA)
Shahid Ahmad, MS (University of Illinois, Urbana Champaign, USA)
Fahad Zulfiqar, MS (University of Sheffield, England)
Muhammad Saqib, MS (NUST, Pakistan)
Sibtul Hassan Shirazi, MS (GIK Institute, Pakistan)
M. Kashif Azeem, MS (NUST, Pakistan))
Asif Ahmed, MS (University of Engineering and Technology, Peshawar)
Mahmood ul Hassan, MS (University of Engineering and Technology, Peshawar)

Faculty (on study leave for PhD)

Rahim Umer (Nanjing University of Aeronautics and Astronautics, China)
Taimoor Ali (University of Oxford, UK)
Naveed Ahmed Azam (Kyoto University, Japan)

Joint Faculty

Khasan Karimov, PhD (Physical Technical Institute S.-Petersburg, Russia)
S. M. Ahmed, PhD (University of Sheffield, UK)
Mohammad Akbar, PhD (Tokyo University, Japan)

Lab Engineers

Muhammad Sadiq, BS (GIK Institute, Pakistan)
Usama Javed, BS (GIK Institute, Pakistan)
Hasnain Amin, MS (NUST, Pakistan)
Hamza Naeem (NUML, Pakistan)
Mohammad Mohsin (GIK Institute, Pakistan)

Graduate Assistants

Asim Ullah, MS (Applied Physics) (GIK Institute, Topi, Pakistan)
Shamsul Arifeen, MS (Applied Mathematics) (GIK Institute, Topi, Pakistan)
Sadaf Shaheen, M.Phil (Mathematics) (NUST, Islamabad, Pakistan)
Shabeela Malik, MS (Applied Mathematics) (GIK Institute, Topi, Pakistan)
Sabiha Qazi, MS (Applied Mathematics) (GIK Institute, Topi, Pakistan)
Nayab Ali, MS (Applied Physics) (COMSATS University, Islamabad, Pakistan)
Mehboob ul Haq, MS (Mathematics) (COMSATS University, Abbottabad)
Ayesha Noreen, BS (Mathematics), (National University of Sciences and Technology (NUST))
Laiq Zada, M.Sc (Mathematics), University of Peshawar
Nayyab Amjad, BS (Physics), (Hazara University, Mansehra)
Saad Rasheed, BS (Physics), (Abbottabad University of Science and Technology, Abbottabad (AUST))
Shazma Ali, BS (Physics), (COMSATS Institute of Information Technology)
Sana Saeed, BS (Physics), (Abbottabad University of Science and Technology, Abbottabad (AUST))
Ahmad Ali, BS (Physics), (University of Peshawar)
Muhammad Waqas, BS (Physics), (Islamia College, Peshawar)
Nadia Anwar, MS (Applied Physics) (GIK Institute, Topi, Pakistan)
Mujeeb Rahman, MS (Applied Physics) (COMSATS University, Islamabad)
Sajid Khan, MS (Applied Physics) (COMSATS University, Islamabad)
Anique Ahmed, MS (Applied Physics) (GIK Institute, Topi, Pakistan)
Rida Zahid, BS (Physics), Abbottabad University of Science and Technology, Abbottabad (AUST)
Alina, BS (Mathematics), (Islamia College, Peshawar)
Zafar Ali, BS (Physics), (Islamia college Peshawar)
Wiqar Hasnain BS (Mathematics), (COMSATS University, Islamabad)
Mehboob Alam, BS (Mathematics), (University of Peshawar)

PS to Dean, FES

Muhammad Safiq, MA English (University of Peshawar, Pakistan)

Program Educational Objectives

The Faculty of Engineering Sciences at GIK Institute has formulated the Program Educational Objectives (PEOs) using the feedback from the stake-holders. There are three PEOs for the ES program.

PEO 1: Graduates having a strong scientific foundation practicing as competent, continuously developing engineers in Engineering Sciences related fields.

PEO 2: Graduates providing leadership in their organizational and technical capacities, working whether as an individual or as part of a team.

PEO 3: Graduates acting as ethical and responsible professionals providing solutions with due consideration to economic, environmental and safety impacts of their work on society.

Program Learning Outcomes

There is a set of twelve Program Learning Objectives (PLOs) of Engineering Sciences program which describe what students are expected to know/perform/attain by the time they graduate from Faculty of Engineering Sciences. These PLOs are set such that all course deliveries encompass these objectives, and are described as follows:

PLO 1: Engineering Knowledge Ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PLO 2: Problem Analysis Ability to identify, formulate, research literature, and analyze complex engineering problems reaching

substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PLO 3: Design/Development of Solutions Ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

PLO 4: Investigation Ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.

PLO 5: Modern Tool Usage Ability to create, select and apply appropriate techniques, resources, and modern engineering and IT

PLO 6: The Engineer and Society Ability to apply reasoning apparatus (PASCO) and Variable "g" pendulum (PASCO). informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.

PLO 7: Environment and Sustainability Ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

PLO 8: Ethics Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PLO 9: Individual and Teamwork Ability to



work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.

PLO 10: Communication Ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PLO 11: Project Management Ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

PLO 12: Lifelong Learning Ability to recognize importance of and pursue lifelong learning in the broader context of innovation and technological developments.

The modern perspective of an engineering curriculum, especially for programs emphasizing OBE, is that it is the most important instrument for grooming the above-mentioned 12 GAs in students. Therefore, it is viewed to consist of a number of Knowledge Profiles (WKs) that inculcate different dimensions of thinking – mathematical, computational, design and creative – among students in Cognitive, Psychomotor and Affective domains. In particular, the institution should ensure that at least the following knowledge profiles are incorporated in the curriculum:

WK1 Natural Sciences: A systematic theory-based understanding of natural sciences applicable to the discipline.

WK2 Mathematics and Computing: The concept-based mathematical thinking, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline.

WK3 Engineering Fundamentals: A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.

WK4 Engineering Specialization: The knowledge of Engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.

WK5 Engineering Design: The Design

Thinking Knowledge that supports engineering design in a practice area of an engineering discipline.

WK6 Engineering Practice: The Knowledge of engineering practices (technology) in different practice areas of an engineering discipline.

WK7 Engineering in Society: A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity i.e. economic, social, cultural, environmental and sustainability

WK8 Research Literature: Engagement with selected knowledge in the research literature of the discipline

Sustainable Development Goals (SDGs):

SDG 1 No Poverty: End poverty in all its forms everywhere.

SDG 2 Zero Hunger: End hunger, achieve food security and improved nutrition and promote sustainable agriculture.

SDG 3 Good Health and Well-Being: Ensure healthy lives and promote well-being for all at all ages.

SDG 4 Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 Gender Equality: Achieve gender equality and empower all women and girls.

SDG 6 Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

SDG 7 Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all.

SDG 8 Decent Work and Economic Growth: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

SDG 9 Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

SDG 10 Reduced Inequalities: Reduce inequality within and among countries.

SDG 11 Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient and sustainable.

SDG 12 Responsible Consumption and Production: Ensure sustainable consumption

and production patterns.

SDG 13 Climate Action: Take urgent action to combat climate change and its impacts.

SDG 14 Life Below Water: Conserve and sustainably use the oceans, seas and marine resources for sustainable development.

SDG 15 Life on Land: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

SDG 16 Peace, Justice and Strong Institutions: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.

SDG 17 Partnerships for The Goals: Strengthen the means of implementation and revitalize the global partnership for sustainable development.

Accreditation:

The BS Degree Program in Engineering Sciences is accredited under Washington Accord Level-II category of Outcome Based Education (OBE) system by the Pakistan Engineering Council (PEC).

Teaching and Research Labs:

Faculty of Engineering Sciences has many teaching and research laboratories.

Teaching Labs:

A brief introduction to the teaching labs in FES is presented below.

PH101L Applied Physics Lab: In this laboratory students perform the experiments related to the vector properties, equilibrium, kinematics, and dynamics of translator motion, two-dimensional motion Work-Energy Theorem, and rotational dynamics. The laboratory also covers fundamentals of electricity and magnetism. The laboratory is equipped with various kits including Mechanics System Kit (PASCO), Air Track Kit (PASCO), Rotational Dynamics Kit (PASCO and PHYWE), Projectile Launcher (PASCO) Gravitational constant apparatus (PASCO), Free Fall apparatus (PASCO) and Variable “g” pendulum (PASCO).

ES211L Circuit Analysis Lab: This lab will give students the ability to analyze AC circuits and Analyze Variable-Frequency Network Performance, analyze circuits using Laplace Transform technique, analyze circuits using Fourier techniques and two port parameter concepts.

ES212L Logic Design Lab: This lab introduces to logic design and basic building blocks used in digital systems. A study of basic and complex digital logic circuit design, and their implementation. Circuit schematic development simulation of digital systems. Experiments explore designs with combinational and sequential logic. Students work through design activities, which include testing, implementing, troubleshooting, and a final lab project.

ES213L Computer Architecture Lab: This lab will give students the ability to simulate



combinational and sequential logic using Verilog HDL as well as to design logic and digital computer systems having RISC Based Architecture.

ES231L Electronics I Lab: This lab will demonstrate will help students to analyze and demonstrate the diode-based circuits in various configurations, the operational principle of circuits for bipolar junction transistor (BJT) and field effect transistor (FET).

ES314L Microprocessor Systems and Interfacing Lab: This lab is meant for the students to learn about typical microprocessor and microcontroller-based systems. It is used in two courses, computer architecture and microprocessor/microcontroller Interfacing. The laboratory is equipped with oscilloscopes, digital trainers, Burners (Programmers), digital multimeters and support electrical and electronics accessories.

ES332L Signals and Systems Lab: This lab is performed in computer simulation lab. All computers are installed with MATLAB software and connected with centralized printer. Student performed signals and systems analysis in frequency and time domain using Signals and Systems toolbox.



Gulfiza Naseem

As the only girl among my classmates, coming all the way from Karachi to Swabi to major in Engineering Sciences (a major most people hardly know anything about) was like an extreme turn of events in my life. However, I believe this to be one of the best decisions I've ever made for myself. The way GIK Institute moulds you as a person is an experience unparalleled and I wouldn't trade it for anything otherwise.

ES334L Foundation of Photonics Lab: Laboratory experiments introducing geometrical and physical optics, characterization of LEDs & Laser diodes, fiber transmission, laser beams, interferometers, optical systems (cameras, scanners, sensors), polarization devices, emission & photoabsorption spectroscopy, demonstration and use of high power laser, demonstration and use of Keithley 4200-SCS Semiconductor Characterization System for study of electronic and photonic devices, modeling and simulation of photonic devices.

ES341L Numerical Analysis Lab: The goal of this lab is to get familiar with discrete Simulation techniques and their uses in various science and engineering applications. It also aims to provide basic knowledge of designing simulation models, simulation algorithms and their implementation on PCs.

ES361L Solid State Electronics Lab: Students are trained to measure material characteristics such as resistivity measurement, conductivity type and carrier concentration using state-of-the-art modeling software. Experiments on Solar Cell I-V characterization and thermoelectric generator are also conducted in this lab. Major equipment includes Hall Effect board (P/nGe), Hall Effect board (Zn/Cu), Universal Measuring Amplifier and support accessories.

ES322L Instrumentation Lab: In this lab students are trained how to interface the physical world with the computer by using the LabView software. The students are given tasks of sensors interfacing including thermal, mechanical, and optical sensors. They also learn how to develop the graphical user interface. At the end of the semester students are also given the open-ended problem of any electro-mechanical system.

ES474L Optoelectronics Lab: Laboratory experiments introduce principles of optical waveguiding, fiber optic communications, optical network analysis, principles of lasers, optical modulators, and WDM component characterization. Laboratory facilities include Michelson interferometer kits, advanced optics kits, spectrometers, DSP lock-in-amplifiers, He-Ne lasers, high power Nd:YAG laser, diode

lasers, laser power meters, PIN diodes, APDs, phototransistors, computers with DAQ cards, Oscilloscopes, analog and digital trainers, photonic device fabrication & characterization, software tools for the modeling & simulation photonic devices and systems, and a wide range of other photonic components and kits.

ES425L Fiber Optic Communication Lab:

In this lab students are trained fiber optics patch cards, optical modulators, WDM and directional couplers. They also learn how to develop the graphical user interface. The facilities include Newport fiber optics kits, fiber optics patch cards, optical modulators, WDM and directional couplers.

ES324L Discrete System Modeling and Simulation Lab:

This lab is used to simulate and analyze different models of System Design and Engineering Management. The lab is equipped with 20 Core i7 PCs running on Windows 10 operating system. These PCs are interconnected via broadband network and students have access to internet, e-mail, and a high-speed laser printer. Different software tools such as MATLAB and Simulink are used to perform simulations of various engineering designs. Arena, SPSS, and Excel packages are used to perform discrete-event simulations and analysis of output data to solve problems of engineering management.

ES471L Model Engineering Lab: The purpose of this lab is to equip students with a set of skills that will enable them to model complex engineering problems as mathematical models. Subsequently, modeling engineering systems as a set of Linear/Nonlinear equations, Differential-Algebraic Equations (DEAs), Differences equations, or stochastic models. In this lab, students will also learn different modeling toolboxes such as Simulink and Mathematic.

ES441L Engineering Optimization Lab:

The goal of this lab is to train students to solve real-world optimization problems. In the planning or operation of an engineering system, engineers have to make technological and managerial decisions at several stages. The ultimate objective of all such decisions is to make optimal actions that result in a minimum effort required or to maximize the

desired outcomes. This lab will introduce optimization tools, which will help students model an engineering design problem as an optimization problem, and then solve them with appropriate optimization methods.

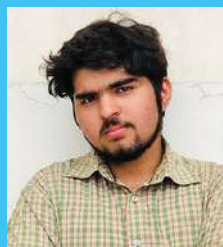
ES442L Machine Learning Lab: Artificial Neural Networks, Support Vector Machines, Overfitting, noisy data, and pruning, Measuring Classifier Accuracy; Linear and Logistic regression; Unsupervised Learning; Hierarchical Agglomerative Clustering, k-means partitional clustering; Self-Organizing Maps (SOM) k-Nearest-neighbor algorithm; Semisupervised learning with EM using labeled and unlabeled data; Reinforcement Learning; Hidden Markov models.

Research Labs:

A brief introduction to the research labs in FES is presented below

Magnetism and Magnetic Materials Lab:

Magnetism and magnetic materials laboratory is equipped with the High-Power vibrating sample magnetometer of Lakeshore. It can generate the power of 11 Tesla. This lab consists of a high-power magnet, power supply and a low temperature probe for measuring the magnetic behavior of the conducting and



Muhammad Abdullah

In the beginning when I decided to choose my bachelor's degree, I was really lost with all the over-whelming options that were available. Undecided about which specific field to pursue, I chose Engineering Sciences at GIKI as a new approach for my learning, and I must say it has to be one of the best decisions I ever made. The information and experience I have learnt while studying this program helped me learn about different fields and make up my mind for the path beyond Bachelors. Education aside, GIKI has helped me grow as an individual and learn some key skills which I never thought I would acquire, and I have the amazing society culture to thank for that. From the management of events at national scale to even the smallest things such as ethics, I have learnt a lot.

superconducting materials.

Thermal Analysis Lab: The thermal analysis laboratory has state-of-the-art equipment purchased from PerkinElmer including Differential Scanning Calorimeter (PerkinElmer DSC-7), Differential Thermal Analyzer (DTA-7), Thermal Gravimetric Analyzer (TGA-7) and Dynamic Mechanical Analyzer (DMA-7). This equipment can be used to investigate the kinetic parameters and change of mass and mechanical properties of various materials with temperature. Moreover, the equipment has direct application for the product development in the paper ceramic, polymer, rubber, glass, and paint industries.

Spectroscopy Lab: Spectroscopy laboratory houses PerkinElmer Fourier Transform Infrared Spectrometer (FTIR System 2000) and UV/VIS/NIR (Spectrometer Lambda-19). Facilities are available for the spectroscopic analysis of liquid, solid and gaseous sample in transmission as well as reflection mode. The equipment has direct application in environmental studies, chemical biochemical and pharmaceutical industries.

Organic Electronics Research Lab: The laboratory has so far produced eight PhDs and several MS students in the investigation of organic semiconductors, conducting polymers and nanoparticles of organic semiconductors for potential applications in organic electronic and photonics devices. The facilities are available for the fabrication and characterization of organic electronic and photonic devices, such as, junction diodes, sensors, etc.

High Power LASER Research Lab: The high-power laser lab at the Faculty of Engineering Sciences is currently engaged in research projects in the fields of laser ablation, laser micromachining and laser materials processing. The laboratory facilities include a Quantel Brilliant B high power Q-switched Nd:YAG Laser of energy of 950 mJ at its fundamental wavelength of 1064 nm.

Advance Photovoltaics Research Labs: The Faculty of Engineering Sciences, GIK Institute is home to the four state-of-the-art Advance Photovoltaics Research Laboratories including:

- Dye-sensitized solar cell fabrication laboratory
- Solar cell I-V and C-V characterization laboratory
- Solar cell transient photo-voltage and photocurrent characterization laboratory
- Laboratory for the design and simulation of molecular systems for efficient solar energy harvesting

The labs were established for the development of next generation solar cell technology and computational design of molecular systems for efficient harvesting of solar energy as part of the collaborative research project entitled "New Approaches for Lower Cost, Longer Stability, and Higher Efficiency of Dye-Sensitized Solar Cells (DSSCs)" between the Faculty of Engineering Sciences, Ghulam Ishaq Khan Institute (GIKI) of Engineering Sciences and Technology and Department of Electrical Engineering, Center for Advanced Photovoltaics, South Dakota State University (SDSU), Brookings, USA under the Pakistan-U.S. Science and Technology Cooperation Program Phase V.

Photolithography Lab: Recently lithography lab has been refurbished in the faculty of Engineering Sciences. This lab is equipped with the MJB3 optical lithographic setup. It provides the resolution up to 2 μm . In this lab we also have the programmable spin coater for the deposition of photoresist and for the characterization we have the lock-In amplifier. In addition to this equipment one temperature controller is also available for the controlled environment characterization. Both the undergrad and graduate students are given hands on photolithography setup. Students from the Faculty of Engineering Sciences, Faculty of Electronics Engineering and Faculty of Materials Science and Engineering come and work on photolithography.

Computational Physics Lab: Graduate students in the lab apply various numerical and simulation methods to understand various physical phenomena from device engineering to signal processing.

Coursework Requirements

A student majoring in Engineering Sciences must complete the following courses:

(a) General Education Requirements (52 Credit Hours)

Course Titles	Course Codes	Credit Hours
Basic Engineering	MM101, MM141, CH161, IF101, IF102	06
Computing	CS101, CS101L, CS112, CS112L	07
English Language	HM101, HM102	06
Humanities	HM211, HM321, HM322	09
Mathematics	MT101, MT102, ES111, MT202, MT203, ES304	18
Sciences	PH101, PH101L, CH101	06

(b) Core Requirements (50 Credit Hours)

Course Titles	Course Codes	Credit Hours
Circuit Analysis I	ES211/EE211	3
Circuit Analysis Lab	ES211L/EE211L	1
Logic Design	ES212/EE221	3
Logic Design Lab	ES212L/ES221L	1
Computer Architecture	ES213/EE222	3
Computer Architecture Lab	ES213L/EE222L	1
Circuit Analysis II	ES214/EE212	3
Data Structures and Algorithms	ES221/CS221	3
Electronics I	ES231/EE231	3
Electronics I Lab	ES231L/EE231L	1
Thermodynamics	ES232/ME231	3
Microprocessor Systems and Interfacing	ES314/EE221/CE324	3
Microprocessor Systems and Interfacing Lab	ES314L/EE221L	1
Signals and Systems	ES332/CS341	3
Signals and Systems Lab	ES332L/CS341L	1
Numerical Analysis	ES341	3
Numerical Analysis Lab	ES341L	1
Engineering Electromagnetics	ES371	3
Instrumentation	ES322/EE213/ME202	3
Instrumentation Lab	ES322L/EE213L/ME243L	1
Senior Design Project Part-I and II	ES481/ES482	6

(c) Specialization Requirement (19 Credit Hours)**1. Photonics and Microelectronics**

Course Titles	Course Code	Credit Hours
Foundations of Photonics	ES323	3
Foundations of Photonics Lab	ES323L	1
Solid State Electronics	ES361	3
Solid State Electronics Lab	ES361L	1
Optoelectronics	ES474	3
Optoelectronics Lab	ES474L	1
Fiber-Optic Communications	ES425	3
Fiber-Optic Communications Lab	ES425L	1
VLSI Design	ES426	3

2. Modeling, Simulation, and Machine Learning

Course Titles	Course Code	Credit Hours
Discrete System Modeling and Simulation	ES324	3
Discrete System Modeling and Simulation Lab	ES324L	1
Advanced Statistics	ES325	3
Model Engineering	ES471	3
Model Engineering Lab	ES471L	1
Engineering Optimization	ES441	3
Engineering Optimization Lab	ES441L	1
Machine Learning	ES442	3
Machine Learning Lab	ES442L	1

(d) Specialization Technical Electives (06 Credit Hours)**1. Photonics and Microelectronics (Any Two)**

Course Titles	Course Code	Credit Hours
Laser Engineering	ES443	3
Geometric Optics	ES444	3
Biophotonics	ES445	3
Imaging & Displays	ES461	3
Semiconductor Devices and Applications	ES465	3
Introduction to ASIC Design	EE422	3
Digital Integrated Circuit Design	EE423	3
Materials Characterization	MM494	3

Electronic and Magnetic Materials	ES463/MM393	3
Nanomaterials and Nanotechnology	ES462/MM391	3
Nanotechnology for Energy	MM499	3
Microelectronics Manufacturing Engineering	ES466	3
Nanosystems and Devices	MM495	3

2. Modeling, Simulation, and Machine Learning (Any Two)

Course Titles	Course Code	Credit Hours
Analysis for Modeling and Simulation	ES467	3
Heat Transfer and Modeling	ES446	3
Financial Engineering Models	ES447	3
Data Science	CS439	3
Big Data Analytics	DS461	3
Web Mining and Social Media Analysis	CS438	3
Web Engineering	CS463	3
Computer Graphics	CS433	3
Cloud Computing	AI408	3
Introduction to Mobile Computing	AI426	3
Cyber Security	CS420	3
Block Chain	CS411	3
Bio-Inspired Computing	CS472	3
Medical Image Processing	AI420	3
Computational Biology	CS476	3
Business and Economic Forecasting	MS446	3
Investment and Portfolio Management	AF453	3
Financial Risk Management	AF454	3
Business Logistics Strategy	SC462	3

(e) Management Electives (Any Two) (06 Credit Hours)

Course Titles	Course Code	Credit Hours
Operations Management	MS492	3
Industrial Safety	MS493	3
Total Quality Management	MS494	3
Maintenance Management	MS495	3
Project Management	MS496	3
Technology Management	MS426	3
Industrial Management	MS449	3
Supply Chain Management	MS491	3
Accounting and Finance	MS447	3

Apart from the recommended list above, any other management course MS4XX may be chosen with the consultation of the advisor.

(f) Inter-Faculty Electives (03 Credit Hours)

These electives have to be chosen from faculties other than that of Engineering Sciences with the consultation of the advisor.

(g) Summer Training/4-8 Weeks Internship (Pass/Fail Grade; Nil Credits)

Every student is required to participate in a program of practical training in industry or an R&D organization and submit a formal written report during the summer of Junior Year.

(h) Total Credit Requirements (136 Credit Hours)

For the BS degree in Engineering Sciences a student is required to complete 136 credit hours

Degree Plan

1st Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH161	Occupational Health and Safety	1	0	1	None	None
	CS101	Computing and AI	2	0	2	None	CS101L
	CS101L	Computing and AI Lab	0	3	1	None	None
	HM101	Communication Skills	1	6	3	None	None
	IF101	Innovation and Makers Lab I	0	3	1	None	None
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	PH101L
	PH101L	Applied Physics Lab	0	3	1	None	None
2nd Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CS112	Object Oriented Programming and Design	3	0	3	CS101	None
	CS112L	Object Oriented Programming and Design Lab	0	3	1	None	CS112
	HM102	Critical Thinking and Expository Writing	2	3	3	None	None
	MM101	Materials and Nanotechnology	2	0	2	None	None
	MM141	Materials Lab	0	3	1	None	MM101
	MT102	Differential Equations and Linear Algebra I	3	0	3	MT101	None
	ES111	Probability and Statistics	3	0	3	MT101	None
	IF102	Innovation and Makers Lab II	0	3	1	IF101	None

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
3rd Semester	ES211/EE211	Circuit Analysis I	3	0	3	MT102	MT202
	ES211L	Circuit Analysis Lab	0	3	1	NONE	ES211
	ES212/EE221	Logic Design	3	0	3	NONE	NONE
	ES212L/EE221L	Logic Design Lab	0	3	1	NONE	ES212/EE221
	ES232	Thermodynamics	3	0	3	MT202	NONE
	HM211	Pakistan and Islamic Studies	3	0	3	NONE	NONE
	MT202	Calculus II	3	0	3	MT101	NONE

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
4th Semester	ES221/CS211	Data Structure and Algorithms	3	0	3	CS112	None
	ES214/EE212	Circuit Analysis II	3	0	3	ES211/EE211	None
	ES213/EE222	Computer Architecture	3	0	3	CS101, ES212	None
	ES213L/EE222L	Computer Architecture Lab	0	3	1	None	ES213/EE222
	MT203	Complex Variables and Transforms	3	0	3	MT202	None
	ES231/EE231	Electronics I	3	0	3	ES211/EE211	None
	ES231L/EE231L	Electronics I Lab	0	3	1	None	ES211/EE211



Specializations (Streams):

1. Photonics and Microelectronics Stream

5th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES304	Linear Algebra II	3	0	3	MT202	None
	ES314/EE323	Microprocessor Systems and Interfacing	3	0	3	ES213	None
	ES314L/EE323L	Microprocessor Systems and Interfacing Lab	0	3	1	None	ES314/EE323
	ES332/CS341	Signals and Systems	3	0	3	ES214/EE212	None
	ES332L/CS341L	Signals and Systems Lab	0	3	1	None	ES332/CS341
	ES323	Foundations of Photonics	3	0	3	PH101	None
	ES323L	Foundations of Photonics Lab	0	3	1	None	ES323

6th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES361	Solid State Electronics	3	0	3	PH101	None
	ES361L	Solid State Electronics Lab	0	3	1	None	ES361
	ES322/EE213	Instrumentation	3	0	3	ES211	None
	/ME202	Instrumentation	3	0	3	ES211	None
	ES322L/EE213L	Engineering Electromagnetics	3	0	3	PH102, MT201	None
	/ME243L	Instrumentation Lab	0	3	1	None	ES322
	ES371	Engineering Electromagnetics	3	0	3	PH101, MT202	None
	ES341	Numerical Analysis	3	0	3	MT202	None
	ES341L	Numerical Analysis Lab	0	3	1	None	ES341
	HM322	Corporate Law and Professional Ethics	3	0	3	None	None

7th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES474	Optoelectronics	3	0	3	ES361	None
	ES474L	Optoelectronics Lab	0	3	1	None	ES474
	ES425	Fiber-Optic Communications	3	0	3	ES323	None
	ES425L	Fiber-Optic Communications Lab	0	3	1	None	ES425
	ES426	VLSI Design	3	0	3	ES361	None
	ES481	Senior Design Project Part I	0	9	3	None	None
	MS4XX	General Management Elective I	3	0	3	-	-

8th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES4XX	Specialization Technical Elective I	3	0	3	-	-
	ES4XX	Specialization Technical Elective II	3	0	3	-	-
	ES482	Senior Design Project Part II	0	9	3	None	None
	MS4XX	General Management Elective II	3	0	3	-	-
	XX4XX	Interfaculty Elective	3	0	3	-	-

2. Modeling, Simulation, and Machine Learning Stream

5th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES304	Linear Algebra II	3	0	3	MT202	None
	ES314/EE323	Microprocessor Systems and Interfacing	3	0	3	ES213	None
	ES314L/EE323L	Microprocessor Systems and Interfacing Lab	0	3	1	None	ES314/EE323
	ES332/CS341	Signals and Systems	3	0	3	ES214/	ES332/CS341
	ES332L/CS341L	Signals and Systems Lab	0	3	1	None	ES332/CS341
	ES324	Discrete System Modeling and Simulation	3	0	3	ES202	None
	ES324L	Discrete System Modeling and Simulation Lab	0	3	1	None	ES324
	HM321	Sociology and Human Behavior	3	0	3	None	None

6th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES325	Advanced Statistics	3	0	3	ES202	None
	ES322/ EE213/ ME202	Instrumentation	3	0	3	ES211	None
	ES322L/ EE213L/ ME202L	Instrumentation Lab	0	3	1	None	ES322
	ES371	Engineering Electromagnetics	3	0	3	PH101, MT202	None
	ES341	Numerical Analysis	3	0	3	MT202	None
	ES341L	Numerical Analysis Lab	0	3	1	None	ES341
	HM322	Corporate Law and Professional Ethics	3	0	3	None	None

7th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES471	Model Engineering	3	0	3	MT202	None
	ES471L	Model Engineering Lab	0	3	1	None	ES471
	ES441	Engineering Optimization	3	0	3	MT202	None
	ES441L	Engineering Optimization Lab	0	3	1	None	ES441
	ES442	Machine Learning	3	0	3	ES325	None
	ES442L	Machine Learning Lab	0	3	1	None	ES442
	ES481	Senior Design Project Part I	0	9	3	None	None
	MS4XX	General Management Elective	3	0	3	None	None

8th Semester	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	ES4XX	Specialization Technical Elective I	3	0	3	-	-
	ES4XX	Specialization Technical Elective II	3	0	3	-	-
	ES482	Senior Design Project Part II	0	9	0	None	None
	MS4XX	General Management Elective	3	0	3	None	None
	XX4XX	Interfaculty Elective	3	0	3	None	None

Course Description

MT101 Calculus I (3-0-3): Functions of one variable, limits and continuity, differentiation of functions of one variable, properties of differentiable functions, differentials and linear approximation, maxima minima and curvature, applied optimization problems of functions of one variable, indefinite integrals and techniques of integration, definite integrals and fundamental theorem of calculus, applications of definite integrals, Infinite Series, Definitions, convergence, properties. Integral test, basic and limit comparison tests, ratio and root tests. Alternating series and absolute convergence. Power series. McLaurin and Taylor series. Applications of power series.

Pre-requisite(s): None

MT102 Differential Equations and Linear Algebra I (3-0-3): Matrix algebra and general properties of matrices, elementary row operations, reduction of matrices into echelon and reduced echelon form, rank of a matrix, determinants and their properties, solution of system of linear algebraic equations, Gaussian elimination and Gauss-Jordan method, eigenvalue and eigenvectors, basic definitions involving differential equations, first order ordinary differential equations and their solution techniques, applications of first order differential equations, second and higher order differential equations, superposition principle, some special types of second order differential equations and their solution techniques, applications of second order differential equations, higher order linear differential equations with constant coefficients, complimentary and particular solutions, solutions by undetermined coefficients and variation of parameters, Cauchy-Euler equations, applications of higher order differential equations, power series solutions and systems of linear differential equations.

Pre-requisite(s): MT102

PH101 Applied Physics (3-0-3): Introduction to engineering mechanics problems, Vector Operations, Force Vectors, Motion in one, two and three dimensions, Newton Laws and its applications, Momentum, Rotational dynamics and kinematics, Moment of Force, Principle of Moments, Moment of a Couple, Work and Energy, Electrostatics, Electric Potential,

Magnetic Field.

Pre-requisite(s): None

ES111 Probability and Statistics (3-0-3): Introduction to Statistics and Data Analysis, Statistical Inference, Samples, Populations, and the Role of Probability. Sampling Procedures. Discrete and Continuous Data. Statistical Modeling. Types of Statistical Studies. Probability: Sample Space, Events, Counting Sample Points, Probability of an Event, Additive Rules, Conditional Probability, Independence, and the Product Rule, Bayes' Rule. Random Variables and Probability Distributions. Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, Means and Variances of Linear Combinations of Random Variables, Chebyshev's Theorem. Discrete Probability Distributions. Continuous Probability Distributions. Fundamental Sampling Distributions and Data Descriptions: Random Sampling, Sampling Distributions, Sampling Distribution of Means and the Central Limit Theorem. Sampling Distribution of variance, T-Distribution, ANOVA, F-test, F Quantile function and Probability Plots. Single Sample & One- and Two-Sample Estimation Problems. Single Sample & One- and Two-Sample Tests of Hypotheses. The Use of p-values for Decision Making in Testing Hypotheses (Single Sample & One- and Two Sample Tests), Linear Regression and Correlation. Least Squares and the Fitted Model, Multiple Linear Regression and Certain, Nonlinear Regression Models. Tools: Excel, R, Stata or similar tools.

Pre-Requisite: None

MT202 Calculus II (3-0-3): Parametric representation of plane curves; Calculus with parametric curves; Polar coordinates; Graphing polar equations; Conic sections in polar coordinates; Areas and arc lengths in polar coordinates. Vectors in three dimensions. Dot and cross product. Lines and planes. Surfaces. Limits, continuity, Partial derivatives. Increments and differentials, Chain rules. Directional derivatives, gradient. Tangent planes and normal lines to surfaces. Taylor series for functions of several variables. Extrema of functions of several variables. Relative extrema, Lagrange multipliers. Double integrals, Definition, and evaluation. Area and volume. Double integrals in polar coordinates. Surface

area. Triple integrals in Cartesian, Cylindrical, and spherical coordinates. Applications to engineering and science.

Pre-requisite(s): MT101

MT203 Complex Variables and Transforms

(3 0 3): Introduction to Complex Number System, Argand diagram, De Moivre's theorem and its Application Problem Solving Techniques, Complex and Analytical Functions, Harmonic Function, Cauchy-Riemann Equations, Cauchy's theorem and Cauchy's Line Integral, Power series, Taylor series, Laurent series, Residual integration, Singularities, Poles, Residues, Contour Integration, Laplace transform definition, Laplace transforms of elementary functions, Properties of Laplace transform, Periodic functions and their Laplace transforms, Inverse Laplace transform and its properties, Convolution theorem, Inverse Laplace transform by integral and partial fraction methods, Heaviside expansion formula, Solutions of ordinary differential equations by Laplace transform, Applications of Laplace transforms, Fourier theorem and coefficients in Fourier series, Even and odd functions, Complex form of Fourier series, Fourier transform definition, Fourier transforms of simple functions, Magnitude and phase spectra, Fourier transform theorems, Inverse Fourier transform, Series solution of differential equations, Validity of series solution, Ordinary point, Singular point, Forbenius

method, Indicial equation, Bessel's differential equation, its solution of first kind and recurrence formulae, Legendre differential equation and its solution, Rodrigues formula.

Pre-requisite(s): MT102

ES202 Engineering Statistics (3-0-3): Review of set algebra and combinatorial analysis, sample space and events, axiomatic definition of probability, rules of calculation of probabilities, conditional probability and probability of the composite random events, independent experiments, discrete and continuous random variables, binomial, Poisson, multinomial, exponential and normal distribution, data analysis and descriptive statistics, introduction to inferential statistics, point estimation, methods of moments and methods of maximum likelihood, confidence intervals, tests of hypothesis, first and second types of errors, tests for mean, proportion and variance, chi-square and student's t-test.

Pre-requisite(s): MT101

ES211/EE211 Circuit Analysis I (3-0-3):

Basic Concepts, resistive circuits, nodal and loop analysis techniques, operational amplifiers, additional analysis techniques such as using superposition, Thevenin's and Norton's Theorems, capacitance, and inductance, first- and second-order transient circuits.

Pre-requisite(s): MT101, MT202 (co-requisite)



ES212/EE221 Logic Design (3-0-3): Number systems, codes, set theory, relations, functions, Boolean Algebra, Logic gates, combinational logic, programmable logic devices, sequential logic, latches, flip-flops, finite state machines, counters, shift registers, pseudorandom sequence generators, memories, adders, subtractors, multiplication, division, comparators, fault detection, introduction to programmable logic devices and implementation of the digital circuit using Verilog/HDL.

Pre-requisite(s): None

ES213 Computer Architecture (3-0-3): Review of Verilog HDL, registers and register transfers, memory basics, computer design basics, instruction set architecture, central processing units, input—output and communication and memory systems.

Pre-requisite(s): CS101, ES212

ES214 Circuit Analysis II (3-0-3): AC steady-state analysis, steady state power analysis, variable-frequency network performance, the Laplace transform and its application to circuit analysis, Fourier analysis techniques and two-port networks.

Pre-requisite(s): ES211/EE211

ES221/CS211 Data Structures and Algorithms (3-0-3): Introduction to data structures and algorithms, arrays, stacks, binary search, queues, linked lists, trees, graphs and operations, algorithm performance, dynamic memory management.

Prerequisite(s): CS112/CS102L

ES221/CSE211 Data Structures and Algorithms (3-0-3): Fundamentals data structures: record, stack, list, queue, tree. Algorithms: sorting and searching. Graph theory.

Pre-requisite(s): CS101, ES221

ES231/EE231 Electronics I (3-0-3): Introduction to electronics, semiconductor diode, diode applications, bipolar junction transistor, transistor configuration, DC biasing, field effect transistor, BJT and FET small signals equivalent circuit models, design of BJT and FET amplifiers and differential amplifiers.

Pre-requisite(s): ES211/EE211

ES232 Thermodynamics (3-0-3): Fundamentals of thermodynamics including work

and heat, laws of thermodynamics, properties of pure substances, energy analysis of closed systems, mass and energy analysis of control volumes, entropy, enthalpy, reversibility, irreversibility, study of some processes and cycles. Pre-requisite(s): MT102

ES304 Linear Algebra II (3-0-3): Matrices algebra, determinants, linear systems and solutions, vectors in 2 space and 3 space, vector algebra and related theorems, vector spaces, subspaces and related theorems, linear combinations and related theorems, linear dependent and independent vectors, basis and related theorems, rank and nullity, Gram-Schmidt Process, inner product spaces, eigenvalues and eigenvectors, diagonalization of matrices and related theorems, linear transformation, kernel and range of linear transformation, applications to engineering and science.

Pre-requisite(s): MT202

ES314/EE221/CE324 Microprocessor Systems and Interfacing (3-0-3): Introduction to microprocessors; general purpose and embedded features, architecture and assembly language programming of typical micro controllers (such as 8051, PIC, AVR, Raspberry Pi), different types of instructions, addressing modes, time delay, crystal oscillator, I/O port and timer/counter programming, serial port programming, interrupts programming, interfacing to external memory, real world interfacing, LCD, ADC, sensors, and keyboard interfacing, interfacing with 8255 and RTC interfacing, motor control. Introduction to Arduino and Raspberry Pi development boards, their interfacing and programming. Pre-requisite(s): ES213

ES332/EE351 Signals and Systems (3-0-3): Introduction to continuous and discrete time systems, analysis of continuous time (CT) system using Fourier and Laplace Transforms, ideal and practical CT filters, sampling analysis of discrete time (DT) systems, difference equations and unit sample response, z-transform, DT Fourier transform and linear feedback systems.

Pre-requisite(s): ES214/EE212

ES334 Foundation of Photonics (3-0-3): Introduction to photonics engineering,

nature and properties of light, light sources and laser safety, basic geometrical optics, basic physical optics, lasers and applications, optical modulation and detection, integrated optics, nonlinear optics, optical waveguides and fibers, fiber optic telecommunication, optical sensors, organic/inorganic and hybrid photovoltaics, biophotonics, nanophotonics, and optical micro-electro-mechanical systems (mems).

Pre-requisite(s): PH101

ES341/CSE342 Numerical Analysis (3-0-3):

Error and computer arithmetic, Root-finding for non-linear equations, interpolation and polynomial approximation, solution of system of linear equations, numerical differentiation and integration and numerical solution of ordinary differential equations.

Pre-requisite(s): MT202

ES361/EE333 Solid State Electronics (3-0-3):

Introduction to semiconductor materials, band theory of solids, carrier transport in semiconductor, Schrodinger's equation and wavefunctions, Fermi-Dirac probability, Kronig-Penny model, E-k diagram, pn-junction and metal-junction devices, metal-oxide semiconductor devices, and bipolar junction transistor.

Pre-requisite(s): PH101

ES371 Engineering Electromagnetics (3-0-3):

Vector analysis, static electric and magnetic fields, Maxwell's equations, electric and magnetic boundary value problems, Poisson's and Laplace's equation, displacement current.

Pre-requisite(s): PH101, MT202

ES445 Computer Simulation Methods (3-0-3):

Introduction to simulation, simulation and modeling, types and uses of simulation, continuous system simulation, simulation schemes, simulation of basic mechanical and electrical systems, transfer functions and their simulation, simulation of combination of systems, discrete event simulation, simulation of basic engineering problems of discrete nature, analysis of output data, simulation of stochastic systems, review of standard probability distributions, Monte-Carlo simulation, simulation of random variates, simulation of deterministic and stochastic inventory problems, simulation and analysis of simple

M/M/k queuing systems.

Pre-requisite(s): ES344

ES325 Advanced Statistics (3-0-3): Statistical methods are used for the analysis of different datasets for forecasting the values, predicting the unknowns, relating the variables for getting deeper insights, and relating data differences with real-world complexities. Data Science extracts knowledge from data on the basis of hidden patterns which can be made explicit by incorporating the statistical algorithms in it. This course is designed to prepare students on statistical techniques with a purview of artificial intelligence and data science.

Pre-requisite(s): ES202

ES324 Discrete System Modeling and Simulation (3-0-3):

This course covers "Discrete-Event" simulation at an introductory level, preparing students for advanced studies in these fields. The main purpose of the course is to provide an introduction to the modeling and simulation of discrete-state, discrete-event systems (DES). This course will provide an investigation of the steps of a DES simulation study, including problem formulation, conceptual model design, simulation model development, input data modeling, output data analysis, verification and validation, and design of simulation experiments.

Pre-requisite(s): ES202

ES441 Engineering Optimization (3-0-3):

Brief review of LP models and simplex algorithm, general transportation model, network models and their tabular representation, transportation and transshipment models, transportation algorithms, assignment models and their various ramifications, Hungarian algorithm, integer linear programming and related models, zero-one programming, standard examples, modeling of various situations occurring in real world, network models, basic terminology of graph theory, spanning tree, minimum path, and maximum flow problems, network optimization algorithms, project management, PERT and CPM, queuing models, distribution of inter-arrival and service times and simple M/M/k systems.

Pre-requisite(s): MT202

ES442 Machine Learning (3-0-3): Introduction to machine learning; concept learning;

General-to-specific ordering of hypotheses, Version spaces Algorithm, Candidate elimination algorithm; Supervised Learning: decision trees, Naive Bayes, Artificial Neural Networks, Support Vector Machines, Overfitting, noisy data, and pruning, Measuring Classifier Accuracy; Linear and Logistic regression; Unsupervised Learning: Hierarchical Agglomerative Clustering, k-means partitional clustering; Self-Organizing Maps (SOM) k-Nearest-neighbor algorithm; Semisupervised learning with EM using labeled and unlabeled data; Reinforcement Learning: Hidden Markov models, Monte Carlo inference Exploration vs. Exploitation Trade-off, Markov Decision Processes; Ensemble Learning: Using committees of multiple hypotheses. Bagging, boosting
Pre-requisite(s): ES325

ES471 Model Engineering (3-0-3): The goal of this course is to develop an understanding of the various modeling paradigms appropriate for capturing system behavior and conducting digital computer simulations of many types of systems. The techniques and concepts discussed typically include mathematical modeling of engineering systems based on Linear System of Equations, Nonlinear System of Equations. The course will also cover modeling of Complex Engineering Systems as a set of Differential-Algebraic-Equations

(DEAs) models, Graphical Models, Integer Programming, and Stochastic Models.
Pre-requisite(s): MT202

ES32/EE213/ME202 Instrumentation

(3-0-3): Precision measurements terminologies principles of different measurement techniques; instruments for measurement of electrical and non-electrical quantities; systems for signal processing and signal transmission; modern instrumentation techniques; static and dynamic responses of instrumentation and signal conditioning; data acquisition systems; principles of operation, construction and working of different analog and digital meters, Advanced Testing & Measuring instruments recording instruments, signal generators, Input and output transducers; types of bridges for measurement of resistance, inductance, and capacitance; power and energy meters; voltage / current measurements. Programmable Logic Controllers (PLC), SCADA and communication are introduced. After learning the principles of developing PLC programs, examples of control systems are presented.

Pre-requisite(s): ES211

ES463 Electronic and Magnetic Materials

(3-0-3): Molecular Collisions, Thermal Fluctuations and Noise, The Crystalline State, Defects and their significance, Drude Model, Temperature dependent Resistivity, Matthies-



sen's Rule, Nordheim's Rule, Thermal Conductivity, Electrical Conductivity of nonmetals etc. Origin of magnetic moment of atoms, theories of magnetism such as molecular theory and electron theory, hysteresis and magnetization curve, magnetic domains, domain walls, methods of observation of domains, classification of magnetic materials according to magnetic properties.

Pre-requisite(s): ES361

ES465 Semiconductor Devices and Applications (3-0-3):

Semiconductor device fabrication, metal-semiconductor and metal-insulator-semiconductor junctions and devices, photonic devices, transferred- electron devices, switching devices, other semiconductor devices, amorphous semiconductors, band models of amorphous semiconductors, electronic applications, optical applications, magnetic applications, super conductive materials, and devices.

Pre-requisite(s): ES361

ES466 Microelectronics Manufacturing Engineering (3-0-3):

Designing of electronic devices and integrated circuits, manufacturing process of electronic devices and integrated circuits, electronic devices processing equipment's and their manufacturing limit, microlithography masking and patterning by UV lithography technique, electron beam lithography: design and patterning, positive and negative resist systems and resist-materials characterization, oxidation, diffusion, ion implantation, metallization and plasma etching processes.

Pre-requisite(s): None

ES425 Fiber Optic Communication:

This course is related to the principles of optical fiber communication systems. The course covers three topics: 1) The optical fiber as a transmission channel. 2) Optoelectronic devices used in transmitters, receivers, and multiplexers. 3) Design of the overall communication system and assessment of its performance. In part 1, step-index and graded-index multimode and single-mode optical fibers are described and their attenuation and dispersion characteristics are determined. The transfer function of the fiber system is determined. Part 2 introduces the basic principles of interaction of light with semiconductor materials,

including absorption and electroluminescence. Light emitting diodes, laser diodes, and photodiodes are introduced as the basic components of optical transmitters and receivers. Semiconductor and fiber optical amplifiers are also introduced. Part 3 deals with the design of the digital fiber communication system, including derivation of the bit error rates for attenuation- and dispersion-limited systems and determination of the maximum data rates possible for a given length. Introductions to wavelength-division multiplexing (WDM) and optical fiber networks are also provided.

Pre-requisite(s): ES334

ES443 Laser Engineering (3-0-3): It is an introductory course on lasers which covers principles of laser amplification and oscillations, design of lasers, and general characteristics of excitation systems. It is suitable for students with backgrounds in physics, electrical engineering, materials and other disciplines who require a fundamental knowledge of lasers and how they operate. The course covers the basic physics of laser operation, and includes understandings of resonator theory, pulsed and continuous wave operation of lasers. Most popular lasers are described, as well as a pulsed techniques such as Q-switching, mode-locking and harmonic generation. The student is also introduced to the exciting types of new lasers being developed.

Pre-requisite(s): ES334

ES444 Geometric Optics (3-0-3): Geometric optics is the study of light in its simplest form by treating light as rays. Light rays travel in straight lines until they encounter an interface (such as a mirror or a lens) where they may be redirected by reflection and refraction. This course describes the physical principles that determine how rays behave at various interfaces. These principles are then used to model simple optical systems with varying degrees of fidelity. Natural optical phenomena (rainbows, mirages, total-internal reflection, etc.) and classic optical systems (prisms, telescopes, cameras, etc.) will be analyzed throughout the course. Linear systems will be introduced to analyze more complex optical systems. This course provides the fundamentals needed for optical system design.

Pre-requisite(s): ES334

ES445 Biophotonics (3-0-3): This course is an introduction to photobiology (interaction of light with biological matter), tissue optics, light-induced cellular processes, optical biosensors, and cellular and molecular imaging. Biophotonics is an emerging multidisciplinary field where light-based methods are utilized to reveal biological mechanisms and diagnose or treat several diseases. This course introduces the basics of biology and photonics and provides the most relevant and important application examples selected from chemistry, biology, pharmacology and medicine. For examples, it includes how to detect and identify new viruses and how to manipulate the brain of mouse with light, etc.

Pre-requisite(s): ES334

ES461 Imaging & Displays (3-0-3): This course introduces the basic principles of two- and three-dimensional imaging systems. It begins with the mathematical description of image formation as a linear system and draws on the student's knowledge of signals and systems to introduce the concepts of point spread function, transfer function, resolution, and restoration. Actual physical imaging systems (such as microscopes, telescopes, and copiers) operating in the gazing and scanning configurations are subsequently modeled and their resolution assessed. Interferometric imaging systems and their applications in metrology are described. Techniques for depth profiling are then introduced including point-by-point scanning (as in laser scanning fluorescence microscopy), echo ranging (as in sonar and radar imaging), and interferometry (as in optical coherence tomography). This is followed by an introduction to computational imaging, including the techniques of computed tomography (CT), range tomography, and magnetic resonance imaging (MRI). Hyperspectral imaging systems and their various configurations are then described including applications in detection (of tumors, for example) and classification (of different targets). Performance measures such as sensitivity and specificity are introduced. Applications for remote sensing, nondestructive testing, and biology and medicine are highlighted.

Pre-requisite(s): ES334

ES462/MM391 Nanomaterials and Nanotechnology (3-0-3): Introduction to Nanoscience and Nanotechnology, Physical chemistry of solid surfaces, surface energy, electrostatic stabilization, steric stabilization, zero-dimensional nanostructures: nanoparticles, quantum dots, one dimensional nanostructures: nanowires and nanorods, template-based synthesis, two dimensional nanostructures. Thin films by physical and chemical methods, three-dimensional nanostructures: nano-carbons, fullerenes, CNTs and graphene, core shell nanostructures, nanomaterials hazards and safety procedures.

Pre-requisite(s): ES361

ES474 Optoelectronics (3-0-3): The course begins with a description of the interaction of light with semiconductor materials in a p-n junction configuration. This includes the phenomena of absorption, electroluminescence, and stimulated emission. The distinction between direct and indirect compound semiconductor materials is noted. Basic devices are then described: photodiodes, light emitting diodes (LEDs), semiconductor optical amplifiers, and laser diodes are then described. Array detectors, including complementary metal-oxide-semiconductor (CMOS) and charge-coupled devices (CCD) arrays, and array LEDs are then introduced. Basic specifications and applications of each of these devices are described, including solar cells, imaging with array detectors, and LED displays.

Pre-requisite(s): ES361

ES467 Analysis for Modeling and Simulation (3-0-3): An introduction to analysis techniques appropriate to the conduct of modeling and simulation studies. Topics include input modeling, random number generation, output analysis, variance reduction techniques, and experimental design. In addition, techniques for verification & validation are introduced. Course concepts are applied to real systems and data.

Pre-requisite(s): ES202

ES446 Heat Transfer and Modeling (3-0-3): Standard partial differential equations of heat transfer in two and three dimensions, transformation to cylindrical coordinates, general solution of heat equation under various boundary conditions, steady and transient

heat conduction in solids, contact heat transfer in heterogeneous materials, heat transfer at phase transformation, free boundary problems and methods of their solutions, heat and mass transfer in low temperature plasma, applications to plasmotrons and electrical relays.

Pre-requisite(s): ES202, ES342

ES447 Financial Engineering Models

(3-0-3): Corporate finance and financial evaluation, financial statements modeling, building a pro forma model, portfolio models, calculating efficient portfolios, efficient portfolios without short sales, portfolio optimization, the binomial option pricing model, the Black-Scholes model, immunizing strategies, modeling the term structure, Monte Carlo methods, simulating stock prices, Monte Carlo simulations for investments, simulating options and option strategies and Monte Carlo methods for option pricing.

Pre-requisite(s): ES441

ES481 and ES482 Senior Design Project Part – I and II (0-18-6):

The aim of this course is to sharpen the skills of the electronic engineering students by participating in projects that are to be identified in collaboration with the industry. Every project will be assigned a faculty advisor. The students may work independently or jointly (in small groups) on the projects. The duration of the project team is one full year. The progress will be monitored through interim presentations and reports. A final report will be due at the end of the term.

Streams Description

Photonics and Microelectronics: Photonics and Microelectronics is located at the crossroads of natural sciences and engineering. Photonics and Microelectronics stream labs are equipped with state-of-the-art facilities ranging from simple diode lasers to high power cutting lasers, optical communication systems to high precision spectrometers. The course contents have been designed to enable the students to gain an insight into the basic working, design, simulation, fabrication and testing of a wide range of photonic and microelectronic devices and systems such as optical fiber communication links, solar cells, sensors, photodiodes, etc. Also, software tools are employed for the modeling and simulation

of numerous photonic and microelectronic systems.

The potential applications of Photonics and Microelectronics are virtually unlimited and include chemical synthesis, medical diagnostics, on-chip data communication, laser defense, and fusion energy, to name several interesting additional examples.

- Consumer equipment: barcode scanner, printer, CD/DVD/Blu-ray devices, remote control devices.
- Telecommunications: optical fiber communications, optical down converter to microwave.
- Medicine: correction of poor eyesight, laser surgery, surgical endoscopy, tattoo removal.
- Industrial manufacturing: the use of lasers for welding, drilling, cutting, and various methods of surface modification.
- Construction: laser leveling, laser range finding, smart structures
- Aviation: photonic gyroscopes lacking mobile parts.
- Military: IR sensors, command and control, navigation, search, and rescue, mine laying and detection.
- Entertainment: laser shows, beam effects, holographic, art.
- Information processing.
- Metrology: time and frequency measurements, range finding.
- Photonic computing: clock distribution and communication between computers, printed circuit boards, or within optoelectronic integrated circuits; in the future: quantum computing.

Career in Photonics and Microelectronics:

The job opportunities for those engineers who will graduate with expertise in photonics currently and during the coming decades are outstanding both in the national and international levels.

- Atmospheric sciences
- Biomedical engineering
- Defense and Security
- Energy
- Instrumentation and measurements
- Materials and nanotechnology
- Microelectromechanical systems (MEMs)
- Optical image processing
- Optical networking and communications

- Chip Design

Modeling, Simulation, and Machine Learning:

Modeling, Simulation, and Machine Learning is a dynamic field that is utilized in engineering, science, health science, business, education, and many other disciplines. This emerging field is based on developments in diverse engineering areas and brings elements of art, engineering, and science together in a complex and unique way that requires domain experts to enable appropriate decisions. This specialization requires designing of mathematical model of actual or theoretical physical systems. executing the model on a computer and analyzing the output. Computer simulations are extensively being used in aerospace industry, automobile systems, financial markets, environment systems and medical sciences. Machine Learning provides a practical treatment of design, analysis, implementation and applications of algorithms, which learn from examples. Modeling, Simulation, and Machine Learning are particularly important because the description of the system behavior

by experimentation might not be feasible due to some of the following reasons.

- Some experiments may be very harmful
- Some experiments might take longer time than expected and may also be very costly
- There might be obstructions during experimentation
- We might not have access to inputs and outputs.

Career in Modeling, Simulation, and Machine Learning:

- Oil and gas industry (e.g., reservoir characterization)
- Space/defense industry (e.g., in national security mission, simulation of universe, space vehicles and missile trajectories)
- Software systems (e.g., simulation software used by Google, IBM)
- Chemical interactions (e.g., paper and pulp industry).
- Engineering Optimization
- Operations Research
- Intelligent Systems
- Logistics and Data Analytics



FACULTY OF MATERIALS AND CHEMICAL ENGINEERING



The Faculty of Materials and Chemical Engineering (FMCE) is one of the six faculties at GIK Institute of Engineering Sciences and Technology. There are two departments in the faculty:

- (i) Department of Materials Science and Engineering
- (ii) Department of Chemical Engineering

The faculty offers both Materials Engineering, and Chemical Engineering programs at undergraduate and graduate. FMCE employs highly qualified teaching faculty from the universities of international repute and state-of-the-art laboratories to provide students with a conducive learning experience. The faculty keeps a history of academic achievements, which is manifested by commitment to excellence in teaching and pursuance of high quality research addressing multidisciplinary challenges.

Faculty Mission

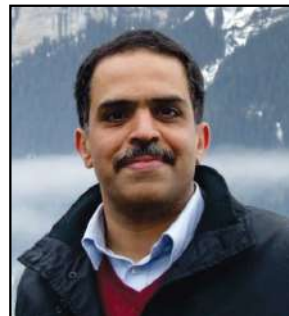
The faculty strives to train and educate students in the fields of Materials Science and Engineering and Chemical Engineering for their future role to contribute in academia, research, business and industry.

Thrust Areas

Materials processing, manufacturing and characterization, surface engineering and coating technology, nanotechnology and nanomaterials, advanced materials, ceramics, polymers and composites, biomaterials, corrosion and degradation

Faculty

Fazal Ahmad Khalid SI, DPhil (University of Oxford, UK)
 Fahd Nawaz Khan, PhD (University of Northumbria at Newcastle, UK)
 Javaid Rabbani Khan, PhD (University of Newcastle Upon Tyne, UK)
 Fida Muhammad, PhD (University of California, USA)
 Muhammad Imran Khan, PhD (University of Tsukuba, Japan)
 Shozab Mehdi, PhD (PIEAS, Pakistan)
 Khurram Imran Khan, PhD (Politecnico di Torino, Italy)
 Muhammad Ramzan Abdul Karim, PhD (Politecnico di Torino, Italy)
 Rashid Ali, PhD (University of Roma Tre, Italy)
 Sajjad Hussain, PhD (University of Sao Paulo (USP) Brazil)
 Muhammad Usman Farooq, PhD (University of Waterloo, Canada)
 Syed Zameer Abbas, PhD (GIK Institute, Topi)
 Hammad Amjad Khan, PhD (Hanyang University, South Korea)
 Shanza Rehan, PhD (University of Science and Technology, South Korea)
 Mohsin Ali Marwat (Huazhong University of Science & Technology, China)
 Tauheed Shehbaz, MS (NUST, Islamabad)



Dean

Fahd Nawaz Khan,
 PhD (University of Northumbria
 at Newcastle, UK)

Hafiz Muzammil Irshad, MS (KFUPM, KSA)
 Ramesha Tariq, MS (UET Lahore)
 Nida Zafar, MS (NUST Islamabad)
 Asim Itaf MS (GIK Institute, Topi)

Faculty on Study Leave

Engr Tahir Sattar

Adjunct Faculty

Peter Humphrey Draper, PhD (Imperial, London)
 Jawad Dar, PhD (QMUL – City University London)
 Tahir I. Khan, PhD (University of Cambridge)

Lab Engineers and Graduate Fellows

Sohail Khan, BS Engr (GIK Institute, Topi)
 Waqas Afridi, BS Engr (GIK Institute, Topi)
 Saad Ali, BS Engr (GIK Institute, Topi)
 Yousaf Haroon, BS Engr (GIK Institute, Topi)
 Kamil Khan, BS Engr (GIK Institute, Topi)
 Abu Bakar Tahir, BS Engr (UET, Lahore)

Graduate Assistants

Rida Batool Naqvi, MS (GIK Institute)
 Waseem Shehzad, MS (UET Lahore)
 Kamran Alam, MS (UET Peshawar)
 Muhammad Umar, (MS GIK Institute)
 Iqra Yasmeen, (MS COMSATS)
 Muhammad Awais (UET Peshawar)
 Ateeq uz Zaman (GIK Institute)
 Muhammad Usman (NFC Institute)
 Talha Faizi (GIK Institute)
 Nishwana Tahir (NFC Institute)
 Sundas Khushnood, (UET Peshawar)
 Muhammad Fawad Khan, (UET Peshawar)
 Zia ud Din, (IST Islamabad)
 Amna Ijaz, (IST Islamabad)
 Wajahat Khalid, (UET Peshawar)

Personal Assistant to Dean FMCE

Mohajir Shah, MA (Peshawar University)

Mission Statement

The mission of the department of materials science and engineering is to develop and disseminate the understanding of structure, property, processing and performance of materials so that our graduates could lead and excel in academia, research, business and industry in ethical and professional manner.

Materials Engineering Program

Materials Engineering is a challenging, rewarding, and highly respected profession and is regarded as one of the broadest engineering disciplines dealing with production, processing, characterization, selection and design of new and exotic materials for micro to nanoscale applications. It encompasses metals and alloys, ceramics, semiconductors, polymers, glasses, composites, biomaterials and recently developed nanomaterials. Typical job functions of materials engineers include selection and designing of various classes of materials, developing innovative structures through advanced manufacturing processes, corrosion and failure analysis and characterization of materials and nanostructures.

Materials graduates are engaged in a very wide range of industries, not only the materials production but also manufacturing industries, where materials are becoming an increasingly important factor in terms of the competitive edge of many advanced applications such as in transportation, health care, energy production, biomedical engineering, and aerospace industry. Mechanical, thermal, electrical, magnetic, optical and chemical properties of materials are continuously being improved by materials engineers globally which in turn leads to improvements in our lifestyle.

Undergraduate Program

The department offers a four-years BS in Materials Engineering degree program with specialization in Nanotechnology or Manufacturing based on theory and laboratory work. The curriculum is specifically designed to commensurate with both the need of local industry and R & D organizations as well as keeping in view the emerging research trends abroad. The first two years in the program are devoted to establishing a sound foundation in mathematics, science and basic engineering. The final two years establish the

structure-processing-properties relationships for all the major classes of materials, i.e., metals and alloys, ceramics, polymers and composites. Specialization tracks and combination of electives allow a student to develop a concentration in a particular area of materials. A sequence of two design courses in the senior year serves as a capstone design experience for the program.

The BS program in Materials Engineering has a very favourable student/faculty ratio and students have significant personal interaction with the faculty. Talented faculty with international qualifications has been hired and state of the art materials synthesis and characterization equipment like Atomic Force Microscope (AFM), Nanoindenter, Thermomechanical Analyzer (TMA), Simultaneous Thermal Analyser (TGA/DSC), Magnetron Sputtering unit, Scanning Electron Microscope (SEM), X-ray Diffractometer (XRD), latest computerised Potentiostat and lab scale Electric Arc furnace, Heat treatment Box and Tube furnaces, Vacuum oven, Hot Iso-Static Press (HIP), Cold Iso-Static Press CIP, Hydrothermal Reactor (~1100 °C) Probe Sonicator, Spin Coater, Programmable dip coater, Doctor Blade Coater/Tape Casting Unit etc. are available to provide practical hands-on experience to support teaching and research in specialised areas of Nanotechnology and Manufacturing of materials. The faculty offers conducive environment for students to learn latest theories and develop practical skills in the labs. To sustain the quality of teaching, outcome-based education (OBE) system has been in place since fall 2014 and the engineering program curriculum has been updated to international benchmarks with collaboration of academia and industry representatives.

Outcome Based Education

Program Educational Objectives (PEOs)

Graduates in Materials Science and Engineering will:

PEO-1: Pursue successful career in industry, research organizations, academia and allied professions.

PEO-2: Demonstrate creativity and innovation in addressing engineering problems in a sustainable way.

PEO-3: Use leadership, entrepreneurial and team working skills in ethical and professional manner.

Program Learning Outcomes (PLOs)

The Learning Outcomes for the BS in Materials Engineering Program are listed below:

PLO-1: Engineering Knowledge

Students shall have the ability to apply knowledge of mathematics, science, engineering fundamentals and materials engineering specialization to the solution of complex engineering problems.

PLO-2: Problem Analysis

Students shall have the ability to identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

PLO-3: Design/Development of Solutions

Students shall have the ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

PLO-4: Investigation

Students shall have the ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.

PLO-5: Modern Tool Usage

Students shall have the ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.

PLO-6: The Engineer and Society

Students shall have ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to

professional engineering practice and solution to complex engineering problems.

PLO-7: Environment and Sustainability

Students shall have the ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

PLO-8: Ethics

Students shall have the ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PLO-9: Individual and Teamwork

Students shall have the ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.

PLO-10: Communication

Students shall have the ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PLO-11: Project Management

Students shall have the ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

PLO-12: Lifelong Learning.

Students shall have the ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and



technological developments.

ASM-TMS International Student Chapter

The International Student Chapter of ASM-TMS at GIK Institute helps students in professional development and training. Its activities include organization of seminars, designing and publishing department newsletter, video shows, discussions, and industrial visits.

Knowledge Areas

Materials engineering courses are linked to the knowledge areas as given in the table below:

	Description	Courses
WK1	Natural Sciences	Applied Physics, Applied Chemistry and Environment, Materials Chemistry.
WK2	Mathematics & Computing	Calculus I, Differential Equations and Linear Algebra I, Calculus II, Object Oriented Programming and Design, Object Oriented Programming and Design Lab, Computing and AI Lab, Probability and Statistics, Engineering Economics, Numerical Analysis.
WK3	Engineering Fundamentals	Materials and Nanotechnology, Applied Electrical Engineering, Thermodynamics of Materials, Materials Evaluation Techniques, Phase Equilibria and Microstructure, Alloy Production, Strength of Materials, Crystallography and X-Ray Diffraction, Heat treatment and Processing, Deformation and Fracture, Polymer and Composite, Ceramics and Glasses, Corrosion Degradation and Protection, Materials Characterization.
WK4	Engineering Specialization	Joining of Materials, Manufacturing Processes - I, Manufacturing Processes - II, Smart and Functional Materials, Nanomaterials and Nanotechnology - I, Nanomaterials and Nanotechnology – II, Electronic and Magnetic Materials, Nano-systems and Devices, Nanotechnology for Energy, Surface Engineering, Advanced Materials, Automobile Engineering and Materials, Biomaterials.
WK5	Engineering Design	Casting Design and Foundry Technology, CAD/CAM, Finite Element Methods, Standards and Quality Assurance.
WK6	Engineering Practice	Applied Physics Lab, Innovation and Makers Lab I, Innovation and Makers Lab II, Occupational Health and Safety, Materials and Nanotechnology Lab (Materials Lab I), Materials Labs II, III, IV, V, VI and VII.
WK7	Engineering in Society	Communication Skills, Critical Thinking and Expository Writing, Pakistan and Islamic Studies, Sociology and Human Behaviour, Corporate Law and Professional Ethics, Entrepreneurship and Marketing.
WK8	Research Literature	Senior Design Project - I, Senior Design Project – II.

Sustainable Development Goals (SDGs)

SDG 1 No Poverty: End poverty in all its forms everywhere.

SDG 2 Zero Hunger: End hunger, achieve food security and improved nutrition and promote sustainable agriculture.

SDG 3 Good Health and Well-Being: Ensure healthy lives and promote well-being for all at all ages.

SDG 4 Quality Education: Ensure inclusive

and equitable quality education and promote lifelong learning opportunities for all.

SDG 5 Gender Equality: Achieve gender equality and empower all women and girls.

SDG 6 Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.

SDG 7 Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all.

SDG 8 Decent Work and Economic Growth:

Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

SDG 9 Industry, Innovation and Infrastructure:

Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

SDG 10 Reduced Inequalities: Reduce inequality within and among countries.

SDG 11 Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient and sustainable.

SDG 12 Responsible Consumption and Production: Ensure sustainable consumption and production patterns.

SDG 13 Climate Action: Take urgent action to combat climate change and its impacts.

SDG 14 Life Below Water: Conserve and sustainably use the oceans, seas and marine resources for sustainable development.

SDG 15 Life on Land: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

SDG 16 Peace, Justice and Strong Institutions: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.

SDG 17 Partnerships for The Goals: Strengthen the means of implementation and revitalize the global partnership for sustainable development.

Accreditation

The Department of Materials Science and Engineering offers a program leading to the Bachelor of Science in Materials Engineering degree that is accredited by the Pakistan Engineering Council (PEC <http://www.pec.org.pk>) under Washington Accord Level-II category of Outcome Based Education (OBE) system.

Graduate Program (MS and PhD)

In addition to educating and training undergraduates, the faculty is actively involved in offering graduate and post graduate programs leading to MS and PhD degrees respectively. The following graduate and post graduate programs can be pursued at Department of Materials Science and Engineering:

- i. MS Materials Engineering
- ii. MS Nanotechnology and Materials Engineering
- iii. PhD Materials Engineering

The graduate engineers with BS degree are enrolled to study at the MS and PhD levels in following research areas with a prescribed course work: Smart Materials, Advanced Coatings, Corrosion & Protection, Biomaterials, Composite Materials, Super Alloys, Functional Materials, Magnetic Materials, Superconducting Materials, Nanomaterials and Nanocomposites, Energy Storage Materials, Aerospace Materials and Failure Analysis of Materials.

Most of the students enrolled for MS and PhD programs are financially supported by various sources including GIK Institute's graduate assistantships during their graduate studies. These students are required to complete the prescribed course work as well as demonstrate their research capability through independently conducted research.

Department of Materials Science and Engineering Laboratories

The department of Materials Science and Engineering houses following laboratories

1. Mechanical Testing Lab
2. Heat Treatment Lab
3. Melting and Casting Lab
4. Corrosion Lab
5. Non-Destructive Testing Lab
6. Metallography Lab
7. Scanning Electron Microscopy Lab
8. X-Ray Diffraction Lab
10. Thermal Analysis Lab
11. Polymer Lab
12. Alloy Making Lab
13. Thin Films Lab
14. Ceramics Lab
15. Atomic Force Microscopy and Nanoindentation Lab
16. Computational Lab
17. Workshop and Welding Practices Lab
18. Nanomaterials Synthesis Lab

Course Work Requirements

For BS in Materials engineering degree student must complete the following requirements as detailed in Table a-g.

(a) General Education Requirement (55 Credit Hours)

Course Title	Course Code	CH
Humanities/Social Sciences	HM101, HM102, HM211, HM322, HM321	15
Basic Engineering Courses	IF101, IF102, MS291, EE213, CH161	9
Mathematics	MT101, MT102, MT202, ES341	15
Sciences	PH101, PH102, CH101, MM201	9
Computer Science	CS101, CS112, CS112L	7

(b) Core Requirement (49 Credit Hours)

Course Title	Course Code	CH
Materials and Nanotechnology	MM102	2
Thermodynamics of Materials	MM231	3
Materials Evaluation Techniques	MM212	3
Phase Equilibria and Microstructures	MM232	3
Strength of Materials	MM222	3
Alloy Production	MM233	3
Crystallography & X-ray Diffraction	MM323	3
Heat Treatment and Processing	MM334	3
Deformation and Fracture	MM324	3
Polymers and Composites	MM365	3
Ceramics and Glasses	MM362	3
Corrosion Degradation and Protection	MM435	3
Materials Labs I to VII		
(Interactive sessions)	MM141L, MM242L, MM243L, MM344L, MM345L, MM446L, MM447L	8
Senior Design Project	MM481, MM482	6

(c) Specialization in Manufacturing (18 Credit Hours)

Course Title	Course Code	CH
Joining of Materials	MM351	3
Manufacturing Processes-I	MM352	3
Smart and Functional Materials	MM353	3
Manufacturing Processes-II	MM451	3
CAD/CAM	MM453/ME418	3
Entrepreneurship and Marketing	MS434	3

(d) Specialization in Nanotechnology (18 Credit Hours)

Course Title	Course Code	CH
Nanomaterials and Nanotechnology-I	MM391	3
Nanomaterials and Nanotechnology-II	MM392	3
Electronic and Magnetic Materials	MM393	3
Materials Characterization	MM494	3
Nanosystems and Devices	MM495	3
Nanotechnology for Energy	MM499	3

(e) Technical Electives (06 Credit Hours)

Course Title	Course Code	CH
Casting Design and Foundry Technology	MM416	3
Surface Engineering	MM436	3
CAD/CAM	MM453/ME418	3
Powder Metallurgy	MM454	3
Finite Element Methods	MM455	3
Nuclear Materials	MM464	3
Nanostructured Materials	MM467	3
Automobile Engineering and Materials	MM469	3
Standards and Quality Assurance	MM472	3
Materials Characterization	MM494	3
Nanosystems and Devices	MM495	3
Advanced Materials	MM496	3
Biomaterials	MM497	3
Electronic and Magnetic Materials	MM393	3



(f) Management Electives (06 Credit Hours)

Course Title	Course Code	CH
Fuel and Energy Management	MM479	3
Technology Management	MS498	3
Operations Management	MS492	3
Industrial Safety	MS493	3
Total Quality Management	MS494	3
Maintenance Management	MS495	3
Project Management	MS496	3
Lean Enterprise Management	MS489	3
Human Resource Management	MS412	3
Supply chain Management	MS491	3
Entrepreneurship and Marketing	MS434	3
Industrial Management	MS449	3

(g) List of Laboratory Courses in Each Semester (08 Credit Hours)

Lab Code	Associated Course	CH
MM141L Materials and Nanotechnology Lab (Lab-I)	MM101	1
MM242L Materials Lab-II	MM212, MM231	1
MM243L Materials Lab-III	MM232, MM233	1
MM344L Materials Lab-IV	MM323, MM324, MM334, MM391	2
MM345L Materials Lab-V	MM352, MM362, MM365, MM392, MM393	1
MM446L Materials Lab-VI	MM435, MM436, MM494	1
MM447L Materials Lab-VII	MM496, MM451, MM495, MM499	1

(h) Summer Internship (Pass/Fail Grade; 0 CH)

Every student is required to participate in summer internship program (eight weeks) during the summer of their third year and submit a formal written report along with a presentation at the end of the internship period.

(i) Total Credit Requirements (134 CH)

For the award of BS degree in Materials Engineering, a student has to complete 134 credit hours.



BS Program in Materials Engineering: Semester-wise Degree Plan

1st Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MT101	Calculus I	3		3	None	None
	PH101	Applied Physics	3		3	None	PH101L
	PH101L	Applied Physics Lab		3	1	None	None
	CS101	Computing and AI	2		2	None	CS101L
	CS101L	Computing and AI Lab		3	1	None	None
	CH101	Applied Chemistry and Environment	2		2	None	None
	HM101	Communication Skills	1	2	3	None	None
	IF101L	Innovation and Makers Lab 1		3	1	None	None
	CH161	Occupational Health and Safety	1		1	None	None

2nd Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CS112	Object Oriented Programming and Design	3		3		
	CS112L	Object Oriented Programming and Design Lab		3	1		
	HM102	Critical Thinking and Expository Writing	3		3		
	MM101	Materials and Nanotechnology	2		2		
	MM141L	Materials and Nanotechnology Lab		3	1		MM101
	MT102	Differential Equations and Linear Algebra I	3		3	MT101	
	ES111	Probability and Statistics	3		3	MT101	
	IF102L	Innovation and Makers Lab II		3	1		

3rd Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MT202	Calculus II	3	0	3	MT102	-
	MM231	Thermodynamics of Materials	3	0	3	-	-
	MM212	Materials Evaluation Techniques	3	0	3	MM101	-
	EE213	Applied Electrical Engineering	3	0	3	MT101	-
	MM242L	Materials Lab II	0	3	1	MM101	-
	HM211	Pakistan and Islamic Studies	3	0	3	-	-

4th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MM232	Phase Equilibria and Microstructures	3	0	3	MM231	-
	MM222	Strength of Materials	3	0	3	-	-
	MM233	Alloy Production	3	0	3	MM101	-
	MM243L	Materials Lab III	0	3	1	MM101	-
	MMxxx	Materials Chemistry	3	0	3	-	-
	MS291	Engineering Economics	3	0	3	-	-

Manufacturing Stream

5th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MM323	Crystallography and X-ray Diffraction	3	0	3	MM101	-
	MM324	Deformation & Fracture	3	0	3	MM222	-
	MM334	Heat Treatment and Processing	3	0	3	MM232	-
	MM344L	Materials Lab IV	0	6	2	MM101	-
	MM351	Joining of Materials	3	0	3	MM212	-
	HM321	Sociology and Human Behaviour	3	0	3	-	-

6th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MM352	Manufacturing Processes-I	3	0	3	MM324	-
	MM353	Smart and Functional Materials	3	0	3	-	-
	MM365	Polymers and Composites	3	0	3	MM101 & CH101	-
	MM362	Ceramics and Glasses	3	0	3	MM231	-
	MM345L	Materials Lab V	0	3	1	MM101	-
	ES341/CS342	Numerical analysis	3	0	3	MT202	-
	HM322	Corporate Law and Professional Ethics	3	0	3	-	-

7th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MM435	Corrosion Degradation and Protection	3	0	3	CH101 & MM231	
	MM453/ME418	CAD/CAM	2	3	3	-	
	MM4XX	MM Technical Elective	3	0	3	-	
	MS4XX	Management Elective	3	0	3	-	
	MM481	Senior Design Project I	3	0	3	-	
	MM446L	Materials Lab VI	0	3	1	MM101	

8th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MS434	Entrepreneurship and Marketing	3	0	3	-	
	MM451	Manufacturing Process-II	3	0	3	MM352	
	MS4XX	Management Elective	3	0	3	-	
	MM482	Senior Design Project II	3	0	3	-	
	MM496	Advanced Materials	3	0	3	MM101	
	MM447L	Materials Lab VII	0	3	1		

Nanotechnology Stream

	Course Code	Course Titles	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
5th Semester	MM323	Crystallography and X-ray Diffraction	3	0	3	MM101	-
	MM324	Deformation & Fracture	3	0	3	MM222	-
	MM334	Heat Treatment and Processing	3	0	3	MM232	-
	MM391	Nanomaterials and Nanotechnology-I	3	0	3	-	-
	MM344L	Materials Lab IV	0	6	2	MM101	-
	HM321	Sociology and Human Behaviour	3	0	3	-	-

	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
6th Semester	MM392	Nanomaterials and Nanotechnology-II	3	0	3	MM391	-
	MM393	Electronic and Magnetic Materials	3	0	3	-	-
	MM362	Ceramics and Glasses	3	0	3	MM101	-
	MM365	Polymers and Composites	3	0	3	CH101 & MM101	-
	MM345L	Materials Lab V	0	3	1	MM101	-
	ES341/CS342	Numerical analysis	3	0	3	MT201	-
	HM322	Corporate Law and Professional Ethics	3	0	3	-	-

	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
7th Semester	MM435	Corrosion Degradation and Protection	3	0	3	CH101 & MM231	-
	MM494	Materials Characterization	3	0	3	MM101	-
	MM4XX	MM technical elective	3	0	3	-	-
	MS4XX	Management Elective	3	0	3	-	-
	MM481	Senior Design Project I	3	0	3	-	-
	MM446L	Materials Lab VI	0	3	1	-	-



8th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	MM495	Nanosystems and Devices	3	0	3	MM391	-
	MM499	Nanotechnology for Energy	3	0	3	MM391	-
	MM4XX	MM Technical Elective	3	0	3	-	-
	MS4XX	Management Elective	3	0	3	-	-
	MM482	Senior Design Project II	3	0	3	-	-
	MM447L	Materials Lab VII	0	3	1	MM101	-
	MM483	MSE Seminar	1	0	0		-

COURSE DESCRIPTION

CH101 Applied Chemistry and Environment

(2-0-2): Importance of Chemistry for Engineers, industrial applications, mass spectrometry, stoichiometry and chemical calculations, Standardization of solution for quantitative titration, electrochemistry, batteries, solid state batteries, corrosion mechanisms, causes and its control, nano-chemistry, deposition of thin films, CVD, PVD, Purification of silicon for electronic chip manufacturing, polymers, fossil fuels, clean combustion processes, integrated gasification combined cycle IGCC, CO₂ capturing from exhaust gasses and Syngas from coal, green chemistry and its twelve principles, chemistry for environmental sustainability, photochemistry, free radicals, interaction of solar radiation with molecules in the atmosphere, acid rain, ozone in the atmosphere, urban smog, treatment of water for domestic and industrial uses, treatment of wastewater to preserve aquatic ecosystem.

Pre Req: Nil

MM102 Materials and Nanotechnology

(2-0-2): Structure of crystalline materials, imperfection in crystalline materials, diffusion in solids and its application in technology, mechanical properties of metals & Alloys, ceramics materials and their applications in engineering, Composites, nano materials and its impact on industry, electrical properties of materials, thermal properties of materials, magnetic properties of materials, optical properties of materials.

Pre Req: Nil

MMxxx Materials Chemistry (3-0-3):

Solutions, colloids, emulsions, surfactants, heating curves, lattice energy, intermolecular

forces, solvents, ionic liquids, extraction of metals, electronic structure of transition metals, inter-metallic compounds, non-stoichiometric compounds, polymerization, biological molecules, nano-chemistry, interaction between radiation and molecules, corrosion, preparation of large single crystal, zone refining, cooling mixtures, quantum theory. Pre Req: Nil

MM212 Materials Evaluation Techniques

(3-0-3): Evaluation and quality assurance, standard specifications of materials, objectives of materials testing and evaluation, destructive testing of materials, tensile and compression test, bend test, micro and macro hardness testing, shear and torsion tests, sheet metal testing, impact testing and fracture mechanics, fatigue testing, creep testing, importance of non-destructive evaluation (NDE) of materials, methods of NDE, visual inspection (VT), liquid penetrant test (LPT), radiographic examinations (RT), magnetic particles inspection (MPI), ultrasonic testing (UT), Advanced NDE techniques, Microscopy (OM, SEM, TEM, AFM). Pre Req: MM101

MM222 Strength of Materials (3-0-3):

Normal and shear stress and strains in materials, factor of safety, stress concentration, simple loading, tension, torsion and bending, true stress & true strain, modulus of elasticity, elastic vs plastic behaviour of a material, comparison of true strain & conventional strain. deformation under axial loading, stresses and deformation in circular shaft, stresses in elastic range, angle of twist, shear force & bending moment diagram, moment of a force & moment of inertia, symmetric members in pure bending, shear force and bending moment diagram, design of prismatic beam

for bending, transformation of stress and strain, principal stresses, Mohr's circle for plane stress, failure theories and their application to failure analysis

Pre Req: Nil

MM231 Thermodynamics of Materials

(3-0-3): First law of thermodynamics, enthalpy, internal energy, heat capacity, relationship between heat and work, reversible and irreversible processes, statistical interpretation of entropy, partition function and its use to derive expression for internal energy and entropy, Second law, entropy, Third law, Enthalpy and Gibb's energy changes, adiabatic flame temperature, heat engines, refrigerators, heat pumps, property relations, Maxwell relations, Gibbs and Helmholtz free energies, Vant Hoff's isotherm, behaviour of gases, partial properties, behaviour of ideal and non-ideal solutions, chemical equilibria, activity/fugacity and chemical potential, gas-solid equilibria, Ellingham diagrams, Phase equilibria (single and multicomponent systems), Gibbs phase rule.

Pre Req: Nil

MM232 Phase Equilibria and Microstructures

(3-0-3): Solid solution, Gibbs phase rule, one-component diagram, phase diagrams of isomorphous and eutectic systems, microstructures development while slow and fast cooling, Eutectoid and peritectic reactions. congruent phase transformations, phase diagram and free energy, Iron-Iron Carbide phase diagram, development of microstructures in iron-carbon alloy, microstructure development during precipitation, ternary phase diagrams, kinetics of phase transformation, homogenous and heterogeneous nucleation, diffusional transformation and types, martensitic transformation, diffusionless transformation characteristics.

Pre Req: MM231

MM233 Alloy Production (3-0-3): Casting, pattern making and materials, types of patterns, core making and materials, testing and control of molding sands, molding processes and materials, casting techniques, gating system design, melting furnaces, Solidification of pure metal and alloys, casting defects and inspection, crushing and grinding, concentration processes, Blast furnace, steel and cast iron, charge calculations and Ellingham

diagram, steel making processes, non-ferrous metals Al, Ni, Ti, Cu, Mg

Pre Req: MM101

MM323 Crystallography and X-ray Diffraction (3-0-3):

Crystals and crystal systems, construction of crystals and packing of layers, introduction to X-rays, safety precautions, Bragg's analysis of X-ray diffraction, two-dimensional patterns, lattices and symmetry, Bravais lattices and crystal systems, plane group symmetry, point group symmetry, space groups, quasi-crystals, zone axis, reciprocal lattices, Ewald's sphere construction, X-ray diffraction of polycrystalline materials, texture analysis, lattice parameter measurements, indexing patterns, identification of unknown phases, measurement of crystallite size, measurement of internal elastic strains, Electron diffraction and its applications, stereographic projection and its uses.

Pre Req: MM101

MM324 Deformation and Fracture (3-0-3):

Types of stress-strain and flow curves, elastic and plastic deformations, strain and stress tensors, Mohr's circle of stress and strain in 2D and 3D, principal stresses and strains, hydrostatic and deviator strain and stress components, generalized Hooke's law, anisotropy of elasticity, crystallographic aspects of plastic deformation, dislocation and its types, mechanisms of deformation, critical resolved shear stress, strain hardening of single crystal FCC, barriers to dislocation glide, strengthening mechanisms, creep and fatigue, theories of fracture, fracture toughness, failure analysis, fracture mechanics in stress corrosion. Characteristics of fracture observed in ductile and brittle material. The crack opening displacement approach and empirical methods for assessing crack propagation in thick sections.

Pre Req: MM222

MM334 Heat Treatment and Processing

(3-0-3): Fe-Fe₃C Phase diagram, transformation temperatures, kinetics of transformation, IT, CCT and TTT diagrams, interphase precipitation, divorced eutectoid structures, formation mechanisms and morphologies of phases in steel, austenite grain size effect on phase diagram, effect of second phase particles, discontinuous grain growth, annealing, normalising, quenching, austempering, martempering,

HT defects and remedies, effect of quenching rates and quenching media, hardenability, surface hardening processes and surface modification, HT processing of HSLA, dual phase and microalloyed steels, stainless steels and tool steels, Heat treatment equipment.

Pre Req: MM232

MM351 Joining of Materials (3-0-3): Joining types and forces involved, joint stresses and efficiency, design aspects of joints, fastener types and their uses, adhesive bonding, types, joints and forces, preparation and testing, principles of soldering and brazing, types, fillers, fluxes, applications, joints and testing, welding process, fusion and non-fusion welding processes, defects and tests, thermal cycles and effect of filler and environment, microstructure and defects in fusion, partially melted, and heat affected zones, variant and hybrid joining processes.

Pre Req: MM212

MM352 Manufacturing Processes-I (3-0-3): Manufacturing principles, manufacturing and processing operation, engineering materials: review and classification and properties



Jawad Arif

Getting admission to GIK is nonetheless, a promising start for a student's career. It's a complete platform of decent opportunities to avail. At first, I contributed my time here and it was certainly worth the effort. Given the quality teachers who are always prepared to direct you at whatever point you need, as a trade-off for your time and hard work. All things considered, everything without a doubt revolves around the climate of GIKI, which grooms you expertly, regardless of whether you perform in academics or in extra-curricular activities. This place offered me a chance to prove my worth in practical life. Having said that, it's totally up to us to change our lifestyle for the good, avail every opportunity this institute provides, and with each step forward, reach the destination of our choice.

of engineering materials in manufacturing, forging, rod and wire drawing, tube drawing processes, rolling, forming methods, shearing, extrusion, drawing of rods, wires and tubes, sheet metal forming, machining of metals and alloys, machining operations and machine tools and fluids, machining centers and turning centers, machining operations for special geometries, high-speed machining.

Pre Req: MM324

MM353 Smart and Functional Materials (3-0-3): Shape memory materials, superelastic materials, magnetic shape memory alloys, photomechanical materials, self-healing materials, ferrofluids, piezoelectric materials, thermo-electric materials, magnetostrictive materials, functionally graded materials, chromogenic systems, electro-active polymers, temperature responsive polymers Self healing materials, hydrophobic materials, functional HEAs and BMGs.

Pre Req: Nil

MM362 Ceramics and Glasses (3-0-3): Silicates and clay minerals, powder making and characterization of powders, traditional ceramics-triaxial porcelain; raw materials processing, forming/shaping, glazing, sintering, RBS, SPS, HIPING and sintering kilns, refractories and their applications, silicon carbide, silicon nitride, sialon, boron nitride, bioceramics, piezo ceramics and biomimetics, cement, glass-ceramics and glasses, glass fibers and optical fibers, processing of glasses, heat treatment and testing of glasses, characterization of ceramics.

Pre Req: MM101

MM365 Polymers and Composites (3-0-3): Polymer types and application, polymerization and kinetics, molecular weight, structure and morphology, crystallization regimes, glass transition and melting point, mechanical properties, processing, composite classification, applications, matrices and reinforcements, rule of mixtures, testing of composites, production of fibers, production of MMCs, CMCs, PMCs, effect of structure on physical and mechanical properties, production, properties and application of carbon-carbon composites, mechanics of composites.

Pre Req: CH101 & MM101

MM391 Nanomaterials & Nanotechnology-I

(3-0-3): Introduction to nano science and nanotechnology, physical chemistry of solid surfaces, surface energy, electrostatic stabilization, steric stabilization, zero dimensional nanostructures: nanoparticles, quantum dots, one dimensional nanostructures: nanowires and nanorods, template-based synthesis, two-dimensional nanostructures. Thin films by physical and chemical methods, three-dimensional nanostructures: nano-carbons, fullerenes, CNTs and graphene, core shell nanostructures, nanomaterials hazards and safety procedures.

Pre Req: Nil

MM392 Nanomaterials & Nanotechnology-II

(3-0-3): Nanotechnology and prospects for business and industry, nano materials characterization methods, nano fabrication methods, nanometrology, nano electronics, nano optics, nano structure and nanofilms, nanocatalysis, nanobiotechnology, biomimetics, medical nanotechnology, environmental nanotechnology, societal implications of nano.

Pre Req: MM391

MM393 Electronic and Magnetic Materials

(3-0-3): Introduction to magnetic materials, diamagnetism and paramagnetism, ferromagnetism, anti-ferromagnetism, ferrimagnetism, domains and the magnetization process, soft magnetic materials, hard magnetic materials, electronic materials, elementary quantum physics semiconductors, semiconductor devices, packaging of devices/ dicing/ wire bonding.

Pre Req: Nil

MM435 Corrosion Degradation and Protection

(3-0-3): Corrosion engineering, corrosion environments and damage, corrosion classification, roles of corrosion engineer, electrochemical theory and thermodynamics of corrosion, Nernst equation, Pourbaix diagrams, emf and galvanic series. Faraday law and corrosion rate determination. Electrode kinetics, polarization and types, Ohmic drop at electrolyte/metal interface, mixed potential theory, passivity. Corrosion failures, factors and corrosion and their prevention. Corrosion testing: salt spray/fog test, electrochemical corrosion testing, corrosion data analysis, Tafel extrapolation. Corrosion protection measures: cathodic/anodic protection, coatings and

inhibitors, synergistic mixtures, design considerations, corrosion of ceramics and degradation of polymers.

Pre Req: CH101 & MM231

MM 436 Surface Engineering (3-0-3):

TLK model, surface crystallography, surface point defects, adsorption, physisorption, and chemisorption, contact mechanisms (Hertz Theory), wear in tribo-contacts, residual stresses, friction surfacing, cold gas spraying, electrolytic and electrophoretic deposition, electroless deposition. Anodizing, phosphating, nitrocarburizing, HVOF, detonation gun, plasma spraying, CVD, PECVD, PVD (evaporation, sputtering), (thermionic ion plating, arc evaporation, magnetron sputtering). Hard coatings, hybrid processes, thickness, fracture, adhesion & scratch testing, residual stress measurements, surface acoustic wave spectroscopy, impact excitation.

Pre Req: MM101

MM451 Manufacturing Processes-II (3-0-3):

Component design for manufacturing, non-traditional and non-conventional



Umema Aftab

Shadowing between humongous Mountains till the creating unbreakable bonds ;I love everything that Giki has given, its places, the people ,the vibe. My experience is an amalgamation of strenuous academics and even more strenuous at extra curriculars. Life at Giki has best of both worlds which grooms you in every aspect and therefore I will keep coming back to Giki after graduation to relive the cherished memories.

machining, manufacturing operations, group technology, abrasive machining, thread and gear manufacturing, chemical machining, electro-chemical machining, electric discharge machining, laser jet machining, electron beam machining, hybrid machining operations, sawing (band, circular, hacksaw), flexible operations, sawing (band, circular, hacksaw), flexible Automation and industrial control technologies, sensors and other control systems, materials handling and identification processes, storage and inventory, quality control systems, SPC (statistical process control) and charts, design and process planning, agile manufacturing, production volume and assembly techniques, rapid prototyping, additive manufacturing, micro and nano fabrication, lithography
Pre Req: MM352

MM453/ME418 CAD/CAM (3-0-3):

Introduction and history, geometric modelling, feature based design, CAD hardware and software, 2D and 3D graphics and transformations, assembly modelling and analysis, concurrent engineering, axiomatic design, DFM, DFA, group technology, CE tools, process planning, manual, variant, generative and hybrid approaches, tolerance charts, manufacturing planning and control, cellular and JIT manufacturing, numerical control, NC programming, CNC, DNC, robotics, computer-integrated manufacturing, Creo based lab sessions related

to design, assembly and manufacturing.

Pre Req:

MM454 Powder Metallurgy (3-0-3):

Production of metallic powders, powder characterization techniques, microstructure control in powder, kinetics of solidification in powders, powder handling and purity modification, shaping and compaction, sintering, characterization of sintered components, finishing operations, structural and porous components, cermets, dispersion strengthened materials.

Pre Req: Nil

MM455 Finite Element Methods (3-0-3):

Mathematical modelling and its applications in thermal, structural and coupled analysis, general steps of the Finite Element Method (FEM), applications and advantages of FEM, computer programs for FEM analysis, introduction of stiffness matrix, boundary conditions, potential energy approach, use of ANSYS for problems, global stiffness matrix, comparison of finite element solutions to exact solution, Galerkins's residual method and its application, axisymmetric elements and their use, practical consideration in modelling, results interpretation, plane stress and plane strain analysis, thermal and structural stress analysis using ANSYS multi-physics, use of ANSYS optimization for structural optimization under thermal-structural coupled analysis, introduction to Finite Difference Methods, use of SolidCast for



casting simulation and mould design.

Pre Req: Nil

MM469-Automobile Engineering and Materials (3-0-3): Introduction to automobile engineering (layout and components, engine, chassis frame and body, wheel and tyre etc.) Materials for consideration and use in automotive body structures, historical perspective and evolving materials technology, advanced materials for lightweight automotive structures (advanced steels, aluminium and magnesium alloys, polymers and composites etc.). Manufacturing and design of lightweight automotive structures (vehicle architecture design and manufacturing; casting, forming and joining processes for automotive components). Corrosion, protection and recycling of the automotive structures.

Pre Req: Nil

MM494 Materials Characterization (3-0-3): Physical characterization of materials, particle size distribution by laser diffraction, dynamic light scattering, centrifugal sedimentation, crystallite size by XRD, BET specific surface area, laser confocal fluorescence microscopy, NSOM and STED, electron microscopy, SEM and types, TEM, sample preparation techniques, Ion milling, FIB, AFM and STM, chemical spectroscopy of materials by XRF, EDX, XPS/AES, FTIR, Raman spectroscopy.

Pre Req: MM101

MM495 Nanosystems and Devices (3-0-3): Microtechnological foundations, clean room technology, components, operation and maintenance, preparation of nanostructures, nanotechnical structures, nanotransducers, technical nanosystems, NEMS and MEMs, nanodiodes, nanotransistors, nanoswitches, nanostructures as optical sensors.

Pre Req: MM391

MM496 Advanced Materials (3-0-3): Review of engineering materials, advanced materials: requirement and applications, shape memory alloys (SMA): thermally activated and magnetic, magnetostrictive and piezoelectric materials, intermetallics, bulk metallic glasses, modern steels, functionally graded materials, super alloys, Ti alloys, advanced Coatings and composites, metallic foams, biomaterials.

Pre Req: MM101

MM497 Biomaterials (3-0-3): Materials for biomedical applications, types of biomaterials, synthesis and fabrication, advanced alloys for production of hip joints, prostheses, and implants, surface properties and cells interaction, Hydroxyapatite (HA) surface coatings, dental materials, biocompatibility, polymers and biocomposites for tissue engineering, next-generation biomaterials and emerging manufacturing technologies.

Pre Req: MM101

MM499 Nanotechnology for Energy (3-0-3): Nanotechnology in clean and renewable energies: nanotechnologies in solar cells and thin film photovoltaics, nanotechnologies in rechargeable batteries: Li-ion batteries, Li-Polymer batteries, energetic materials, nanotechnologies in thermoelectricity, fuel cells and supercapacitors, nanotechnology in hydrogen production and storage, energy sustainability, Green nanofabrication, safety and economics.

Pre Req: MM391

Laboratory Courses:

MM141L Materials and Nanotechnology Lab (Materials Lab-I) (0-3-1): The complementary laboratory course to the MM102 lectures. Experiments and demonstrations to give a basic understanding of the structure and properties of materials and an introduction to their fabrication and testing.

Pre Req:

MM242L Materials Lab-II (0-3-1): Experiments using some of the main techniques for the destructive and non-destructive evaluation of materials.

Pre Req: MM101

MM243L Materials Lab-III (0-3-1): Experiments concerning casting of different metallic materials and optical microscopic techniques for the study and evaluation of materials.

Pre Req: MM101

MM344L Materials Lab-IV (0-6-2): Exercises concerning crystallography, and experiments demonstrating the use of XRD. Experiments designed to examine the effect of processing parameters and phase transformations on

the properties of steels, alloys. Also includes experiments related to joining of materials using various techniques. Synthesis and characterization of nanomaterials is also a part of this lab course.

Pre Req: MM101

MM345L Materials Lab-V (0-3-1):

Experiments related to polymers, composites and their properties, mechanical working, synthesis of nanomaterials and thin films are included in this lab.

Pre Req: MM101

MM446L Materials Lab-VI (0-3-1):

Experiments related to surface engineering, characterization of materials using advanced techniques, corrosion testing & analysis and case studies.

Pre Req: MM101

MM447L Materials Lab-VII (0-3-1):

Experiments related to powder metallurgy,

advanced materials, semiconductor and energy storage devices.

Pre Req: MM101

Careers in Materials Engineering

Majority of our graduates are hired by national and multinational research/academic institutions and industry within one year of graduation. They are also well prepared for graduate work so that they can choose to continue their education leading to MS and PhD degree. More specifically, our engineers are employed in a broad range of technical areas such as national research organizations and industries, multinational companies and academia. Private organizations and engineering consultancy companies also hire a significant number of GIKI materials graduates for their ongoing projects. A considerable number of our graduates have chosen to become successful entrepreneurs of Pakistan in their respective fields.



Department of Chemical Engineering

Introduction:

Rapid commercialization of conventional and modern, man-made products gave birth to process industry. The dynamics of the industry requires group of highly trained professionals from almost all engineering disciplines. However chemical engineer organizes his/her coordination at the process plant and thus deemed as process brain. Such responsibility demands basic knowledge of all conventional trades of engineering in addition to in-depth knowledge of large-scale industrial dynamics. Continuous and safe process operation is an exclusive responsibility of this trade, in addition to design, problem investigation and troubleshooting. Well-versed chemical engineer, during his/her career, usually encounters a diverse field of application in thermodynamics dictating unit processes.

Department of Chemical Engineering is endeavouring to achieve excellence as per requirements of Outcome Based Education (OBE) system to enhance the capabilities of its graduates. The department offers a 4-years degree program in chemical engineering detailing basic principles and mathematics of process operations in the first two years. Third & final year deal with the advanced level of the trade closely selected to cope the industrial requirements. The newly established laboratories are the prime feature, providing state of the art equipment. Most of the laboratories are designed having conventional features imitated by the more sophisticated and risk-free digital equipment. Experiments are designed to trigger the thinking of students and not just mere data logging.

Mission

The mission of the department of chemical engineering is to develop and disseminate the understanding of designing and operations of chemical processes so that our graduates can excel in academia, research, business, and industry to contribute ethically for the humanistic development of the society.

Thrust Areas

Process & Equipment computer aided design, Heat transfer and Thermodynamic behavior, Reaction Kinetics and Catalysis, Separation Processes, Process Modeling and Simulation Process Optimization, Pollution Abatement and Control, Polymer based materials & Membrane Technology.

Faculty:

Prof. Dr. Javaid Rabbani Khan (HoD), PhD Chemical Engineering (University of New Castle upon Tyne, U.K.)

Muhammad Shozab Mehdi, PhD Chemical Engineering (PIEAS, Pakistan)

Khurram Imran Khan, PhD Chemical Engineering (Politecnico di Torino, Italy)

Sajjad Hussain, Ph.D (University of Sao Paulo, Brazil),



HoD

Prof. Dr. Javaid Rabbani Khan

Post-Doctorate (State University of Sao Paulo, Brazil).
Muhammad Usman Farooq, PhD Chemical Engineering, University of Waterloo, Waterloo ON Canada).

Hammad Amjad Khan, PhD Environmental Engineering (Hanyang University, South Korea)

Ramesha Tariq, MS Chemical Engineering (UET Lahore, Pakistan)

Abdul Wahab, MS Chemical Engineering (GIK Institute, Pakistan)

Nida Zafar, MS Chemical Engineering (NUST, Pakistan)

Faculty on Study Leave:

Faraz Saeed Butt

Imran Abbas

Lab Engineers:

Furqan Yousafzai, BS, GIK Institute, Pakistan

Fazal Wahab, BS, GIK Institute, Pakistan

Hezb Ullah, BS, GIK Institute, Pakistan

GAs:

Iqra Yasmin, MS, CUI, Lahore, Pakistan

Sundas Khusnood, MS, NUST, Pakistan

Kamran Alam, MS, US-PCASE, UET Peshawar, Pakistan

Muhammad Umar, MS, GIK Institute, Pakistan

Shafiq Uz Zaman, MS, CUI, Lahore, Pakistan

Sajjad Asghar, BS, GIK Institute, Pakistan

Muhammad Usman, BS, NFC IET Multan, Pakistan

Nishwana Tahir, BS, NFC IET Multan, Pakistan

Amina Bibi, BS, UET Peshawar, Pakistan

M. Arslan Akhtar, BS, NFC IET Multan, Pakistan

Abdul Rafay, BS, UET Peshawar, Pakistan

Muhammad Danish Javed, BS, NFC IET Multan, Pakistan

Muhammad Aqif, BS, UET Peshawar, Pakistan

Nabeel Hussain, BS, GIK Institute, Pakistan

Personal Secretary to HoD:

Mr. Asad ur Rehman

Program Educational Objectives (PEOs)

Program Educational Objectives (PEOs) are extensive statements that define what graduates are likely to achieve within three to four years of graduation.

PEO-1: Exerting for carrier growth in Industry, Consultancy, R&D or Academia for sustainable development of society.

PEO-2: Contributing as persistent work force to develop strategies by addressing engineering problems for maintaining quality assurance.

PEO-3: Possessing entrepreneurial and communication skills to conduct and contribute in professional and ethical manner while exhibiting teamwork.

Program Learning Outcomes (PLOs)

There is a set of twelve Program Learning Outcomes (PLOs) of Chemical Engineering program which describe what students are expected to know/perform/attain by the time they graduate from Department of Chemical Engineering. The PLOs are given below:

PLO-1: Engineering Knowledge: Ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PLO-2: Problem Analysis: Ability to identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PLO-3: Design/Development of Solutions: Ability to design solutions for complex

engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

PLO-4: Investigation: Ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.

PLO-5: Modern Tool Usage: Ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.

PLO-6: The Engineer and Society: An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.

PLO-7: Environment and Sustainability: An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

PLO-8: Ethics: Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PLO-9: Individual and Teamwork: Ability to work effectively, as an individual or in a team, on multifaceted and/or multidisciplinary settings.

PLO-10: Communication: Ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PLO-11: Project Management: Ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

PLO-12: Lifelong Learning: Ability to recognize importance of and pursue lifelong learning in the broader context of innovation and technological developments.



Knowledge Profiles (WK):

A Washington Accord Programme provides:

	Description	Courses
WK1	Natural Sciences	Applied Physics, Applied Chemistry and Environment, Materials and Nanotechnology, Inorganic and Organic Chemistry.
WK2	Mathematics & Computing	Calculus I, Calculus II, Differential Equations and Linear Algebra I, Computing and AI Lab, Object Oriented Programming and Design, Object Oriented Programming and Design Lab, Probability and Statistics, Engineering Economics, Numerical Analysis.
WK3	Engineering Fundamentals	Applied Electrical Engineering, Physical and Analytical Chemistry, Chemical Process Industries, Energy Engineering, Chemical Engineering Principles, Particle Technology, Chemical Engineering Thermodynamics I, Chemical Engineering Thermodynamics II, Fluid Mechanics I, Fluid Mechanics II, Environmental Engineering, Transport Phenomenon, Instrumentation and Process Control.
WK4	Engineering Specialization	Food Technology, Petroleum Refinery Engineering, Pharmaceutical Engineering, Nuclear Engineering, Water Treatment and Purification, Biochemical Engineering, Technology, Product Design and Development, Heterogeneous Catalysis, Piping Design, Environmental Impact Assessment, Fuel and Clean Technology, Industrial Waste Management.
WK5	Engineering Design	Heat Transfer, Mass Transfer, Reaction Kinetics and Reactor Design, Simultaneous Heat and Mass Transfer, Process Modelling & Simulation, Chemical Engineering Plant Design
WK6	Engineering Practice	Applied Physics Lab, Innovation and Makers Lab I, Innovation and Makers Lab II, Occupational Health and Safety, Applied Chemistry and Process Industries Lab, Energy and Thermodynamic Lab, Fluid Mechanics Lab, Heat and Mass Transfer Lab, Particle Technology Lab, Environment and Reaction Engg. Lab, Instrumentation and Process control Lab, Separation Processes Lab, Process Modelling and Simulation Lab.
WK7	Engineering in Society	Communication Skills, Critical Thinking and Expository Writing, Pakistan and Islamic Studies, Sociology and Human Behaviour, Corporate Law and Professional Ethics.
WK8	Research Literature	Chemical Engineering Project Design I, Chemical Engineering Project Design II.

AIChE-GIKI Student Chapter

The International Student Chapter of AIChE at GIK Institute helps students in professional development and training. Its activities include organization of seminars, video shows, discussions, and industrial visits.

Accreditation

The BS Degree in Chemical Engineering is accredited under level II by the Pakistan Engineering Council (PEC).

Laboratories

The Department of Chemical Engineering houses state of art labs of core chemical engineering courses, Industrial products quality testing labs, Particle Technology, Thermodynamics, Reaction Engineering, Heat and Mass Transfer Operations, Fluid Mechanics, Environmental Engineering, Instruments and Process Control, Process Simulation & modelling and Mechanical Workshop.

Course Work Requirements

For BS in Chemical Engineering degree student must complete the following requirements as per details given in Table a-f.

(a) General Education Requirements (55 Credit Hours)

Course Title	Course Code	CH
Mathematics	ES111, ES341, MT101, MT102, MT202.	15
Sciences	CH101, PH101, PH101L.	06
Comp. System Eng.	CS101, CS101L, CS112, CS112L.	07
Basic Engineering Courses	CH161, EE201, IF101L, IF102L, MM101, MM101L.	09
Humanities/Social Sciences/ Management	HM101, HM102, HM211, HM321, HM322 , MS291.	18

(b) Core Requirements (67 Credit Hours)

Course Title	Course Code	CH
Inorganic and Organic Chemistry	CH201	3
Physical and Analytical Chemistry	CH202	2
Chemical Process Industries	CH211	2
Energy Engineering	CH212	3
Chemical Engineering Thermodynamics I	CH214	3
Chemical Engineering Principles	CH231	3
Fluid Mechanics I	CH241	3
Heat Transfer	CH311	3
Mass Transfer	CH313	3
Chemical Engineering Thermodynamics II	CH321	3
Reaction Kinetics and Reactor Design	CH322	3
Process Modelling & Simulation	CH331	2
Particle Technology	CH341	3
Fluid Mechanics II	CH342	2
Environmental Engineering	CH361	2
Simultaneous Heat and Mass Transfer	CH411	3
Transport Phenomenon	CH412	3
Instrumentation and Process Control	CH415	3
Chemical Engineering Plant Design	CH441	3
Chemical Engineering Project Design	CH481, CH482	6
Chemical Engineering Labs	CH251L, CH252L, CH253L, CH351L, CH352L, CH353L, CH451L, CH452L, CH453L.	9
Chemical Engineering Project Design	CH481, CH482	6

(c) Technical Electives (06 Credit Hours)

Course Title	Course Code	CH
Food Technology	CH413	3
Petroleum Refinery Engineering	CH414	3
Pharmaceutical Engineering	CH417	3
Nuclear Engineering	CH418	3
Water Treatment and Purification	CH419	3
Biochemical Engineering	CH420	3
Product Design and Development	CH421	3
Heterogeneous Catalysis	CH422	3
Piping Design	CH442	3
Environmental Impact Assessment	CH461	3
Fuel and Clean Technology	CH462	3
Industrial Waste Management	CH471	3
Polymers and Composites	MM365	3
Nanomaterials and Nanotechnology	MM391	3
Biomaterials	MM497	3
Corrosion Degradation and Protection	MM435	3

(d) Engineering Management Electives (06 Credit Hours)

Course Title	Course Code	CH
Maintenance Engineering and Industrial Management	CH371	3
Operation Management	MS492	3
Industrial Safety	MS493	3
Total Quality Management	MS494	3
Maintenance Management	MS495	3
Project Management	MS496	3

(e) Summer Training (Pass/Fail Grade, NIL Credit)

Every student is required to complete an industrial internship or training program (eight weeks) during the summer of third year and submit a formal written report.

(f) Total Requirement (134 Credit Hours)

1st Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	CH161	Occupational Health and Safety	1	0	1	None	None
	CS101	Computing and AI	2	0	2	None	None
	CS101L	Computing and AI Lab	0	3	1	None	CS101
	HM101	Communication Skills	1	2	3	None	None
	IF101L	Innovation and Makers Lab 1	0	3	1	None	None
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	None
	PH101L	Applied Physics Lab	0	3	1	None	PH101

2nd Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CS112	Object Oriented Programming and Design	3	0	3	None	None
	CS112L	Object Oriented Programming and Design Lab	0	3	1	None	None
	ES111	Probability and Statistics	3	0	3	MT101	None
	HM102	Critical Thinking and Expository Writing	3	0	3	None	None
	IF102L	Innovation and Makers Lab II	0	3	1	None	None
	MM101	Materials and Nanotechnology	2	0	2	None	None
	MM101L	Materials and Nanotechnology Lab		3	1	None	MM101
	MT102	Differential Equations and Linear Algebra I	3	0	3	MT101	None

3rd Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH202	Physical and Analytical Chemistry	2	0	2	None	None
	CH211	Chemical Process Industries	2	0	2	CH101	None
	CH231	Chemical Engineering Principles	3	0	3	None	None
	CH251L	Applied Chemistry and Process Industries Lab.	0	3	1	None	CH211, CH202
	EE201	Applied Electrical Engineering	3	0	3	PH101	None
	HM211	Pakistan and Islamic Studies	3	0	3	None	None
	MT202	Calculus II	3	0	3	MT101	None

4th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH201	Inorganic and Organic Chemistry	3	0	3	CH211	None
	CH212	Energy Engineering	3	0	3	None	None
	CH214	Chemical Engg. Thermodynamics I	3	0	3	None	None
	CH241	Fluid Mechanics I	3	0	3	None	None
	CH252L	Energy and Thermodynamic Lab.	0	3	1	None	CH212, CH214
	CH253L	Fluid Mechanics Lab.	0	3	1	None	CH241
	MS291	Engineering Economics	3	0	3	None	None

5th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH311	Heat Transfer	3	0	3	None	None
	CH313	Mass Transfer	3	0	3	CH214	None
	CH321	Chemical Engg. Thermodynamics II	3	0	3	CH214	None
	CH341	Particle Technology	3	0	3	None	None
	CH351L	Heat and Mass Transfer Lab.	0	3	1	None	CH311, CH313
	CH352L	Particle Technology Lab.	0	3	1	None	CH341
	HM321	Sociology and Human Behaviour	3	0	3	None	None

6th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH322	Reaction Kinetics and Reactor Design	3	0	3	CH321	None
	CH331	Process Modelling & Simulation	1	3	2	None	CH322
	CH342	Fluid Mechanics II	2	0	2	CH241	None
	CH353L	Environment and Reaction Engg. Lab.	0	3	1	None	CH322, CH361
	CH361	Environmental Engineering	2	0	2	None	None
	ES341/CS342	Numerical Analysis	3	0	3	MT202	None
	HM322	Corporate Law and Professional Ethics	3	0	3	None	None



7th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH411	Simultaneous Heat and Mass Transfer	3	0	3	CH311, CH313	None
	CH415	Instrumentation and Process Control	3	0	3	None	None
	CH451L	Separation Processes Lab.	0	3	1	None	CH411
	CH453L	Instrumentation and Process Control Lab.	0	3	1	None	CH415
	CH481	Chemical Engg. Project Design-I	0	9	3	None	None
	XXXXX	Technical Elective I	3	0	3	None	None
	XXXXX	Management Elective I	3	0	3	**	**

8th Semester	Course Code	Course Title	Lec. Hrs	Lab. Hrs	CH	Pre-req	Co-req
	CH412	Transport Phenomena	3	0	3	CH411	None
	CH441	Chemical Engg. Plant Design	3	0	3	CH411, MS291	None
	CH452L	Process Modelling and Simulation Lab.	0	3	1	CH331	None
	CH482	Chemical Engg. Project Design-II	0	9	3	None	None
	XXXXX	Technical Elective II	3	0	3	None	None
	XXXXX	Management Elective II	3	0	3	None	None

Course Description

CH101 Applied Chemistry and Environment (2-0-2): Importance of Chemistry for Engineers and Its Application in Industries, Mass spectrometry, Determination of atomic masses, stoichiometry and chemical calculations, Standardization of solution for quantitative analysis, Electrochemistry, Batteries, Solid state batteries, Corrosion mechanism, causes and its control, Nano-chemistry, deposition of thin films, CVD, PVD, Purification of silicon for electronic chip manufacturing, Basic Chemistry of Polymers, Fossil fuels, clean combustion processes, Integrated gasification combined cycle IGCC, CO₂ capturing from exhaust gasses and Syngas from coal, Chemistry for Environmental sustainability and green chemistry including its twelve principles, Photochemistry, Free radicals, Interaction of solar radiation with molecules in the atmosphere, Acid rain, ozone in the atmosphere, Urban smog, Treatment of water for domestic and industrial uses, treatment of wastewater to preserve aquatic ecosystem.

CH161 Occupational Health and Safety (1-0-1): Environment, Health and Safety Management, Chemical and biological hazards, safety and controls, Mechanical/ Machine safety and control, Electrical Safety and controls, Construction hazards including silica and Cementous fume inhalation, Safety and control, Fire hazards, Safety, and controls, Workplace hazards and controls, Physical and psychological Health Hazards and Risk controls, Workplace accident investigation procedures and case history, Permit to work system in industry, Environmental toxicant; including Radon from building materials, exposure, and measurements, Responsibilities of employee and employer, First Aid procedures and cardiopulmonary resuscitation (CPR), Climate change, industrialization and controls.

CH201 Inorganic and Organic Chemistry (3-0-3) Pre-requisite(s): CH211: Inorganic Chemistry: Chemistry for chemical engineering, Atomic, ionic and molecular solids, Atomic structure and Transition elements, Multiple oxidation states and shielding effect, Redox reactions, Coordination compounds, Isomerism, Catalysis, Chemistry in

non-aqueous solvents, radioactivity, Organic Chemistry: Shape and structure of organic compounds, Hydrocarbons and their derivatives, Functional groups, Carbonyl compounds, Conformation, Isomerism and chirality, Polarity, Inductive effect and acidity in organic compounds, Reaction mechanism in organic reactions and structure-reactivity relationships, Macromolecules and polymers, Biologically important compounds such as amino acids and polypeptides, Bio-chemical processes, Carbohydrates: di and poly-saccharides, Nucleic acids and DNA.

CH202 Physical and Analytical Chemistry

(2-0-2): The rates of reactions, order reactions and methods for determination of order of reactions. Solution Chemistry: Ideal and non-ideal solutions, Raoult's law, vapor pressure, boiling point, and freezing point, osmotic pressure, and Henry's law. Dynamic Equilibrium and Thermodynamics: Equilibrium reactions in different phases, extent of reactions and equilibrium constants, Gibbs energies of formation, van't Hoff equation, Le-Chatelier's principle. Colloid and Surface Chemistry: adsorption and adsorption isotherms, Freundlich and Langmuir adsorption isotherms, Colloidal solutions, Polymer Chemistry, Instrumental methods of chemical analysis, Chromatography, and spectroscopy.

CH211 Chemical Process Industries (2-0-2)

Pre-requisite(s): CH101: Various well established chemical manufacturing processes; Latest trends in process industries, Fertilizer manufacturing, Insecticides production, Soaps and detergents, Sugar industry, Cement industry, chemical manufacturing units; Soda ash, Caustic soda, Chlorine, Sulphuric acid, Water treatment plants, Fermentation and Food processing industry, Tannery processing, Pulp and paper and Basic pharmaceutical industries.

CH212 Energy Engineering (3-0-3):

Classification of Conventional energy resources, Origin, characterization and taxonomy of available fuels, Principles of combustion, Combustion of solid, liquid, and gaseous fuels. Fluidized Bed. Combustion calculation; energy requirements and combustion efficiency of Industrial Burners. Fuel economy measures, excess air, heat distribution,

temperature control, draft control and Flame stability. Fuel quality and combustion efficiency enhancement. Importance of Alternate Energy Resources; Solar, Wind, Wave, Tidal, geothermal, Nuclear and Hydel, Densification and calorific value up-gradation of biomass, Carbonization and Gasification of biomass, Energy generation from municipal waste, Development of fuel cells. Environmental and global impact of energy resources, Industrial fuels and selection criteria, Energy audit, conservation waste heat recovery.

CH214 Chemical Engineering

Thermodynamics I (3-0-3): Thermodynamic systems and processes, Reversible and Irreversible Processes, The first law of thermodynamics, Energy balance for open systems with and without reaction; Ideal/non-ideal gas model, Equation of state and Property relations Second law of thermodynamics, Entropy and Entropy balance for closed and open systems, Isentropic efficiencies of turbines, Nozzles, Applications of thermodynamics to flow processes, Nozzles, Turbines, Compressors, Heat engines, Refrigeration, Air conditioning and Liquefaction of gases.

CH231 Chemical Engineering Principles

(3-0-3): Units, dimensions and conversions, Temperature and Pressure scales, Composition of mixtures, Principles of stoichiometric combination, Nature of balances; Concept of a balance, Input-output relationships, Steady-state considerations, Sub-systems and interconnections, Mass balance diagrams and tables, Mass balances for items of plant, Choice of basis/datum for balances, Overall and component balances, Limiting and excess reactants,



Balances for systems with recycle, purge and by-pass streams, Mass balances for reactive processes, Mass balances for unit operations, Tie components, Balances for batch and continuous plant. Concepts of Energy balance, Mass and energy balances for reacting systems, Balances for combustion processes.

CH241 Fluid Mechanics I (3-0-3): Concept, Properties and types of fluids, Stress analysis of static fluids, Newton's Law of viscosity, Introduction to non-Newtonian fluids. Estimation and measurement of Pressure and Pressure gradient, Manometry, Buoyancy and Stability, Basic physical Laws in Fluid Mechanics, Conservation of Mass, Continuity equation, Linear Momentum, Angular Momentum and Energy, The Bernoulli's Equation and its application, Dimensional Analysis and Similitude, Viscous Flow in internal flows, Concept of Laminar and turbulent flow, Concept of friction and pressure drop in flowing fluids, Friction factor in laminar and turbulent flows in pipes, Concept of equivalent diameter, Pipe sizing problems, Flow measuring devices such as Bernoulli devices, Notches and Weirs. Concept of Boundary layer and its importance in fluid mechanics, brief introduction to external flows.

CH311 Heat Transfer (3-0-3): Basic concepts of thermal energy Modes of heat transfer, steady and unsteady state conduction in different co-ordinates, convective heat transfer with and without phase change. Correlations for forced and natural convection. Analogy between momentum and heat transfer. Radiation heat transfer, Heat transfer in extended surfaces. Industrial applications, categorization, selection criteria and design



of numerous heat transfer equipment: Heat exchangers, Waste heat recovery boilers, Evaporators, Condensers, Boilers. Heat transfer augmentation techniques.

CH313 Mass Transfer (3-0-3): Pre-requisite(s): CH214: Diffusion in Fluids Molecular and Eddy diffusion in a gas and liquid, Steady state diffusion under stagnant and laminar flow condition, Diffusion measurement and calculations, Ordinary diffusion in multicomponent gaseous mixtures, Diffusion in solids, Interface mass transfer, Theory of mass transfer, Concept of mass transfer coefficient, Overall mass transfer coefficient, Analogies between momentum and mass transfer coefficients, Absorption theories of gas absorption, Design of absorption towers, Absorption with chemical reaction, Concept of NTU and HTU, Adsorption, Ion-Exchange, Extraction Processes, Liquid-Liquid extraction, Leaching; General principles, Factors influencing the rate of extraction, Mass transfer in leaching operations.

CH321 Chemical Engineering Thermodynamics II (3-0-3): Pre-requisite(s): CH214: Advanced principles of thermodynamics focusing on phase equilibria, Maxwell relationships, Gibbs Duhem's theorem, Thermodynamics of separation processes, Two component systems, Liquid-vapor equilibria, Ideal and non-ideal solutions, Composition of vapor in equilibrium with liquid, Fractional distillation, Azeotropes, mixing, Liquid-solid equilibria. Thermodynamic analysis of power plants, Liquefaction and refrigeration systems, Chemical reaction equilibrium.

CH322 Reaction Kinetics and Reactor Design (3-0-3) Pre-requisite(s): CH321: Equilibrium and effect of heat on reactions, Rate of reactions, Molecularity and order of reaction, Reaction mechanism, conversion and reactor sizing, rate laws and stoichiometry for a single and multiple reactions/reactors and its applications to steady-state isothermal and non-isothermal reactors along with pressure drop effect. Collection and analysis of rate data, catalytic reactors and catalysis.

CH331 Process Modelling & Simulation (2-0-2): Pre-requisite(s): CH322, CH241: Introduction to MATLAB, Linear algebra

applications including Eigen Values, Symbolic and numerical differentiation (ODE solution), Calculation of integrals, First and second order transfer function, Plots for stability analysis, Mathematical modelling related to chemical engineering. Solution of mathematical models on MS Excel and MATLAB. Pinch Analysis, Optimization, Development of process flow diagrams for various process industries and de bottlenecking using simulation software such as ASPEN, Economic evaluation of processes.

CH341 Particle Technology (3-0-3): Introduction to particle technology, Fundamentals of solid handling, Transportation (conveying) and storage, Size reduction (crushing and grinding) and enlargement crystallization, Pelletization and granulation, Screening, Sieving, Coagulation, Flocculation, Fluidization, Mixing, Filtration, Agitation, Particles settling, Statistics of particle size and their distribution and classification, Various solid handling and Processing equipment, Emulsions, Coagulants and Powder technology. Powder characterisation and handling for pharmaceutical industries.

CH342 Fluid Mechanics II (2-0-2): Pre-requisite(s): CH241: Centrifugal pumps; Concept Characteristics; NPSH and its application in chemical engineering, Concept of specific speed, Similarity laws in centrifugal pumps. Pumps in series and parallel. Positive displacement pumps; classification, characteristics, selection and matching system characteristics with pump characteristics. Turbines; classification and selection. Compressible flow and its application in chemical engineering, Concept of choked flow in CD nozzle, Compressors; classification, characteristics, and selection.

CH361 Environmental Engineering (2-0-2): Introduction to environment and ecology, Pollution concept, types of pollution, Environmental policy and standards, Environmental Monitoring (Air, Water & Soil), Objectives of sampling and monitoring program, Design and types of samples; Pre-sampling requirements/information, Sampling and design purposes, Air pollution control technologies, Water pollution control technologies, Water treatment technologies, Soil pollution control technologies, Noise pollution control technologies, Biotechnology for

environment, Industrial pollution control, Solid Waste management.

CH371 Maintenance Engineering and Industrial Management (3-0-3): Maintenance: Preventive, predictive, Break down and total productive maintenance, Individual versus group replacement, Internal versus external maintenance, Scheduling of maintenance, computerized maintenance, Inspection techniques, Non-destructive testing techniques, Basics of rigging and lifting, Lubrication and lubricants, Industrial management, Process layout analysis and comparison, Material handling considerations in layout, Production planning methods, Capacity planning and control; Production control systems, Job shop scheduling, Quality Control, Production control charts, Scheduling techniques, Software for project management, Purchasing and procurement, Inventory control. Organizational structure, Human resource management, Project management principles, PERT/CPM, Total quality management, Labour and engineering laws, Prevention, and settlement of disputes.

CH411 Simultaneous Heat and Mass Transfer (3-0-3): Pre-requisite(s): CH311, CH313: Various industrial distillations: Binary distillation, Extractive distillation, Molecular distillation, Azeotropic distillation, Steam distillation, Reactive distillation, Multi-component distillation. Key components in multi-component mixtures and recovery fraction. Continuous flash distillation with heat balancing, Column Design, Tray design, hydraulics and performance, Batch distillation, Drying: Diffusion and Capillary theory of



drying, Classification, and selection of dryers. Freeze drying, Flash drying, Partial-recycle dryers, The drying of gases, Humidification and Cooling Towers, Crystallization, Operation, and equipment.

CH412 Transport Phenomena (3-0-3)

Pre-requisite(s): CH411: Transfer processes; A review of the mechanisms of momentum, Energy and mass transport, Momentum transport; Derivation of equations of continuity and motion (Navier-Stoke's equation) at molecular level, Equations of change both for isothermal, non-isothermal and multi component systems, Velocity distribution, Application in laminar and turbulent flow problems. Energy transport: Derivation of energy equation, Mechanism of energy transport at molecular level, Temperature distributions in flow, Application to heat transfer problems involving conduction, Forced and free convection, Application in laminar and turbulent flow problems. Mass transport: Derivation of species conservation equations for binary and multicomponent mixtures, Application to mass transfer problems with and without chemical reaction, Application in laminar and turbulent

flow problems.

CH413 Food Technology (3-0-3):

Application of biotechnology in chemical Industry, Biodegradation, Bio mass productivity and activity, Aerobic and anaerobic processes, Bio-chemical processes involved in the production of food products, Beverages, Organic Acids, Industrial solvents, Various pharmaceutical products and antibiotic and commercial enzymes, Fermentation Industries: Industrial alcohol, Biodiesel and industrial solvents, Waste treatment from Food and pharmaceutical Industry, Bioremediation, Food preservation, Health hazards, Hygiene and sanitation, Important food industries in Pakistan and Food legislation.

CH414 Petroleum Refining Engineering (3-0-3):

Introduction, Origin, Formation and composition of petroleum, Indigenous and world resources, Refinery products, Properties, Standard tests, Characterization and evaluation of crude oil stocks, Crude pre heating and preliminary treatment, Pipestill heaters, Desalting, atmospheric and vacuum distillation, Steam stripping, Arrangement of towers, Calculation of number of trays, Types of reflux employed, Packie's approach, Processing plans, schemes and product patterns of refineries, Modern separation, Conversion and treatment processes, Thermal and catalytic cracking and reforming, Hydrocracking, Auxiliary processes and operations, refinery corrosion and metals, Blending plants, Product design and marketing, Use of linear programming techniques to solve refinery blending and production problems, Overview of petroleum act.

CH415 Instrumentation and Process Control (3-0-3):

Instrumentation: instrument's terminologies and performance, P&I diagram, Instrumentation and sensors, Mathematical modelling of process control. Transfer functions. Dynamic behaviour of chemical processes. Control loops and its components, Feedback control. Dynamic behaviour of closed-loop systems. Stability analysis. Frequency response analysis. Controller design and tuning. Introduction to computer control. Laboratory and simulations applications Cascade Control, Ratio Control, Split Range Control, Feed Forward Control.



Ali Nawed Abbasi
3rd year Student

Being a subject to GIKI's Bachelors, was one of my extensive wishes. Having immense educational significance was the main reason why I admired the institute. But the way, GIKI has changed my outlook, perspective and approach is unreasonably tremendous.

Being located in the East of KPK, makes it even more intriguing. Arrival upon GIKI, the diverse environment I experienced, astonished and scared me simultaneously, initially. But as the degree progressed, it has played a very significant role in how it groomed me personally and professionally. The people associated to the institute have always made it easier for me to adjust, modify and improve. The quality of education, the environment it provides, the diversity it introduces you to, the sense of responsibility it makes one incorporate, have enabled me to encompass countless skills on and off academics.

CH417 Pharmaceutical Engineering (3-0-3): Engineering principles to pharmaceutical and life sciences related to industries, Process engineering in the drug discovery, High throughput characterization and optimization of new chemical entities, Solid-state engineering, and intelligent pharmaceutical manufacturing systems.

CH418 Nuclear Engineering (3-0-3): Role and importance of nuclear energy, Nuclear reactors cross-sections, Reaction rates, Nuclear fission and chain reaction, Critical conditions, Conversion and breeding, Reactor components and their characteristics, Classification and design features, Production and power reactors, Fast and fusion reactor systems, Fuel cycles, Uranium enrichment, Fabrication of fuel, Reprocessing of irradiated fuel, Fuel cycle performance, In-core fuel management and fuel management strategies and handling of nuclear waste.

CH419 Water Treatment and Purification (3-0-3): Primary and secondary treatment of the fresh feed water to the plant, Clarification, Sedimentation, Flocculation and Coagulation, Filtration, ion exchange, Membrane separation and reverse osmosis, Advance technologies, Cooling water treatment, Use of biocides and shock dosage, bacterial count and their importance, Treatment of sea water, Desalination, Treatment of various waste waters and biological wastewater treatment.

CH420 Biochemical Engineering (3-0-3): Basics of Microbiology, Enzyme classification, Enzyme reaction kinetics (Single- substrate Reactions) and energy patterns in biological system, Enzyme Inhibition, Non-ideal enzyme kinetics, isolation of enzymes and immobilized enzyme technology, Applications of enzyme catalysis (Biocatalysis), Transport phenomenon in microbial system, Design and analysis of biochemical reactors (fermentations), Anaerobic and aerobic metabolism photo-synthesis and bio-synthesis, Biochemical and microbiological application to commercial and engineering. Role of Biochemical Engineering as a future technology.

CH421 Product Design and Development (3-0-3): Product design and development. Opportunity generation, Innovation challenges

to develop new products and processes, Opportunity generation, Creating and selecting exceptional opportunities, User needs analysis, Concept selection manufacturing methods, Prototyping exercise, Industrial product development. The strategies of product development. Product planning. Managing customer and technical specifications. Advanced product modeling and modeling in 3D-CAD. Simulation of product dynamics with futuristic approach, animation, photo rendering, top-down-design and generating drawings. The software used is SolidWorks with associated modules.

CH422 Heterogeneous Catalysis (3-0-3): Introduction and basic concepts, Adsorption, Rates and kinetic models of catalytic reactions, Catalyst preparation and manufacture, Characterization of physicochemical properties, Surface characterization, Supported metal catalysts, Acid-base catalysts and zeolites, metal oxide catalysts and catalytic oxidation, Examples of important heterogeneous catalytic reactions.

CH441 Chemical Engineering Plant Design (3-0-3) Pre-requisite(s): CH411, MS291:



Laiba Iqbal
2nd year Student

Coming GIKI has been one of the most difficult choices I have made in my life. Being this distant from home truly puts one out of their comfort zone however as somebody who was passionate about Chemical Engineering and Chemistry, I am one of the fortunate ones to be pursuing my dream and that too in an institute like GIKI. In spite of the undeniable male to female proportion, I have consistently felt like I was given the equivalent opportunities and similar platform as my male peers and I have always been included in everything equally. This institute chooses us and nourishes us in a manner that sets us up for our professional lives and with there being such countless societies and activities blended in with studies, life in GIKI never gets boring. It takes a lot of guts, and you must have to be incredibly fortunate to have the option of packing your bags and enter the gates of GIKI. As once you're in, you will be a different and better version of yourself.

Process design and development. General design considerations, Health and safety, HAZOP study, Contingency plans, Design codes & standards, Economics and optimization, Materials selection for various services, Fabrication of desired component to facilitate processes, Vessel design; Low, medium and high-pressure storage and transportation vessels, Cryogenic vessels. Design of mass transfer equipment: Material transport and Material handling. Heat transfer equipment including furnaces and refrigeration units, Piping and pipeline design, Basic Concepts of Optimization, Optimization of Unconstrained Functions, Linear Programming Applications, Non-Linear Programming with Constraints, and Application of computer aided design, Engineering Ethics, Local, and Global Impact Analysis.

CH442 Piping Design (3-0-3): Process plant layout and equipment, Oil & gas pipeline design per ASME B31.4 / B31.8, Piping stress analysis, Process piping drafting, Liquid pipeline hydraulics, Fire safety piping and use of various design software related to the field e.g., Piping Systems Fluid Flow.

CH461 Environmental Impact Assessment (3-0-3): Principles and purposes of IEE and EIA and its significance for the society, Cost and benefits of EIA, Main stages in EIA process, public consultation and participation in EIA process, EIA methods and techniques for impact prediction and evaluation.

CH462 Fuel and Clean Technology (3-0-3): Fossil fuel processing towards green technologies, CO₂ sequestration, Pollution monitoring, BIOMASS (Biomass resources and potential contribution, Biomass and waste technologies, Biomass heat digestion, Gasification, Anaerobic digestion, Biofuel technologies, Bio harvesting, Environmental issues, Economics and costs), Wind energy, Solar and Tidal Energy, Geothermal energy, Hydroelectric Power plants, Clean technology applications and materials fabrication for Clean tech. (aero gels and sol gels etc.)

CH471 Industrial Waste Management (3-0-3): Environmental management ISO 14001, EMAS, Environmental auditing, Environmental Policies and regulations,

Different types of eco-labelling, Pollution prevention and potential implementation strategies, Material Recycling, Waste characterization, Physical, Chemical & Biological methods to treat liquid waste streams, Treatment of solid waste including separation, Incineration & composting. Production of biogas, Anaerobic digestion and other stabilization methods, Dewatering, Drying, Air pollution Management.

Final Year Project Design

CH481 Chemical Engineering Project Design-I (0-9-3): Chemical Engineering Project Design is the practical demonstration of student's theoretical knowledge. Groups of students are assigned a project of industrial scope and importance under the supervision of faculty member, which includes literature survey for process selection and feasibility, development of flow sheet, material and energy balances, design and sizing of different units or equipment, instrumentation and control, materials selection and cost estimation, economic analysis of plant, safety aspects/HAZOP study etc. The progress will be monitored through interim presentations and reports.

CH482 Chemical Engineering Project Design -II (0-9-3): Student will continue work on the Engineering Chemical Engineering Project Design-I (CH481). Students may require the fabrication of small unit for hands on experience. The progress will be monitored through interim presentations and report. A final report will be due at the end of term.

Laboratory Courses

CH251L Applied Chemistry and Process Industries Lab (0-3-1) Co-requisite(s): CH211, CH202: The main purpose of this laboratory is to provide the students an appreciation for the potential applications and limitations of analytical methods of analysis. It is also aimed to provide the students an opportunity to develop their abilities in the laboratory skills required for accurate and precise chemical analyses. Some experiments in this laboratory are designed to demonstrate quantitative analysis of water, milk and soap samples.

CH252L Energy and Thermodynamic Lab (0-3-1) Co-requisite(s): CH212, CH214:

Experiments related to fuel and its properties and chemical engineering thermodynamics and energy engineering.

CH253L Fluid Mechanics Lab (0-3-1)

Co-requisite(s): CH241: The experiments in this laboratory are designed to demonstrate various phenomena of fluid flow.

CH351L Heat and Mass Transfer Lab (0-3-1)

Co-requisite(s): CH311, CH313: The laboratories for this course are equipped with heat and mass transfer experimental benches.

CH352L Particle Technology Lab (0-3-1)

Co-requisite(s): CH341: Experiments related to the particle technology and vapor liquid equilibrium.

CH353L Environmental and Reaction Engineering Lab (0-3-1)

Co-requisite(s): CH322, CH361: The laboratories for this course are equipped chemical reactors and environmental engineering.

CH451L Separation Process Lab (0-3-1)

Co-requisite(s): CH411: Experiments in this lab are related to various separation processes such as absorption, adsorption, extraction, distillation, drying, crystallization, humidification etc.

CH452L Process Modelling and Simulation Lab (0-3-1)

Pre-requisite(s): CH331: Introduction of Aspen HYSYS and MATLAB SIMULINK for chemical engineering process simulation.

CH453L Instrumentation and Process control Lab (0-3-1)

Co-requisite(s): CH415: Lab is designed to practice experiments associated

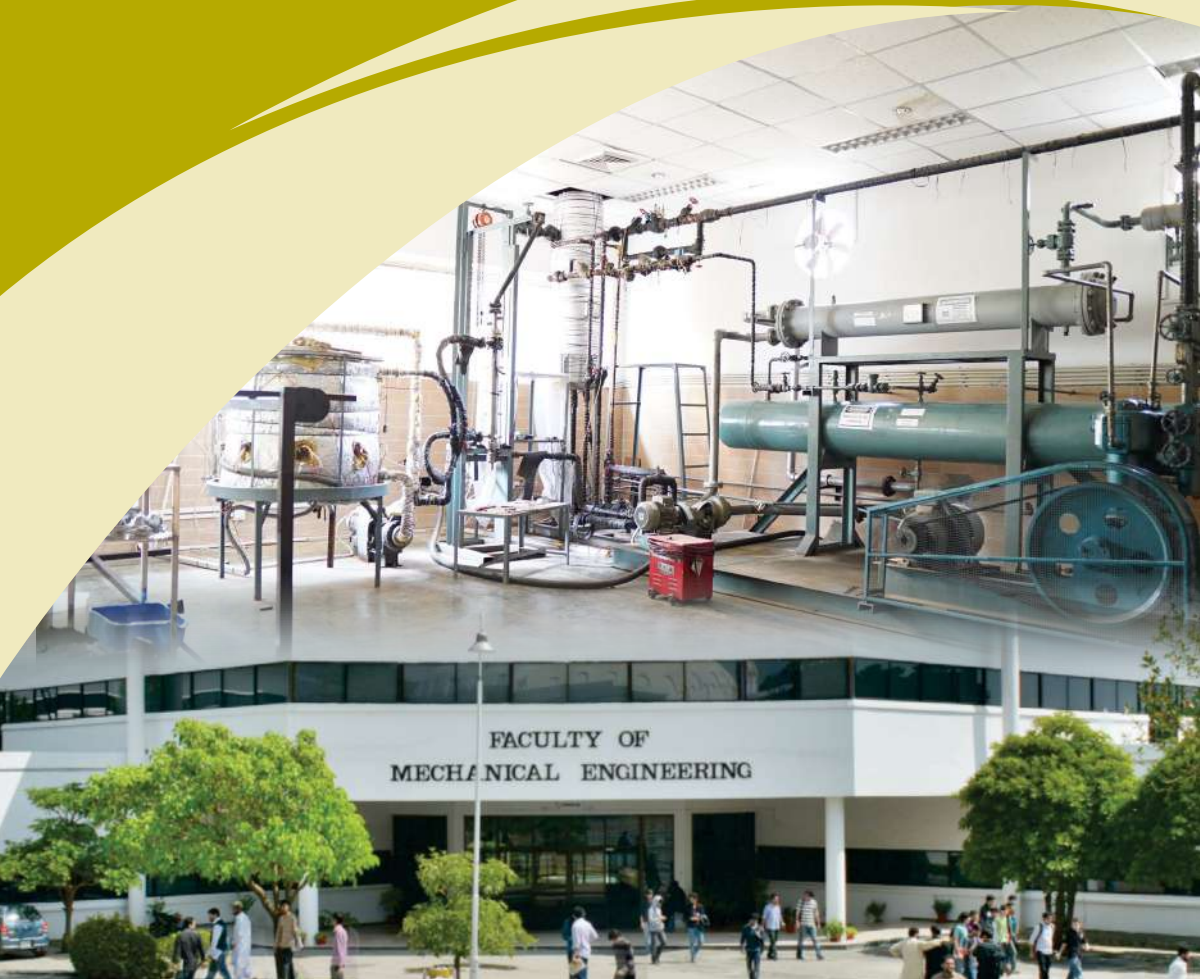
with instruments used to measure process variables. Also practice controlling industrial processes while implementing different control loops.

Careers in Chemical Engineering

Quality of modern living standards has encouraged the mass production of various utilities, necessities, and amenities. Since the birth of process and processing industry, after 18th century, there is a dramatic increase in its volume. Population trends and chain of never ending new/modern products ensures the growth in this sector. Furthermore, struggling third world countries like Pakistan are now focusing to process their raw materials in their own facilities. When it comes to realization, chemical engineers become an essential part of the team to chart the layout and erection of the new production line. Existing plants also require chemical engineers not only supervising and ensuring their smooth operation but also for troubleshooting, demanding interaction between the engineers and scientists from various other fields. Resources at the faculty are designed to inculcate the necessary knowledge, practices and behavioural aspects into the graduates, prerequisites for the responsibilities of professional life. Chemical engineers find their utility in various industries including chemical and petrochemical, nuclear, energy, oil & gas, food, pharmaceutical, cosmetics, and in various defence sectors, in addition to emerging research & innovation fields. Furthermore, these engineers are equipped to collaborate with different resources at the plant including management, utility engineers and above all with the technicians and plant operators as they will be their observing eyes in the field.



FACULTY OF MECHANICAL ENGINEERING



Introduction

Mechanical engineers confront diverse and challenging engineering problems which requires integration of science, engineering and socio-economic knowledge. They are trained to solve real-world engineering problems arising in energy production, heating, ventilation, air conditioning and manufacturing. They design complex machines, manufacturing processes and translate them into real products that addresses societal issues. However, with recent advances, mechanical systems are increasingly integrated with electronics, sensors, actuators, micro-controllers and computers. Mechanical engineering education thus needs to gear up to keep up with the fast-changing technology development.

The Faculty of Mechanical Engineering (FME) has a modern curriculum designed by experts and practitioner of engineering profession. Drawing on from disciplines such as mechanics and materials, thermal and fluid sciences, design and manufacturing, vibration and control systems, the curriculum also reflects the latest advances in finite element methods, computational fluid dynamics, computer-aided design and additive manufacturing. To cope with modern electro-mechanical systems courses and labs on robotics, instrumentation, electrical circuits and devices, electrical machines and drives, study of embedded systems, micro-controllers and mechatronics are also tightly integrated in the curriculum. The courses are delivered by research active faculty members who infuse life into classroom through real-world examples and stimulating complex engineering problems. Covering a gamut of applications ranging from tiny printed electronics, micro-fluidics, composite materials, advanced manufacturing, computer-controlled machines, natural fluid refrigeration, power plants, high speed rotating machines, frictionless bearings to industrial robots and unmanned air, underwater and ground vehicles. The curriculum is tightly knitted with modern well-equipped labs that takes students beyond the realm of theory and displays marvels of mechanical engineering in action. Inspiring enthusiastic young enquiring minds to unravel the mysteries, discover, explore and grow.

Thurst Area

- Design and Manufacturing Engineering
- Thermo-Fluid Engineering
- System Dynamics and Controls





Dean

Khalid Rehman,

PhD, JEJU National University, South Korea

Faculty

Wasim Ahmed Khan, PhD, University of Sheffield, UK, Chartered Engineer, FIMechE

S. M. Ahmad, PhD, University of Sheffield, UK, Chartered Engineer, MIMechE

Taqi Ahmad Cheema, PhD, Kyungpook National University, South Korea

Muhammad Asif, PhD, Hanyang University, South Korea

Ahmad Abbas, PhD, GIK Institute, Pakistan

Massab Junaid, PhD, GIK Institute, Pakistan

Ali Turab Jafry, PhD, Sungkyunkwan University, South Korea

Abid Imran, PhD, Hanyang University, South Korea

Arsalan Arif, PhD, Hanyang University, South Korea

Asif Khan, PhD, Dongguk University Seoul, South Korea

Muhammad Bilal Khan, PhD, Beijing Institute of Technology, China

Faheem Ahmad, MS, INHA University, South Korea

Malik Hassan, MS, GIK Institute, Pakistan

Sadia Bakhtiar, MS, UET Peshawar, Pakistan

Muhammad Qasim Zafar, MS, UET Lahore-Pakistan

Rizwan Ullah, MS, GIK Institute Topi

Faculty on study leave for PhD

Ghulam Hussain, PhD, Nanjing University of Aeronautics & Astronautics, China

Lab Engineers

Samar Abbas, BS, University of WAH

Syed Usman Ali Bukhari, BS, GIK Institute

Ridah Afzal, BS, UET Peshawar

Muhammad Fahad Zahid, BS, NUST

Wasim Shah, MS-GIK Institute

Graduate Assistant

Muhammad Sohail Gohar (PhD Scholar) (MS GIK Institute)

Asif Ullah (PhD Scholar) (MS NUST)

Salman Ahmad (PhD Scholar) (MS GIK Institute)

Muhammad Hammad (PhD Scholar) (MS UET Lahore)

Muteeb ul Haq (PhD Scholar) (MS NUST)

Babar Ashfaq (PhD Scholar) (MS II.U.I Islamabad)

Atif Muzaffar (PhD Scholar) (MS GIK Institute)

Hafiz Muhammad Rizwan (PhD Scholar) (MS UET Lahore)

Hamza Qayyum, (BS, PIEAS Islamabad)

Muhammad Sulaiman, (BS, IST Islamabad)

Muhammad Usman Umer (BS, II.U.I Islamabad)

Shahvaiz Khan, (BS, UET Peshawar main campus)

Masroor Khan, (BS, GIK Institute)

Jawad Rabbi, (BS, UET Peshawar main campus)

Muhammad Ahmed Rabbani, (BS, UET Lahore)

Minhaj Ullah, (BS, UET Peshawar main campus)

Syed Mir Hamza, (BS, UET Peshawar main campus)

Wajeeda Bibi, (BS, UET Peshawar main campus)

Muhammad Hasnain Tariq (BS, UET Rachna Campus)

Hamna Malik, (BS, UET Lahore Faisalabad Campus)

Hassan Ali, (BS, UET Peshawar, Jaloza campus)

Salman Khan, (BS, UET Peshawar main campus)

Mubashar Ali (BS, UET Lahore)

Usman Ahmad (Institute of Space Technology (IST), Islamabad)

Personal Secretary to Dean

Nizakat Ali Khan, MA, Peshawar University, MBA, Virtual University.

Muhammad Tofiq, Students Section

Faculty Mission

The faculty is aimed at producing professionals with sound knowledge-base, leadership qualities, and social rectitude. They are capable to intelligently respond and adapt to technological advancements in the field of Mechanical Engineering.

Outcome Based Education (OBE)

FME has adopted a student centric OBE system that leads to enhanced learning outcomes for students and relies heavily on continuous quality improvement. Moreover, curriculum is continuously revised keeping in view the country's industrial needs as well as adopting best international practices. Our students are thus equipped with twelve key graduate attributes or Program Learning Outcomes. These attributes or learning outcomes groom students with sound intellectual, theoretical and practical experiences that qualify them to address a variety of societal needs ethically. Thus, enabling graduating students to step with confidence into industry, research organizations or in the domain of higher education. FME firmly believes that the education we provide will enable our graduates to stand out from the crowd and has the potential to rise and shine.

Undergraduate Program

FME offers a well-structured 4-year BS degree program in mechanical engineering. As a cross-disciplinary program, it gives its students a sound foundation of engineering principles and promotes communication and practical skills that are the need of the present and future knowledge driven industry. Classroom theory is reinforced through extensive laboratory work, problem-based learning, and complex engineering problem. FME is focused on design and manufacturing, thermo-fluids as well as system dynamics and control, thus, enabling students to pursue their final year design projects as per their future career aspirations.

Program Educational Objectives (PEO)

- PEO_1** Graduates practicing in a variety of Mechanical engineering and allied disciplines.
- PEO_2** Graduates performing in a responsible, professional and ethical manner as an individual and as part of a team.
- PEO_3** Graduates advancing their knowledge and excelling in their chosen domain.

Program Learning Outcomes (PLOs):

Program Learning Objectives are given below:

Engineering Knowledge: An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

Problem Analysis: An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

Design/Development of Solutions: An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

Investigation: An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.

Modern Tool Usage: An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.

The Engineer and Society: An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.

Environment and Sustainability: An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

Individual and Teamwork: An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.

Communication: An ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project Management: An ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

Lifelong Learning: An ability to recognize importance of and pursue lifelong learning in the broader context of innovation and technological developments.

Knowledge Areas:

WK1 Natural Sciences: A systematic theory-based understanding of natural sciences applicable to the discipline.

WK2 Mathematics and Computing: The concept-based mathematical thinking, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline.

WK3 Engineering Fundamentals: A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.

WK4 Engineering Specialization: The knowledge of engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.

WK5 Engineering Design: The Design Thinking Knowledge that supports engineering

design in a practice area of an engineering discipline.

WK6 Engineering Practice: The Knowledge of engineering practices (technology) in different practice areas of an engineering discipline.

WK7 Engineering in Society: A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity i.e. economic, social, cultural and environmental and sustainability

WK8 Research Literature: Engagement with selected knowledge in the research literature of the discipline.

Sustainable development goals (SDGs):

Students are encouraged to apply the following SDGs in the senior design projects.

- Quality Education,
- Clean Water and Sanitation
- Affordable and Clean Energy
- Industry, Innovation and Infrastructure
- Sustainable Cities and Communities
- Responsible Consumption and Production
- Climate Action

Re-Accreditation

The BS degree program in Mechanical Engineering is re-accredited by Pakistan Engineering Council (PEC) under level II i.e., OBE.

Laboratories

In order to reinforce the classroom learning environment, FME arranges laboratory sessions for its students within various courses. The purpose of these laboratories is to make sure that the theory and principles learnt during the lecture hours are practically verified. Each lab session is comprised of 3 hours. The labs are supervised by the course instructor and conducted by expert lab engineers. The various labs at FME are:

1. Computational Mechanics Lab
2. Fluid Mechanics Lab
3. Heat Transfer, Refrigeration and Air Conditioning Lab
4. Heat Engine Lab
5. Solid Mechanics Lab
6. Sub and Super-Sonic Wind Tunnel Labs
7. CNC Training Lab
8. CNC Industrial Lab
9. Mechanical Workshop
10. Mechanical Vibration and Control Systems



Lab

11. Electronics and Instrumentation Lab
12. Mechatronics Design Lab
13. Composite Structures Lab

Student Chapters:

FME encourages the participation of students in different national and international competitions and events. FME hosts a number of student chapters of various technical societies which organize different events and student participation activities. Details are given below.

- IMechE organizes International Workshop on Functional Reverse Engineering (WRE) and other technical events
- AIAA GIKI Chapter organizes annual

Design, Build, & Fly Competition/Airex Innovation Challenge

- ASME GIKI Chapter organizes annual All Pakistan Event International Mechanical Engineering Convention (IMEC).
- SMEP organizes technical events/ TechFest & Resistance.
- ASHRAE organizes Thermocon and other technical events.

Course Work Requirements

A student majoring in Mechanical Engineering must complete the following courses:

(a) General Education Requirements (52 Credit Hours)

Course Titles	Course Code	Credit Hours
Computer Science & Engineering	CS101, CS101L, CS102L	4
Humanities	HM101, HM102, HM211, HM321, HM322	15
Basic Engineering	CH101, CH161, MM141, MM102, ME101, ME102, MS291	13
Mathematics	MT101, MT102, MT201, ES202, ES341/CS342	15
Sciences	PH103, PH104, PH102L	5

(b) Core Requirements (69 Credit Hours)

Course Titles	Course Code	Credit Hours
Circuits and Electronic Devices Electronics and Instrumentation Lab	ME203, ME243	4
Electrical Machines and Drives Mechatronics Lab	ME403, ME447	4
Measurement and Instrumentation	ME202	2
Engineering Mechanics (Statics, Dynamics)	ME211, ME212	6
Mechanics of Solids	ME213, ME314	6
Thermodynamics	ME231, ME232	6
Fluid Mechanics	ME321, ME322	6
Theory of Machines	ME313	3
Heat Transfer	ME333	3
Design of Machine Elements	ME261, ME364	5
Manufacturing Processes	ME353	3

Mechanical Vibrations	ME315	3
Design Project	ME481, ME482	6
System Dynamics & Control	ME464	3
Finite Element Analysis	ME467	3
Mechanical Eng. Lab. Courses	ME244, ME342, ME346, ME347, ME446, ME447	6

(c) Technical Electives (9 Credit Hours)

Design and Manufacturing

Course Titles	Course Code	Credit Hours
CAD/CAM	ME418	3
Introduction to Automobile Engineering	ME465	3
Introduction to Finite Element Methods	ME466	3
Fundamental of Composite Materials	ME419	3
Mechanical Engineering Design	ME468	3
Additive Manufacturing	ME453	3
Micro & Nano Fabrication	ME454	3
Stress Analysis	ME416	3
Microelectromechanical Systems	ME454	3

Thermo Fluids

Course Titles	Course Code	Credit Hours
Introduction to Computational Fluid Dynamics	ME423	3
Gas Dynamics	ME424	3
Combustion	ME434	3
Refrigeration & Air-conditioning	ME439	3
Power Plants	ME471	3
Gas Turbine	ME473	3
I.C. Engines	ME474	3
Energy Management & Conservation	ME475	3
Thermo-Fluids Systems Design	ME425	3

System Dynamics and Control

Course Titles	Course Code	Credit Hours
Robotics	ME452	3
Design of Experiments in Mechanical Engineering	ME469	3
Industrial Automation	ME493	3
Flight Dynamics and Control	ME494	3
Artificial Intelligence for Mechanical Engineers	ME404	3

(d) Management Elective (At Least 6 Credit Hours)

Course Titles	Course Code	Credit Hours
Operation Management	MS492	3
Industrial Safety	MS493	3
Total Quality Management	MS494	3
Maintenance Management	MS495	3
Technology Management	MS426	3
Project Management	MS496	3
Industrial Management	MS449	3
Supply Chain Management	MS491	3
Accounting and Finance	MS447	3
Entrepreneurship and Marketing	MS434	3
Macro and International Economics	MS448	3

(e) Summer Internship (Pass/Fail grade; NIL Credit).

Every student is required to participate in a compulsory industrial training of 4-6 weeks during the summer of Junior Year and submit a formal written report/presentation.

(f) Total Credit Requirements

For the B.S. degree in Mechanical Engineering, a student has to complete 136 credit hours.



Degree Plan

1st Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	CH161	Occupational Health and Safety	1	0	1	None	None
	CS101	Computing and AI	2	0	2	None	None
	CS101L	Computing and AI Lab	0	3	1	None	None
	HM101	Communication Skills	1	6	3	None	None
	IF101	Innovation and Makers Lab I	0	3	1	None	None
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	None
	PH101L	Applied Physics Lab.	0	3	1	None	None
Total			12	15	17		

2nd Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	CS112	Object Oriented Programming and Design	3	0	3	CS101	None
	CS112L	Object Oriented Programming and Design Lab.	0	3	1	CS101L	None
	ES111	Probability and Statistics	3	0	3	None	None
	HM102	Critical Thinking and Expository Writing	2	3	3	HM101	None
	IF102	Innovation and Makers Lab II	0	3	1	IF101	None
	MM101	Materials and Nanotechnology	2	0	2	None	None
	MM141	Materials Lab I	0	3	1	None	None
	MT102	Differential Equations and Linear Algebra I	3	0	3	MT101	None
Total			13	12	17		



3rd Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	MT202	Calculus II	3	0	3	MT101	None
	MS291	Engineering Economy	3	0	3	None	None
	ME211	Statics	2	0	2	PH101	None
	ME204	Engineering Graphics	1	3	2	None	None
	ME202	Measurement and Instrumentation	2	0	2	None	None
	ME251	Manufacturing Processes	2	0	2	None	None
	ME245	Manufacturing Processes Lab	0	3	1	None	ME251
	ME231	Thermodynamics I	3	0	3	MT101	None
Total Credit Hrs.			16	6	18		

4th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	ME212	Dynamics	3	0	3	PH101	None
	ME213	Mechanics of Solids I	3	0	3	ME211	None
	ME203	Circuits and Electronic Devices	3	0	3	PH101	None
	ME243	Electronics and Instrumentation Lab	0	3	1	None	ME203
	ME244	Statics and Dynamics Lab.	0	3	1	ME211	ME212
	HM211	Pak. & Islamic Studies	3	0	3	None	None
	ME232	Thermodynamics II	3	0	3	ME231	None
Total Credit Hrs.			15	6	17		

5th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	ME321	Fluid Mechanics I	3	0	3	MT101 ME212	None
	ME363	Design of Machine Elements I	2	0	2	ME245	None
	ME313	Theory of Machines	3	0	3	ME212	None
	ES341/ CS342	Numerical Analysis	3	0	3	MT102	None
	ME346	Thermo-Fluid Lab-I	0	3	1	ME232	ME321
	HM322	Corporate Law and Professional Ethics	3	0	3	None	None
	ME314	Mechanics of Solids-II	2	0	2	ME213	None
	ME348	Mechanics of Solids and Machines Lab	0	3	1	ME213	ME313 ME314
Total Credit Hrs.			16	6	18		

6th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	ME322	Fluid Mechanics II	3	0	3	ME321	None
	ME333	Heat Transfer	3	0	3	ME231 ME321	None
	ME315	Mechanical Vibration	3	0	3	MT102	
	ME212	None					
	ME364	Design of Machine Elements-II	2	0	2	ME363	None
	ME347	Thermo-Fluid Lab-II	0	3	1	None	ME322 ME333
	HM321	Sociology and Human Behavior	3	0	3	None	None
	ME316	Computer Aided Engineering	1	1	2	ME251	None
Total Credit Hrs.			15	4	17		

7th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	MS49x	Management Electives I	3	0	3	**	**
	ME4xx	Technical Elective I	3	0	3	**	**
	ME464	System Dynamics and Control	3	0	3	MT102	None
	ME446	Mechanical Vibration and Control Systems Lab	0	3	1	ME315	
	MT102	ME464					
	ME481	Senior Design Project-I	0	9	3	None	None
	ME403	Electrical Machines and Drives	3	0	3	ME203	None
	ME447	Mechatronics Lab	0	3	1	ME203	ME403
Total Credit Hrs.			12	15	17		



8th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	MS49x	Management Elective II	3	0	3	**	**
	ME467	Finite Element Analysis	2	3	3	None	None
	ME4xx	Technical Electives II	3	0	3	**	**
	ME4xx	Technical Elective III	3	0	3	**	**
	ME482	Senior Design Project-II	0	9	3	ME481	None
Total Credit Hrs.			11	12	15		

Total Credit Hours: 136



Course Description

ME101 WORKSHOP PRACTICE (0-3-1):

Principles and practice of machine tools of the mechanical engineering metal shop, measurements, filing and fitting; drilling; welding; bench work, grinding and sheet metal operations, conventional turning and milling operations are included.

ME102 ENGINEERING GRAPHICS (1-3-2):

Introduction to Engineering graphics, drawing instruments, projection theory, orthographic projections, projection of points and lines, dimensioning and tolerance, engineering geometry, sectioning, orthographic reading and writing, engineering curves, development of surfaces, fastening method and connectors, production engineering drawing, Computer-aided drawing.

ME202 MEASUREMENT AND INSTRUMENTATION (2-0-2):

Introduction to measurement standards, major blocks of a measurement system, introduction to instruments (characteristics and calibration), statics and dynamic characteristics of instruments, analog and digital signals, active and passive filters, temperature measurement, pressure measurement, position measurement, hall-effect sensing and application, piezoelectric sensors and applications, design of a sensor, data acquisition systems and post processing/ data analysis, introduction to microcontroller, introduction to IoT, IoT implementation and case studies.

ME203 CIRCUITS AND ELECTRONIC DEVICES (3-0-3):

Introduction to electrical circuits, KVL and KCL, Norton Thevenin theorems, AC and DC characteristics of RL, RC and RLC circuits, Introduction to diodes,

structure, properties, types and applications, Silicone controlled Rectifier (SCR), Introduction to BJTs, structure, types and applications, Introduction to MOSFETs, structure, types and applications, Introduction to IGBTs, Number systems, operations, conversion, 1's & 2's complements, Digital logic gates, Boolean algebra, Karnaugh mapping.

Pre-requisite(s): PH104.

ME211 STATICS (2-0-2): Review of basic concepts of scalars, vectors, Newton's law, forces, rectangular components, moment, couple, resultants. Equilibrium: System isolation and the free-body diagram, equilibrium conditions in 2-D and 3-D. Structures: Plane trusses, method of joints, method of sections, space trusses, frames and machines. Distributed Forces: Center of mass, centroids of lines, areas, and volumes, composite bodies and figures. Beams: External effects, internal effects, flexible cables. Friction: Introduction, types of friction, dry friction, wedges.

Pre-requisite(s): PH101

ME212 DYNAMICS (3-0-3): Introduction to Dynamics: Basic concepts, Newton's laws, units, solving problems in dynamics, rectangular coordinates (x-y), normal and tangential coordinates (n-t), polar Coordinates (r- Θ), and space. Types of Motion: Curvilinear motion, relative motion (translating axes), constrained motion of connected particles, force, mass, and acceleration, Newton's second law, equation of motion. Work and Energy: Work, kinetic energy, potential energy. Impulse and Momentum: Linear impulse and linear momentum, angular impulse and angular momentum, special applications, impact, central-force motion. Kinetics of Systems of Particles: Introduction, generalized Newton's second law, work-energy, impulse-momentum, conservation of energy and momentum, steady mass flow, variable mass. Plane Kinematics of Rigid Bodies: Introduction, rotation, absolute motion, relative velocity, instantaneous center of zero velocity, relative acceleration, and motion relative to rotating axes.

Pre-requisite(s): ME111

ME213 MECHANICS OF SOLIDS I (3-0-3): Concepts: Normal and shear stress, strain, material, factor of safety, stress concentration, pressurized thin-walled cylinder,

simple loading tension, torsion and bending, deflection with simple loading, superposition techniques, statistically indeterminate member, thermal stresses, combined stresses, Mohr's circle, combined loading

Pre-requisite(s): ME111

ME231 THERMODYNAMICS-I (3-0-3):

Introduction to Thermodynamics: System and boundary, specific volume, pressure and temperature, equilibrium state, processes, methods to solve thermodynamics problems. Understanding Heat and Work Interactions: First law of thermodynamics and its applications, energy balance of closed system, energy analysis of power, refrigeration and heat pump cycles. Phase and Pure Substance: Phase change processes, p-v-T relation, property diagrams, equation of state, specific heats, compressibility poly-tropic process relation. Conservation of Mass for Control Volume: Evaluating mass rate balance, 1-D flow, conservation of energy for control volume, applications to practical devices. Introduction to Second Law: Spontaneous and non-spontaneous processes, thermodynamic cycles, irreversible and reversible process, Carnot cycle, Calusius inequality. Entropy: Entropy change, T-s diagram, entropy generation, increase of entropy principle, entropy rate balance of closed systems and control volumes, isentropic efficiencies.

Pre-requisite(s): MT101.

ME232 THERMODYNAMICS-II (3-0-3):

Review of Thermodynamics I: Energetics, efficiency. Vapor Power Systems: Modeling and analyzing vapor power systems, superheat and reheat, regenerative vapor power cycle, other vapor cycle aspects. Gas Power Systems: Air-standard-Otto cycle, diesel cycle, dual cycle, Brayton cycle, regenerative gas turbines with reheat & inter cooling, gas turbines for aircraft propulsion, combined cycles, Ericsson and Stirling cycle. Refrigeration and Heat Pump Systems: Vapor compression refrigeration systems, cascade and multistage systems, absorption refrigeration, heat pump systems, gas refrigeration systems. Ideal Gas Mixtures: Mixture composition, p-V-T relations for ideal gas mixtures, U, H, S and specific heats for ideal gas mixtures. Psychometric Principles and Psycho-meters: Psychometric charts, analyzing air-conditioning processes, cooling

towers. Reacting Mixtures and Combustion: Combustion process, conservation of energy in reacting systems, adiabatic flame temperature, thermodynamic relations, equations of state, important mathematical relations.
Pre-requisite(s): ME231.

ME251 MANUFACTURING PROCESSES

(2-0-2): Traditional Machining Operations, Cutting Tool Technology. Non-Traditional Machining Operations: Mechanical, Electrical & Thermal Processes. Metal Forming Technology: Extrusion, Drawing, Sheet Metal Forming. Welding Technology: Fusion Welding Processes and Solid-State Welding Processes. Processing of Polymers and Composites, 3D Printing.
Pre-requisite(s): -

ME313 THEORY OF MACHINES (3-0-3):

Introduction to Theory of Machines: Linkage synthesis and analysis (Graphical and Analytical position, velocity and acceleration analysis). Cams: Terminology, S V A J diagrams, double dwell cam design, single dwell cam design. Mechanical drives: chains drives and belt drives. Balancing: Static and dynamic balancing, introduction and types of Governors. Gyroscopes: Gyroscopic couple, effect of gyroscopic couple on navigation.
Pre-requisite(s): ME212

ME314 MECHANICS OF SOLIDS II (2-0-2):

Analysis of stress in two and three dimensions, Principal stresses, Mohr's circle for stress, Thick-walled pressure vessels, Symmetrical and asymmetrical loading, Introduction to fracture mechanics, Impact loading, Fatigue and creep,

Virtual work, Theories of elastic failure, Theory of columns.

Pre-requisite(s): ME 213

ME315 MECHANICAL VIBRATIONS (3-0-3):

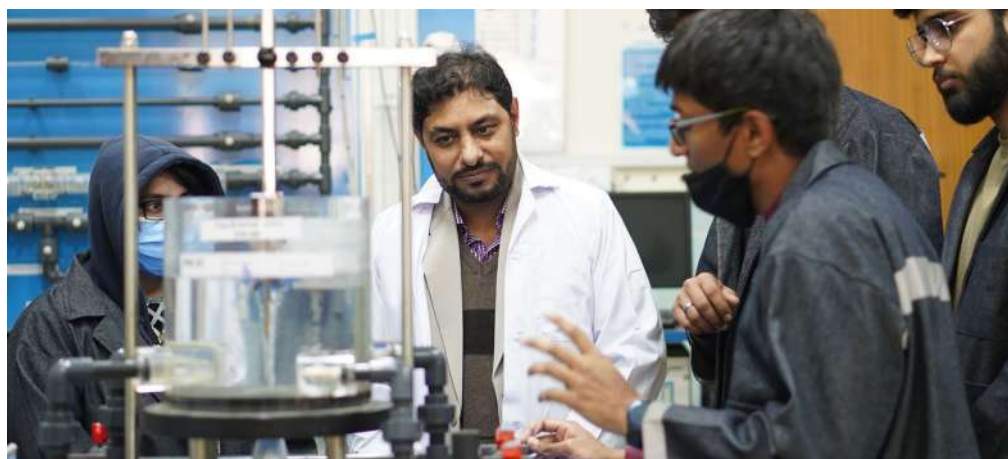
Introduction to Vibrations: Harmonic motion, damping, modeling and energy methods, stiffness, measurement, design considerations, stability. Harmonic Excitation: Un-damped system, damped system, alternative representations, base excitation, rotating unbalance, measurement devices, damping. Impulse Response: Response to arbitrary input, response to arbitrary periodic input. Transformation Methods: Random inputs, shock spectrum, measurements, stability. 2 DOF System: Eigenvalues and natural frequencies, modal analysis. Multiple DOF: Viscous damping, modal analysis, Lagrange equations, acceptable levels of vibrations, vibration isolation, vibration absorbers, damping, optimization, viscoelastic damping, critical speed, active suppression.
Pre-requisite(s): MT201, ME212.

ME316 COMPUTER AIDED ENGINEERING

(1-3-2): Engineering design and manufacturing; Introduction to CAD/CAM; Conventional Vs. non-conventional machine tool; Geometric modelling; Feature based design; 2D and 3D graphics and transformations; assembly modelling; Computer Numerical Control; Machine Tool Configuration ISO-841; CNC Programming and Digital Manufacturing.
Pre-requisite(s): ME251

ME321 FLUID MECHANICS-I (3-0-3):

Introduction to Fluids: Fluid behavior and



properties, specific weight, viscosity, compressibility, vapor pressure, surface tension. Fluid Statics: Pressure at a point, pressure variation in a fluid at rest, measurement of pressure, hydrostatic forces on plane and curved surfaces, buoyancy, pressure variation in fluids with rigid body motion. Fluids Dynamics: Newton's second law along and normal to a streamline, static, stagnation, dynamic and total pressure. Bernoulli's Equation: The energy line and hydraulic grade line, velocity and acceleration field, control volume and system representation, Reynolds transport theorem. Conservation Equations: The continuity equation, conservation of momentum: Newton's second law, the energy equation. Dimensional Analysis: Buckingham Pi Theorem, determination of Pi terms. Pipe flow: Fully developed laminar and turbulent flow, dimensional analysis of pipe flow. Pre-requisite(s): MT 101, ME 212

ME322 FLUID MECHANICS II (3-0-3):

Introduction: Fluid mechanics I review. Differential Analysis of Fluid Flow: Fluid element kinematics, conservation of mass and linear momentum. Inviscid and Potential Flows: Uniform flow, source and sink, vortex, doublet, superposition of basic potential flows, half body, Rankine ovals, flow over circular cylinder. Viscous Flow: Navier-Stokes equation, flow between fixed parallel plates, Couette flow, steady, laminar flow in circular tubes, flow in annulus. Flow Over Immersed Bodies: Boundary layer characteristics, Prandtl/Blasius



Ahmad Hassan
Junior Year FME

You'll have to learn to strike a balance between social and academic life. I've come across remarkable Instructors who have fuelled my passion for engineering, and I am forever indebted to them. With this strong foundation that GIKI has helped me build, I feel motivated and confident in my capabilities to make my dreams a reality.

solution, momentum integral estimates, turbulent boundary layer flow, effects of pressure gradient, friction drag, pressure drag, lift. Compressible Flow: Ideal gas relationships, Mach number, types of compressible flow, isentropic flow of an ideal gas, converging-diverging duct flow, nonisentropic flow of ideal gas, Fanno and Rayleigh flow, normal shock waves. Turbomachines: Basic energy and angular momentum consideration, centrifugal pumps, pump performance curves and pump selection, dimensional parameters and similarity laws, Introduction of Machine Learning in Turbomachines with examples, impulse turbines, reactive turbines, compressors, compressible flow turbines, windmills. Pre-requisite(s): ME 321

ME333 HEAT TRANSFER (3-0-3):

Introduction of Heat Transfer: Physical origins of conduction, convection and radiation, The conservation of energy. Conduction: The conduction rate equation, steady-state conduction in the plane wall and radial systems, conduction with thermal energy generation, finned surfaces, 2-D conduction: Shape factor, numerical methods, transient conduction: The lumped capacitance method. Convection: Boundary layers, laminar and turbulent flow, external flow convection, internal flow convection, the energy balance, free and forced convection. Heat Exchangers: Types, use of the LMTD method, the effectiveness-NTU method. Radiation: Processes, radiation intensity, blackbody radiation, emission, absorption, reflection, and transmission by real surfaces, radiation exchange between surfaces, multi-mode heat transfer. Diffusion Mass Transfer: Physical origins and rate equations, mass transfer in non-stationary media, The stationary medium approximation, conservation of species for a stationary medium, homogeneous chemical reactions. Pre-requisite(s): ME231, ME321.

ME363 DESIGN OF MACHINE ELEMENTS I (2-0-2):

Philosophy and concept of engineering design, Design codes and standards, determination of permissible and actual stresses, factor of safety, Design of keys, cotter, and couplings, Design of brakes and clutches, Flywheel, Design of welded, riveted and bolted joints, Design of translation screws, Mechanical springs, Flexible mechanical elements.

Pre-requisite(s): ME245

ME364 DESIGN OF MACHINE ELEMENTS II (2-0-2):

Design of shafts, tolerances, standards of fits & tolerances, Design of rolling contact bearings, hydrodynamic theory of lubrication, journal bearings, Gear trains, Design of spur, helical, bevel and worm gears: kinematics, force analysis, AGMA equations

Pre-requisite(s): ME261

ME403 ELECTRICAL MACHINES AND DRIVES (3-0-3):

Introduction to fundamental laws of electromagnetism, magnetization, magnetic and electric circuit analogy and analysis, energy conversion principles, rotary and linear machines, AC and DC machines, Introduction to Microcontroller Programming especially the features to drive and control the electrical machines, design and specifications of solenoids and transformers, Introduction to DC motors, construction, operating principle, Introduction to half and full-bridge circuits, types of stepper motors, construction and drive, BLDC motors and drives, linear motors and actuators, rotary to linear motion conversion mechanisms, types of linear motors, advantages of linear motors, construction Introduction to AC machinery, single phase and polyphase machines, synchronous and asynchronous machines, construction, types and characteristics, single and three phase inverters, Introduction to generators.

Pre-requisite(s): ME 203

ME404 ARTIFICIAL INTELLIGENCE FOR MECHANICAL ENGINEERS (3-0-3):

Introduction, Optimization in engineering, optimization fundamentals, Graphical optimization, Linear programming and Numerical techniques, Fuzzy Logic, Algebras of Fuzzy Sets, Introduction to Fuzzy Control, Artificial Neural Networks, Introduction to Artificial Intelligence Systems, fundamentals of neural networks, neural network control, Hands-on session with Matlab/Simulink.

Pre-requisite(s): MT 201, ES202

ME416 STRESS ANALYSIS (3-0-3): Review of mechanics of materials, Stress transformations, general 3D stress state, Mohr's circle in 3D, strain transformations, generalized stress-strain relationship, equilibrium and compatibility, introductory topics from theory of

elasticity, Airy stress functions, Prandtl's stress functions for torsion, shear flow, torsion of thin-walled tubes, bending of unsymmetrical beams: stress & deflection, bending of thin flat plates, axisymmetric circular plates in bending, thick-walled cylinders & rotating disks, contact stresses. Overview of Experimental Stress Analysis, Stress analysis –Experimental approaches, Specific domain of these approaches, Advantages and disadvantages.

Pre-requisite(s): ME213, ME314

ME418 CAD/CAM (2-3-3): Introduction and history, geometric modeling; feature and design, CAD hardware and software; 2D and 3D graphics and transformations; assembly modeling and analysis, concurrent engineering; axiomatic design; DFM; DFA; Taguchi method; group technology; value engineering; CE tools, process planning; manual, variant, generative and hybrid approaches; tolerance charts, manufacturing planning and control, cellular and JIT manufacturing; MRP II. Numerical control; NC programming; CNC; DNC, robotics, computer- integrated manufacturing.

Pre-requisite(s): ME101, CS101.

ME419 FUNDAMENTALS OF COMPOSITE MATERIALS (3-0-3):

History and introduction, Nomenclature and classification, Fundamental equations, Symmetric, asymmetric and other characteristic layering setups, Classical lamination theory, Failure criteria, laminated structures



Erum Iftikhar

Junior Year FME

Every day at GIKI was a new challenge. It did get a bit overwhelming at times, but all this has helped me prepare for my professional life. Being a part of its diverse culture has helped me grow on a personal level as well. The intense academic schedule and multiple society activities have immensely improved my time management and have given me the ability to deal with any situation.

ME423 INTRODUCTION TO COMPUTATIONAL FLUID DYNAMICS

(2-1-3): Governing Equations, Finite Difference Method, Truncation Error, Finite Volume Method, Conversion of Governing Equations to Algebraic Equations, Numerical Solutions to Algebraic Equations, Solution Analysis, Consistency, Stability, Convergence, Residuals and Convergence Tolerance, Accuracy, Sources of Errors, Controlling the Solution Errors, Efficiency, Case Studies from internal and external flow, heat transfer and turbulence modeling

Pre-requisite(s): ME333, ME322.

ME424 GAS DYNAMICS (3-0-3): Flow of compressible fluids; one-dimensional flows including basic concepts; isentropic flow; normal and oblique shock waves; Rayleigh line; Fanno flow and simple waves; multi-dimensional flows; small perturbation theory for linearized flow; method of characteristics for nonlinear flows.

Pre-requisite(s): ME321, ME322

ME425 THERMO-FLUIDS SYSTEMS DESIGN

(3-0-3): Engineering Design Process, Design of Thermal Systems, Design for Environment, Safety and Reliability, Air Distribution Systems, Liquid Piping Systems, Heat Exchanger Selection and Design, Power Generation, Refrigeration and HVAC Systems, Mathematical Modeling of Thermal Equipment and Systems, System Simulation and Computer-Aided Design, Design Optimization and System Performance Evaluation, Exergy and Thermo-economic Analysis, Life Cycle Cost, Cost Estimation.

ME434 COMBUSTION (3-0-3): Combustion

thermodynamics; chemical kinetics; reaction rate; explosion in gases; detonation; laminar and turbulent flames in pre-mixed gases; diffusion flames; liquid droplet combustion; theory of thermal ignition; combustion of particles; propellant and rocket propulsion.

Pre-requisite(s): ME232, ME333

ME439 REFRIGERATION & AIR-CONDITIONING

(3-0-3): Psychrometric principles and design of air-conditioning equipment and ducts; consideration of human comfort in heating and cooling; heating and cooling calculations and design; principles of refrigeration; cycles; refrigerants; absorption refrigeration; multi-pressure systems.

Pre-requisite(s): ME232, ME333

ME452 ROBOTICS (3-0-3): An overview of robotics; forward kinematics; inverse kinematics; Denavit-Hartenberg coordinate transformations; motion kinetics; force/torque relations; trajectory planning, Lagrange equations; position control; PID control; inverse dynamics feed forward control; nonlinear control; Applications of Artificial Intelligence (AI) in Robotics.

Pre-requisite(s): ME212, ME313

ME453 ADDITIVE MANUFACTURING

(3-0-3): Introduction and Basic Principles of various additive manufacturing techniques, history, current development and fundamental engineering aspects. Techniques, printing mechanisms, advantages and limitations of Polymer, Metal, Ceramic and other additive manufacturing technologies. Principles and strategies for additive manufacturing process. Additive manufacturing applications. Future of additive manufacturing.

ME454 MICRO AND NANO FABRICATION

(3-0-3): Introduction to micro and Nano fabrication, Basic Micro fabrication Techniques (lithography, etching), Thin film deposition (Chemical vapor deposition (CVD), sputtering, Physical vapor deposition (PVD), Surface modification, Micromachining (drilling, milling, electric discharge machining, turning), Micro fabrication using 3D Printing and Photo catalytic reaction, Micro joining (solid-state bonding, soldering and brazing, fusion micro-welding, adhesives), Assembly and automation, Applications of micro



fabrication, Introduction to Nanofabrication and Materials, Nanofabrication Techniques (E-Beam Nanofabrication, Scanning Probe Techniques, rapid prototyping, X-ray lithography, nano grooving), Nano joining (Bonding using nanoparticles, Focused ion beam machining, growth patterning, welding using electron beam, Indirect joining by SEM and TEM, Ion beam, resistance, ultrasonic, laser), Self-Assembly and Template Manufacturing, Applications of nanofabrication and materials.

ME455 MICROELECTROMECHANICAL SYSTEMS (3-0-3): Overview of MEMS and microsystems, Working principles of microsystems, Engineering science for microsystems design and fabrication, Engineering mechanics for microsystems design, Thermo-fluid engineering and microsystems design, Scaling laws in miniaturization, Materials for MEMS and microsystems, Microsystems fabrication processes, Microsystems design, Assembly, packaging, and testing of microsystems
Pre-requisite(s): Nil

ME464 SYSTEM DYNAMICS AND CONTROL (3-0-3): Introduction to Control Systems: dynamic systems, modeling and simulation, utility and application. Mathematical Models of Systems and Simulation: Review of mathematical modeling techniques, modeling of mechanical-electrical, electro-mechanical and process control systems, linearization, case studies. System Response Analysis: Time response of dynamical systems, classical solution of ODEs, Time domain solution of ODEs, frequency response. Feedback Control Characteristics: Why feedback, error signal

analysis, disturbance signals in a feedback control system, control of transient response, steady-state error. Performance of feedback control system: Introduction to second-order systems, effect of poles and zeros, s-plane root location, steady-state error of feedback control system. Stability of linear system: Routh-Hurwitz stability criteria for dynamical systems, Introduction to feedback control System: Performance specifications, design of different classical control laws/algorithms to control a dynamical system, performance analysis and improvement. Frequency response analysis of linear systems.
Pre-requisite(s): MT201

ME465 INTRODUCTION TO AUTOMOBILE ENGINEERING (3-0-3): Introduction, layout and components; power generation (engine, engine systems and testing), transmission, wheel and tire, chassis frame and body, suspension system, control systems (steering, brake); vehicle design (performance, axle loading, chassis design, vehicle mechanics); ergonomics, legislation, automobile industry in Pakistan.

ME466 INTRODUCTION TO FINITE ELEMENT METHODS (2-3-3): Introduction; stress analysis by FEM; direct stiffness method, energy, variational principles and Ritz method; co-ordinate transformation; iso-parametric formulation; solution of Eigenvalue, boundary value, discretized time dependent problems.

ME467 FINITE ELEMENT ANALYSIS (2-1-3): Introduction, Direct Stiffness method, Variational methods/Weighted Residual methods, Formulation of equations for 1D and



2D elements, Shape Functions, Iso parametric formulation, FEA applications in Structural mechanics, Fluid mechanics and Heat transfer using commercial code.

Pre-requisite(s): NIL

ME468 MECHANICAL ENGINEERING DESIGN (3-0-3):

Philosophy and concept of engineering design, engineering creativity, phases and procedures in design, management of engineering projects, computer-aided design (CAD), case studies in design with emphasis on system modeling, optimization and reliability, application of industrial design codes.

Pre-requisite(s): ME261.

ME469 DESIGN OF EXPERIMENTS IN MECHANICAL ENGINEERING (3-0-3):

Introduction and importance experimental design, Statistics, Random variables and probability distributions, Distributions of sampling statistics, Hypothesis testing and decision making for single and multiple samples, Regression/building empirical models, Analysis of Variance (ANOVA), Introduction to uncertainty and error analysis of experimental data causes and types of experimental errors, Choice of sample size in designed experiments, Factorial Designs

ME471 POWER PLANTS (3-0-3): Energy and environment, gas power plants, hydroelectric power plants, vapor power plants, nuclear reactors, fuels, combustion, turbines, compressors, pumps, boilers, exhaust analysis, renewable energy resources (geothermal, wind, biomass, solar, etc.), waste water treatment, environmental impacts, feasibility, cost analysis.

Pre-requisites: ME321, ME232, ME333



ME473 GAS TURBINES (3-0-3):

Thermodynamic analysis and analytical design of gas turbine engines; topics in combustion, internal compressible flow, boundary layer, thrust determination for ramjets and turbojets, axial and centrifugal compressor, axial and centripetal turbines.

Pre-requisite(s): ME321, ME232

ME474 INTERNAL COMBUSTION ENGINES (3-0-3):

Fundamentals of internal combustion engines; study of fluid flow, thermodynamics, combustion, heat transfer, friction phenomenon, and fuel properties relevant to engine power, efficiency and emissions; examination of design features and testing characteristics of different types of engines.

Pre-requisite(s): ME321, ME232

ME481 SENIOR YEAR DESIGN PROJECT – I (0-9-3):

Senior Design Project is offered to the students during the final year and is a two semesters long activity performed in a group of not more than four members. Each group decides a project under the supervision of a PhD faculty member. The scope of the first part of the project includes introduction, literature review, theoretical design, and analysis of the intended system/product/process with or without the help of computational software.

ME482 SENIOR YEAR DESIGN PROJECT-II (0-9-3):

In the second phase, physical model (prototype) development and testing is carried out by the students. The results of the experiments/testing are reported, analyzed, and discussed. The students are also required to analyze the socio-economic impact of their project with environment and sustainability considerations.

ME475 ENERGY MANAGEMENT & CONSERVATION (3-0-3):

Energy and environment, fuels and materials, energy auditing and surveying, energy consumption in manufacturing, heat transfer, heat balance and energy flow charts, heat recovery, energy technologies, instrumentation and measurements, sustained reductions in energy use, economics, waste heat recovery.

ME493 INDUSTRIAL AUTOMATION (3-0-3):

Automation, Analog and Digital Electronics, ASCII, Unicode, Number Systems, Sensor

Transducers & Actuators, Microprocessors and Microcontrollers, Computer Interfacing, electronic communication, Electrical Power, Pneumatics, Hydraulics, Machines and Processes, CNC, PLC, SCADA, Reliability Availability Maintainability and Safety (RAMS), Industry 4.0, ERP, e-Commerce

ME494 FLIGHT DYNAMICS AND CONTROL

(3-0-3): Systems of axes and notation, Static equilibrium and trim, the equation of motions, longitudinal dynamics, Lateral dynamics, Flight stability, Flight control.

Lab Courses

ME243 ELECTRONICS AND INSTRUMENTATION LAB (0-3-1):

Laboratory experiments related to circuits, electronic devices and instrumentation.

Co-requisite(s): ME202, ME203

ME244 STATICS AND DYNAMICS LAB (0-3-1):

Laboratory experiments related to Statics and Dynamics.

Pre-requisite(s): ME111; Co-requisite(s): ME212

ME245 MANUFACTURING PROCESSES LAB (0-3-1):

Principles and practice of machine tools of the mechanical engineering metal shop, measurements, filing and fitting; drilling; welding; bench work, grinding and sheet metal operations, conventional turning and milling operations.

Co-requisite(s): ME251

ME346 THERMO-FLUID LAB-I (0-3-1):

Laboratory experiments related to Thermodynamics, and Fluid Mechanics I.

Pre-requisite(s): ME232; Co-requisite(s): ME321

ME347 THERMO-FLUID LAB-II (0-3-1):

Laboratory experiments related to Heat Transfer and Fluid Mechanics II.

Co-requisite(s): ME333, ME322

ME348 MECHANICS OF SOLIDS AND MACHINES LAB (0-3-1):

Laboratory experiments related to mechanics of solids and theory of machines.

Pre-requisite(s): ME213, Co-requisite(s): ME313, ME314,

ME447 MECHATRONICS LAB (0-3-1):

Laboratory experiments related to Mechatronics.

Pre-requisite(s): ME203; Co-requisite(s): ME403

ME446 MECHANICAL VIBRATIONS AND SYSTEM DYNAMICS AND CONTROL LAB (0-3-1):

Laboratory experiments related to Mechanical Vibrations and System Dynamics and Controls.

Pre-requisite(s): ME315, MT201; Co-requisite(s): ME464

Careers in Mechanical Engineering

Mechanical Engineering involves application of mechanics and thermal-fluid sciences to design, manufacture, Operate, Maintain, various processes and plants. Mechanical Engineers are the backbone of many industries and can have a career in a wide spectrum of fields including shipbuilding, aerospace, railroad works, automotive, turbomachinery, pharmaceutical, petroleum, chemical, process and energy industries, just to name a few. FME graduates are well sought after by R&D/defense organization, fertilizers and cement companies, power plants and petrochemical industries, automobile manufacturers as well as by various multi-national companies. Re-accreditation of BS program by Pakistan Engineering Council (PEC) under Level-II category i.e., Washington Accord (WA) accredited engineering degree, implies that the BS degree is recognized as equivalent to a degree from WA signatory countries. This provides an additional mobility to our graduates across the world as it makes Easier, Engineers to gain professional registration in other countries.

DEPARTMENT OF CIVIL ENGINEERING



Introduction

The ever green field of Civil Engineering is a necessity for the people dwelling all around the globe. The increase in population calls for more structural construction but lesser vegetative destruction. Be it water sanitation, building construction, travelling through canyons, over, under and around the mountains, Civil Engineering is an integral part of all developments.

Being a multi-dimensional field, the Civil Engineering program at GIK Institute offers six major thrust areas to its students. The program trains its graduates to participate effectively in the development challenges by getting involved in multiple projects of national and international scopes.

The department aims to train its graduates to gain practical understanding of the theoretical knowledge obtained in class by applying them in the department's immaculate laboratories. The campus of GIK Institute provides necessary field areas for understanding the application, and integration of field and laboratory outcomes.

Thrust Areas

- Water Resources Engineering
- Geotechnical Engineering
- Transportation Engineering
- Structural Engineering
- Environmental Engineering
- Construction Management and Engineering
- Geo-informatics Engineering





HOD

M. Ashraf Tanoli

PhD (Tottori University, Japan)

Faculty

Muhammad Ashraf Tanoli
 Khawar Rehman
 Shamsher Sadiq
 Shiraz Ahmed
 Muhammad Waseem
 Hafiz Ahmed Waqas
 Mehtab Alam
 Muhammad Farjad Iqbal
 Muhammad Naveed

PhD (Tottori University, Japan)
 PhD (Hanyang University, Republic of Korea)
 PhD (Hanyang University, Republic of Korea)
 PhD (University of Hasselt, Belgium)
 PhD (University of Rostock, Germany)
 PhD (University of Tokyo, Japan)
 PhD (University of Chinese Academy of Sciences, China)
 MS (Shanghai Jiao Tong University, China)
 MS (Polimi, Italy)

Lab Engineers

Muhammad Shahkar
 Kaleem Afzal
 Muhammad Ilyas
 Arbaz Khan

BS Eng. (COMSATS, Pakistan)
 BS Eng. (UET Peshawar, Pakistan)
 BS Eng. (UET Peshawar, Pakistan)
 BS Eng. (UET Peshawar, Pakistan)

PS to HoD

Ms. Sehrish Mazhar

Department Mission

The department aims to produce competent engineers who can work in the society, both as field professionals and academic researchers. The department also aspires to produce engineers who are able to integrate theories and practices of civil engineering being exposed to the contemporary issues in the industries.

OBE

BS Civil Engineering at GIK Institute is one the first programs across Pakistan to have started on OBE guidelines from day one.

Program Educational Objectives (PEOs)

Main emphasis of Civil Engineering Faculty is to produce graduates with the following credentials.

PEO1: Graduates become professional engineers to work in leading national and international organizations to address challenging issues.

PEO2: Graduates engage in profession, academics, and research to keep abreast with latest developments and sustainable practices in the field.

PEO3: Graduates demonstrate the fundamentals of the professionalism, ethics, and quality performance that will enable them to be leaders and contributors to the society.

Program Learning Outcomes (PLOs)

At the time of graduation, students must possess the following attributes:

PLO 1: Engineering Knowledge: An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PLO 2: Problem Analysis: An ability to identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

PLO 3: Design/Development of Solutions: An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and

environmental considerations.

PLO 4: Investigation: An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data and synthesis of information to derive valid conclusions.

PLO 5: Modern Tool Usage: An ability to create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of limitations.

PLO 6: The Engineer and Society: An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.

PLO 7: Environment and Sustainability: An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

PLO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PLO 9: Individual and Team work: An ability to work effectively as an individual or in a team, on multifaceted and/or multidisciplinary settings.

PLO 10: Communication: An ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective report and design documentation, make effective presentations and give and receive clear instructions.

PLO 11: Project Management: An ability to demonstrate management skills and apply engineering principles to one's own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.

PLO 12: Lifelong Learning: An ability to recognize importance of and pursue lifelong learning in the broader context of innovation and technological developments.

Knowledge Areas

	Description	Courses
WK1	Natural Sciences	Applied Physics, Applied Chemistry and Environment, Engineering Geology
WK2	Mathematics and Computing	Calculus I, Calculus II, Differential Equations and Linear Algebra I, Probability and Statistics, Numerical Analysis, Finite Element Analysis, Computing and AI, Object Oriented Programming and Design.
WK3	Engineering Fundamentals	Mechanics of Solids I, Basic Surveying, Construction Materials and Concrete Technology, Fluid Mechanics I, Transportation Engineering, Advanced Surveying, Mechanics of Solids II, Basic Structure Analysis, Soil Mechanics, Indeterminate Structural Analysis, Fluid Mechanics II, Hydrology and Water Management, Civil Engineering Drawing and Graphics, Quantity Surveying and Cost Estimation
WK4	Engineering Specialization	Structural Engineering, Design of Structures, Pre-stressed Concrete, Introduction to Earthquake Engineering, Hydraulic Engineering Design, Applied Hydrology, Hydropower Engineering, Traffic Engineering, Pavement Materials and Design, Geometric Design of Highways, Contract Management, Environmental Impact Assessment, Environmental Engineering II, Solid Waste Management, Remote Sensing, Watershed Modeling Using GIS, Climate Change, Machine Learning Applications in Civil Engineering, Introduction to Computational Fluid Dynamics, Rock Mechanics, Slope Stability
WK5	Engineering Design	Geotechnical and Foundation Engineering, Reinforced Concrete Design I, Reinforced Concrete Design II, Environmental Engineering I, Highway Engineering, Design of Steel Structures, Tunneling and Underground Space Structures
WK6	Engineering Practice	Innovation and Makers Lab I, Innovation and Makers Lab II, Mechanics of Solids Lab, Basic Surveying Lab, Construction Materials and Concrete Technology Lab, Fluid Mechanics I Lab, Advanced Surveying Lab, Soil Mechanics Lab, Fluid Mechanics II Lab, Geotechnical and Foundation Engineering Lab, Hydrology and Water Management Lab, Civil Engineering Drawing and Graphics Lab, Geo Informatics Lab, Reinforced Concrete Design lab, Environmental Engineering Lab, Quantity Surveying and Cost Estimation Lab, Highway Engineering Lab
WK7	Engineering in Society	Occupational Health and Safety, Pakistan and Islamic Studies, Corporate law and Professional Ethics, Civil Engineering Economics, Communication Skills, Critical Thinking and Expository Writing
WK8	Research Literature	Senior Design Project I, Senior Design Project II

Sustainable Development Goals (SDGs)

Civil Engineering being a diverse field directly caters to society and environment in their development and conservation needs, therefore it targets all the SDGs defined by the UN. The curriculum of BS Civil Engineering program at GIK Institute is set to enable its graduates to identify the potential for achievement of SDGs in their respective professions.

Accreditation

The PEC has granted formal approval to initiate BS civil Engineering.

Laboratories

Theory of Structures Lab

This lab provides latest facilities to idealize structural response, function of individual members and the behavior of structures under different scenarios and loading conditions. The equipment can be used to observe the elastic behavior of structural elements and the strain energy that a member can absorb at the ultimate loading.

Transportation Lab & Highway Lab

This lab is used for the testing of pavement materials, consisting of both the binder and the aggregates. Lab facilities allow bitumen testing which includes penetration and grading, flash and fire point, and ductility tests. Equipment is also utilized for aggregate testing such as impact, abrasion, fatigue resistance, flakiness

index, and the elongation numbers. The lab also has its own plate load testing and California Bearing Ratio (CBR) equipment.

Geotechnical Engineering Lab

This lab has one of the most advanced Tri-axial testing machine and direct shear test machine. The lab also has the sufficient gear to analyze the properties of soil, their gradation, moisture values and the Atterberg limits. The installed equipment can determine permeability of soil, verify Darcy's law, and calculate the dry density as well as the bulk density of the soil for the optimum moisture content.

Mechanics Lab

The mechanics lab helps the students to visualize the actions and reactions of the forces and the portions of an object responsible for producing resistance to these forces. The lab clearly describes the changes occurring in the objects due to difference in their materials and dimensions. The center of gravity apparatus, the polygon of forces apparatus, the friction value determination apparatus, the moment balancing and the virtual work apparatus etc. are few of the equipment from this lab.

Materials Engineering and Concrete Lab

The materials lab has a very high quality Universal testing machine (UTM) that is computer controlled and also gives output in the form of digital data. The Machine is latest and highly efficient. It can perform the tensile



as well as the compression tests on both; concrete and steel specimens. The lab also contains the equipment for testing the properties of fresh and hardened concrete. In this lab, different tests on aggregates necessary for mix proportioning are carried out as well.

Surveying Lab

This lab covers both basic and advanced surveying techniques for recording measurements, including precision steel taping methods to calculate horizontal measurements, digital theodolites to perform angular measurements, and automatic levels for elevation measurements. In addition, the labs are also equipped with numerous total stations, which enable horizontal, vertical, and angular measurements to be made in a single operation.

Environmental Engineering Lab

This lab is well equipped for conducting water chemistry and environmental microbiology experiments. In addition to standard laboratory equipment such as pH meters, turbidity meters, dissolved oxygen meters, and ovens, there are several key analytical instruments available for research. The lab has testing facility for both fresh and waste water contamination and can be utilized to assess the physical, chemical and biological contaminants. Furthermore, commercial testing can also be rationally carried out in the lab.

Fluid Mechanics and Hydraulics Lab

This lab has equipment including state of the art 10 m long hydraulic flume with advanced experimental capabilities for studying open channel and coastal engineering processes. The lab is equipped with pressure gauges, venturi flume, Bernoulli's theorem apparatus, turbines and pumps to conduct all basic and depth level fluid mechanics experiments. The

equipment is digitized to accurately measure the head losses in different cases including various materials and types of bends.

Hydrology and Water Management Lab

The laboratory has facilities to study the natural hydrological cycle. The equipment in the laboratory can be used to simulate hydrologic conditions by using an advanced hydrology system, a precipitation simulator, a hydrograph tank, and a drainage/water table management tank. Furthermore, the seepage flow, ground water flow and water table can also be observed in the lab.

Drawing and Computer Aided Design Lab

The drawing lab enhances the skills of the students by interaction with architectural models and advanced instruments. The CAD lab teaches the students to use different software to draw, model and design multiple components. The basic drawing and graphics courses are incorporated and practically demonstrated to the students using AutoDesk suite. The students are also taught to design water supply channels using EpaNET and GIS, Structural designing using SAP 2000, ETABS and SAFE, managerial design by Primavera and geotechnical hazard assessment using OpenQuake.

Student Societies

The Department of Civil Engineering has an active student society called 'Institution of Civil Engineers'. The society attracts active participation from students who arrange trainings, workshops, and seminars to work together on diverse contemporary issues such as digital automation, green construction, sustainability, and AI in Civil Engineering. The objective of the society is to promote personal and academic growth of students.



A. General Education requirements (56)

Subject Area	Course Code	CH
Computer Sciences	CS101, CS101L, CS112, CS112L	7
Humanities	HM101, HM102, HM211, HM322	12
Management Sciences	CV323, CV323L, CV407, CV452	8
Natural Sciences	MT101, MT102, MT201, ES111, ES341, CH101	18
Basic Engineering	PH101, PH101L	4
Interdisciplinary Engineering	MM101, CH161, MM141, IF101, IF102, MM101	7

B. Core requirements (75)

Subject Area	Course Code	CH
Surveying and Geo Informatics	CV201, CV201L, CV202, CV202L, CV305, CV305L	7
Geotechnical Engineering	CV230, CV231, CV231L, CV332, CV332L	10
Structural Engineering	CV211, CV212, CV212L, CV213, CV313, CV315	15
Concrete Engineering	CV215, CV215L, CV314, CV314L, CV414, CV414L	11
Hydraulic Engineering	CV221, CV221L, CV322, CV322L, CV425	10
Transportation Engineering	CV241, CV442, CV442L	7
Construction Engineering	CV351, CV452	2
Environmental Engineering	CV361, CV361L	4
Civil Engineering Drawing	CV210L	1
Quantity Surveying and Cost Estimation	CV403L	2
Project	CV481, CV482	6

C. Technical electives (6)**1. Structural Engineering**

CV413	Structural Engineering	3
CV417	Design of Structures	3
CV416	Pre-stressed Concrete	3
CV418	Introduction to Earthquake Engineering	3

2. Water Resources Engineering

CV422	Hydraulic Engineering Design	3
CV424	Applied Hydrology	3
CV426	Hydropower Engineering	3

3. Geotechnical Engineering

CV435	Rock Mechanics	3
CV433	Slope Stability	3
CV434	Design and Construction of Earthen Dams	3
CV436	Tunneling and Underground Space Structures	3

4. Highway and Transportation Engineering

CV443	Traffic Engineering	3
CV444	Pavement Materials and Design	3
CV445	Geometric Design of Highways	3

5. Construction Management and Engineering

CV453	Architectural and Building Information Modeling	3
MS426	Project Management	3
MS434	Entrepreneurship	3
MS362	Supply Chain Management	3

6. Environmental Engineering

CV464	Environmental Impact Assessment	3
CV465	Environmental Engineering II	3
CV466	Solid Waste Management	3

7. Geo Informatics Engineering

CV472	Remote Sensing	3
CV473	Watershed Modeling Using GIS	3
CV474	Climate Change	3

8. Interdisciplinary

ME467/CV419	Finite Element Analysis	3
ME423/CV427	Introduction to Computational Fluid Dynamics	3
CV408	Machine Learning Applications in Civil Engineering	3

E. Summer Internship (Pass/Fail Grade; NIL Credit)

Every student is required to complete a compulsory training program of cumulative 6 to 8 weeks after the 6th semester (3rd academic year).

F. Survey Camp (Pass/Fail Grade; NIL Credit)

A survey camp after 4th semester (2nd academic year) is mandatory for all BS Civil Engineering students.

G. Total Requirements (137 Credit Hours)

For the BS degree in Civil Engineering, a student has to complete 137 credit hours of course work, survey camp and a cumulative six to eight weeks' internship after 6th semester.

CIVIL ENGINEERING – SEMESTER WISE BREAKDOWN

1st Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	MT101	Calculus I	3	0	3	None	None
	PH101	Applied Physics	3	0	3	None	None
	PH101L	Applied Physics Lab	0	3	1	None	None
	CS101	Computing and AI	2	0	2	None	None
	CS101L	Computing and AI Lab	0	3	1	None	CS101
	CH101	Applied Chemistry and Environment	2	0	2	None	None
	CH161	Occupational Health and Safety	1	0	1	None	None
	HM101	Communication Skills	1	2	3	None	None
	IF101	Innovation and Makers Lab 1	0	3	1	None	None

2nd Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	MT102	Differential Equations and Linear Algebra I	3	0	3	None	None
	ES111	Probability and Statistics	3	0	3	None	None
	CS112	Object Oriented Programming and Design	3	0	3	None	None
	CS112L	Object Oriented Programming and Design Lab	0	3	1	None	None
	MM101	Materials and Nanotechnology	2		2	None	None
	MM141	Materials Lab 1	0	3	1	None	MM101
	HM102	Critical Thinking and Expository Writing	3	0	3	None	None
	IF102	Innovation and Makers Lab II	0	3	1	None	None



3rd Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	MT202	Calculus II	3	0	3	MT101	None
	CV201	Basic Surveying	2	0	2	None	None
	CV201L	Basic Surveying Lab	0	3	1	None	CV201
	CV210L	Civil Engineering Drawing and Graphics	0	3	1	None	None
	CV211	Mechanics of Solids I	3	0	3	None	None
	CV215	Construction Materials and Concrete Technology	2	0	2	None	None
	CV215L	Construction Materials and Concrete Technology Lab	0	3	1	None	CV215
	CV221	Fluid Mechanics I	2	0	2	None	None
	CV221L	Fluid Mechanics I Lab	0	3	1	None	CV221
	CV230	Engineering Geology	2	0	2	None	None

4th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	CV202	Advanced Surveying	1	0	1	CV201	None
	CV202L	Advanced Surveying Lab	0	3	1	None	CV202
	CV212	Mechanics of Solids II	2	0	2	CV211	None
	CV212L	Mechanics of Solids Lab	0	3	1	None	CV212
	CV213	Basic Structure Analysis	3	0	3	None	None
	CV231	Soil Mechanics	3	0	3	None	None
	CV231L	Soil Mechanics Lab	0	3	1	None	None
	CV241	Transportation Engineering	3	0	3	None	None
	HM211	Pakistan and Islamic Studies	3	0	3	None	None

5th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	ES341	Numerical Analysis	3	0	3	None	None
	CV313	Indeterminate Structural Analysis	3	0	3	CV213	None
	CV322	Fluid Mechanics II	3	0	3	CV221	None
	CV322L	Fluid Mechanics II Lab	0	3	1	None	CV322
	CV332	Geotechnical and Foundation Engineering	3	0	3	CV231	None
	CV332L	Geotechnical and Foundation Engineering Lab	0	3	1	None	CV332
	CV323	Hydrology and Water Management	2	0	2	None	None
	CV323L	Hydrology and Water Management Lab	0	3	1	None	CV323

6th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	CV305	Geo Informatics	1	0	1	None	None
	CV305L	Geo Informatics Lab	0	3	1	None	CV305
	CV314	Reinforced Concrete Design I	3	0	3	CV215	None
	CV314 L	Reinforced Concrete Design I Lab	0	3	1	CV215	None
	CV315	Design of Steel Structures	3	0	3	None	None
	CV351	Construction Engineering	2	0	2	None	None
	CV361	Environmental Engineering I	3	0	3	CV221	None
	CV361L	Environmental Engineering Lab	0	3	1	None	CV361
	HM322	Corporate law and Professional Ethics	3	0	3	None	None

7th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	CV403L	Quantity Surveying and Cost Estimation	0	3	1	None	None
	CV407	Civil Engineering Economics	2	0	2	None	None
	CV414	Reinforced Concrete Design II	3	0	3	CV314	None
	CV414L	Reinforced Concrete Design II Lab	0	3	1	None	CV414
	CV442	Highway Engineering	3	0	3	None	None
	CV442L	Highway Engineering Lab	0	3	1	None	CV442
	CVxxx	Technical Elective I	3	0	3	None	None
	CV481	Senior Design Project I	0	9	3	None	None

8th Semester	Course Code	Course Title	Class Hrs	Lab Hrs	Credit Hrs	Pre-requisite	Co-requisite
	CV408	Machine Learning Applications in Civil Engineering	3	0	3	None	None
	CV425	Hydraulics & Irrigation Engineering	3	0	3	None	None
	CV452	Construction Management	3	0	3	None	None
	CVxxx	Technical Elective II	3	0	3	None	None
	CV482	Senior Design Project II	0	9	3	CV481	None

COURSE DESCRIPTION

CV201 Basic Surveying: (2-0-2) The introduction and importance of survey in civil engineering. Calculation of Area and Volumes (earth work calculation, DMD method, Simpson, and trapezoidal rule). Traversing (traversing with a prismatic compass and, computations, and adjustments of traverse). Measurement of horizontal and vertical angles. Leveling (Type of levels, principle of leveling, classification of leveling, errors in leveling). Contouring (characteristics of contour line, locating contours, interpolation of contours, and plotting of contours using Modern tools). Global Positioning System (types, accuracy of GPS, factors affecting GPS). Modern Tools usage in surveying.

CV201L Basic Surveying Lab: (0-3-1)

Introduction to AutoCAD for surveying sketches and Reconnaissance Survey. To measure and range a survey line when the stations are inter-visible and not inter-visible, to measure the horizontal distance along a slope by stepping, to set out a baseline and perpendicular lines/offsets in the field. Triangulation, open, closed and GPS Traversing. Leveling and its types like fly leveling and profile leveling. Contouring with grid method and plotting with modern tools.

CV202 Advanced Surveying: (1-0-1)

Tacheometry (Principles of tangential and stadia methods. Field observations and Calculations for measurement of horizontal distance and height). Theodolite Traversing (Measurement of angles using a theodolite, Checks in traversing, Traverse computations). Independent and consecutive coordinates using Gale's table. Triangulation (Classification, steps and operations involved in triangulation survey, selection of base line and triangulation stations, adjustment of angles, satellite station) Photogrammetry (Introduction, classification, applications, types of aerial photographs, geometry and scale of vertical aerial photograph, flight planning, relief displacement). Hydrographic surveying. Tunnel Surveying. Geographic Information System. Remote Sensing. Curves (Different types of curves and their components, Computations and setting out by different methods)

CV202L Advanced Surveying Lab: (0-3-1) To determine height of a building when base is accessible. To determine height of a building when base is inaccessible. Theodolite Triangulation, its

adjustments and plotting. Theodolite Traversing and Plotting of Traverse. Setting out simple curve by Rankine's Method. Setting out simple curve by offset from long chord. Setting out simple curve by offset from tangent. Setting out compound curve by Rankine's Method. Setting out transition curve. Building Layout. Demonstrate working on Total Station.

CV211 Mechanics of Solids I: (3-0-3) Simple Stress and Strain (Hooke's law, Moduli of elasticity, Lateral strain, Volumetric strain, Poisson's ratio, Temperature stresses). Shear Force and Bending Moment Diagrams. Stresses in beams (Theory of simple bending, Applications of flexure formula, Computation of shear stresses in beams, Shear center and shear flow). Columns and Struts (Axially loaded columns, Euler's formula, Rankine Gordon formula for short and intermediate columns, Slenderness ratio). Circular shafts. Springs (Open coil springs, Closed coil springs, Leaf springs). Strain Energy (Strain energy due to direct loads, shear force, bending moment and torque; Stresses due to impact loads; Application of strain energy, its minimization and equilibrium).

CV212 Mechanics of Solids II: (2-0-2) Stress Analysis (Analysis of stresses and strains due to combined effect of axial force, shear force and bending moment. Mohr's circle for stresses & strains. Strain rosettes). Theories of Failure. Unsymmetrical Bending. Thin and Thick Cylinders (Introduction to cylindrical pressure vessels, Stresses in Thin cylinders, Strains in Thin cylinders, Thick Cylinders, stresses in thick cylinders). Analysis of Curved Beams (Curved Beams and Stresses in Curved Beams, Circumferential Stresses in Curved Beams, Location of Neutral Axis in Curved Beams, Radial Stresses in Curved Beams). Eccentrically Loaded Columns (Deflections and bending moment in eccentrically loaded column, The Secant Formula).

CV212L Mechanics of Solids Lab: (0-3-1) To determine young's modulus of elasticity of steel bar. To perform young's modulus of elasticity for concrete. To perform compression test on wood sample of 2"x2"x2" size, with load parallel and perpendicular to grains. To perform impact test on metals. To perform bending test on 1.5"x3"x30" specimen. To investigate the relationship between the deflections and the applied loads and the effect of variations in length and cross-sectional dimensions on the beam deflection. To study the effect of

combined bending & torsion. To study the effects unsymmetrical bending of beams. To determine and compare the modulus of rigidity for different materials. To determine the crippling load for struts of varying slenderness ratios and end fixing conditions.

CV213 Basic Structure Analysis: (3-0-3)

Redundancy and stability of structures. Analysis of Determinate pin Jointed Structures. Analysis of Statically Determinate Rigid Jointed Plane Frames. Influence lines for reactions, shear force and bending moment in statically determinate beams and paneled girders, influence lines for member forces in pin jointed frames, Calculation of maximum stress function in these structures. Three Hinged Arches, Cables and Suspension Bridges: Basic considerations in analysis and design. Moving loads on three hinged arches and suspension bridges. Moment area method, conjugate beam method, double integration method, Castiglano's second theorem. Rotation and deflection of plane trusses and frames. Principle of virtual work, unit load method, graphical method.

CV215 Construction Materials and Concrete Technology (3-0-3)

Tiles and Bricks, Plastics, Glass, Wood composites, Paints and varnishes. Concrete and Concrete Materials (cement, sand, and gravel). Workability of Concrete. Admixtures. Mixing, Transporting, Placing & Compaction of Concrete. Curing of Concrete (Curing of concrete, Maturity of concrete. ACI recommendations). Mix Design. Testing of hardened concrete (Compression test, flexural test, creep), Strength of Concrete (Nature of strength of concrete, factors affecting the strength). Durability of Concrete (Physical and Chemical).

CV215L Construction Materials and Concrete Technology Lab (0-3-1)

Tests on Cement and Aggregates, Measurement of the fineness of cement, setting time of cement, specific gravities of fine and coarse aggregate, fineness modulus of sand, measurement the moisture content in coarse and fine aggregate, the water absorption in coarse and fine aggregate. soundness of coarse aggregate, organic impurities in sand, Slump Test, compaction factor Test, VeBe Test, Mix design by ACI method. Compressive strength and, Measurement of air content in concrete, Effect of curing on the strength of concrete, Effects of water cement ratio on the strength of concrete, Effect of age on the strength of concrete.

CV221 Fluid Mechanics I: (2-0-2) Properties of Fluids (Shear stresses in moving fluids, specific weight, specific gravity, specific volume. Dynamic and kinematic viscosity; Surface tension; Capillarity; Vapor pressure). Pressure and Head (Pascal's law for pressure at a point, Variation of pressure in a static fluid under gravity, Absolute and gauge pressure). Pressure measuring instruments Hydrostatics, Kinematics of Flow (Uniform and steady flow; Real and ideal fluids; Compressible and incompressible flow; One, two and three-dimensional flow; Streamline; Laminar and turbulent flow; Velocity variation over a section, Mean velocity; Discharge). Fundamental Equations of Fluid Motion. Applications of the Energy equation (Flow through a tapering pipe, Venturi meter, Notches, Orifices, Pitot tube, Sharp-crested weir).

CV221L Fluid Mechanics I Lab: (0-3-1)

Demonstration of various parts of hydraulic bench and measure of discharge. Calibration of pressure gauge using dead weight tester. Determination of the center of pressure of a submerged plane surface. To investigate the stability of a floating body. Demonstration of Bernoulli's theorem. Study of flow through Venturi meter. Determination of the coefficients of contraction, velocity and discharge for an orifice. To study the impact of jet on a flat plate and a hemispherical cup. Demonstration of Reynolds number apparatus. Demonstration of flow visualization on laminar flow table. Determination of the coefficient of discharge for a V-shaped and a rectangular notch

CV230 Engineering Geology: (2-0-2)

Introduction to Geology Rocks and Minerals (main groups, igneous, sedimentary and metamorphic rocks, importance of minerals and ores, rock cycle). Weathering and Erosion (agents of



weathering, weathering classification, cycle of erosion, erosion types, land forms, mass wasting, formation of meanders and ox-bow lakes). Structural Features (dip, strike, folds, faults, joints, unconformities, effects of structural features on civil engineering projects). Earthquakes (definition and technical terms, causes, classification, measuring of earthquake intensity), Landslides, Glaciers and Glaciations. Tunneling. Engineering Applications.

CV231 Soil Mechanics: (3-0-3) Basic Properties of Soil: (Soil formation and nature of soil constituents, Physical properties of soil, Weight volume relationships, Mechanical analysis of soil Hydrometer analysis, Consistency limits of soil). Soil Classification Systems: (AASHTO and Unified soil classification system). Soil Compaction: (Moisture density relationships, Properties and structures of compacted soils, Factors effecting compaction). Permeability and Seepage: (Hydraulic gradients, Darcy's law, Coefficient of permeability, Factors affecting permeability, Field determination of permeability), Capillarity, shrinkage and swelling, Frost heave and collapsible soils), Settlement and Consolidation, In-situ tests; Types of soil samples, samplers and soil sampling.

CV231L Soil Mechanics Lab: (0-3-1) To conduct the Sieve Analysis of Soil, to determine the Moisture Content of Soil by; Oven Drying Method, Speedy Tester Method, to find the Specific Gravity of Soil. Atterberg Limits, Hydrometer Analysis to measure and quantify the size distribution of the clay and silt content in soil, to conduct the Permeability Test of soil using Constant Head Apparatus and Variable Head Apparatus, to find the Optimum Moisture Content (OMC) for compacted soil by, Standard

Proctor Test, Modified Proctor Test, to determine the in Situ Density of Soil by Core Cutter Method and by Sand Replacement Method, to conduct the consolidation test to find settlement of soil.

CV241 Transportation Engineering: (3-0-3)

Introduction to Transportation Systems. Airport Engineering, Aircraft Characteristics and Computation of Runway Length, Airport Configurations, Geometric Standards and Obstruction Clearance Criteria, Railway Engineering, Track, Rail Gauges, Rails and Rail Fastenings, Sleepers, Ballast and Formation (Function and type of sleepers; Sleeper material, density, spacing and stiffness of tracks; Material for ballast and renewal), Creep and Rail-bed Soil, Station and Yards, Construction and Maintenance. Coastal Engineering, Ports and Harbor, Effects of Tides, Winds and Sea waves, Ports and Harbor Structures, Maintenance and Modernization (Coastal protection; Dredging, sluicing).

CV210L Civil Engineering Drawing and Graphics: (0-3-1)

Types of Civil Engineering drawing. Set of the working drawings of a residential building. Preparation of plans, elevations and sections Use of computer graphics software. Introduction of building information modeling (BIM). Application of architectural planning and rendering in Civil Engineering projects. Basic Concepts of AutoCAD. Building Drawing, Structural Drawing (Elements of structural drawing and detailing: Preparation of foundation plan, structural framing, slab details, staircase details, water tanks, beam and column elevations and sections mostly pertaining to reinforced concrete structures. Details of steel roof truss, connection details and fabrication drawings,



Plumbing and electrical detailing pertaining to small residential units).

CV305 Geo Informatics: (2-0-2) Introduction to Geo informatics; Resources of information; photogrammetric surveying, Satellite System, Aerial and Satellite photogrammetry. Geographic Information System (GIS): Fundamentals of GIS, Spatial Data types and acquiring consideration. Data models and structures. Coordinate System, Datum and map projection and their transformation. Attribute-based operation, Introduction to Spatial Analysis. Remote Sensing (RS): Basic Concepts. Physicals basis of Remote Sensing, Earth Resources Satellites/ Platforms, Sensors, Types of Resolutions, Geo-Referencing, Image Processing Techniques. Classification. Global Positioning System (GPS): Navigational Satellites, Positioning Systems, Fundamentals and Elements of GPS.

CV305L Geo Informatics Lab (0-1-1) Tutorial on Spatial Data Generation, Modeling, Analysis and Applications. On Satellite Image Dereferencing Enhancement, And Filtration, Transformation, Classification and Accuracy Assessment and Application. Laboratory Session Evolving Use of State of The Art GIS. Use Image Processing Software. Handling and Analysis Spatial Datasets Including Satellite Images Reading. To Digitize MAPs.

CV313 Indeterminate Structural Analysis: (3-0-3) Method of Consistent Deformations (Analysis of statically indeterminate beams, frames, Introduction to Castigliano's second theorem, Analysis of Indeterminate Trusses, Application of virtual work method to deflection of trusses, Analysis of trusses with redundant support viscosity. Determination of the coefficient of drag of a sphere falling through a column of water. Determination of head loss through a straight pipe of uniform cross section. Determination of head loss through different pipe fittings. To study cavitation phenomenon. To study characteristics of a centrifugal pump. To study characteristics of a Francis turbine. To study the characteristics of a reciprocating pump.

CV323 Hydrology and Water Management: (2-0-2) Surface water hydrology (Hydrological cycle, Hydrologic equation and Importance and practical applications of hydrology). Precipitation, Water Losses, Evaporation, Transpiration, Evapotranspiration. Runoff & Hydrograph (Runoff

& Factors Affecting Runoff, Computation of Runoff, Characteristics of hydrograph, Components of a hydrograph, Unit Hydrograph, S-curve and Discharge estimation by probabilistic method). Stream Flow Measurement. Ground water hydrology (Aquiclude, Aquifuge, Aquifer & Types of Aquifer). Well Hydraulics (Steady Radial Flow to Well in Confined & Unconfined Aquifers – Dupuits Theory, Assumptions & Limitations of Dupuits Theory, Well Losses, Capacity of Well, Interference among Wells).

CV323L Hydrology and Water Management Lab: (0-3-1) Measurement of precipitation and Computation of Average Rainfall over a Basin, Stream Flow Measurement, Methods of Stream Gauging and Measurement of Stream Flow by Current Meter,

CV332 Geotechnical and Foundation Engineering: (3-0-3) Shear Strength of Soil, Bearing Capacity of Soil (Terzaghi's bearing capacity theory; Meyerhoff's method; Vesic's method Factors affecting bearing capacity; Bearing capacity from field tests; Effect of GWT on bearing capacity Stresses due to Point Load, Strip Load etc., Types of Foundations. Geotechnical design of shallow foundation, Deep Foundations, Lateral Earth Pressure (Active, at rest and Passive earth pressures; Rankine and Coulomb's theories; External stability of gravity and cantilever retaining walls). Stresses in soil, Slope Stability: Types of slopes, Factors affecting stability and remedies, Types of failure Methods of analysis; Swedish circular method, Taylor's slope stability number and Bishop's Methods.

CV332L Geotechnical and Foundation Engineering Lab: (0-3-1) To find the geotechnical parameters cohesion (c) and angle of internal friction (ϕ) of soil by Direct Shear Test Apparatus, To find the geo-technical parameter cohesion (c) and angle of internal friction (ϕ) by Unconfined Compression Test Apparatus, To conduct the Consolidated Drained Test of soil by using Tri-axial Compression Test Apparatus, To conduct the Consolidated Undrained Test of soil by using Tri-axial Compression Test Apparatus, To conduct the Unconsolidated Undrained Test of soil by using Tri-axial Compression Test Apparatus, To conduct the Standard Penetration Test (SPT) to find the bearing capacity of Soil, To conduct the Cone Penetration Test (CPT) to find the bearing capacity of Soil, To conduct the plate load test to find out the bearing capacity of Soil.

CV351 Construction Engineering: (2-0-2)

Construction projects, Project goals and objectives, Project categories, Building permits, codes and construction standards, Sustainability. Construction Equipment, Investment costs, Maintenance and repair costs. Productivity and cost effectiveness. Site selection and orientation of building, Grading considerations, Layout techniques with special reference to buildings. Construction Methodologies, Methodologies for Excavation in different types of soils, In-situ and pre-cast concrete construction of buildings. Design and use of formwork. Methods of concreting. Construction joints, Mass concreting, Process of site reactions). Method of least work (Application to beams and trusses). Slope-Deflection Method. Moment Distribution Method. Rotation Contribution Method (Analysis of Indeterminate beams and frames). Three-Moment Equation. Analysis of Arches. Influence lines for indeterminate beams. Column Analogy Method (Analysis of prismatic and non-prismatic beams and frames, Evaluation of stiffness factors and carry-over factors).

CV314 Reinforced Concrete Design I: (3-0-3)

Basic principles of reinforced concrete design and associated assumptions, Behavior of reinforced concrete members in flexure, Design philosophy. Working stress method. Serviceability criteria and checks for deflection, crack width, and crack spacing. Ultimate Strength Design method, Analysis of one-way solid and ribbed slabs with general discussion on other slab systems, Shear stress in reinforced concrete sections, Design for diagonal tension. Design and detailing for bond, anchorage, development length, laps and splices.

Short Columns (Axial capacity within elastic range and at ultimate loads bending of short columns about single axis). Stairs, Analysis and Design of Various Types of Stairs and Staircases.

CV314L Reinforced Concrete Design I

Lab: (0-3-1) Flexural Test, Split Cylinder Test, Compression Test for Cylinder and Cube, Non-Destructive testing (Ultrasonic Pulse Velocity and Schmidt Hammer Test), Bar Bending Schedule, Preparing the Complete layout of load-bearing residential building and masonry wall footing.

CV315 Design of Steel Structures: (3-0-3)

Introduction to Structural Steel Design, Stress-strain Relationship in Structural Steel, Introduction to ASD (Allowable Stress Design) and LRFD (Load and Resistance Factor Design) methods, Fabrication and erection methods of steel structures. Tension Members. Compression Members. Design of Beams. Connection Design (Riveted connections, Types of rivets). Bolted connections. Welded Connections. Plastic Analysis and Design (Theory of Plastic analysis and Introduction to plastic design). Design of Roof Trusses (Roof trusses-introduction, Design loads and load combinations, Roof truss analysis and Design of roof trusses, using AISC specification), and Introduction to Relevant Software Packages.

CV322 Fluid Mechanics II: (3-0-3)

Dimensional Analysis and Similitude (Geometric, kinematic and dynamic similarities; Different dimensionless numbers and their significance). Laminar Flow Through Pipes, Turbulent Flow Through Pipes (Turbulent flow; Darcy Weisbach equation; Velocity



distribution in fully turbulent flow; Shear stresses in pipes; Universal velocity distribution; Turbulent fluctuating velocity components; Moody chart and Colebrook-White formula; Separation losses). Steady, Incompressible Flow in Pipelines, Steady, Uniform Open-Channel Flow. Incompressible flow around a body, Fluid Machines (Velocity diagrams for fluid machines; turbines, Centrifugal pump, construction features and applications; Reciprocating pump, construction features and applications; Cavitation in Fluid Machines).

CV361 Environmental Engineering I: (3-0-3)

Sources of Water Supply (Types of well construction, Yield of well, Test for yield of well etc). Water Quantity Population Forecast (Water uses & consumption, Types and variations in demand), Rural Water Supply & Urban Water Supply. Collection and Conveyance of Water (Intake, Types of intakes, pipes, pipe corrosion). Water Distributions (Layout and design of water transmission works and distribution networks, service reservoirs, Fixtures and their installation, Service Reservoirs, Fixture and their installation, tapping of water mains). Water Quality (Water impurities and their health significance, Water quality standards, Water quality monitoring). Water Supply Sampling and Testing. Diseases (Water borne, and Vector borne diseases). Estimation of Sewage Quantities, Characteristics of Sewage (Sampling techniques and Examination of waste water). Design, construction and maintenance of sewage system, Separate and combined Systems, Types, Shapes, Size and materials of Sewers, Sewer appurtenances, Pipe Strengths and Tests, Construction and Maintenance of Sewer System and Analysis, diameter and Gradient, Sewer.

CV361L Environmental Engineering Lab:

(0-3-1) To find out the Turbidity of a given sample of waste water by Nephelometric method, To determine the pH value of a given sample of water, To find out the suspended solids (SS), dissolved solids (DS) and total solids (TS) in the given sample of water, To determine the Carbonate, Bicarbonate and Hydroxide Alkalinity in the given sample of water, To find out the concentration of Chlorides and residual chlorine in the given sample of water, To find out the Hardness in the given sample of water by standard EDTA method, To determine the optimum dosage for the turbid water by Jar test, Estimation of Chemical Oxygen Demand (COD) in water To determine the Biochemical Oxygen Demand (BOD) in water, To Determine the

Dissolved oxygen in a sample by Azide Modification method, To find out Hydrogen sulfide and E-coli in the given sample of water.

CV403 Quantity Surveying and Cost Estimation:

(0-3-1) Introduction (Cost estimating process, Design process and Types of cost estimates). Budget Estimates (Budget estimates, Design budget estimates and Budget estimate accuracy). Preconstruction Services Estimate (Development of the preconstruction services estimate, Preconstruction services contract and Process interactions). Pre-Estimate Activities (Estimating process, Invitation for bid, Pre-bid meeting, Bid date and time, Bid document revision, Deciding whether or not to tender a bid, Work breakdown structure, Estimating team, Scheduling the estimating work, Subcontractors and major suppliers, Data resources, estimating forms, files and electronic spreadsheets). Estimating Sub-Contractor Works.

CV407 Civil Engineering Economics (2-0-2)

Fundamentals of Engineering Economics: Basic concepts and principles of Economics, Micro-economics theory. Capital Financing and Allocation: Capital Budgeting, allocation of capital among independent projects, Financing with debt capital, Business Organization and Industrial Relationship: Linear Programming: Mathematical statement of linear programming problems, Graphic solution simplex procedure, Duality problem Depreciation and Taxes: Depreciation concept, Economic life, Methods of depreciation, Profit and returns on capital, productivity of capital, Gain (loss) on the disposal of an asset, Depreciation as a tax shield. Selection between Alternatives: Time value of money and financial rate of return, present value, future value and annuities, Rate of Return Analysis, Incremental analysis, Cost-benefit analysis, Payback period. Cost estimating: Types of Estimates, Approximate estimates – Unit estimate, Factor estimate, Cost indexes, parametric estimate, and Life cycle cost. Financial management: Construction accounting, Chart of Accounts, Financial statements – Profit and loss, Balance sheets, Financial ratios, Working capital management.

CV413 Structural Engineering: (3-0-3)

Virtual displacement principle and displacement method. Element stiffness matrix for bar, beam and plane frame element. Virtual force principle and flexibility method. Flexibility of bar, beam and general flexural elements. Compatibility and equilibrium.

Coordinate transformation. Assembly of structure stiffness matrix. Analysis of 2D trusses, beams and frames by matrix method. Analysis of 2D framed structures with temperature, support settlement and lack of fit. Introduction to SDOF, MDOF and Equation of motion. Derivation of SDOF considering forced vibration of constant and sinusoidal forces with and without damping. Introduction to finite element methods. Computer applications of matrix methods of structure analysis using interactive computer programs and reliability of computer results.

CV414 Reinforced Concrete Design II: (3-0-3)

Two-way slab systems (Introduction, Design of two way slabs by Co-efficient method). Analysis and design of flat plate, flat slabs and waffle slabs, for flexure and shear under gravity loading. Design of biaxial column. Analysis and design of slender columns subjected to combined flexure and axial loading. Guidelines for design of shear walls-an, over view. Design of Different Types of Foundations: Analysis and design of combined footing, strap, strip and mat footings. Pre-stressing Principles & Design Philosophy. Introduction to earthquake resistant design of structures. Design of cantilever retaining walls.



Fahad Iqbal

Getting into GIKI is like a dream that comes true. The diverse community of the Institute makes it a very special place to live and study. Surrounded by dedicated and helpful professors, friends you can always trust, a great dormitory, life never makes you homesick not even for an instant.

GIKI will push you to the limit of your true potential, the experience is highly challenging, but at the end It gives you the feeling of accomplishment. GIKI is home to outstanding academics and excellent laboratories. It provides you the environment to become a responsible person.

In short, GIK has become a place for me where I feel at home and able to be myself.

CV414L Reinforced Concrete Design II Lab: (0-3-1)

Design of a residential building: Design and detailing of slabs, beams, stairs, columns, and footings. Comparison of manual design with the results of FE software (SAP). Design of multi-story buildings using interactive computer programs (ETABS, SAFE etc.): Structural layout of beams, columns, slabs, shear walls, staircase, lift well using ETABS. Defining and assigning load scenarios. Analysis and design of various structural components. Design against seismic and wind loadings. Design of raft or mat footing on SAFE. Reinforcement detailing of structural components. Design of retaining walls: comparison of manual design with SAP results.

CV416 Prestressed Concrete: (3-0-3)

Introduction to pre-stressing, Differences between Prestressed concrete and Reinforced concrete, Advantages and disadvantages of Prestressed Concrete, Typical Prestressed concrete Structures, Pre-stressing Systems, Pre-tensioning and Post-tensioning. Properties of pre-stressing materials, Pre-stressing principles and design philosophy, Response of prestressed members to Axial Loading, Response of prestressed Members to Flexural loading, prestressed Members subjected to Shear, Analysis and design of Prestressed concrete members to resist flexure, Design of Prestressed concrete members to resist shear, Camber and deflection in Prestressed members, Pre-stressing losses, creep and shrinkage, short and long term effects of pre-stressing.

CV417 Design of Structures: (3-0-3)

Introduction to basics of structural systems, design methods and application of Software. Manual design of masonry residential building and examples. Design example of RC building using software. Design of water retaining structures and examples. Design and Analysis concrete Bridges. Manual and software design of industrial building and examples. Design and Analysis of Steel Composite Bridges.

CV418 Introduction to Earthquake Engineering: (3-0-3)

Introduction to earthquake phenomena. Types of seismic waves and introduction to some major earthquakes in the Pakistan and world. Introduction to parameters of seismic response of structures. Introduction to Seismic hazard assessment. General seismic design considerations: common mistakes in practice, regularity, lateral force resisting mechanisms and ductility).

Earthquake excitation, equation of motion of SDOF system for earthquake excitation. Generalized SDOF system, rigid body assemblages, and lumped mass system: shear building. Equation of motion for SDOF Lumped mass system: shear building and solution of various examples. Strong Ground Motion (Equivalent lateral force procedure and Calculation of base shear for given building frame system. Seismic design of reinforced concrete columns, according to provisions of ACI. Seismic design of concrete Structures. Seismic design of steel structures. Detailing of reinforced concrete structures for earthquake resistance as per Code).

CV422 Hydraulic Engineering Design: (3-0-3)

The Flow Resistance (The resistance equation, The Chezy equation, Behavior of Chezy coefficient C , The Manning's equation, The uniform flow). The Gradually Varied Flow computations (Solution of GVF method for uniform channels by direct integration, Step method-distance calculated from depth, Step method-depth calculated from distance, Extension of the methods of GVF equation to irregular channels), Channel Controls (Sharp-crested weirs, The overflow spillway, The drop structure, The underflow gates, Drowned outflow, Broad crested weir and The Parshall flume). The Basics of River Engineering (The natural river, the dominant discharge, Braided and meandering channel, Width-discharge-slope variations, River training and control, Scour around bridge piers).

CV425 Hydraulics and Irrigation Engineering (3-0-3)

Indus Basin Irrigation System. Canal Irrigation (Elementary concept about canal head works. Weirs and barrages). Components and functions. Measures adopted to control silt entry into canals. Design of weirs on permeable foundations. Sheet piles and cut off walls. Crop water requirements. Design of irrigation channels. Comparison of various methods. Dams Engineering (Selection of dam sites, Components of dams, Layout of dams, Types of storage dams, Forces on dams, Design of gravity dams, Reservoir engineering and Operation and regulation of storage reservoirs). Sedimentation. Canal head regulators. Falls. Flumes. Cross drainage works, types and functions. Water Logging and Salinity.

CV426 Hydropower Engineering (3-0-3)

Introduction and importance of Hydropower development. Comparison of hydropower with other energy sources, stages. Main components of low,

medium and high head power schemes. Hydraulic transient and its control through Surge tanks, air chambers, valves, etc. Hydro-mechanical, electro-mechanical components. Auxiliary equipment. Hydropower Project layout and sizing, alternative project layout. Powerhouse. Power Canal. Intake Waterways. Pressure Relieving Structures and dewatering of Powerhouse Pit. Hydro-mechanical components.

CV433 Slope Stability: (3-0-3) Introduction, General Principles of the Behavior of Soft Ground, Field Instrumentation for Soft Ground, Examples of Instrumentation, Applications for Projects in Soft Ground, Behavior of Clay Foundation Soils, Drained and Undrained analysis, Different types of slope failures, their identification, Different method for slope stability analysis, Factor of Safety, Limit Equilibrium Analysis of slopes using software, Methods of Construction, Strategy for Design Studies, Fundamentals of Geosynthetic, Soil-Geosynthetic Interaction, Geosynthetic for Retaining Walls, Geosynthetic for Embankments, Geosynthetic for Shallow Foundations, Geosynthetic for Slopes, Geosynthetic for Earth Dams, Geosynthetic applications general aspects and selected case studies.



Sara Fatima Qazi

Four years ago, I was just a meek girl fresh out of the small town of Hyderabad, having scarce knowledge about anything outside of the constraints of Sindh. Four years later I can affirmatively say that not only have I flourished academically, thanks to a very research-oriented faculty, but I have also grasped multitude of cultures the GIKI's student body has to offer and learnt how to micromanage the tenacious academic pressure alongside the buzzing social life of GIKI. GIKI is not just an institute, it's an experience, with a serene landscape to balance out the distressing days.

CV434 Design and Construction of Earthen Dams: (3-0-3) General Design Criteria, Theoretical Aspects of Seepage (Flow net for earth dam, Use of SEEP/W computer software for generating flow-net). Control of Seepage Through Embankments (Adverse effects of seepage, Methods of seepage control, Provision of core, Design of transition filters, Drainage of embankments, Drainage of upstream face and Control of seepage on downstream face). Control of Seepage Through Foundations. Section Detail related special problems (Section details, Cracking and its control, Dams in fault zones, River diversion, Conduits through earth dams and Dispersive and expansive soils). Quality Control of Earthen Dams (Compaction, Placement control, Field tests, Borrow area control, Foundation preparation and treatment and Contact treatment).

CV435 Rock Mechanics: (3-0-3) Stress and strain: force and stress; principal stresses and stress invariants; plane problems and biaxial stress; displacement and strain; geomechanics conventions. Rock mass structure and characterization: types of structural features. Geo-mechanical properties of discontinuities; structural data collection and presentation; rock mass classification. Rock strength and deformability: concepts and definitions; isotropic an anisotropic rock behavior (uniaxial and multiaxial); behavior of discontinuous rock masses; shear strength of discontinuities. Factors influencing the in-situ state of stress; in situ stress determination and presentation of results. Stress analysis. Closed- form solutions for simple excavation shapes; boundary element, finite element, distinct element, and finite difference analysis methods for continuous rock. Excavation design with case study.

CV436 Tunneling and Underground Space Structures: (3-0-3) Types of Underground Excavations: Tunnel, audit, decline, shaft; parameters influencing location, shape and size; geological aspects; planning and site investigations. Tunneling Methods. Shallow tunnels. mucking and transportation equipment selection. Tunneling by Tunnel Boring Machines. Supports in Tunnels: Principal types of supports and applicability. Ground Treatment in Tunneling. Tunnel Services: Ventilation, drainage and pumping. Methods of Sinking Shafts. Tunneling Hazards

CV442 Highway Engineering: (3-0-3) Pavement Engineering, Pavement Materials (Road bed soils, Stone aggregates, gradation and batch mixing problems, Composition, types and characterization of bituminous materials, Bituminous binder testing, Marshall mix design). Pavement Structural Design Geometric Design, Highway components, typical cross sections. Elements of Road Cross-section (Pavement characteristics; Drainage channels, curves and traffic barriers; Right of way). Horizontal and Vertical road alignment. Road Construction and Maintenance. Traffic Engineering, Properties of Traffic Engineering Elements (Vehicle characteristics, Human factor and driver's characteristics and Road characteristics), Traffic Flows, Traffic Facilities, Traffic Safety and Management.

CV442L Highway Engineering Lab: (0-3-1) Penetration test of bituminous materials. Ductility test of bituminous materials. Softening point test of bituminous materials. Flash and fire point of bituminous materials. Aggregate gradation test. Lab and Field CBR Test. Compaction Test. Blending of aggregates – Mix Design. Marshall Stability and flow test – Mix Design.



CV443 Traffic Engineering: (3-0-3) Traffic Engineering (Elements of traffic engineering, Traffic characteristic, Traffic survey and Mass transit and rapid transit). Traffic Flow Characteristics (Nature of traffic flow, Parameters connected with traffic flow, Categories of traffic flow, Analysis of speed flow and density relationship, Traffic stream characteristics, Interrupted and uninterrupted traffic flow models and Queuing theory and shock wave theory). Traffic Safety (Traffic regulation and control, Traffic signs, Clear roadside recovery areas, Guardrail design, Median barriers, Crash cushions and Highway lighting, Measures of safety and identification of hazardous location and Safety considerations in highway design), Traffic Signals (Signal design/control for arterial roads and delays at isolated traffic signals)

CV444 Pavement Materials and Design: (3-0-3) Materials and Characterization, Asphaltic Materials (Chemical composition of asphalt binders, Asphalt binder properties, Asphalt grades, Sampling and handling, Marshall mix design, Asphalt concrete properties, Batch mixing, transporting and handling of asphalt), Portland Cement Concrete (PCA mix design, Batch mixing, transporting and handling of concrete). Asphalt and Concrete Batching Plants, (Material calculations, Layout and material handling. Practicals: (Standard viscosity test for bituminous materials, bituminous materials extraction test, Marshall stability and flow test, Preparation of job mix and PG binder testing). Pavement Systems (Pavement types, Wheel loads, Design factors, Layered system concept). Pavement Design (ASSHTO design methods - Flexible and Rigid).

CV445 Geometric Design of Highways: (3-0-3) Principles of geometric design. Geometric Design of Highways. Design of Curves (Horizontal curves, Transition curves, Vertical curves, Super-elevation, Analysis of super-elevation, Steps in design of super-elevation, Extra widening of road on horizontal curves). Highway Capacity Manual (Introduction, Usage of manual, making profiles for draftsman/AutoCAD), Geometric Design of Railway Track (Necessity of geometric design of track, Gradient and grade compensation, Ruling gradient, Momentum gradient, Pusher gradient, Gradient in station yards, Grade on curves, Super-elevation or cant). Practicals: (Making horizontal and vertical profiles using CAD, Cross sections of roads on CAD and Introduction to geometric design software).

CV452 Construction Management: (3-0-3) Project, Project Life cycle, Construction Constraints, Engineering Costs and Cost Estimating, Life-Cycle Cost, Cash Flow Diagram, Economic Decision-Making Process, Project Delivery methods, Value Engineering, CPM, PERT, Network Analysis Techniques, Schedule Crashing, Contract Management, Procurement and its types, Contract and its types, Terminologies regarding Procurement and Contract, MRS, MRS Analysis, Technical Specifications, Construction Safety, Analyzing construction Hazards, Unsafe Acts and Conditions, Risk Management in Construction, Risk Classification, Human Resource management in Construction, Challenges of managing people in Construction industry, Conflict and Workplace Relationship, Emotional intelligence, Corporate Social Responsibility (CSR), Quality Management, Innovative Trends in Construction Industry (BIM), Project Management tools Primavera P6 and MS Project.

CV453 Architectural and Building Information Modeling: (3-0-3) Concept of integrated building design. Significance of building system integration at planning and design stage of an architectural project. Building management system. Sustainable architecture. Impacts of architectural design on energy consumption in buildings. Architectural design and planning of appropriate: Structural System, Mechanical System, Electrical Systems, Fire Safety Systems, Communication and Security systems, Lighting, and Illumination Systems. Introduction to Building information modeling and usage of its tool. Software Application: Advanced Application of Autodesk Revit Architecture, Revit MEP, Navis Works.

CV464 Environmental Impact Assessment: (3-0-3) Environmental management, National environmental policy. Environmental legislation, Environmental Impact Assessment (EIA) process, Environmental Impact Prediction and Evaluation during construction & operation of projects, Mitigation measures, Modeling, Environmental monitoring & auditing, Environmental management issues, Methods of impact analysis. Environmental Decision Making. Writing Impact Statement. Water Quality (Water quality and impact of project on water quality). Future of Environmental Impact Assessment (Future of environmental management, Environmental issues and Future of environmental impact assessment). Mitigation of environmental impacts. World

Bank (WB) and Asian Development Bank (ADB) guidelines.

CV465 Environmental Engineering II (3-0-3)

Water Treatment: Treatment of surface and ground water; Screening, Sedimentation, Coagulation; Coagulants and dosages; Filtration, design aspects of slow and rapid filters; Filtration rates, Operation, Fluoridation. Water softening Methods. Water Disinfection and Chemicals. Introduction to Relevant Software Packages. Testing of Sewers Municipal and Industrial Wastes, water Pollution, Causes and Control Parameter, Effluent disposal guidelines and Standards, Pakistan National Environmental Quality Standards NEQS. Primary Treatment System. Secondary Treatment Systems. Trickling filter, classification and efficiency of trickling filters; Design of trickling filters, final clarifier design, Suspended growth process principle, activated sludge process details, loading criteria, sludge volume index; Aeration processes, mixing techniques, analysis and operational problems; Design of aeration tank, secondary clarifier, Oxidation ponds & aerobic, anaerobic and facultative ponds). Sludge Treatment (Amount and characteristics of sludge, sludge conditioning methods; Anaerobic and aerobic digestion).

CV466 Solid Waste Management: (3-0-3)

Pakistan National Environmental Quality Standards-NEQS), Solid Waste Management (Types, characteristics, sources and quantities of solid waste; Collection disposal and recycling), Sewage Disposal Sludge Treatment (Amount and characteristics of sludge, sludge conditioning methods; Anaerobic and aerobic digestion).

CV472 Remote Sensing: (3-0-3) Basic Principles of Remote Sensing, Optical Remote Sensing, Thermal Infrared Remote Sensing, Radar and Sonar Remote Sensing, Digital Processing of Remote-Sensing Imagery, Applications 1-Meteorology, Oceanography, and Environment, Applications 2-General Land Use and Land Cover, Applications 3-Geology, Applications 4-Vegetation: Forestry and Agriculture, Comparison of Sensors and Image Types, Hyper-spectral Remote Sensing, Digital Image Processing and Geographic Information Systems.

CV473 Watershed Modelling Using GIS (3-0-3)

Concepts of watershed modelling, Common GIS applications in hydrology (exercise), Calibration and validation of watershed models, Build your own simple "bucket" model (Matlab), Semi-distributed modelling (exercise), Fully-distributed modelling (TOPKAPI model, exercise), Modelling case studies.

ME467/CV419 Finite Element Analysis (2-1-3)

Matrix theory, Direct Stiffness method, Energy method, Development of truss and beam elements, Plane stress and strain elements, Isoparametric formulation, Application of FEM in major streams of mechanical engineering with the aid of examples. Errors during FEM, Use of computational commercial codes (Matlab, ANSYS or APEX) to apply on different case studies.

ME423/CV427 Introduction to Computational Fluid Dynamics (2-1-3)

Applications, Advantages and Future of CFD. Continuity Equation, Momentum equations, Energy Equation, Turbulent Flow .CFD-Techniques: Finite Difference Method, Truncation Error, Finite Volume Method, Structured/



Unstructured Mesh, Finite Element Method, Conversion of Governing Equations to Algebraic Equations, Finite difference Method, Finite Volume Method, Comparison of Finite difference and Finite Volume Method, Steady convection-diffusion process, Unsteady convection-diffusion process, Numerical Solutions to Algebraic Equations, Direct Method, Interactive Method, Jacobi-Gauss Siedel Method, Pressure –Velocity Coupling, SIMPLE Scheme. CFD Solution Analysis: Consistency, Stability, Convergence, Residuals and Convergence Tolerance, Convergence difficulty and using Under-Relaxation, Accelerating Convergence, Accuracy, Sources of Errors, Controlling the Solution Errors, Efficiency. Case studies.

CV408 Machine Learning Applications in Civil Engineering (3-0-3) Introduction to data driven modeling, Introduction to big data analytics: preprocessing, imputation, outlier detection. Introduction of exploratory data analysis and visualization: estimates of location, variability, data distribution, binary and categorical data, correlation, visualizations. Regression and classification problems. Supervised Learning: k-Nearest Neighbors, Support vector machine, Decision trees Bagging and boosting, Artificial neural networks: 1-layer and multi-layer networks, backpropagation and stochastic gradient descent algorithms, convolutional neural networks. Unsupervised Learning: Clustering. Case studies. Optional: Cross validation and optimization.

CV481 Senior Design Project I: (0-9-3) To acquaint students to have experience to design, fabricate, assess, evaluate and present their final year project.

CV474 Climate Change (3-0-3) Greenhouse Gases, Global Warming and Environmental Change, Climate Scenarios and Representative Concentration Pathways (RCP's), Climate Change Projections, Statistical and Dynamic Downscaling, Integrated Assessments, Risk and Vulnerability Analysis, Methods in Adaptation, Methods in Mitigation, Disaster Risk Reduction and Climate Change, Costing Climate Change and Adaptation, Communicating Climate Change.

CV482 Senior Design Project II: (0-9-3) To prepare the students to finalize their final year project and present it in a reasonable and respectable manner.

Careers in Civil Engineering

The role of a Civil Engineer is the development of civil infrastructure and environmental protection procedures following sustainable practices. Civil Engineers accomplish this via stakeholders' feedback, survey reports, cost estimations, implementing standard test specifications and code of practices, and utilizing modern tools and software. The potential job market for the students of Civil Engineering encompasses infrastructure development such as Buildings and construction Enterprises, Highway Authorities, Water and Public Health Administrations, Railways and Airports, Tunneling and Dams Constructions, Environmental Agencies, Coastal and Marine Works, and Energy and power plant Production etc.



SCHOOL OF MANAGEMENT SCIENCES



Introduction

Today, the world of business is being transformed by the technological innovations, unique leadership models and the power of analytics, all defining features GIKI Department of Management Sciences. These aspects are drawing attention of the business community and corporate stakeholders towards revolutionizing the conventional academic practices by offering creative and interdisciplinary solutions. With our campus wide strengths, we are positioned to provide practical solutions to the emerging problems in the business world and educate the next generation of business executives to transform the existing structure of the corporate world.

The Bachelor's in Management Sciences, Program will focus on exploring business models from a societal and economic perspective, with particular emphasis on national, regional and international Supply Chain Management systems, and innovative entrepreneurial models in order to enable students to become pioneering icons in the corporate world. With so much attention being given to environmental concerns by world leaders and industry giants, another niche of this program will be enlightening the students with recent developments and discussions about sustainability in a long-term business context with special reference to Pakistan and South Asia.

The unique Program of BS in Management Sciences at Ghulam Ishaq Khan (GIK) Institute of Engineering Sciences and Technology would offer students once in a lifetime experience of interdisciplinary study, experiential learning and extraordinary co-curricular activities. Combining a solid academic foundation in management with multiple courses in the development of leadership and interpersonal

skills in cross cultural contexts will ensure that students become successful in adapting personally and professionally to changes in the global economy.

Our Program stands out since it focuses on the integration of courses from the domains of business leadership and sustainability with a technological perspective. The students will excel in making economic, societal, ecological and legal decisions from a managerial perspective once they become a part of local and international markets.

Some noteworthy features of our B.S program are as follows:

- Extensive contact with instructors
- Intellectual environment
- Case Study Methodology
- Individual attention
- Executive grooming

Innovative Features

Our Program stands out, since it effectively blends the academic aspects with experiential learning. Academically, the Program has a business development component and a number of specially designed courses emphasizing the role of technology and environment. The high-level curricular programming and the experiential learning embedded in the Program are particularly strong and also unique in the context of undergraduate management education in Pakistan. No other Program lays such an importance on innovation, sustainability and entrepreneurship, while simultaneously maintaining a quantitative and analytical focus.

Thrust Area's/Specializations

- Supply Chain Management
- Accounting and Finance
- Entrepreneurship and Marketing



Faculty Members

Sami Farooq, PhD (University of Nottingham, U.K)
 Maazullah Khan, PhD (Erasmus University Rotterdam)
 Yousaf Ali Khan, PhD (University of Macerata, Italy)
 Kashif Ullah Khan, PhD (University of Science and Technology of China, Hefei, China)
 Abid Ullah, PhD, (Ural Federal University Russia)
 Bushra Sarwar, PhD (University of Science and Technology, Beijing)
 Shahid Rasool, PhD (University of Malaysia Sarawak, Malaysia)
 Bilal Ahmed Memon, PhD (Jiangsu University, China)
 Muhammad Zeshan, PhD (University of Paris1, France)
 Sohaib, PhD (KoÇ University, Turkey)
 Izhar Ali, Juris Doctorate, (Ohio State University, USA)
 Waqas Rehman, MS (University of West of Scotland, UK)
 Hassaan Tariq, MSc (University of Warwick, UK)
 Umme Rabab Syed, M.S. (Sodertorn Hogskola, Stockholm, Sweden)
 Atta ur Rehman Jadoon, MS (Foundation University, Islamabad)
 Hira Ahad, M.Phil. (Forman Christian College, Lahore)
 Abrar Ahmad, MS (International Islamic University, Islamabad)
 Haseeb Ahsan, M.Phil. (National University of Modern Languages, Islamabad)
 Muhammad Shahzeb Fayyaz, MS (IMSciences Peshawar)
 Haroon-ur-Rashid, MS (Riphah International University, Islamabad)



Dean
Sami Farooq

Research Officers

Sabahat Orakzai MS COMSATS, Abbottabad, Pakistan)
 Saadia Ayaz Khan MS (UET, Taxila, Pakistan) – On leave

Graduate Assistants

Ms. Hina Shabbir
 Muhammad Ammad Cheema
 M. Junaid Tayyab
 Muhammad Salman Hassan
 Muhammad Ahmad Sethi
 Muhammad Hamza Farooq
 Habib Tariq
 Abdullah Khan
 Ahmed Owais Durrani
 Zeeshan Alam
 Azfar Ud Din
 Khuzaima Dar
 Ishaq Alam
 Fatima Anjum

P.S to Dean

Mr. Noor ul Bashar

Program Mission

- Educate and Develop such Business graduates who create value for their stakeholders and society at large.
- Build such a conducive environment for faculty and students which is diverse and inclusive, led by learning, entrepreneurial, and research orientation

Mode of Delivery

The courses will be delivered in the format normal to all GIK undergraduate courses. That is, typically, courses will be taught in small lecture sections permitting active involvement of students. Many of the courses will require group assignments and group work. In a number of courses, the cases will be used to complement normal teaching by topics. That is, students will be presented with cases in advance of the classes, and these cases will integrate what they have learned both in the course in question and in other courses.

Students will come prepared to the class for the discussion of the cases. Alumni and guest speakers will be invited to deliver lectures in-order to augment the elements of relevance and real-life applications. Some courses will provide students with the experience of actively engaging in situations that are simulated to reflect a variety of different work cultures. The simulations will require the students to incorporate theoretical knowledge gained in their current course and from other courses in the Program.

Program's Educational Objectives

The objectives of the Program are as follows:

- 1) Producing graduates with analytical and critical skills who can research and analyze complex business problems, design feasible alternatives and make efficient and effective decisions by blending theory with practice.
- 2) Producing professionally competent graduates who can take entrepreneurial initiatives, advance their knowledge and excel in their profession.
- 3) Producing graduates who understand group dynamics and can play their role as an effective team member.
- 4) Producing young business experts who understand the importance of ethical and sustainable practices and their

responsibility towards corporate social responsibilities in local and global markets.

- 5) Producing business executives with strong communication and interpersonal skills possessing command on composing and communicating effective business reports and presentations so as to add value to their job performance.

Program Learning Outcomes

- 1) Analysis - Able to utilize analytical and critical skills to research and understand complex business problems and apply data-driven solutions in a highly dynamic business environment.
- 2) National and International Business Understanding - Able to understand the local, regional and international risk factors arising out of cultural values, attitudes and belief systems and integrate them into strategic decision making process for SMEs and large scale organizations.
- 3) Teamwork and Leadership - Able to take advantage of problem-solving, negotiation, and team building skills and in turn reflect high degree of productive leadership.
- 4) Ethics - Able to identify ethical issues within an organization and recommend solutions in the light of ethical theories.
- 5) Communication - Able to compose and deliver audience centered presentations and business documents individually and as a team member.

Knowledge Area's

- Humanities
 - Language and Communication
 - Humanities
 - Social Sciences
- Management
 - Mathematics and Statistics
 - Law and Business Ethics
 - Management
 - Entrepreneurship and Marketing
 - Economics and Environment
 - Accounting and Finance
 - Supply Chain Management
 - Research Project

Sustainable Development Goal's

- **SDG 4 – Quality Education:** The school

is committed to provide Quality education for a diverse range of students. We take pride in providing equal opportunities to every student.

- **SDG 5 – Gender Equality:** School of Management Sciences believes in gender inclusiveness. Either its recruitment or student selection process, we always aim to maintain gender equality.
- **SDG 9 – Industry, Innovation and Infrastructure:** As a business school we believe that Innovation is the essence to any business/management education. Our efforts to promote Innovation are not only reflected in our coursework rather we also lead in providing students the opportunity to innovate through participation in various events.
- **SDG 13 – Climate Action:** At the School of Management Sciences we have introduced various courses that are directly related to sustainability. These courses ensure our commitment towards Climate and Environment. The courses include Sustainability in Supply Chains, Sustainable Development, Environmental Science, Corporate Social Responsibilities and many more.

Degree Nomenclature

- a) The Undergraduate Program in Management Sciences is of 4-years

duration, spread over 8 regular semesters, and consisting of 135 credit hours after completing twelve years of higher secondary school certificate or equivalent.

- b) The BS (Management Sciences) degree would require successful completion of a minimum of 42 courses (3 or 2 cr. hr. each) picked from the following streams. It specifies the minimum and maximum range in each category.
- c) Summer internship: Every student is required to participate in a compulsory internship/training program during the summer of junior year and submit a formal written report.



Compulsory Courses (27 Cr Hr)

Course Title	Course Code	Cr. Hr
Communication Skills	HM101	1
Communication Skills Lab	HM101	2
Critical Thinking and Expository Writing	HM102	2
Critical Thinking and Expository Writing Lab	HM102	1
Islamic Studies/Ethics	HM111	2
Pakistan Studies	HM112	2
Business Communication	HM203	3
Advance Oral Communication	HM404	3
Business Mathematics	MS101	3
Business Statistics	MS102	3
Computing and AI	CS101	2
Computing and AI Lab	CS101	1
Innovation and Makers Lab 1	IF101	1
Innovation and Makers Lab 2	IF102	1

General Courses (27 Cr Hr)		
Course Title	Course Code	Cr. Hr
Sociology	HM324	3
History	HM213	3
International Relations and Currents Affairs	HM221	3
Psychology	HM222	3
Anthropology	HM223	3
Business Ethics	MS311	3
Logic	MS312	3
Pakistan Economy	MS343	3
Environmental Science	MS444	3

Discipline Specific Courses (42-48 Credit Hrs)		
Course Title	Course Code	Cr. Hr
Fundamentals of Management	MS121	3
Human Resource Management	MS122	3
Principles of Marketing	MS131	3
Microeconomics	MS141	3
Introduction to Accounting	MS151	3
International Business	MS224	3
Marketing Management	MS232	3
Macroeconomics	MS242	3
Financial Accounting	MS252	3
Business Finance	MS253	3
Cost and Management Accounting	MS254	3
Business Research Methods	MS303	3
Business Policy	MS223	3
Consumer Behavior	MS333	3
Financial Management	MS355	3
Money and Banking	MS356	3



Elective/Specialization Courses (27-39 Cr. Hrs)			
Course Title	Course Code	Credit Hours	Pre-requisites
Management Information System (MIS & DSS)	CS302	3	CS201
Business Law	MS313	3	MS311
Organizational Behaviour	MS325	3	MS121
Financial Econometrics	MS357	3	MS102
Corporate Finance	MS358	3	MS253
Legal & Taxation Issues	MS359	3	None
Operations and Production Management	MS361	3	None
Supply Chain Management	MS362	3	None
Corporate Social Responsibility	MS414	3	MS311
Project Management	MS426	3	None
Entrepreneurship	MS434	3	MS131
Business Process Design and Analysis	MS435	3	MS131
Industrial Economy	MS445	3	MS141
Business and Economic Forecasting	MS446	3	MS102
Technology Management	MS463	3	None
Senior Year Project I	MS471	3	MS303
Senior Year Project II	MS472	3	MS471

*15 Credit Hours of Elective Courses will be taught.

Specialization Requirement (12 Credit Hours)

1). Accounting and Finance		
Course Title	Course Code	Credit. Hours
Accounting Information Systems	AF451	3
Auditing	AF452	3
Investment and Portfolio Management	AF453	3
Financial Risk Management	AF454	3

2). Entrepreneurship and Marketing		
Course Title	Course Code	Credit Hours
Business Plan for New Ventures	EM431	3
Digital Marketing	EM432	3
Corporate Entrepreneurship and Innovation	EM433	3
Services Marketing	EM434	3

3.) Supply Chain Management

Course Title	Course Code	Credit Hours
Planning and Control System	SC461	3
Business logistic Strategy	SC462	3
Sustainability in Supply chain Management	SC463	3
Lean & Six Sigma in Supply Chain Management	SC464	3

Semester Wise Breakdown

1 st Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Communication Skills	HM101	1	None
	Communication Skills Lab	HM101	2	None
	Islamic Studies/Ethics	HM111	2	None
	Computing and AI	CS101	2	None
	Computing and AI Lab	CS101	1	None
	Innovation and Makers Lab 1	IF101	1	None
	Business Mathematics	MS101	3	None
	Fundamentals of Management	MS121	3	None
	Introduction to Accounting	MS151	3	None

2 nd Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Critical Thinking and Expository Writing	HM102	2	HM101
	Critical Thinking and Expository Writing Lab	HM102	1	HM101
	Innovation and Makers Lab 2	IF102	1	IF101
	Pakistan Studies	HM112	2	None
	Business Statistics	MS102	3	None
	Human Resource Management	MS122	3	None
	Principles of Marketing	MS131	3	None
	Microeconomics	MS141	3	None

3 rd Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Business Communication	HM203	3	HM101
	History	HM213	3	None
	International Relations & Current Affairs	HM221	3	None
	Business Policy	MS223	3	MS121
	Macroeconomics	MS242	3	None
	Financial Accounting	MS252	3	MS151

4 th Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Psychology	HM222	3	None
	Anthropology	HM223	3	None
	International Business	MS224	3	MS121
	Marketing Management	MS232	3	MS131
	Business Finance	MS253	3	None
	Cost and Management Accounting	MS254	3	MS151

5 th Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Business Ethics	MS311	3	None
	Logic	MS312	3	None
	Pakistan Economy	MS343	3	MS242
	Financial Management	MS355	3	MS253
	Elective I	MS3XX	3	
	Elective II	MS3XX	3	

6 th Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Business Research Methods	MS303	3	MS102
	Sociology	HM324	3	None
	Consumer Behaviour	MS333	3	MS131
	Money and Banking	MS356	3	None
	Elective III	MS3XX	3	
	Elective IV	MS3XX	3	

7 th Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Environmental Sciences	MS444	3	
	Specialization Course I	AF/EM/SC4XX	3	
	Specialization course II	AF/EM/SC4XX	3	
	Elective V	MS4XX	3	
	Research Project I	MS471	3	MS303

8 th Semester	Course Title	Course Code	Credit Hours	Pre-requisite
	Advance Oral Communication	HM404	3	HM101
	Specialization Course III	AF/EM/SC4XX	3	
	Specialization Course IV	AF/EM/SC4XX	3	
	Research Project II	MS472	3	MS471

Total Requirements

For the award of BS in Management Sciences along with the options taken from specializations a student must complete 135 credit hours.



Description of Courses

HM101 Communication Skills (1-2-3): The course aims to equip the students with the necessary language and communication skills to cope with their academic and professional needs. The module effectively integrates the four basic skills of language i.e. reading, writing, listening and speaking. The oral communicative competence of the students is enhanced by focusing on the phonological aspects of language. The students are motivated to take part in the classroom sessions where they are encouraged to take the dynamics of stress and intonation in consideration while speaking. The learners are also introduced to the principles of effective writing from the sentence level to full-length texts with special emphasis on logical organization of materials.

HM101 Communication Skills (L): The lab offers opportunities to students to enhance their verbal and non-verbal communication and provide the platform to improve their receptive and productive skills.

CS101 Computing and AI (2-1-3): This course introduces the concepts of computing and programming to a freshman, starting from definitions and descriptions of software, hardware, their types and their history, memory and types of memory, and networks and their types. This is then followed by introducing the basic concepts of computer programming, algorithms, and software programming within the broader domain of computing. The course uses C/C++ as the programming language for demonstration and learning but is general

enough to be applied to any language.

CS101 Computing and AI (L): This lab introduces the concepts of computing and programming to a freshman, starting from definitions and descriptions of software, hardware, their types and their history, memory and types of memory, and networks and their types. This is then followed by introducing the basic concepts of computer programming, algorithms, and software programming within the broader domain of computing. The course uses C/C++ as the programming language for demonstration and learning but is general enough to be applied to any language.

HM111 Islamic Studies (2-0-2): The course of Islamic Studies presents Islam as a balanced mode of life by incorporating basic aspects of human rights and rule of law, brotherhood. Students are also taught the concepts relative to respect of other religions, equality of mankind and harmony between religion and practical aspects of life.

MS101 Business Mathematics (3-0-3): Business Mathematics presents mathematical skills and knowledge that students can apply to solve financial problems. The course provides step-by-step guidance through sample problems and solutions related to banking, credit, basic finance, and investment. Students will also gain an understanding of financial instruments and terminology used in business finance such as compound interest, annuities, and promissory notes. The course will cover topics like elements of Algebra; functions and their graphs; ratios; proportions and percentages; interest and annuities; basic statistical measures; and stocks and bonds.

MS121 Fundamentals of Management (3-0-3): Fundamentals of Management is an introductory course about the management of organizations. It provides guidelines on the principles of management that are applicable to all types of enterprises; basic management philosophy and decision making; principles involved in planning, organizing, leading, and controlling; global environment; managing change and innovation; human resource management introduction; organizational behavior; power politics, conflict, and stress; leading with influence; communication and



information technology; control systems; and operations management. The contents learned in this course will allow students to work effectively with others in an organization. The course will also encourage students to explore the applicability of western management principles and theories in local settings.

MS151 Introduction to Accounting (3-0-3):

This course is designed to provide the introduction to financial accounting and reporting to the students. The course is intended to teach the students to read, understand and analyze financial transactions. Then they will learn how to record these transactions in books of accounts. The desired results from these recordings will be calculated in the form of final accounts.

HM102 Critical Thinking and Expository Writing (2-1-3):

The course develops critical thinking and offers expository writing opportunities in order to produce technical and scientific report writing competence amongst the students. The students are effectively introduced to the underlying mechanics and conventions of technical or professional writing through a series of professional correspondence. The module effectively covers the topics of introduction to communication in technical and intercultural workplaces; identification of purpose of writing, techniques for the preparation for writing a document such as brainstorming outlining, drafting, editing and proofreading; technical writing style and strategies; use of brevity, politeness and accuracy in writing ; formatting and activities. It also equips the students with the techniques of tailoring the content of technical documents to the needs of various kinds of situations and audiences. The course focuses on the dynamics of designing technical reports and writing documents; writing emails, letters, memos, short reports, formal reports, executive summaries, abstracts, progress reports, white papers, and proposals; and presentation of information in oral and written format. The module effectively blends speaking and writing skills as the students are motivated to present their written reports orally in class at the end of the semester.

HM102 Critical Thinking and Expository Writing (L):

The lab provides learners a

platform to present their technical reports and get the feedback from the course instructor to improve their expository writing and develop critical thinking.

HM112 Pakistan Studies (2-0-2): This is an introductory course for examining the political behavior, processes, and government institutions. The course aims to give students an awareness of political ideas, theories, national systems, and public policies. The course effectively lays emphasis on critical analysis of the political issues in national and international level.

MS102 Business Statistics: The course is designed to introduce students to the major concepts and tools for collecting, analyzing, and drawing conclusions from the data. Data and information are integral to the operation and planning of all businesses, and as businesses grow and develop there is an increasing need for the use of formalized statistical methodology to answer business related questions. This course will cover topics like; types of variables and data; frequency distributions; data organization and presentation; numerical measures; measures of dispersion; index numbers, regression and correlation analysis; concepts of probability and sampling methods.

MS122 Human Resource Management (3-0-3):

This course is designed to provide the students with understanding of key HRM functions, designed to help them understand if western human resource management theories and practices have any relevance to local settings. The course will also discuss



the Islamic perspective of managing human resource. The students will also be encouraged to compare and contrast the human resource practices suggested in their textbooks and the practices critical for achieving success from indigenous perspective. The course contents include human resource planning; strategic human resource management; recruitment and selection; interviewing candidates; training and development; performance appraisal and performance management; career development, retention and voluntary/involuntary turnover; compensation; and global human resource management.

MS131 Principles of Marketing (3-0-3): This course will focus on developing an understanding of key marketing concepts aimed at improving the conceptual knowledge of marketing as applicable to decision making process with a focus on tactical marketing mix decisions. Furthermore, it will provide the students with a comprehensive framework to evaluate marketing decisions and to create successful marketing initiatives. The contents included in the course would be definition, evolution and future of marketing; marketing strategies and elements of marketing mix; elements and analysis of marketing environment; ethics and social marketing; strategic marketing planning; sales forecasting; and designing marketing plan.

MS141 Microeconomics (3-0-3): The course would provide an understanding of the principles of microeconomic analysis of business decisions in competitive and noncompetitive markets. The main topics in this course include supply and demand analysis; free markets; scarcity; production possibilities; the price system; government policy; labor markets;

capital, and natural resource markets, and externalities.

HM203 Business Communication (3-0-3):

The main objective of this course is to give students practical awareness of activities such as interacting, informing, instructing and persuading within the business community. It would bring in them the personalities of businesspeople as communicators, who can speak, write and interact with others effectively and professionally. The main topics of the course include introduction to communication, types of internal and external communications, types of formal and informal communication upward, downward and horizontal communication, use of technology in business communication; perception, adaptation and selection of appropriate words; writing emails, memos; and listening and speaking skills in business environments. This course aims at giving students an advanced understanding of the concepts and principles of professional business communication.

HM213 History (3-0-3): The course will help to inculcate historical consciousness about the human past and analytical and interpretative approach towards historical facts amongst the students. It will provide an in-depth, critical introduction to the theoretical, conceptual and epistemological foundations of the academic discipline. The students will be acquainted with the main political events, cultural and intellectual debates; religious movements and social issues. The students will also delve into the cultural heritage in South Asia and the world at large and develop an awareness of the political, constitutional and historical development in Pakistan before and after 1947. It will train the students to use historical knowledge to resolve the socioeconomic, political and intellectual problems of state and society.

HM221 International Relations and Current Affairs (3-0-3):

International relations course is designed to meet the needs of all those students who wish to enhance their understanding of the subject focusing on the changing political, economic and socio-cultural relations within the international system of the modern era. The subject explores the underlying global, regional, and domestic factors that influence relations between actors



on the world stage. It seeks to provide students with the knowledge of the global system tools to function effectively in the present, and the ability to respond to future developments. The course concocts the element of current affairs to establish a productive knowledge of the socio-political happenings in Pakistan and abroad. The course also aims to develop a cogent political thought amongst the students by focusing on the strategies that affect the countenance of national and international politics. It also delves into the elements of constitutions, governments, human rights and economic development from a sociopolitical point of view. Diplomatic relations along with the critical analysis of the dynamics of indigenous and international politics are accentuated in the subject.

MS242 Macroeconomics (3-0-3): The main objective of this course is to give students an understanding of the working of socialist, capitalist and mixed economy at the aggregate level. The basic themes are extended to explore the disciplines of national income, public finance, macroeconomics in closed and open economies, macroeconomic stabilization policies, money and banking link up with conventional macroeconomics.

MS252 Financial Accounting (3-0-3): This course is built upon the basic concepts of Financial Accounting and integrates theoretical and practical aspects of financial accounting and reporting. The course is

designed to introduce the students to the regulatory framework governing the preparation and publication of financial statements of a limited company and groups of companies. It mainly focuses upon the company accounts and their understanding in the context of IFRS and companies' ordinance 1984.

HM222 Psychology (3-0-3): This course surveys the major sub disciplines of the field, including such topics as the brain and neuroscience, behavioral genetics, cognitive and social development, perception, learning, memory, decision-making, language, consciousness, emotions, motivation, psychological disorders, social identity, interpersonal interactions and group and cultural processes. The course is designed to provide students with a well-rounded knowledge of the domains of psychology where the theories and practical aspects of the subject are introduced to them. It focuses on enhancing the knowledge human cognition, behavior and development. Furthermore, students will develop an understanding of the impact of cultural, social and environmental factors that affect mental health and well-being of an individual.

HM223 Anthropology (3-0-3): This course is design as an introduction to the discipline of anthropology as a whole. It presents students with a theoretical grounding in the major subfields of Cultural and linguistic anthropology. The main topics of this course includes the introduction to anthropology, principles of



anthropology, the cultural change and development, political anthropology & social problems, and the economic anthropology. In this class the emphasis is on the holistic nature of the discipline.

Students will be able to explore the history of the discipline and profession. The course objectives are set to learn the basic methods and theories, and the political and ethical dimensions of modern practice in the domain of anthropology. Similarly, the students will learn to examine and interpret evidence using specific examples, from artifacts to sites to regions. The course frequency is set for the business students at every year, there is no pre-requisite course required for this course.

MS224 International Business (3-0-3): This course aims to provide the students with an overview of the unique problems faced by firms engaging in international activities; the importance of understanding the foreign economic, social, political, cultural and legal environment; the mechanics of importing and exporting; joint venture, franchising and subsidiaries; international dimensions of management, marketing and accounting, international financial management. The course also delves into the special problems of multi-national corporations; recent problems of the international economic system; country-risk analysis; the increasing use of counter trade.

MS232 Marketing Management (3-0-3): This course is geared towards providing an understanding of the rationale for marketing decisions from a managerial perspective. Emphasis will be on including the knowledge of integrating theory and practice amongst the students. Students will have to apply analytical techniques they have learned in this

course, to make strategic marketing decisions. The course contents include introduction to marketing; marketing environment and market analysis, market research; buyer or consumer behavior; consumer decision making; marketing research and analysis; market segmentation, targeting and positioning; product concepts; product management; service and nonprofit marketing; pricing strategy; placing strategies; wholesaling and Industrial distribution; retailing; promotional strategies; and intercultural and international marketing.

MS253 Business Finance (3-0-3): The objective of this course is to introduce concepts and techniques of finance and build the foundations for all subsequent finance courses and provide basic knowledge of the analytical tools required by every student. This course is a rigorous introduction to the basic principles of finance and their applications to usual finance issues and decision making. The course provides an overview of business finance management and lays emphasis on the financial statement analysis, time value of money, and management of cash flow, risk and return and source of financing.

MS254 Cost and Management Accounting (3-0-3): This course focuses on the use of accounting information to report managerial performance and to facilitate business decisions. It covers the preparation and use of cost and management accounting information in planning, budgeting, break-even analysis, income determination, product costing, process control, corporate financing and capital structure decision making.

MS311 Business Ethics (3-0-3): It introduces students to ethical and moral issues, conflicts and decisions confronting citizens, groups and communities of Pakistan. The course aims at highlighting the necessity and importance of good character conduct and moral life as manifested in major world religions. The students are also motivated to appreciate the ethical and moral dimensions of Pakistani culture. The course contents include introducing and defining business ethics; social responsibility and development of ethical issues in business; application of moral philosophies to business ethics; ethical decision-making framework;



organizational influences on ethical decision making; influence of significant others in organizations; role of opportunity and conflict; development of effective ethics; and international business ethics.

MS312 Logic (3-0-3): Logic is fundamental to the way humans communicate. Our public debates and private reasoning are shaped by logical principles, even though most of us would struggle to spell them out. Introduction to Logic will teach the basics of formal logic, which provides symbolic methods for assessing and representing the logical form of arguments to the students. The students will develop an understanding of symbolic language and logic, as well as familiarity with precise models of deductive reasoning.

MS313 Business Law (3-0-3): No business enterprise can function without following the country's legal processes as well as its own. If it wishes to expand beyond local borders, then international laws and treaties also come into play. This course will look at company laws, contract laws, buying & selling goods & services, law relating to intellectual property, law of agency, company financial reports, elements of employment law, and management and employment ethics. Prudential Regulations of the State Bank of Pakistan and major regulatory frameworks under the Securities & Exchange Commission of Pakistan will also be discussed.

HM324 Sociology (3-0-3): The purpose of the course is to familiarize the students with the definition of sociology, founders of early sociology, three theories given by early sociologist, definition of culture and its development, elements of culture, cultural integration and cultural variation. The course also elaborates the components of society, social interaction, groups, social role, role of conflict institutions, verbal and non-verbal communication, social group bureaucracy, deviance, conformity; and, national and international social problems.

MS343 Pakistan Economy (3-0-3): This course aims to make students understand the key sectors of economy of Pakistan and contemporary issues in agriculture, industry and financial and social sector. It intends to give students a comprehensive knowledge

about the outlook and comparison of developing and developed economies with specific reference to Pakistan. To make students understand current policies in trade, commerce, fiscal/monetary policy, industry and agriculture. It is designed to provide students with critical information and knowledge about Pakistan economic environment. Important Components which contribute to the development and progress of the economy of country are included here.

MS325 Organizational Behavior (3-0-3): The knowledge of individuals' perceptions, motivational attitudes and behavior enable students to not only understand themselves better, but also to adopt appropriate managerial policies and leadership styles to increase their effectiveness. The focus of instruction will move progressively through the individual, group and organizational levels of behavior and will examine the interrelationships of behavioral phenomena among these levels. Specific topics include leadership, motivation, teamwork, career issues, work roles, job enrichment, employee participation, and work and non-work integration.

MS355 Financial Management (3-0-3): The main objective of this course is to develop a foundation of financial management concepts. This will enable the students to understand how corporations make important investment and financing decisions and establish working capital policies. The course also lays a foundation for more complex financial topics that arise in additional elective courses in finance. This course introduces the students to fundamental principles of finance i.e., time value of money and relations between risk and return, MM theories of irrelevance and efficient markets. Through exposure to these fundamental



concepts, it is expected that the students will learn how to apply their knowledge the valuation of stocks and bonds, financial planning, capital budgeting, long-term financing, capital structure, dividend policy, working capital management, and risk management.

MS303 Business Research Methods (3-0-3):

The objective of this course is to expose students to the principles and methods of business research and encourage them to explore application of theories that have been predominantly developed in Western cultures by using different research techniques. An understanding of the relevance of Western research for local practice would help students to explore various business-related problems and their plausible solutions from an indigenous perspective. Topics would include: introduction to research methods in business and management disciplines; literature searching strategies; literature review; research paradigms and approaches; theory and research; introduction to SPSS; quantitative research design; internal and external validity; survey-based research; case study research; quantitative and qualitative data analysis; and writing research proposals and thesis.

MS223 Business Policy (3-0-3): The course focuses on the formulation and implementation of corporate policies. The knowledge and techniques learned in earlier courses will be applied in an integrated fashion to the process of strategic decision making and organizational change. The topics considered in the course will be relationships of organizations to their environments, the hierarchy of organizational objectives, structured as well as informal approaches to strategic planning,

the integration of business functions, organizational structure, and policy implementation and evaluation. A significant aspect of the course is devoted to assessing the competitive dynamics of firms.

MS333 Consumer Behavior (3-0-3): This course is designed to enhance students' understanding of how and why consumers purchase (or do not purchase) goods and services. It will combine both the theoretical concepts of consumer behavior and its application for marketing strategies related to private, public and non-profit sections. At the conceptual level, it will seek to present an integrated framework around which major areas of consumer behavior can be understood and applied. This course will explore and identify market identities and various sources of influence with the way consumers think and learn from market related information. The knowledge and understanding gained from this course can be utilized in the marketplace to make rational decisions to satisfy consumer needs and wants and remain loyal to products. Specific topics to be covered include cognition process, consumer motivation, emotions; consumer cultural theories; consumption, meaning and identities; and role of market and consumer ideologies.

CS302 Management Information Systems (MIS and DSS) (3-0-3): Information systems began as automation of office systems and have grown into systems that assist managers to make decisions, systems that model successful business practices, and systems that transform the modern business into knowledge-based enterprise. New types of infrastructure and applications are developed and utilized such as ERP (enterprise resource planning), IOS (inter-organizational systems), RIFD (radio frequency identification), and CRM (customer relationship management) to name a few. This course helps students see the connection between information systems (IS) and business performance and explores current information systems concepts and technologies. Students will learn how information systems give a business competitive edge by providing technologies that help managers plan, control and make decisions. Included in course are topics such as hardware and software components of Information systems, e-business concepts and



implementation and survey of common information systems used today.

MS356 Money and Banking (3-0-3): The course will cover both theory and practice of money and banking. The first part of the course will cover commercial and central banking. Topics on commercial banking will include economics of banking; the role of banks in the presence of asymmetric information; examining bank's balance sheet operations with their need to balance; asset, liability, liquidity, and risk and return. The second part will focus on State Bank of Pakistan's role in monetary policy and supervision of banking sector through prudential regulations.

MS357 Financial Econometrics (3-0-3): This course focuses on techniques for estimating regression models, problems encountered in estimating such models, and interpreting their output. The goal of the course is to teach students the basics of the theory and practice of econometrics and to give them experience in estimating econometric models with actual data. The course covers the topics like single equation regression models, regression analysis, two-variable and multiple regression analysis, econometrics modeling, and time series econometrics.

MS358 Corporate Finance (3-0-3): This important module aims to develop the analytical skills for making corporate investment with regards to financial decisions and risk analysis. This course will examine various theories including the concept of present value, the opportunity cost of capital, discounted cash flow analysis, a consortium of valuation techniques, issues between short & long term financial management, risk and return, capital asset pricing model, capital budgeting, corporate capital structure and financing decisions, dividend policy, investment and financial decisions in the international context, including exchange rate/interest rate risk analysis, and issues of corporate governance and control. This course also explores the very patterns of corporate finance that has shaped the familiar yet complex terrain of today's global economy.

MS359 Legal and Taxation Issues (3-0-3): This course is designed to give students a familiarity of various forms of organizations

and the rights and responsibilities of its officers, employees, and shareholders; taxation of the various organizational forms; patents and other forms of intellectual property issues; contract law particularly as it applies to licensing, leases, employees and insurance; and ways to mitigate various forms of risk.

MS361 Operations and Production Management (3-0-3): This course introduces the theory and practice of operations and production management as a functional area in the management of business enterprise. This course will discuss the principles, concepts and basic problems affecting manufacturing and non-manufacturing firms. Topics covered will explore Tangible & intangible functions of production, Discrete & continuous manufacturing processes & systems, Conventional & system approach to Design, Production planning & Control, Process planning, Quality control, Quality assurance, Assembly methods, Packaging, Production work measurements, Production standards, Production philosophies, Operations Strategy, Managing Processes, Process Strategy, Process Analysis, Process Performance & Quality, Constraint Management, Process Layout, managing value chains, Forecasting, Sales & Operations planning, Resource planning, Linear Programming, Scheduling.

MS362 Supply Chain Management (3-0-3): Supply Chain Management includes the materials and information flow among all firms that contribute significantly to a product, from the point of scratch to final product. Elements of supply chain management have been studied and practiced for some time in marketing, logistics, and operations management. This



course will integrate different perspectives from various functions of management to develop a broad understanding of how to manage a supply chain. Topics include Value Chains, Supply chains, Supply chain lifecycle, Supply chain strategy, Resource planning, Procurement, Inventory models, Inventory management, Automated Inventory Tracking System, Sales & Operations Planning, Forecasting, Scheduling, logistics, Contracts, Supply Chain Technology, Distributed Requirement Planning.

HM404 Advance Oral Communication (3-0-3):

The course of Oral Communication has been designed to help students devise oral, interpersonal and physical strategies required to confidently and effectively interact with a variety of audiences. In this module, the students are taught to effectively eclecticize the oral rhetoric with the paralinguistic features and visual aids to deliver successful oral presentations. Apart from other skills students will also achieve excellence in creativity and flair; oral business communication; and problem solving. The major skills targeted in the course are Tutorial Presentations, Poster presentation, Seminar Presentations, Conference Presentations, thesis defense, viva Skills and Negotiation Skills.

MS414 Corporate Social Responsibility (3-0-3):

This introductory course will include the principles of private businesses supporting communities and people. The challenge is often to find a balance between doing good and leveraging these practices to benefit business community and its constituents. This course covers CSR methods, tools, principles,

and practices at the organization and society level. The course is designed to give students a general knowhow of CSR, its general implementation and management in an organization. The contents to be covered in course are legal and economic perspectives on CSR; ownership theory; market and stakeholder's analysis; contemporary public and social issues involving business; global natural environmental issues; technological issues influencing economy and society; community relations and strategic philanthropy; role of government in CSR; and social audit.

MS426 Project Management (3-0-3):

The module focuses on the topics of fundamental principles, Project life cycle, Project organizations and human resource management, PM planning, Work breakdown structure, Estimating time and cost, Precedence relationships, Project scheduling and control technique, Project risk analysis, Time compression and resource levelling, Computerized project management, Special issues in software projects.

MS434 Entrepreneurship (3-0-3):

This course focuses on identifying business opportunities and developing them into a business. The management functions of accounting, finance, and marketing as well as legal and economic considerations are catered in the subject. Students are taught to take business responsibilities and initiatives as business strategies are created. Through the process of developing a business plan, students acquire various necessary skills to operate a successful business. Topics of discussion will include history of entrepreneurship, idea generation, technology and ideas, sources of finance, elevator pitch, patents and formulation of business plan.

MS435 Business Process Design and Analysis (3-0-3):

This course includes identification, development, analysis, controlling, enhancement and management of business processes. Examples from different industries and functional areas within firms would be employed in the course to identify similarities and differences of well-run processes.

MS444 Environmental Sciences (3-0-3):

The environment impacts our way of life in many aspects (e.g. food and fiber production,



resources for building shelter and infrastructure, and water supplies). Adverse impacts to this environment affect the well-being of humans and other living organisms. Therefore, the broad topics covered will include natural environmental systems, physical and social causes of environmental problems, and strategies to mitigate or manage these issues. The course will also address issues like sustainable communities and sustainable development.

MS445 Industrial Economy (3-0-3):

Industrial Economy focuses on the study of firms, industries and markets. When analyzing decision making at the levels of individual firm and industry, Industrial Economics helps in understanding issues like; the levels at which capacity, output and prices are set, the extent to which the products are differentiated from each other, how much firms invest in research and development (R&D), how and why firms advertise. Industrial economy will cover the topics of size and structure of firms, separation of ownership and control, short-run price competition, dynamic price competition, entry deterrence and entry accommodation, product differentiation and non-price competition, price discrimination, vertical relations, the determinants of market structure, competition and industrial policy regulations.

MS446 Business and Economic Forecasting (3-0-3):

This course will examine a more rigorous approach to various financial, econometric and time series approaches for predicting the effects of future corporate planning decisions and policies. The course would include topics of forecasting methods with single equation models, predicting with quantitative as well as qualitative choice models, and simulation with single and multi-equation models. These techniques are used to predict product sales, economic variables, and financial indicators. The course will be useful for recipients and users of forecasts and the ones involved in conducting business and economic forecasts.

MS 463 Technology Management (3-0-3):

This course introduces the students to the concepts of Industrial Networks, Fundamentals of product and process development, Business Community and new generation of managers, Practical Skills, knowledge and experience

in commercialization of new technological innovations. The module also delves into the core ideas of use of multi-disciplinary science based knowledge, Problem-solving, Team work, Outreach activity, Major steps in proof of concept to intellectual property protection, Proto-type development, Fabrication and assembly routes, Materials procurement and Identification and Creation of new markets, Development of Business plan, Appropriate technology and marketing, Distribution and Financing, Routes and strategies for specific technology under development.

EM431 Business Plan for New Ventures (3- 0-3):

Developing a business plan for a new venture and the entrepreneurial process of executing the first phases of new venture creation can be daunting. This course will explore areas like idea conception, entrepreneurship, business planning, market research, entrepreneurial opportunities and strategies, venture analysis and strategy, industry and competitor analysis, marketing plan and risk assessment. Emphasis is placed on high growth business opportunities. The final deliverable will be a complete business plan for a high growth venture and formal presentation of the plan to mock investors. Some individual off campus travel will be required.

EM432 Digital Marketing (3-0-3):

The evolution of traditional mass media has highlighted the importance of digital advertising. This course is designed to give students a thorough understanding of assessment and evaluation of the businesses in the new technological era. The course will also delve into the details for preparing a digital marketing plan for entrepreneurial and business expansion opportunities. Furthermore, it will help the students to plan and develop digital platforms for online businesses and e-commerce solutions.

EM433 Corporate Entrepreneurship and Innovation (3-0-3):

This course focuses on the processes by which teams within an established company conceive, foster, launch and manage a new business that is distinct from the parent company. Following the market development lifecycle, students will examine the management of eight types of innovation: disruptive, application, product, process, experiential, marketing, business model and

structural. The ability to simultaneously integrate and differentiate between a company's existing and new business is crucial to the success of any corporate entrepreneurship effort. As an advanced course, emphasis will be placed on skills needed to promote and manage corporate entrepreneurship including opportunity recognition, selling an idea and conflict management. Furthermore, in this course, we will examine the apparent contradiction in the term "corporate entrepreneurship" and discuss managerial techniques and organizational structures that promote entrepreneurial behavior in a corporate context. Attention will also be given to the difficulties inherent in the process of assessing entrepreneurial performance within a corporation and control mechanisms put in place by corporate managers to restrict the economy of corporate entrepreneurial initiatives.

EM434 Services Marketing (3-0-3): The objective of this course is to give students an understanding of the strategic and managerial issues adherent to marketing services in industries with high service components. It will also provide the learners with an insight to the distinguishing aspects of service marketing thereby advancing their ability to apply marketing strategies to create, communicate and deliver customer value in the service economy.

AF451 Accounting Information System (3-0-3): This course applies the practical application of accrual-basis accounting using accounting software. Students will gain experience in integrated software designed to

handle general ledger, accounts payable, accounts receivable, financial statement analysis, fixed assets, sales order processing, inventory, and payroll. Students will explore various topics in AIS to understand and use technologies in making decisions in specialized areas of the accounting profession, such as managerial accounting, financial accounting, auditing, and tax accounting. Topics include types of AIS applications and systems, technologies and database concepts, internal control issues, audit issues, and systems development issues and current trends in AIS.

AF452 Auditing (3-0-3): This course provides managerial understanding of the auditing process and its importance for effective decision making. The course will cover auditing theory and practice; generally accepted auditing standards; code of ethics; systems of internal controls and its evaluation; compliance; and integrity of information. The course will motivate and prepare students to earn prestigious and globally recognized 'Certified Internal Auditor' certification.

AF453 Investment and Portfolio Management (3-0-3): The focus of this course is on financial theory and empirical evidence that are useful for investment decisions and provide a comprehensive, analytical approach to modern theory of investments. Topics covered include mean variance analysis, Markowitz type portfolio analysis, portfolio construction, asset pricing theory, market efficiency and anomalies, hedge funds and investment funds performance evaluation. Topics include modern portfolio theory,



fundamental and technical analysis of equities, concentrated equities positions, fixed income analysis, benchmarking, capital markets, the appropriate use of mutual and importance of asset allocation.

AF454 Financial risk management (3-0-3):

The course will focus on variety of risks faced by financial managers and tools available for managing these risks. Particularly, it focuses on credit risk, interest rate and liquidity risk, market risk, foreign exchange risk and country risk. The students will learn about the tools and techniques for managing these risks such as future contracts, option contracts, swaps, value at risk, and other standard risk hedging techniques and methods of measuring volatility. Students attending this course are expected to have studied a basic course of investment and portfolio management and have a good understanding of asset pricing models. This course also examines theoretical and practical aspects of risk management with emphasis on the effective use of future options and other financial derivatives to control market risk exposure. Reviews no- arbitrage methods and options, including the Black-Scholes model and binomial tree numerical methods.

SC461 Planning and Control Systems (3-0-3):

This course includes the design and management of planning and control systems within the organization and across the supply chain. It covers business planning; master production scheduling; material requirements planning; just-in-time and theory of constraints, Enterprise resource planning (ERP) and business-to-business (B2B) systems; impact of information technologies on planning and control systems.

SC462 Business Logistics Strategy (3-0-3):

Logistics and distribution are the core components of supply chain management. Logistics management plans, implementation and controls efficient, effective forward and backward flow, storage of goods and services. It also predicts and circulates timely related information between the point of origin and the points of production, purchase and consumption in order to meet customers' requirements. Logistics decisions are typically classified into 1) strategic: dealing with decisions that have a long-lasting effect on the firm; 2) tactical:

including decisions that are updated anywhere between once every quarter and once every year; 3) operational: referring to day-to-day decisions. For this course, the focus will be on strategic and tactical decisions in logistics management.

SC463 Sustainability in Supply Chain Management (3-0-3):

Sustainability in Supply chain Management is a modern concept of management practices attempting to integrate environmental concerns to all stages up and down the supply chain. In a globalized market, the environmental performance criteria extend beyond the single firm to its entire supply chain network across national borders. Topics covered will include closed-loop supply chains; reverse logistics systems; carbon foot printing; water foot printing; life-cycle analysis; and supply chain sustainability strategy.

SC464 Lean and Six Sigma in Supply Chain Management (3-0-3):

Today's world-class companies have been successful in eliminating process inefficiencies and streamlining management hierarchies to cut costs, improve quality and become more responsive to customers. It's more important than ever to increase organizational effectiveness through engagement of people, elimination of waste and cultivation of innovative teams who are constantly striving to improve. These are all part of a management concept known as Lean management. The aim of this course is to go through the entire Lean and six sigma Model, to reinforce critical concepts and implementation methods. By the end of course, participants will have a solid base of knowledge of Lean Management and six sigma concepts and how to build the right behavior in their organization to increase business performance.



Service Courses for Engineers

Introduction: Minors in Management for Engineers:

Globalization has brought new challenges of sustainability, health, and environmental protection; therefore, a new breed of managers is required by companies and organizations to cope up with these issues. Ghulam Ishaq Khan Institute of Engineering and Technology offers Minors in Management, geared towards helping engineers/technologists develop planning, decision making and managerial skills while receiving advanced technical knowledge. It is intended to prepare graduates with the management skills needed to provide engineering leadership in today's multi-disciplinary business environment. The primary focus of the program is on management and application of business skills to engineering leadership situations. The Management Sciences outlook and approach is interdisciplinary within the variety of engineering pursuits.

Educational Objectives: The courses have been specifically designed to:

1. Prepare managers and leaders for engineering and industrial organizations by exposing students to modern concepts of economics, production sciences, and enable them to manage important human as well as financial resources within the enterprises.
2. Familiarize students with the fundamental principles of manufacturing, risk management, project management, and maintenance management.
3. Teach them innovative techniques which can be utilized to manage modern industries.

4. Instill the spirit of entrepreneurship, which will enable them to forge new avenues in the modern economy, and provide them with better foresight, and greater financial flexibility.
5. Equip students with English Language and communication Skills with special emphasis on business communication and technical writing. Reintroduce them to their history, religion and culture.

Professional Outcomes: Courses prepare students in effectively managing the financial, human, and physical resources within the modern economy. They are designed to impart strategic, tactical and operational level knowledge to students in order to enable them to be better managers, analysts, entrepreneurs, and business executives.

The courses also prepare them for academic reading and accurate professional writing. Their presentation skills are improved through class seminars and group discussions which in turn would help them exchange their views and communicate their experience in research with professional colleagues and potential employers.

Seminars: A series of seminars dealing with wide-ranging issues of topical significance are organized at the institute where the students are provided with the platform to hold brain storming sessions and interact with eminent from scholars from various disciplines. The seminars aim at arousing interest of students in current problems, helping them form enlightened opinions about them, and develop skills for rational discourse and argumentation.



Course Title	Course Code	CH
Communication Skills	HM101	1
Communication Skills Lab	HM101	2
Critical Thinking and Expository Writing	HM102	2
Critical Thinking and Expository Writing Lab	HM102	1
Pakistan and Islamic Studies / Ethics / Impact of Science and Technology on Society	HM211	3
Engineering Economics	MS291	3
Sociology and Human Behavior	HM321	3
Corporate Law and Professional Ethics	MS322	3
Human Resource Management	MS412	3
Technology Management	MS426	3
Entrepreneurship and Marketing	MS434	3
Accounting and Finance	MS447	3
Macro and International Economics	MS448	3
Industrial Management	MS449	3
Lean Enterprise Management	MS489	3
Supply Chain Management	MS491	3
Operations Management	MS492	3
Industrial Safety	MS493	3
Total Quality Management	MS494	3
Maintenance Management	MS495	3
Project Management	MS496	3

Description of Courses

HM101 Communication Skills (1-2-3): The course aims to equip the students with the necessary language and communication skills to cope with their academic and professional needs. The module effectively integrates the four basic skills of language i.e. reading, writing, listening and speaking. The oral communicative competence of the students is enhanced by focusing on the phonological aspects of language. The students are motivated to take part in the classroom sessions where they are encouraged to take the dynamics of stress and intonation in consideration while speaking. The learners are also introduced to the principles of effective writing from the sentence level to full-length texts with special emphasis on logical organization of materials.

HM101 Communication Skills (L): The lab offers opportunities to students to enhance

their verbal and non-verbal communication and provide the platform to improve their receptive and productive skills.

HM102 Critical Thinking and Expository Writing (2-1-3): The course develops critical thinking and offers expository writing opportunities in order to produce technical and scientific report writing competence amongst the students. The students are effectively introduced to the underlying mechanics and conventions of technical or professional writing through a series of professional correspondence. The module effectively covers the topics of introduction to communication in technical and intercultural workplaces; identification of purpose of writing, techniques for the preparation for writing a document such as brainstorming outlining, drafting, editing and proofreading; technical writing style and strategies; use of brevity, politeness and accuracy in writing ; formatting and activities. It

also equips the students with the techniques of tailoring the content of technical documents to the needs of various kinds of situations and audiences. The course focuses on the dynamics of designing technical reports and writing documents; writing emails, letters, memos, short reports, formal reports, executive summaries, abstracts, progress reports, white papers, and proposals; and presentation of information in oral and written format. The module effectively blends speaking and writing skills as the students are motivated to present their written reports orally in class at the end of the semester.

HM102 Critical Thinking and Expository Writing (L): The lab provides learners a platform to present their technical reports and get the feedback from the course instructor to improve their expository writing and develop critical thinking.

HM211 Pakistan and Islamic Studies (3-0-3): The course introduces students to the origins and development of Muslim nationalism in South Asia and the struggle for freedom in the wider historical perspective. It also examines the political, socio-cultural and economic aspects of the state and society of Pakistan with reference to the ideals and concepts of its founding fathers.

Islamic Studies presents Islam as a rational code of life with emphasis on Islamic perspectives on fundamental human rights, rule of law, brotherhood and equality of mankind, empirical and rational basis of knowledge and harmony between the religious and the scientific domains of experience.

HM211 Ethics (3-0-3): The course is offered to Non-Muslim students in place of Islamic Studies. It introduces students to ethical and moral issues, conflicts and decisions confronting the citizens, groups and

communities of Pakistan. The course aims at highlighting the necessity and importance of good character, conduct and moral life as manifested in major world religions. The students are also enlightened to appreciate the ethical and moral dimensions of Pakistani culture.

HM211 Impact of Science and Technology on Society (3-0-3): This course, too, is offered to non-Muslims students in place of Islamic Studies as a second option. They are given sociological perspectives on cultural changes and the role played by scientific and technological innovations affecting such changes. The main topics dwelt upon are the holistic character of cultures, the phenomenon of cultural lag and the resultant socio-cultural dislocations, the social order and technology nexus, and the impact of technological advancements on social institutions.

MS291 Engineering Economics (3-0-3): This course delves into the areas of cost concepts, Money time relationships, Measures of worth, Performance analysis from final accounts, Decision-making, Brief introduction of the quantitative techniques and of the behavioural aspects.

HM321 Sociology and Human Behavior (3-0-3): The purpose of the course is to familiarize students with Pakistani parameters with factors that shape a society, theories about personality development, cultural change, socialization, functioning of normative systems, cultural diffusion, social mobility, sub-cultures and counter-cultures, cultural relativism, social stratification, and social institutions.

Organizational behavior, Management and employment, HRD (Human Resource Development)



MS322 Corporate Law and Professional Ethics (3-0-3): The course introduces student to the ethical and moral issues they are likely to confront as Engineers such as the vital impact their work has on health, safety and welfare of people. Major theories of moral development and code of ethics, prescribed by the professional bodies and case studies illustrating ethical and moral dilemma engineers have to cope with. The legal component deals with the constitutional provision regarding fundamental human rights, principles of natural justice, basic aspects of contract law, arbitration, partnership, evidence law, labour laws, and drafting legal documents used in contractual transactions.

MS412 Human Resource Management (3-0-3): The course focuses on the topics of Design and execution of Human resource management strategies, Systematic and strategic thinking about aspects of managing an organization's human assets, Implementation of policies to achieve competitive advantages, Reward systems, Performance management, High-performance human resource systems, Training and development, Recruitment, Retention, Equal employment, Opportunity laws, Work force diversity, and Union management relationships.

MS426 Technology Management (3-0-3): The course focuses on the topics of Industrial networks, Fundamentals of product and process development, Business community and new generations of managers, Practical skills, Knowledge and experience in commercialization of new technological innovations, Use of multidisciplinary science based knowledge, Problem-solving, Teamwork, Outreach activity, Major steps in proof of concept to intellectual property protection, Prototype development, Fabrication and assembly routes, Materials procurement, Identification and Creation of new markets, Development of business plan, Appropriate technology and marketing, Distribution and financing, Routes and strategies for specific technology under development.

MS434 Entrepreneurship and Marketing (3-0-3): The module focuses on the topics of Industrial economic strategy, Preparation of a business plan for new ventures and financing

options for start-up business, Barrier to entry, Corporate governance, Mergers information gained through environmental scans on new business opportunities, Case studies, Sharing the experiences of entrepreneurs and investors, Consulting for inventing start-up or entrepreneurial businesses and for professionals.

MS447 Accounting and Finance (3-0-3): The module focuses on the topics of Financial reporting, Financial Statements, Financial statements as management planning tool, Statements of cash flows, Revenue and expense reorganization, Account receivables, Inventories, Tangible and intangible assets, Liabilities, Bonds, Income taxes, Shareholder's equity, Accounting control, EVA, LIFO, FIFO.

MS448 Macro and International Economics (3-0-3): The module focuses on the topics of International fiscal policies, Macro-environment for firms and organizations, Basic tools of macro-economic management, monetary policy, and Exchange rate policy. Evaluation of the different strategies for economic development including Trade policy, Industry policy, and Natural resource policy. Market crises, Risk management and strategies for future. Major challenges in developed and underdeveloped countries for global integration, Inequality and asset price bubble.

MS449 Industrial Management (3-0-3): The course deals with the principles of industrial management. It focuses on effective and innovative ways of managing physical, human, financial and time resources of industrial and business organizations. It aims at preparing the students to develop a greater awareness of the contemporary trends in organizational management. The course makes an attempt to equip the students with theoretical knowledge and practical skills necessary for a good manager.

MS489 Lean Enterprise Management (3-0-3): The module addresses some of the important issues involved with planning, development, and implementation of lean enterprises. The dimensions of People, Technology, Process, and Management of an effective lean manufacturing company are considered in a unified framework. Particular emphasis is on the integration of these dimensions across the

entire enterprise, including Product development, Production, and Extended supply chain. Analysis tools as well as future trends and directions are explored. A key component of this subject is a team project.

MS491 Supply Chain Management (3-0-3):

Supply Chain Management includes the materials and information flow among all firms that contribute significantly to a product, from the point of scratch to final product. Elements of supply chain management have been studied and practiced for some time in marketing, logistics, and operations management. This course will integrate different perspectives from various functions of management to develop a broad understanding of how to manage a supply chain. Topics include Value Chains, Supply chains, Supply chain lifecycle, Supply chain strategy, Resource planning, Procurement, Inventory models, Inventory management, Automated Inventory Tracking System, Sales & Operations Planning, Forecasting, Scheduling, logistics, Contracts, Supply Chain Technology, Distributed Requirement Planning.

MS492 Operations Management (3-0-3):

This course will provide the students with the necessary knowledge of the basics of managing, manufacturing and Service organization, Strategic decision making, Facility location and layout, Job design and work

compensation, Demand forecasting, Capacity and material planning, Scheduling in various environments, Emerging trends in managing operations, focus on selection and use of quantitative management tools after introducing the fundamental concepts.

MS493 Industrial Safety (3-0-3):

The course aims to focus on the topics of safety regulations and safety management, office safety and manual handling safety of chemical, Fire safety, Radiation safety, Shop floor safety, Machine guarding and robotics safety, Construction safety, Electrical and pressure safety, Environmental protection, Occupational health, First aid basics, and Risk evaluation and management.

MS494 Total Quality Management (3-0-3):

The course contents include Fundamental principles of quality, Standards, Techniques for quality analysis and improvements, Statistical methods to measure quality, and SPC (Statistical Process Control). Acceptance sampling; QFD (Quality Function Deploying), Value engineering, Cross functional management, and benchmarking. ISO-9000 application, clauses and implementation issues.

MS495 Maintenance Management (3-0-3):

The module aims to develop an underlying knowledge of the organization and control of maintenance systems, Maintenance



policies and strategies, Preventive maintenance, Predictive maintenance and condition improvement, Total productive maintenance, Reliability and failure analysis, Scheduling maintenance, Unique challenges of software maintenance, Maintenance performance measure benchmarking and improvement.

MS496 Project Management (3-0-3): The module focuses on the topics of fundamental principles, Project life cycle, Project organizations and human resource management, PM planning, Work breakdown structure, Estimating time and cost, Precedence relationships, Project scheduling and control technique, Project risk analysis, Time compression and resource levelling, Computerized project management, Special issues in software projects.

Career Opportunities with a BS degree in Management Sciences

Our recent graduates have been placed in well

known Multi-National and National companies. Graduates with a BS Management Sciences degree are highly sought in the Industry. The degree opens up multiple avenues for students where they can place themselves. Apart from student startups, our graduates have been hired by the following companies.

- Philip Morris International
- Pakistan Telecommunication Company (PTCL)
- Mitchel's
- Foodpanda
- KraveMart
- AirLift
- Johnny and Jugnu
- Pfizer
- Kerry Logistics, KSA
- Careem
- Dastagyr
- NatWest Bank, UK
- Sanofi
- And many more



OFFICE OF STUDENT AFFAIRS



Dean Student Affairs

Dr. Muhammad Imran Khan
Ph. D. (University of Tsukuba, Japan)

Sabahat Hussain, Assistant Director Student Affairs
Sumira Siddique, Student Wellness Counselor
Arshia Shehzadi, Warden

The Student Affairs Office serves as a liaison between students, faculty and administration. The primary function of Student Affairs Office is to be of assistance to students in any way possible. It leads, directs, and administers overall functions of student counseling, hostel residence, student societies and discipline. The important function of Student Affairs Office is to enhance the quality of student life both in and outside of the classroom.

Overseen by Dean Student Affairs, this office is a central place for students where they can express and get help about any situation they encounter on campus whether it is academic, personal or emotional.

The Student Affairs Office has a full-time Assistant Director Student Affairs, who is readily available to students for advice and help. The students are encouraged to have recourse to him for advising and guidance. Further, guidance is provided as to how our students can spend an enriched all-round intellectual life on campus because we prepare students not only for hours they spend in their classrooms but also for the many more hours of their leisure time for their personal

lives. Outside of classrooms, thoughtful efforts are made to groom our students and provide enriching experience.

The Student Affairs Office functions as a friend and guide of students, it administers their needs from the time they step in the Institute for admission, to the time they graduate from the Institute. It provides forums for cultivation of their literary and artistic potentialities and furnishes them healthy outlets in sports and games which make their stay on campus, a rich experience of multi-dimensional growth.

The student Affairs Office provides proactive support and capacity building services to promote co-curricular activities in the Institute which could enrich our graduates at every stage of their life as well as they could build strong relationships with their peers, faculty, administration and other stakeholders.

Student Counseling Services

The Office of Student Affairs promotes a sense of community and belongingness among students. The student counselor works regularly with students to help them in their personal issues. Further, they are motivated in their academic matters and encouraged how to nourish their life with plentiful intellectual activities. Also, relaxation methods are offered to deal with academic pressures if any. We assure the students the complete confidentiality of their concerns.

Societies and Clubs

The institute is situated in calm and tranquil countryside. The campus of the institute is a self-contained cosmos. Here the students generate a fullness of life for themselves, and their superabundant energies find outlets in



a host of socio-cultural activities. Through generous financial allocations and faculty supervision, the institute supports all modes of constructive self expression for the students. There is thus much on the campus by way of recreational facilities, which mitigate the rigors of very demanding academic engagements. To nurture the intellectual and recreational activities, students are encouraged to join student societies in GIK Institute which are managed by students with the support of faculty and administration. Each society is assisted by a faculty advisor. Different kinds of technical and recreational events are organized by these societies which range from scientific to art competitions.

Student Societies at GIK Institute

GIK Institute recognizes a wide variety of student societies to facilitate diverse interests of the student body. Office of Student Affairs is the governing body for all societies that are administered by student representatives. Societies are granted certain privileges to promote student participation in a variety of physical, intellectual and recreational activities. There are many student societies in GIK Institute which support contemporary scientific and engineering perspectives as well, and are devoted to advance theory and practice of their respective fields. Few of student societies which have contributed to GIK Institute, are discussed below:

1 Science Society

The GIKI Science Society is one of the biggest

and oldest society in the Institute, working towards promotion of interest in all general sciences in a predominantly technical environment. It has a full calendar of events comprising of an intra event at GIK Institute, Science Marathon, All Pakistan Science Fair and International Science and Youth Symposium annually, with the ultimate mission of scientific prosperity and social development in the country. GIK Science Society annually publish a Science Magazine called 'Aurora', which keeps the students updated to the enthralling news and developments, especially in the world of science.

2. AIAA - (American Institute of Aeronautics and Astronautics) GIK Chapter

The American Institute of Aeronautics and Astronautics (AIAA) is a prominent technical society at GIK Institute. Formerly known as the GIKI Aerotech Club, it has made its mark by projecting an image that makes this technical society appealing to the masses. The society is a platform for Aerospace enthusiasts and it conducts various events, seminars and workshops each year. AIAA also maintains a fleet of Radio Controlled Aircraft which are used for both technical and recreational purposes.

3. ASM/TMS – (The Mineral, Metal & Material Society/American Society of Materials) GIK Chapter

The mission of ASM/TMS is to promote the science and engineering professions concerned with minerals, metals and materials.





The objective of this TMs chapter is to make students aware of the role of materials and metallurgical engineering in today's international market place. TMS also helps students from other disciplines of engineering to better understand and perform in their disciplines of engineering materials.

4. ACM - (Association for Computing Machinery) GIK Chapter

ACM is a worldwide professional organization devoted to advancing the theory and practice of computer science. Its student chapter at GIKI is involved in activities ranging from arranging students workshops, special courses and introductory seminars to holding software and computer games competitions.

5. Women Engineering Society (WES)

The Women Engineering Society has undertaken the task of exploring the reasons behind the low representation of women in the field of engineering and seeking remedial measures to make this profession more viable for women. To fulfill this need, a student society (Women Engineering Society) has been established, which is wholly managed by female students.

It has successfully arranged two national symposiums to discuss the problems faced by women in this field and also the problems faced by the industry in hiring and retaining of women engineers. The feedback has been enormously encouraging and WES plans to keep working at raising awareness about the viability of engineering profession for women.

6. Literary and Debating Society (LDS)

The society holds debates, declamation contests and literary evenings, and sessions of poetry recitation. It also arranges participation of the GIKI teams in debates and declamation contests held by other universities. LDS is the oldest society of the institute. A special feature of its agenda is the student teacher discussion usually held in the auditorium. These collective sessions prove invaluable for a critical evaluation of the diverse aspects of the campus life, and go a long way in promoting rapport between the students and the faculty. The LDS also organizes and participates in international events.

7. Cultural, Dramatic, & Entertainment Society (CDES)

This is one of the most active and widely applauded societies which add colors to the campus life. It creates entertaining and healthy diversion throughout the year by organizing musical concerts, drama festivals, skit competitions, picnics, and bonfires. The activities of the society help in identifying artistic talent among students and nourish students to participate in art competitions. The society also celebrates the much popular "Basant Festival" every spring.



8. SOPHEP- (Society for the Promotion of Higher Education in Pakistan)

In this modern day and age, engineers and scientists are not isolated from the management of the firms they work in. As a matter of fact, they themselves can become senior managers later on. Due to this, companies are perpetually on the look-out for people with solid communication skills. The implication for students is that besides studying hard, they must strive to develop solid presentation and communication skills as well. One of the major problems facing today's Pakistani students is the lack of interaction between the Pakistan's academic and professional communities. As a result of this, young undergraduates lack sufficient knowledge of the professional world and companies remain unaware of the great potential of these future workers.

In this regard, SOPHEP bridges the gap among these two vital communities. SOPHEP holds workshops that are conducted by successful GIKI alumni that help students to refine their people-skills and professional grooming.

9. Project Topi

Project Topi is a student based society which works for the social uplift of the people living in and around the GIK Institute. This society began back in 2000 with the aim of academic uplift of the people of Topi and other parts of Swabi. Over the Years the domain stretched to areas like medical initiatives, blood camps, and woman/men empowerment, micro-financing and continual support of some very poor families as well.

10. NAQSH

Naqsh Art Society is another emerging and popular student society. It organizes the much awaited 'All Pakistan Art Gala' every year, featuring various artistic competitions on a national scale. The society promotes aesthetic sense and faster artistic touch among the prospective engineers.

11. Media Club

To promote creative expression amongst the students, the GIKI Media Club caters for the taste of all. It comprises of Photography Club, Desktop Publishing and vision Club. Emphasis is laid both on still photography and video. The members are supposed to compile and compose the newsletters/magazines. The

media club is responsible for the coverage of various events within the Institute and also organizes the very popular annual movie competition at the campus.

12. Sports Society

Sports Society comprises of the coordinators of various sports clubs of the Institute. It promotes and regulates sports and games on the campus. The existing facilities include a sports complex, which houses Swimming Pool, Squash, Basketball, Volleyball & Badminton Court and a Gymnasium. Outdoor facilities include cricket, hockey, football fields, tennis courts, beach soccer, and beach volleyball. Students' hostels have ample provision for the indoor games, such as table tennis, carom and chess. The Society organizes friendly matches throughout the year culminating in annual interfaculty tournaments and competitions. The sports society also organizes fixture tournaments with other educational institutions and provides the forum for sportsmen to take part in various national sports events.

Following students societies are functional in the Institute:-

1. ACM GIK Chapter: Association of Computing Machinery
2. ASME GIK Chapter: American Society of Mechanical Engineers
3. ASM/TMS GIK Chapter: American Society of Materials/the Materials, Mineral and Metal Society
4. ASHRAE GIK Chapter: American Society of Heating, Refrigerating and Air Conditioning Engineers
5. CDES: Cultural, Dramatic and Entertainment Society



6. GMS: GIKI Mathematic Society
7. GSS: Graduate Student Society
8. IET GIK Chapter: Institute of Engineering and Technology
9. IEEE GIK Chapter: Institute of Electrical and Electronic Engineering
10. LDS: Literary and Debating Society
11. Media Club: Includes GIKI Vision, Photography and Desktop Publishing
12. Naqsh Art Society: NAQSH promotes and propagates art in GIK Institute.
13. Netronix: NETRONiX is the sole caretaker of GIKI's hostel network, one of the largest in Pakistan & perhaps the only one run by Undergraduate students. Over the past years, NETRONiX has not only maintained its core objective of handling the whole LAN of GIKI but also provided the students a platform to learn and implement their skills. In addition to this, NETRONiX also provides various other services to the students e.g p2p sharing, IPTV, PLEX, gaming servers etc & is the only dedicated gaming community in GIKI which hosts intra-GIKI and All Pakistan gaming events annually.
14. Project Topi: Project Topi is a student run volunteer society which runs many projects for the welfare of Topi community.
15. Web Team: The GIKI Webteam is an in-house team of students that voluntarily design and manage the GIKI website and its related affairs, with their services being officially recognized by the Institute.
16. Science Society: Science society deals in contemporary developments in scientific field.
17. SOPHEP: Society for the Promotion of Higher Education in Pakistan
18. Sports Society: Sports society maintains the sports facilities and organizes events with wide participation.
19. SPIE GIK Chapter: Society for Photo-Optical Instrumentation Engineer
20. SMEP GIK Chapter: Society of Mechanical Engineers of Pakistan
21. WES: Women Engineers Society
22. AIAA GIK Chapter: American Institute of Aeronautic & Astronomic.
23. AIESEC: Its agenda primarily includes sending youth for exchange programs, belonging to different cultures to other countries. It is an attempt to develop and consolidate friendly ties between countries.
24. AIChE GIK Chapter: American Institute of Chemical Engineers, GIKI Chapter.
25. CBS: Character Building Society
26. ImechE: institute of Mechanical Engineering.
27. LES: Leadership and Entrepreneurial Society.
28. ICE: Institution of Civil Engineers



Open House & Career Fair is organized at the GIK Institute Campus every year. Its objective is to invite senior representatives from the industry to visit the Institute, have a view of the facilities and firsthand information on the Institute's academic and research activities. It also provides a forum for faculty members and senior management from the industry to exchange views and discuss matters of mutual interest, such as, sponsoring research projects and reviewing the academic curricula to meet the needs of industry. Another important

aspect of the Open House & Career Fair is the employment for our graduating students. The Open House & Career Fair provides an excellent opportunity for the prospective employers to judge the caliber of the graduating engineers through discussions and interviews. Poster presentations of senior year projects are also held in conjunction with the Open House & Career Fair in which the students have an opportunity to display their projects to the prospective employers.



Office of Research, Innovation and Commercialization - ORIC

Muhammad Amin Qureshi, Deputy Director
Syed Abbas Gillani Deputy Director
Memoona Assistant Director

The Institute has established Office of Research, Innovation and Commercialization (ORIC) to promote research and commercialization and to undertake following assignments and programs:

- Organizing Industrial Open House (IOH)
- Supporting the Institute's strategic research direction and policies
- Increasing and diversifying external research funding
- Improving recruitment and retention of top faculty
- Improving integration of research and education at all level of the Institute
- Improving translation of research into the public benefit
- Strengthening Institute-Industry relations
- Promoting entrepreneurship, technology-transfer and commercialization activities that energize and support the local and national industry
- Promoting and enhancing cross-cutting and multi-disciplinary research initiatives

The office program and activities are supervised by the Pro-Rector (Academic) and officers including Director, Manager University Industrial Linkages and Technology Transfer, Manager Research Operation and Development and Research Associate.



Quality Enhancement Cell (QEC)

The Quality Enhancement Cell (QEC) at GIK Institute is taking necessary steps to achieve the highest level of quality in education and ensure the effective learning experience of students. Under the supervision of Rector, GIK has advanced towards the External Quality Assurance.

International Rankings:

QEC GIK institute for the 1st time submitted data in international GreenMetric Rankings and was ranked 3rd in KPK region, 9th in private sector universities, 22nd among National universities and 532 globally.

Purpose of participation in UI GreenMetric is to aid the university's efforts in internationalization and recognition by getting its sustainability efforts on the global map. The world faces unprecedented global challenges such as population trends, global warming, over exploitation of natural resources, oil-dependent energy, water and food shortages and sustainability. We realize that higher education has a crucial role to play in addressing these challenges. UI GreenMetric leverages the crucial role that HEIs can play in raising awareness by doing an assessment and comparing efforts in education for sustainable development, sustainability research, campus greening, and social outreach. GIK has the same mission and goal to hit those sustainable targets and keep progressing. The key parameters of GreenMetric Rankings are:

- Setting and Infrastructure (SI)
- Energy and Climate Change (EC)
- Waste (WS)
- Water (WR)
- Transportation (TR)
- Education and Research (ED)



The Institute has established Faculty Gold Medals for best academic performance in each faculty. Two Institute Gold Medals have also been established for students with outstanding performance in the BS Degree. The Ghulam Ishaq Khan Gold Medal is awarded for the best academic performance among all the graduates of the Institute. The Quaid e Azam Gold Medal is awarded for the best overall performance among all the graduates of the Institute.

Quaid-e-Azam Medal**MINAHILL TARIQ****GIK & Faculty Medal ME****SHEIKH ABDUL MAJID****Faculty Medal CS****MUHAMMAD RIAZ UL HAQ****Faculty Medal CE****USMAN FAROOQ****Faculty Medal EEE****MUHAMMAD USMAN UMER****Faculty Medal EEP****MINAHILL JAMSHAD****Faculty Medal ES****MUHAMMAD SHEHRYAR****Faculty Medal CME****HUZAIFA KAMRAN****Faculty Medal MSE****DANYAL HAKEEM JOKHIO****Faculty Medal MGS****AFIFA DILAWAIZ CHEEMA**



Taimur Inam Uppal

BATCH 11

Studying BS Computer Systems Engineering at GIK helped me develop the critical thinking skills I use each day when making recommendations to management regarding the financial implications of attracting new customers, developing new products, or utilizing new business strategies.

I delivered multi-million dollar projects with leading Telecom companies incl. Ericsson, Huawei and Nokia across different regions in EMEA and Asia. I would like to thank my teachers and institute for this.



Mr. Gul Wiqar Khan

Prestigious Bell Labs (DMTS) award, 2022

Mr. Gul Wiqar Khan won the prestigious Bell Labs (DMTS) award, in 2022. The DMTS award is conferred by Nokia annually to honor technical excellence and sustained R&D contributions to the company and industry.

Mr. Gul Wiqar graduated from GIK Institute from FEE and completed his MS in Telecommunication from Helsinki University of Technology Finland.



Muhammad Nasrullah

(Batch 10, FCSE) Cofounder and CEO at Integrity

Ever since I was in school, I had my heart set on going to GIKI and becoming a sort of a mad scientist/engineer. You can imagine my delight when I was admitted in Fall of 2000. GIKI was a place where there was so much opportunity to grow.

The quiet and serene environment made me focus on my passion of creating software programs to solve problems. Over time I worked with lots of brilliant soon-to-be engineers on problems ranging from automation to artificial intelligence.

That clear-headedness and focus helps me to this day where we work in the cutting edge of the software industry under immense pressure. Things move at a rapid pace and you need to have focus and the know-how to keep up with the changes. GIKI has taught me to work under pressure and be creative at the same time. The lessons you learn here always stay with you no matter how many years pass.

Farid Rahman

President SOPREST

Shakil Durrani

Executive Director SOPREST

Irfan Ahmad

Secretary SOPREST and BoG



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Secretary to Rector

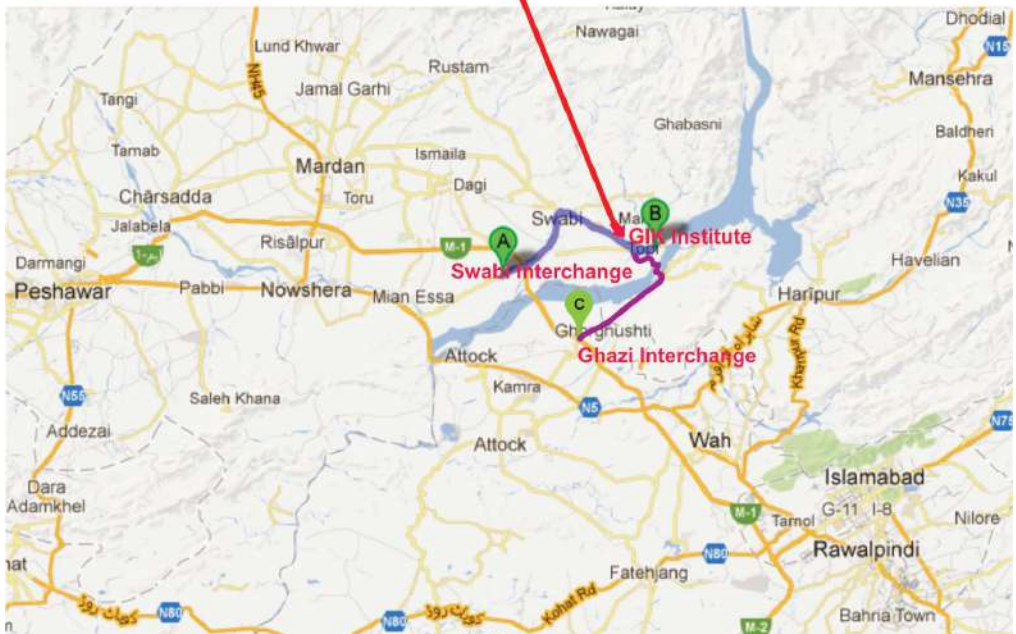
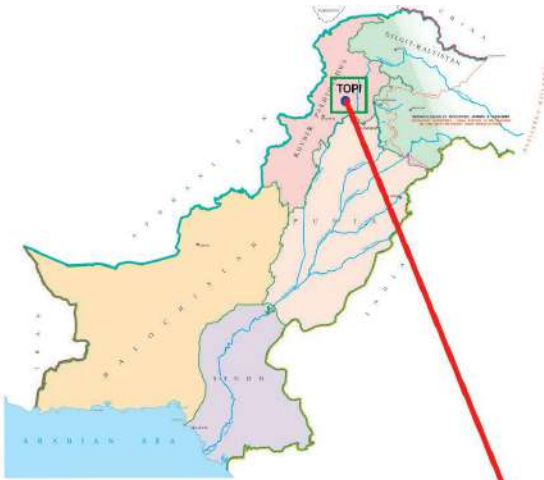
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Asstt. Director (HR/Facilitation)

Mr. Sher Ali Shah
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HOW TO GET TO GIK INSTITUTE

- The Institute is less than one and half hour drive from Islamabad and Peshawar.
- Starting from Islamabad, exiting Islamabad-Peshawar Motorway M1(Point C) at Ghazi Interchange, follow road towards Tarbela and reach Campus after passing through Ghazi Barrage.
- Starting from Peshawar, you should follow the Motorway M1 (Point A) till you reach Swabi Interchange. From there come to Topi and the campus via Swabi.



DISCLAIMER

While every effort has been made to ensure the accuracy of the information in this Prospectus, the Institute can accept no responsibility for any errors or omissions. The Institute reserves the right to amend, offer delete or discontinue course(s) or amend admission requirement whenever it sees fit and prospective and registered students should enquire as to the up to date position should they need to know. The Institute takes all reasonable steps to provide educational services in the manner set out in the Prospectus and in other documents that will be issued to you if you are accepted as a student of the Institute. Should certain circumstances beyond the control of the Institute interfere with its ability to provide educational services, the Institute will take all reasonable steps to minimise the resultant disruption to educational services.

Should you become a student of the Institute, this notice shall be incorporated as a term of any contract between you and the Institute. Any offer of a place at the Institute is made on the basis that in accepting such an offer, you signify your consent to compliance with registration procedures, to observance of the Act, Guidelines, Rules and Regulations of the Institute.

Undergraduate Admission Committee

Prof. Dr. Wasim Ahmad Khan (Convener)

Mr. Muhammad Faheem Akhtar

Dr. Muhammad Imran Khan

Dr. Raja Hashim Ali

Dr. Ahmad Kamal Hassan

Dr. Muhammad Usman

Dr. S. Zameer Abbas

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Dr. Massab Junaid

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Mr. Hassaan Tariq

Director Finance

Director IT

Mr. Muhammad Waqas Malik

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